

Original Mixed Cohomology Periods, Kernels and Arithmetic Control

Split-Zero research programme

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Complete proof edition. The chapters below retain the original periods, amplitude units, primary multiplicities, cyclic maps, source masses and four endpoint metrics. Each chapter supplies its full calculation and proof. Earlier finite bounds remain recorded in their original form; subsequent chapters give their proved refinements. In particular KAF retains the factorial denominator of OKV and improves the absolute kernel bound, while AMT evaluates the source differences left in OMV.

Sources and scope. The complete preceding Gamma and observation proofs, including their historical analytic providers, are supplied in the source closure. The original all-multiplicity cyclic maps remain in OCS. The evaluated five-member quadric orbit, exact kernel dimension, and its individual volume conclusions state their simple-quartet period domain explicitly. The full all-multiplicity operator estimate is given in KAF. OCF, OKM and OKA calculate the actual kernel frame, its original metrics and the boundary action. OMG proves the full all-multiplicity Segre–Veronese geometry and retains the actual cyclic stabilizer.

Current refinement. AKS sharpens the complete kernel bound to $32 \log(25 \log 3 / (4\pi))$ at the kq scale. AKJ evaluates the original residue inverse and conductor degree flag. AKP proves the precise theta representative-change and minimum-section maps. RUI now proves all-rank exterior control of the original finite conductor, and CEP carries it through the exact source ideals, four endpoint quotients and original compressed covariances. The resulting conductor error is $o(kq)$ for each fixed actual period. CTV now gives the exact outer-root quotient and its controlled norm tails. ROQ calculates the actual residual observation, all its positive increments and full graph metric. OSP proves a full-norm comparison with explicit moments of the original Gamma sources. OAR and PMR carry these finite errors into the arithmetic return. OFV now evaluates the outer contribution at the kq scale, using the full EIQ equilibrium and QLG finite partition estimate supplied below. FMC constructs fixed original coefficient frames with their exact Gamma and polynomial-degree maps. OVR propagates the evaluated outer term into the original kernel, residual observation, arithmetic boundary and signed mixed return. Its remaining analytic quantity is the original residual Gamma compression. All preceding chapters remain in full, with their proved refinements.

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1 The literal cyclic sectors and the original mixed kernel

This calculation evaluates the cyclic projector already present in the original period observation. Its coefficients retain the original period integrals, amplitude unit, and all primary nilpotents. The result includes an exact finite matrix for the actual rank, as well as unconditional rank bounds obtained from the original symmetric tensor geometry.

1.1 Original data and the Fourier basis of the literal matrix

Use the following data of OPG1–5, including the same contours and their orientations:

$$\begin{aligned} \alpha_{\epsilon,\eta} &= \frac{1}{2} + \epsilon\delta + i\eta\gamma, \quad \epsilon, \eta \in \{+1, -1\}, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \\ h(z_1) &= \prod_{\epsilon,\eta} (z_1 - \alpha_{\epsilon,\eta})^m, \quad n = 4m, \quad N = n + 1, \quad E_h = \mathbb{C}[z_1]/(h), \\ k &= 4l + 1, \quad l, m \geq 1, \quad c = k/2, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \\ \beta_{a,b} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \beta_{a,b})^e, \quad E = \mathbb{C}[S]/(\chi). \end{aligned} \tag{OCS1}$$

Here A_h and A are multiplication by z_1 and S on their respective algebras, and $Y = (A - cI)/i$. The original amplitude unit is $U = M_{j_h v_h}$, with $v_h = (2\xi)/h$ and every original $v_h(\alpha) \neq 0$. For the specified nonzero u and its branch, $\Phi'_h = h$, $\Phi_h(0) = 0$, and

$$\Gamma_j = -\ell_0 + \ell_j, \quad \arg \ell_j = \frac{\arg u + \pi + 2\pi j}{N}, \quad (\Pi[P])_j = \int_{\Gamma_j} e^{\Phi_h(z_1)/u} \text{rem}_h P(z_1) dz_1. \tag{OCS2}$$

The representative has degree less than n . The convergence and invertibility of this original period map are the complete period theorems retained in OPG4. No period value or unit derivative is chosen in the present calculation. The original injection and observation are

$$\begin{aligned} \iota[P] &= [P(z_1 + \cdots + z_k)], \quad \mathcal{V} = \Pi^{\otimes k} U^{\otimes k} \iota, \\ P_k &= \frac{1}{N} \sum_{a=0}^{N-1} (C^{-a})^{\otimes k}, \\ L &= P_k \mathcal{V} = j\Lambda, \quad B = \text{im } L, \quad K = \ker \Lambda. \end{aligned} \tag{OCS3}$$

The map $j : B \hookrightarrow (\mathbb{C}^n)^{\otimes k}$ is the original image inclusion. The injection ι has precisely the defining polynomial in OCS1: on each ordered primary tensor the sum of its k independent nilpotents has index $e = 1 + k(m - 1)$, its top coefficient being $(k(m - 1))!/((m - 1)!)^k \neq 0$, and its distinct eigenvalues are the displayed $\beta_{a,b}$. Thus $\mathcal{V}$ is injective.

On a period-coordinate column $x \in \mathbb{C}^{N-1}$ the literal matrix in OPG4–5 is

$$(Cx)_j = x_{j+1} - x_1 \quad (1 \leq j < N - 1), \quad (Cx)_{N-1} = -x_1. \tag{OCS4}$$

Set $\zeta = \exp(2\pi i/N)$ and, for $1 \leq t \leq N - 1$, let

$$(f_t)_j = \zeta^{jt} - 1, \quad F = [f_1 \ \cdots \ f_{N-1}], \quad (F^{-1}x)_t = \frac{1}{N} \sum_{j=1}^{N-1} \zeta^{-tj} x_j. \tag{OCS5}$$

The last expression is the exact inverse. Indeed $\sum_{j=1}^{N-1} \zeta^{-tj} (\zeta^{uj} - 1) = N \mathbf{1}_{t=u}$ for $t, u \neq 0$ modulo N , by the finite geometric sum. The first $N - 2$ rows of OCS4 give $\zeta^{(j+1)t} - \zeta^t = \zeta^t (\zeta^{jt} - 1)$, and the last row gives $1 - \zeta^t = \zeta^t (\zeta^{(N-1)t} - 1)$. Consequently $Cf_t = \zeta^t f_t$, proving the orientation of every inverse power in OCS3, as well as $C^N = I$. The original Euclidean Gram of this basis is

$$F^* F = N(I_{N-1} + \mathbf{1}\mathbf{1}^*). \tag{OCS6}$$

To verify it, expand $\sum_j (\zeta^{-jt} - 1)(\zeta^{ju} - 1)$ and use the same geometric sums; its value is $N(\mathbf{1}_{t=u} + 1)$. Thus the complete coefficient and metric transformations are both specified.

1.2 The ordered-to-symmetric map and the exact sector projector

Let

$$\mathcal{A}_k = \{\nu \in \mathbb{Z}_{\geq 0}^{N-1} : \sum_t \nu_t = k\}, \quad \text{wt}(\nu) = \sum_{t=1}^{N-1} t\nu_t \pmod{N}, \quad \mathcal{I}_{k,\tau} = \{\nu \in \mathcal{A}_k : \text{wt}(\nu) = \tau\}. \quad (\text{OCS7})$$

For each ν define the following unscaled symmetric tensor:

$$S_\nu = \sum_{\substack{(t_1, \dots, t_k) \in \{1, \dots, N-1\}^k \\ \#\{j: t_j = t\} = \nu_t \text{ for each } t}} f_{t_1} \otimes \dots \otimes f_{t_k}. \quad (\text{OCS8})$$

Every distinct ordered word occurs once. Since the ordered tensors in the f_t form a basis, their disjoint word supports show that the S_ν are linearly independent. A permutation-invariant tensor has equal coefficients on each word orbit, so they span exactly $\text{Sym}^k(\mathbb{C}^{N-1})$, realized as that invariant subspace of the full ordered tensor product. In particular, for $x = (x_1, \dots, x_{N-1})$,

$$(Fx)^{\otimes k} = \sum_{\nu \in \mathcal{A}_k} x^\nu S_\nu, \quad (C^{-a})^{\otimes k} S_\nu = \zeta^{-a \text{wt}(\nu)} S_\nu, \quad P_k(Fx)^{\otimes k} = \sum_{\nu \in \mathcal{I}_{k,0}} x^\nu S_\nu. \quad (\text{OCS9})$$

The finite average in OCS3 is exactly one on weight zero and zero on every other weight, by its geometric sum. No multinomial coefficient is omitted: the multiplicity $k! / \prod_t \nu_t!$ is already the number of summands in OCS8.

Write $\mathcal{F}_k : \mathbb{C}^{\mathcal{A}_k} \rightarrow \text{Sym}^k(\mathbb{C}^{N-1})$ for the basis isomorphism $e_\nu \mapsto S_\nu$. Let J_0 be the coordinate inclusion of $\mathbb{C}^{\mathcal{I}_{k,0}}$ into $\mathbb{C}^{\mathcal{A}_k}$, π_0 the coordinate restriction in the opposite direction, and let $\mathcal{F}_0 = \mathcal{F}_k J_0$, with its target the original ordered tensor space. The exact projector factorization is

$$P_k \mathcal{F}_k = \mathcal{F}_0 \pi_0. \quad (\text{OCS10})$$

All images of $\mathcal{V}$ are symmetric: polynomial evaluation at a sum is invariant under every factor permutation, and the same map ΠU is applied in all factors. Therefore OCS10 applies on the entire original image $\mathcal{V}(E)$.

For completeness, the full inherited metric is retained in these coordinates. Put $G = F^* F$ as in OCS6. The Gram of the basis OCS8 is

$$\mathcal{G}_{\nu, \mu} = \sum_{\substack{a_{tu} \geq 0 \\ \sum_u a_{tu} = \nu_t, \sum_t a_{tu} = \mu_u}} \frac{k!}{\prod_{t,u} a_{tu}!} \prod_{t,u} G_{tu}^{a_{tu}}. \quad (\text{OCS11})$$

Indeed the coefficient of $\bar{x}^\nu y^\mu$ in $(\bar{x}^T G y)^k$ is the right side by the multinomial theorem. Taking the inner product of the two pure tensors in OCS9 gives the same coefficient as $\langle S_\nu, S_\mu \rangle$. This proves OCS11, including its off-diagonal entries; it is positive definite because OCS8 is a basis. On the full symmetric coordinates the projector matrix is $J_0 \pi_0$, and its adjoint for this exact Gram is $\mathcal{G}^{-1}(J_0 \pi_0)^* \mathcal{G}$. Thus the Fourier-sector computation makes no unproved orthogonality or norm assertion about the original cyclic average.

1.3 Every primary period coefficient of the projected curve

For each original root α let $E_{\alpha,a}$, $0 \leq a < m$, be the degree-less-than- $4m$ primary representatives of OPG10. Explicitly, if $h_\alpha = h/(z_1 - \alpha)^m$, put

$$E_{\alpha,a}(z_1) = \text{rem}_h \left[h_\alpha(z_1) \sum_{j=0}^{m-1} \frac{(h_\alpha^{-1})^{(j)}(\alpha)}{j!} (z_1 - \alpha)^{j+a} \right], \quad b_{j;\alpha,a} = \int_{\Gamma_j} e^{\Phi_h(z_1)/u} E_{\alpha,a}(z_1) dz_1. \quad (\text{OCS12})$$

The four primary factors are coprime. The Taylor inverse gives constant one on its own factor and zero on the other three; thus the classes $[E_{\alpha,a}]$ form the full primary basis with all m nilpotent degrees at every root. Define the exact finite Fourier and unit coefficients

$$\widehat{b}_{t;\alpha,a} = \frac{1}{N} \sum_{j=1}^{N-1} \zeta^{-tj} b_{j;\alpha,a}, \quad f_{t;\alpha,d} = \frac{i^{-d}}{d!} \sum_{a=d}^{m-1} \widehat{b}_{t;\alpha,a} \frac{v_h^{(a-d)}(\alpha)}{(a-d)!}, \quad 0 \leq d < m. \quad (\text{OCS13})$$

Let $\omega_\alpha = (\alpha - 1/2)/i$ and set

$$\mathbf{v}(w) = \Pi U \exp(w(A_h - \frac{1}{2}I)/i)[1], \quad x(w) = F^{-1}\mathbf{v}(w), \quad x_t(w) = \sum_{\alpha} e^{\omega_\alpha w} \sum_{d=0}^{m-1} f_{t;\alpha,d} w^d. \quad (\text{OCS14})$$

To verify the final equality, in the primary algebra at α multiply the Taylor series of v_h by the terminating series of $\exp(w(z_1 - \alpha)/i)$. The coefficient on $[E_{\alpha,a}]$ is $e^{\omega_\alpha w} \sum_{d=0}^a v_h^{(a-d)}(\alpha) (w/i)^d / ((a-d)!d!)$. Apply the original polynomial period map and the inverse in OCS5; this gives precisely OCS13–14. In particular, the unreduced entire function has passed through its full primary remainder before any period is integrated.

For $\nu \in \mathcal{I}_{k,0}$, $0 \leq a, b \leq k$ and $0 \leq D < e$, define $c_{\nu;a,b,D}$ by the following finite multinomial rule. Sum over integers $n_{t,\alpha,d} \geq 0$ satisfying

$$\begin{aligned} \sum_{\alpha,d} n_{t,\alpha,d} &= \nu_t \quad \text{for each } t, & \sum_{t,\alpha,d} d n_{t,\alpha,d} &= D, \\ \sum_{t,d,\alpha: \operatorname{Re} \alpha = 1/2 + \delta} n_{t,\alpha,d} &= a, & \sum_{t,d,\alpha: \operatorname{Im} \alpha = +\gamma} n_{t,\alpha,d} &= b. \end{aligned}$$

For these indices put

$$c_{\nu;a,b,D} = \sum_{\text{stated indices}} \left(\prod_{t=1}^{N-1} \frac{\nu_t!}{\prod_{\alpha,d} n_{t,\alpha,d}!} \right) \prod_{t,\alpha,d} f_{t;\alpha,d}^{n_{t,\alpha,d}}. \quad (\text{OCS15})$$

Empty index sets give zero and zero exponents give factors one. Every original primary coefficient is present in this finite rule. The ordinary multinomial expansion of OCS14 now proves

$$x(w)^\nu = \sum_{a,b=0}^k e^{\omega_{a,b} w} \sum_{D=0}^{e-1} c_{\nu;a,b,D} w^D, \quad \omega_{a,b} = (\beta_{a,b} - c)/i = (2b - k)\gamma - i(2a - k)\delta. \quad (\text{OCS16})$$

Indeed $D \leq k(m-1) = e-1$ and the counts of each original sign give exactly the displayed exponent. The sums with different (a,b) are distinct since their real and imaginary parts are independently distinct.

1.4 The actual finite observation matrix and its complete kernel

Define the primary coefficient isomorphism $\mathcal{H} : \mathbb{C}^q \rightarrow E$ by sending coordinate (a,b,D) to $\varepsilon_{a,b}(S - \beta_{a,b})^D$, where $\varepsilon_{a,b}$ is obtained from χ by the same truncated Taylor-inverse construction as OCS12, now of order e . The proof by the coprime factors is identical and shows that this is a basis of all q primary directions. Set

$$\mathfrak{A}_{\nu;(a,b,D)} = i^D D! c_{\nu;a,b,D}, \quad \mathfrak{A} : \mathbb{C}^q \longrightarrow \mathbb{C}^{\mathcal{I}_{k,0}}. \quad (\text{OCS17})$$

Then, on the original spaces and with the original ordered target,

$$\boxed{L\mathcal{H} = \mathcal{F}_0 \mathfrak{A}, \quad K = \mathcal{H}(\ker \mathfrak{A}), \quad \operatorname{rank} \Lambda = \operatorname{rank} \mathfrak{A}.} \quad (\text{OCS18})$$

Here is a proof giving every factorial and phase. In the primary sum algebra,

$$e^{wY}[1] = \sum_{a,b} e^{\omega_{a,b}w} \sum_{D=0}^{e-1} \frac{i^{-D} w^D}{D!} \varepsilon_{a,b} (S - \beta_{a,b})^D.$$

The exact original tensor identity of OPG8 gives $\mathcal{V}e^{wY}[1] = \mathbf{v}(w)^{\otimes k}$: the centre $c = k/2$ splits into the k original centres $1/2$, while U commutes with multiplication by z_1 . Applying the literal projector and then OCS9 and OCS16 gives the right-hand curve with coefficients $\mathcal{F}_0 c_{a,b,D}$. Exponential polynomials $w^D e^{\omega_{a,b}w}$ with these distinct exponents are linearly independent. To prove this directly, apply $\prod_{(a',b') \neq (a,b)} (\partial_w - \omega_{a',b'})^e$ to any zero linear combination. It kills all the other exponents. On the selected polynomial factor, each remaining operator is $\partial_w + (\omega_{a,b} - \omega_{a',b'})$, invertible on the polynomials of degree less than e because its constant term is nonzero and differentiation is nilpotent there. The surviving polynomial must therefore vanish. Doing this for every pair proves the asserted independence. Equating coefficients now gives $L\mathcal{H}e_{a,b,D} = \mathcal{F}_0(i^D D! c_{a,b,D})$. This proves the first equality in OCS18 on a basis. The other two follow from injectivity of $\mathcal{F}_0$ and j .

In particular the isomorphism from the actual sector image to the original boundary is

$$\mathcal{Q} : \text{im } \mathfrak{A} \xrightarrow{\sim} B, \quad \mathcal{Q}x = j^{-1} \mathcal{F}_0 x, \quad E/K \xrightarrow{\sim} \text{im } \mathfrak{A}, \quad [z] \mapsto \mathfrak{A} \mathcal{H}^{-1} z. \quad (\text{OCS19})$$

Both inverses exist on precisely their displayed images by OCS18. These are full coefficient maps; the inherited Gram of $\mathcal{F}_0$ is $J_0^* \mathcal{G} J_0$ from OCS11.

For a completely explicit finite rank prescription, define

$$\Delta_r(\mathfrak{A}) = \sum_{\substack{I \subseteq \mathcal{I}_{k,0}, \ J \subseteq \{(a,b,D)\} \\ |I|=|J|=r}} |\det \mathfrak{A}_{I,J}|^2, \quad \Delta_0 = 1. \quad (\text{OCS20})$$

Every entry of every minor is the finite expression OCS12–17 in the actual periods and amplitude derivatives. Gaussian elimination proves that a matrix has rank at least r exactly when one of its r by r minors is nonzero: pivot rows and columns supply such a minor, and a nonzero minor supplies r independent columns. Hence the actual rank is exactly $\max\{r : \Delta_r(\mathfrak{A}) > 0\}$. This evaluates its finite deciding data without prescribing generic values or asserting that any unexamined maximal minor is nonzero.

The Gamma vectors used in the existing endpoint increments have the equally explicit coordinates

$$j\lambda_d = \mathcal{F}_0 \hat{\lambda}_d, \quad (\hat{\lambda}_d)_\nu = \frac{1}{\sqrt{h_d^{(s)}}} \sum_{a,b=0}^k \sum_{D=0}^{e-1} c_{\nu;a,b,D} p_d^{(D)}(\omega_{a,b}), \quad h_d^{(s)} = (2\pi)^{s/2} d!(s/2)_d, \quad s \in \{1, k\}. \quad (\text{OCS21})$$

Taylor expansion of $p_d(Y)[1]$ in the sum primary basis gives the coefficient $i^{-D} p_d^{(D)}(\omega_{a,b})/D!$. Multiplication by OCS17 cancels exactly these factorials and phases, proving OCS21. This also proves that the generating formula of OPG9 becomes $(1+z^2)^{-s/4} \mathcal{F}_0(x(\arctan z)^\nu)_{\nu \in \mathcal{I}_{k,0}}$, with the original scalar exponent and both existing sources.

1.5 Exact invariant dimensions and unconditional forced losses

Put $d_{N,k,\tau} = |\mathcal{I}_{k,\tau}|$. By OCS7–10 this is the dimension of the full symmetric sector, and its exact finite character formula is

$$d_{N,k,0} = [T^k] \frac{1}{N} \sum_{h|N} \varphi(h) \frac{1-T}{(1-T^h)^{N/h}}. \quad (\text{OCS22})$$

To prove it, the trace generating series for C^a on symmetric tensors is $\prod_{t=1}^{N-1} (1 - T\zeta^{at})^{-1}$, by summing the monomial basis OCS8. If ζ^a has order h , the list ζ^{at} with $0 \leq t < N$ contains every

h th root N/h times. Their product is $\prod_{t=0}^{N-1} (1 - T\zeta^{at}) = (1 - T^h)^{N/h}$. Removing the $t = 0$ factor gives the series $(1 - T)/(1 - T^h)^{N/h}$. There are exactly $\varphi(h)$ indices a of that order, so averaging the traces of the original finite projector proves OCS22. The inverse powers give the same average by $a \mapsto -a$. In ordinary finite binomial terms, for $k \geq 1$ this is

$$d_{N,k,0} = \frac{1}{N} \sum_{h|N} \varphi(h) \left\{ \mathbf{1}_{h|k} \binom{k/h + N/h - 1}{N/h - 1} - \mathbf{1}_{h|(k-1)} \binom{(k-1)/h + N/h - 1}{N/h - 1} \right\}. \quad (\text{OCS23})$$

The binomial expansion of $(1 - T^h)^{-N/h}$ follows by counting weak compositions, proving the coefficient extraction used here. It follows from the actual matrix OCS17 that

$$\text{rank } \Lambda \leq \min(q, d_{N,k,0}), \quad \dim K \geq \max(0, q - d_{N,k,0}). \quad (\text{OCS24})$$

These inequalities apply to the original map, since its full image factorization was proved in OCS18; the ambient dimension has not been equated with its actual rank. There is also an unconditional nonzero observation for the original $k \geq 5$. Each entire function $x_t(w)$ in OCS14 is nonzero. Otherwise its nonzero coordinate covector would annihilate every derivative at zero of $\Pi U \exp(w(A_h - 1/2)/i)[1]$. Those derivatives span the full space, since $[1]$ is cyclic for multiplication by z_1 and ΠU is invertible. A product of finitely many nonzero entire functions is nonzero. The zero sector is nonempty: take the weights $1, 1, N - 2$ and then $(k - 3)/2$ pairs $1, N - 1$. Their sum is a multiple of N . Its monomial curve in OCS9 is nonzero, so $\text{rank } \Lambda \geq 1$.

For $m = 1$, $N = 5$, and every $k \geq 0$, write

$$D_k = \binom{k+3}{3}, \quad \epsilon_k = \begin{cases} 1 & k \equiv 0 \pmod{5}, \\ -1 & k \equiv 1 \pmod{5}, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{OCS25})$$

$$d_{5,k,0} = \frac{D_k + 4\epsilon_k}{5}, \quad d_{5,k,\tau} = \frac{D_k - \epsilon_k}{5} \quad (\tau \neq 0).$$

For each of the four nonidentity powers of C , the trace series is $(1 - T)/(1 - T^5)$, whose coefficient is ϵ_k . The weight- τ projector introduces the Fourier multiplier; the sum of its four nontrivial fifth roots is 4 when $\tau = 0$ and -1 otherwise. This proves both formulas. For the requested original orders the exact counts and resulting kernel lower bounds are

k	D_k	$d_{5,k,0}$	$q = (k+1)^2$	$\max(0, q - d_{5,k,0})$
5	56	12	36	24
9	220	44	100	56
13	560	112	196	84
17	1140	228	324	96
21	2024	404	484	80
25	3276	656	676	20
29	4960	992	900	0

(OCS26)

These are integer evaluations of OCS25. The strict inequality $d_{5,k,0} < q$ holds for $k = 4l + 1$, $l \geq 1$, exactly for the first six orders in the table. They exhaust $k \leq 25$ in this progression. For $k \geq 29$,

$$D_k - 5(k+1)^2 = \frac{(k+1)(k^2 - 25k - 24)}{6} \geq 460 > 4,$$

where $k^2 - 25k - 24$ is positive and increasing from $k = 29$. Even the most negative correction $4\epsilon_k \geq -4$ therefore gives $d_{5,k,0} > q$. This proves the threshold without extrapolating the finite table.

1.6 The actual simple-quartet quadric and a stronger rank bound

Continue with $m = 1$. Order the four roots as $(++)$, $(+-)$, $(-+)$, $(--)$ and let $\mathcal{E}_{\text{prim}} : \mathbb{C}^4 \rightarrow E_h$ be the primary basis isomorphism whose columns are the four $[E_{\alpha,0}]$. Define the actual invertible matrix and the original exponential quartet by

$$\mathbf{H} = F^{-1} \Pi U \mathcal{E}_{\text{prim}}, \quad t_{\epsilon, \eta}(w) = e^{(\eta\gamma - i\epsilon\delta)w}, \quad x(w) = \mathbf{H}t(w). \quad (\text{OCS27})$$

Every column of $\mathbf{H}$ is OCS13 with $d = 0$, namely the Fourier transform of its original primary period times its actual nonzero $v_h(\alpha)$. Each displayed map is invertible, so $\mathbf{H}$ is invertible without a genericity assumption. Put $\mathcal{R}_k = \mathbb{C}[x_1, x_2, x_3, x_4]_k$, the homogeneous polynomials of degree k , and define

$$Q_0(t) = t_{++}t_{--} - t_{+-}t_{-+}, \quad Q(x) = Q_0(\mathbf{H}^{-1}x). \quad (\text{OCS28})$$

This is the literal transformed quadratic relation of the original curve; its coefficients are determined by the periods and unit through OCS27. In particular $Q \neq 0$.

The kernel of homogeneous restriction to that curve is exactly

$$\ker\{\mathcal{R}_k \rightarrow \text{Hol}(\mathbb{C}), f \mapsto f(x(w))\} = Q\mathcal{R}_{k-2} \quad (k \geq 2). \quad (\text{OCS29})$$

Here is a complete verification. First work in the t coordinates. A degree- k monomial has exponent $(2b - k)\gamma - i(2a - k)\delta$, where a is its count of first sign $+$ and b its count of second sign $+$. These $(k + 1)^2$ complex exponents are distinct: equality of their real parts forces b equal and equality of their imaginary parts forces a equal. Equivalently, the two frequencies 2γ and $-2i\delta$ admit no nonzero integer relation, by their real and imaginary parts. Thus the exponentials have no extra polynomial relation of this homogeneous type; their linear independence follows, for example, by differentiating $0, \dots, (k + 1)^2 - 1$ times and using the Vandermonde determinant of the distinct exponents. Every pair (a, b) occurs: the count r_{++} may be any integer between $\max(0, a + b - k)$ and $\min(a, b)$, a nonempty interval, and the other three counts are $a - r_{++}$, $b - r_{++}$, and $k - a - b + r_{++}$.

Within any fixed pair (a, b) , two monomial exponent vectors differ by an integer multiple of $(1, -1, -1, 1)$. After their common monomial is factored out, their difference is therefore the difference of equal powers of $t_{++}t_{--}$ and $t_{+-}t_{-+}$. The identity $A^r - B^r = (A - B) \sum_{j=0}^{r-1} A^{r-1-j} B^j$ shows that this difference is divisible by Q_0 . Any polynomial vanishing on the curve has coefficient sum zero within each pair (a, b) , by the proved exponential independence. It is thus a sum of these monomial differences and is divisible by Q_0 . Its quotient is homogeneous of degree $k - 2$. Conversely $Q_0(t(w)) = 0$, since each of its two products is one. This proves the assertion in t coordinates, and the invertible substitution OCS27 proves OCS29. In particular $\dim(\mathcal{R}_k / Q\mathcal{R}_{k-2}) = (k + 1)^2 = q$; no information about the original curve has been replaced by a freely chosen quadric.

Let $\mathcal{I}_k \subseteq \mathcal{R}_k$ be the span of the weight-zero monomials x^ν , using the weights $1, 2, 3, 4$ from the exact Fourier basis. The coordinate functions of the original projected curve in OCS9 are precisely these monomials restricted to $x(w)$. The span of the vector curve $Le^{wY}[1]$ equals $\text{im } L$: its derivatives at zero contain $LY^d[1]$, $0 \leq d < q$, and $Y^d[1]$ is a basis because Y is an invertible affine change of multiplication by S in the degree filtration. The rank of that vector curve equals the dimension of the span of its coordinate functions, by the equality of row and column ranks of its finite exponential coefficient matrix. Hence OCS29 proves the exact equality

$$\text{rank } \Lambda = d_{5,k,0} - \dim(\mathcal{I}_k \cap Q\mathcal{R}_{k-2}). \quad (\text{OCS30})$$

Fix any multiplicative monomial order, for instance lexicographic order, and write x^μ for the leading monomial of the actual Q , with $|\mu| = 2$. Choose a row-echelon basis of the finite-dimensional intersection in OCS30 in descending monomial order. Its leading monomials are distinct, all have weight zero, and all are divisible by x^μ : the leading monomial of Qf is the product of those of Q and f because the order is multiplicative and the coefficient field has no zero divisors. Thus each such leading monomial is $x^\mu x^\eta$ for a distinct degree- $k - 2$ monomial of weight $-\text{wt}(\mu)$. Counting this set and using OCS30 gives

$$d_{5,k,0} - d_{5,k-2,-\text{wt}(\mu)} \leq \text{rank } \Lambda \leq \min(q, d_{5,k,0}). \quad (\text{OCS31})$$

The left side depends only on the actual leading monomial of the actual period quadric, and remains valid whatever its other coefficients are. A uniform consequence retaining the strongest count independent of that monomial is

$$r_k^- := d_{5,k,0} - \max_{\tau \in \mathbb{Z}/5} d_{5,k-2,\tau} = \left\lfloor \frac{(k+1)^2}{5} \right\rfloor + \mathbf{1}_{5|k} \leq \text{rank } \Lambda. \quad (\text{OCS32})$$

For the equality, $D_k - D_{k-2} = (k+1)^2$, as direct subtraction of the two cubic binomials shows. Formula OCS25 then gives

$$r_k^- = \frac{(k+1)^2 + 4\epsilon_k - \max(4\epsilon_{k-2}, -\epsilon_{k-2})}{5}.$$

For k congruent respectively to $0, 1, 2, 3, 4$ modulo five, its numerator correction is respectively $4, -4, -4, -1, 0$. The corresponding residues of $(k+1)^2$ are $1, 4, 4, 1, 0$. These five cases prove OCS32 for every $k \geq 2$. For the original orders in OCS26 the complete unconditional intervals are therefore

k	r_k^-	$\min(q, d_{5,k,0})$	$\max(0, q - d_{5,k,0})$	$q - r_k^-$
5	8	12	24	28
9	20	44	56	80
13	39	112	84	157
17	64	228	96	260
21	96	404	80	388
25	136	656	20	540
29	180	900	0	720

(OCS33)

The middle two rank columns bound $\text{rank } \Lambda$; the last two columns bound $\dim K$ from below and above, respectively. No endpoint of either interval has been assigned as the actual value without its original minors.

1.7 A second exact finite matrix for the simple-quartet kernel

The quadric calculation also gives the actual kernel a polynomial quotient model. Write $Q(x) = \sum_{|\mu|=2} q_\mu x^\mu$. Let T_Q be the full multiplication matrix $\mathcal{R}_{k-2} \rightarrow \mathcal{R}_k$ in the monomial bases:

$$(T_Q)_{\nu,\eta} = q_{\nu-\eta}, \quad |\nu| = k, \quad |\eta| = k-2, \quad q_{\nu-\eta} = 0 \text{ if an entry of } \nu - \eta < 0.$$

Let T_{out} be its rows of nonzero cyclic weight. Every coefficient q_μ is explicitly fixed by $Q_0(\mathbf{H}^{-1}x)$, and the inverse matrix may be computed as $\text{adj } \mathbf{H} / \det \mathbf{H}$, with nonzero denominator proved in OCS27. Multiplication by Q is injective. The condition that Qf belongs to $\mathcal{I}_k$ is exactly $T_{\text{out}}f = 0$. Thus OCS30 gives the further exact rank identities

$$\begin{aligned} \text{rank } \Lambda &= d_{5,k,0} - D_{k-2} + \text{rank } T_{\text{out}}, \\ \dim K &= D_k - d_{5,k,0} - \text{rank } T_{\text{out}}. \end{aligned} \quad (\text{OCS34})$$

The ranks of this concrete matrix are decided by its finite minors, by the same elimination proof as OCS20. This is an alternative exact calculation from the actual periods, not an assumption about the quadric's coefficients.

To give the typed map behind the kernel equality, let E^* denote the complex linear dual. Define $\Theta : \mathcal{R}_k \rightarrow E^*$ as follows: $\Theta(f)$ is the unique covector such that

$$\Theta(f)(e^{wY}[1]) = f(x(w)) \quad \text{for all } w. \quad (\text{OCS35})$$

For $m = 1$ every function on the right is a linear combination of the q distinct exponentials in OCS16. The primary expansion of $e^{wY}[1]$ gives exactly those exponentials as a basis of covector functions. This proves existence and uniqueness. The proof of OCS29 proves surjectivity and

$\ker \Theta = Q\mathcal{R}_{k-2}$. Moreover $\Theta(\mathcal{I}_k) = \text{im } L^\vee$, where $L^\vee$ denotes the complex linear dual map: the monomial coefficient functionals in the basis OCS8 pull back along L to exactly OCS35 for x^ν of weight zero. These functionals span the full dual of the actual image, since every linear functional on a subspace of a finite-dimensional space extends to the whole space. Restriction of E^* to K has kernel $\text{im } L^\vee$: a covector vanishing on K factors uniquely through $E/K \simeq \text{im } L$. Thus the exact maps give

$$\boxed{\text{coker } T_{\text{out}} \simeq \mathcal{R}_k / (\mathcal{I}_k + Q\mathcal{R}_{k-2}) \xrightarrow{\Theta|_K} K^*} \quad (\text{OCS36})$$

For the first isomorphism, use the direct monomial splitting of $\mathcal{R}_k$ into its zero and nonzero sectors; after quotienting by $\mathcal{I}_k$, multiplication by Q becomes precisely T_{out} . For the second, the surjectivity and kernel were just proved. This identifies the actual lost directions by a full quotient map.

1.8 All primary action directions remain in the mixed calculation

For arbitrary original $m \geq 1$, retain the full primary matrix $\mathfrak{A}$ in OCS17. In the basis $\mathcal{H}$ the original sum action is exactly

$$J e_{a,b,D} = \omega_{a,b} e_{a,b,D} + \begin{cases} i^{-1} e_{a,b,D+1} & D < e-1, \\ 0 & D = e-1, \end{cases} \quad Y\mathcal{H} = \mathcal{H}J. \quad (\text{OCS37})$$

This follows by multiplying $\varepsilon_{a,b}(S - \beta_{a,b})^D$ by $(S - c)/i = \omega_{a,b} + (S - \beta_{a,b})/i$ in its original primary algebra. Consequently the actual possible action leaving the cyclic kernel is the fully specified map

$$\ker \mathfrak{A} \longrightarrow \text{im } \mathfrak{A}, \quad x \longmapsto \mathfrak{A}Jx, \quad \Lambda Y\mathcal{H}x = \mathcal{Q}\mathfrak{A}Jx. \quad (\text{OCS38})$$

The codomain assertion follows from $\mathfrak{A}Jx$ being a value of the same matrix $\mathfrak{A}$ on Jx . For the original inclusion $I : K \hookrightarrow E$, this is the OPG22 defect transported through the exact kernel isomorphism in OCS18. In particular none of the discarded sectors has been declared invariant under Y .

The part of the kernel undetected by all original iterates is exactly computed by the stacked matrix

$$\mathfrak{D} = \begin{bmatrix} \mathfrak{A} \\ \mathfrak{A}J \\ \vdots \\ \mathfrak{A}J^{q-1} \end{bmatrix}, \quad \mathcal{K}_\infty = \mathcal{H} \ker \mathfrak{D}, \quad \dim \mathcal{K}_\infty = q - \text{rank } \mathfrak{D}, \quad \dim(K/\mathcal{K}_\infty) = \text{rank } \mathfrak{D} - \text{rank } \mathfrak{A}. \quad (\text{OCS39})$$

Indeed $\mathfrak{A}J^a x$ is the exact sector-coordinate value of $\Lambda Y^a \mathcal{H}x$ by OCS18 and OCS37. The polynomial $\chi(c + iT)$ annihilates J and has degree q with leading coefficient i^q . It expresses every higher power in the span of the first q , so the stacked kernel is precisely the intersection over all iterates and is J -invariant. Rank-nullity proves both dimension formulas. Its quotient identifies the directions that the original one-step observation loses and subsequent original iterates detect.

Equations OCS17–21, OCS34–39 supply actual finite maps and deciding minors for the original observation and its kernel. The forced kernel losses in OCS26 and the rank intervals OCS33 concern this unchanged projector and these unchanged periods. The endpoint volume calculation still uses the full positive sums of the actual vectors λ_d in OPG16–18 and their exact inherited coordinates OCS11 and OCS19. No volume growth estimate is deduced from these dimension bounds alone.

2 The original quadric orbit and an explicit invariant lifting map

This calculation uses the original simple-quartet construction, with $m = 1$, $k = 4l + 1$, $l \geq 1$, $q = (k + 1)^2$, $0 < \delta < 1/2$, and $\gamma > 2$. The period parameter, its branch, the polynomial contours, the complete amplitude unit, and the observation are those of OPG and OCS. The degree-eight polynomial below is the product of four actual orbit equations. Its degree and all of its coefficients are derived from that orbit; no source order is selected to match an estimate.

2.1 The actual coefficient ring and the original observation

Order the original roots as $(++)$, $(+-)$, $(-+)$, $(--)$, with $\alpha_{\epsilon, \eta} = 1/2 + \epsilon\delta + i\eta\gamma$. Let $\mathcal{E}_{\text{prim}}$ have columns the four original primary idempotents in $E_h = \mathbb{C}[z_1]/(h)$. For the original matrices Π, U, F in OCS2, OCS5, OCS27 put

$$\begin{aligned} \mathbf{H} &= F^{-1}\Pi U \mathcal{E}_{\text{prim}}, & t_{\epsilon, \eta}(w) &= e^{(\eta\gamma - i\epsilon\delta)w}, \\ x(w) &= \mathbf{H}t(w), & Q(x) &= Q_0(\mathbf{H}^{-1}x), & Q_0(t) &= t_{++}t_{--} - t_{+-}t_{-+}. \end{aligned} \quad (\text{OQC1})$$

Every matrix in $\mathbf{H}$ is the specified original isomorphism; in particular its nonzero amplitude factors are the actual $v_h(\alpha)$, and its period entries are the original convergent polynomial integrals. Thus $\det \mathbf{H} \neq 0$ by the included period-isomorphism and primary-basis proofs. Define

$$\mathcal{P}_d = \mathbb{C}[x_1, x_2, x_3, x_4]_d, \quad R_d = \mathcal{P}_d / (Q\mathcal{P}_{d-2}), \quad \mathcal{P}_d = R_d = 0 \quad (d < 0). \quad (\text{OQC2})$$

The matrix of $Q_0(t) = t^T M_0 t$ has entries $(M_0)_{14} = (M_0)_{41} = 1/2$ and $(M_0)_{23} = (M_0)_{32} = -1/2$, all others zero. Hence

$$M = \mathbf{H}^{-T} M_0 \mathbf{H}^{-1}, \quad \det M = \frac{1}{16(\det \mathbf{H})^2} \neq 0. \quad (\text{OQC3})$$

In particular Q is irreducible. Indeed, a reducible homogeneous quadratic over $\mathbb{C}$ is a product of two linear forms, whose symmetric coefficient matrix is a sum of two rank-one matrices and has rank at most two. This contradicts OQC3. The polynomial ring over a field is a unique factorization domain, by the repeated one-variable Gauss lemma, so its irreducible element Q is prime. Consequently $R = \mathbb{C}[x_1, x_2, x_3, x_4]/(Q)$ is an integral domain.

For clarity, the complete homogeneous restriction calculation has the precise kernel $Q\mathcal{P}_{d-2}$. In the t coordinates a degree- d monomial restricts to

$$\exp(((2b - d)\gamma - i(2a - d)\delta)w), \quad 0 \leq a, b \leq d. \quad (\text{OQC4})$$

Every pair occurs: the number of $(++)$ factors can range from $\max(0, a + b - d)$ to $\min(a, b)$. The exponents are distinct by their real and imaginary parts. Their exponentials are linearly independent, since differentiating orders zero through $(d + 1)^2 - 1$ gives a Vandermonde matrix with nonzero determinant. Two monomials giving the same pair have exponent vectors differing by a multiple of $(1, -1, -1, 1)$. After their common monomial is removed, their difference is divisible by $t_{++}t_{--} - t_{+-}t_{-+}$ through $A^r - B^r = (A - B) \sum_{j=0}^{r-1} A^{r-1-j} B^j$. A vanishing polynomial has coefficient sum zero within each pair by the independence just proved, and is therefore a sum of these differences. Conversely $Q_0(t(w)) = 0$. Transport by $\mathbf{H}$ proves the stated kernel and

$$\dim R_d = (d + 1)^2 \quad (d \geq 0). \quad (\text{OQC5})$$

Write $\zeta = e^{2\pi i/5}$ and $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$. The literal Fourier calculation gives $CF = FD$. On polynomials define the original induced action and its average by

$$(gf)(x) = f(D^{-1}x), \quad g^5 = 1, \quad \mathbf{R}f = \frac{1}{5} \sum_{a=0}^4 g^a f, \quad \mathcal{I}_d = \mathcal{P}_d^{(g)}. \quad (\text{OQC6})$$

Thus $\mathcal{J}_d$ consists of the weight-zero monomials, since $gx^\nu = \zeta^{-\sum j\nu_j} x^\nu$. This is the dual of the original average of $(C^{-a})^{\otimes d}$; both the inverse powers and the factor $1/5$ are retained.

For the original algebra E_d of the d -fold sum, define $\Theta_d : R_d \xrightarrow{\sim} E_d^\vee$ by

$$\Theta_d([f])(e^{wY_d}[1]) = f(x(w)), \quad Y_d = (A_d - dI/2)/i. \quad (\text{OQC7})$$

Here $\vee$ denotes the complex-linear dual. The primary expansion of the vector on the left gives all distinct exponentials OQC4; therefore existence and uniqueness of this covector and bijectivity follow from the preceding complete restriction calculation. The original observation $\Lambda_d : E_d \rightarrow B_d$ obeys

$$J_d := \text{im}(\mathcal{J}_d \rightarrow R_d), \quad \Theta_d(J_d) = \text{im } \Lambda_d^\vee, \quad \dim J_d = \text{rank } \Lambda_d, \quad R_d/J_d \xrightarrow{\sim} (\ker \Lambda_d)^\vee. \quad (\text{OQC8})$$

Indeed the coordinate functions of the projected pure tensor in OCS9 are precisely $x(w)^\nu$ of weight zero; their pulled-back covectors span the dual of its actual image. A covector vanishing on $\ker \Lambda_d$ factors through that image, which proves the last isomorphism as well. OQC8 specifies the exact original image; it has not been replaced by the whole invariant tensor space.

2.2 The complete orbit decision from the actual coefficients

Put $Q_a = g^a Q$ for $0 \leq a < 5$. The orbit of the line $\mathbb{C}Q$ has size one or five because five is prime. Moreover a size-one orbit has $gQ = Q$, with no other possible scalar. To prove this, write $gQ = \lambda Q$. Then $\lambda^5 = 1$, while OQC3 and the determinant of D , which is $\zeta^{10} = 1$, imply

$$\det(D^{-T}MD^{-1}) = \det M = \lambda^4 \det M. \quad (\text{OQC9})$$

Thus $\lambda^4 = 1$ and $\lambda^5 = 1$, forcing $\lambda = 1$. If any nonidentity power fixed the line, its powers generate the same cyclic group, so the preceding argument applies. This proves the entire orbit classification.

There is an exact coefficient test, requiring no genericity premise. Write the actual $Q = \sum_{|\mu|=2} q_\mu x^\mu$ and put

$$\mathfrak{e}(Q) = \sum_{\substack{|\mu|=2 \\ \sum j\mu_j \not\equiv 0 \pmod{5}}} |q_\mu|^2. \quad (\text{OQC10})$$

All coefficients here are the entries of OQC3, including the full inverse of the original period matrix. Since the monomials are a basis and each summand is nonnegative, $\mathfrak{e}(Q) = 0$ exactly when Q is invariant. The only degree-two weight-zero monomials are x_1x_4 and x_2x_3 . Hence the size-one case is exactly $Q = ax_1x_4 + bx_2x_3$ with $ab \neq 0$; nonvanishing of the two coefficients follows from OQC3. Positive $\mathfrak{e}(Q)$ gives the five distinct irreducible orbit equations. These are exhaustive outcomes of a test on the given periods, not assumptions on them.

In the size-one outcome, multiplication by Q preserves every weight. If Qf is invariant, then $Q(gf - f) = 0$ in the polynomial ring, and $gf = f$. Therefore

$$\ker(\mathcal{J}_k \rightarrow R_k) = Q\mathcal{J}_{k-2}, \quad \text{rank } \Lambda_k = d_{5,k,0} - d_{5,k-2,0}, \quad \dim \ker \Lambda_k = q - d_{5,k,0} + d_{5,k-2,0}. \quad (\text{OQC11})$$

This also proves that $J_k = R_k^{(g)}$: an invariant class can be lifted to a polynomial, and averaging that lift gives an invariant representative. The count itself follows from

$$d_{5,k,0} = \frac{\binom{k+3}{3} + 4\epsilon_k}{5}, \quad \epsilon_k = \mathbf{1}_{k \equiv 0 \pmod{5}} - \mathbf{1}_{k \equiv 1 \pmod{5}}. \quad (\text{OQC12})$$

For a proof, the identity element has symmetric-power generating series $(1-z)^{-4}$; each of the other four powers has series $\prod_{j=1}^4 (1 - z\zeta^j)^{-1} = (1-z)/(1-z^5)$. Averaging their coefficients gives

OQC12. Subtraction and $\binom{k+3}{3} - \binom{k+1}{3} = (k+1)^2$ evaluate the rank in OQC11, for every $k \geq 2$, as

$$\left\lfloor \frac{(k+1)^2}{5} \right\rfloor + \mathbf{1}_{k \equiv 0 \text{ or } 3 \pmod{5}}. \quad (\text{OQC13})$$

The five numerator corrections before division by five are 4, -4, -4, 4, 0, respectively; the residues of $(k+1)^2$ are 1, 4, 4, 1, 0. This verifies every case in OQC13.

2.3 An explicit invariant lift for the five-member orbit

For the size-five outcome determined by OQC10, retain the full actual products

$$\mathcal{H}_Q = \prod_{a=0}^4 Q_a \quad (\deg \mathcal{H}_Q = 10), \quad \mathcal{A}_Q = \prod_{a=1}^4 Q_a \quad (\deg \mathcal{A}_Q = 8). \quad (\text{OQC14})$$

The action permutes all five factors cyclically, so $g\mathcal{H}_Q = \mathcal{H}_Q$. None of the four factors of $\mathcal{A}_Q$ is a scalar multiple of Q . Since each is irreducible and Q is prime, multiplication by $[\mathcal{A}_Q]$ is injective on the integral domain R .

For every homogeneous $f \in \mathcal{P}_d$ the exact trace identity is

$$[5R(\mathcal{A}_Q f)]_{(Q)} = [\mathcal{A}_Q f]_{(Q)}. \quad (\text{OQC15})$$

Indeed $g^a \mathcal{A}_Q$ is the product of all orbit factors except Q_a . For $a \neq 0$ it contains $Q_0 = Q$. Those four terms vanish in the displayed quotient, and the $a = 0$ term is exactly the right side. This proves OQC15 with its original average and full factors, including every period-dependent coefficient.

The lift also has a well-defined typed quotient map. Let $\mathcal{U} = \mathbb{C}[x_1, x_2, x_3, x_4]/(\mathcal{H}_Q)$; the invariant equation $\mathcal{H}_Q$ makes the cyclic action well defined on this quotient. Then

$$\begin{aligned} \mathfrak{L}_d : R_d &\longrightarrow (\mathcal{U}_{d+8})^{(g)}, & [f]_{(Q)} &\longmapsto [5R(\mathcal{A}_Q f)]_{(\mathcal{H}_Q)}, \\ \text{res } \mathfrak{L}_d([f]) &= [\mathcal{A}_Q f]_{(Q)}, & \text{res} : \mathcal{U} &\rightarrow R. \end{aligned} \quad (\text{OQC16})$$

If f changes by Qb , its proposed image changes by $5R(\mathcal{H}_Q b) = 5\mathcal{H}_Q Rb$, hence by zero in $\mathcal{U}$. This proves well-definedness. The image is invariant because it has an invariant polynomial representative; OQC15 proves the second line. Injectivity follows from the injectivity of multiplication by $[\mathcal{A}_Q]$ on R . All maps in OQC16 are thus established, including their kernels.

The actual original invariant image consequently contains

$$[\mathcal{A}_Q]R_{k-8} \subseteq J_k \subseteq R_k. \quad (\text{OQC17})$$

The first inclusion follows by taking the invariant polynomial representatives OQC15, and uses no dimension inference. Therefore, for every $k \geq 8$ in this outcome,

$$\text{rank } \Lambda_k \geq (k-7)^2, \quad \dim \ker \Lambda_k \leq (k+1)^2 - (k-7)^2 = 16k - 48. \quad (\text{OQC18})$$

These inequalities follow from the proved injection and OQC5. They may be combined with the unconditional actual-matrix bounds in OCS24, OCS31–33 by taking the stronger lower and upper bounds. For the original degrees $k = 9, 13, 17, 21, 25, 29$, the displayed upper bounds on the original kernel are respectively 96, 160, 224, 288, 352, 416; at the first two degrees the earlier unconditional bounds may be stronger. No such upper bound is assigned to the size-one outcome, whose exact rank is OQC11. Together OQC10–18 compute the complete orbit alternatives from the original coefficients. Evaluating $\mathfrak{e}(Q)$ in the original period family is a separate calculation on those explicit coefficients; no outcome is postulated here.

2.4 The induced original dual map and its complete action defect

At $k \geq 9$ the degree $d = k - 8$ is positive, and the original d -fold sum algebra is defined using the same h, Π, U . For the size-five outcome, OQC7, OQC8 and OQC17 give the uniquely typed injective map

$$\begin{aligned} \mathfrak{J}_k : E_{k-8}^\vee &\hookrightarrow B_k^\vee, \\ \Lambda_k^\vee \mathfrak{J}_k &= \Theta_k M_{[\mathcal{A}_Q]} \Theta_{k-8}^{-1}, \quad \mathfrak{J}_k^\vee : B_k \twoheadrightarrow E_{k-8}. \end{aligned} \quad (\text{OQC19})$$

The right side of the second line has image in $\text{im } \Lambda_k^\vee$ by OQC17, and $\Lambda_k^\vee$ is injective because Λ_k is onto its actual image. This gives existence and uniqueness. Each of the three right-hand maps is injective, so $\mathfrak{J}_k$ is injective. Finite-dimensional duality then proves the stated surjection, using the canonical bidual identifications. The new degree $k - 8$ records contraction against the degree-eight polynomial derived in OQC14. It does not replace either original Gamma source of order 1 or k .

The surviving action and its defect are fully determined. Put

$$\Omega = \text{diag}(\gamma - i\delta, -\gamma - i\delta, \gamma + i\delta, -\gamma + i\delta), \quad G = H\Omega H^{-1}, \quad (\mathcal{D}f)(x) = \sum_{j=1}^4 (Gx)_j \partial_j f(x). \quad (\text{OQC20})$$

Here $x'(w) = Gx(w)$ by OQC1. In the original t coordinates, the weights of each product in Q_0 sum to zero, so $\mathcal{D}Q = 0$. Thus this derivation acts on each R_d . Differentiating OQC7 proves exactly

$$\Theta_d \mathcal{D} = Y_d^\vee \Theta_d. \quad (\text{OQC21})$$

Both sides evaluated at $e^{wY_d}[1]$ are the derivative of $f(x(w))$, and these vectors span E_d by the distinct exponentials; hence equality holds as maps, not only on a curve. Leibniz's rule gives the complete defect

$$\begin{aligned} Y_k^\vee \Lambda_k^\vee \mathfrak{J}_k - \Lambda_k^\vee \mathfrak{J}_k Y_{k-8}^\vee &= \Theta_k M_{[\mathcal{D}\mathcal{A}_Q]} \Theta_{k-8}^{-1}, \\ \mathcal{D}\mathcal{A}_Q &= \sum_{a=1}^4 (\mathcal{D}Q_a) \prod_{\substack{1 \leq b \leq 4 \\ b \neq a}} Q_b. \end{aligned} \quad (\text{OQC22})$$

The codomain in the first line is $E_k^\vee$. Its image has not been asserted to lie in $\text{im } \Lambda_k^\vee$. Restriction to $\ker \Lambda_k$ supplies the exact remaining map into $(\ker \Lambda_k)^\vee$, through OQC8. In particular OQC19 is not used as an unproved intertwining of boundary actions. OQC22 retains every term that can leave the observed dual subspace.

Finally, the failure of the cyclic average to commute with the original action is explicitly calculable before any quotient:

$$([\mathcal{D}, g^a]f)(x) = (\nabla f)(D^{-a}x)^T (D^{-a}G - GD^{-a})x, \quad [\mathcal{D}, R] = \frac{1}{5} \sum_{a=0}^4 [\mathcal{D}, g^a]. \quad (\text{OQC23})$$

The chain rule applied separately to $f(D^{-a}x)$ and to $(\mathcal{D}f)(D^{-a}x)$ proves both identities. Thus all action comparisons in the invariant lift have explicit original coefficient maps. The rank estimates in OQC18 concern exactly the kernel used in the original Gamma and arithmetic quotient Grams, because those constructions use the same Λ_k . An estimate for its volume still requires those original metrics; no volume conclusion has been inferred solely from its dimension.

3 Exact original rank from the complete quadric orbit

Retain every original object and map in OQC1–23. In particular $m = 1$, $k = 4l + 1$, $l \geq 1$, $q = (k + 1)^2$, $\mathcal{P} = \mathbb{C}[x_1, x_2, x_3, x_4]$, and $gf(x) = f(D^{-1}x)$ for the original Fourier matrix $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$. The polynomial $Q = Q_0(H^{-1}x)$ has its actual original period and amplitude coefficients. Write $\mathcal{J} = \mathcal{P}^{(g)}$, $R = \mathcal{P}/(Q)$ and $J = \text{im}(\mathcal{J} \rightarrow R)$, with the grading and the actual observation identification of OQC8. For negative indices put $\mathcal{J}_d = 0$ and $d_{5,d,0} = 0$.

The orbit decision is the exhaustive test OQC10. The size-one case already has the exact rank OQC11–13. The following calculation completes the other case using the same original coefficients. It does not assert that a period parameter belongs to that case before its coefficient test has been evaluated.

3.1 Every invariant vanishing polynomial contains the full orbit

In the size-five case, let $Q_a = g^a Q$ and retain exactly $\mathcal{H}_Q = \prod_{a=0}^4 Q_a$ from OQC14. Each Q_a is irreducible, because an invertible linear substitution preserves factorizations and Q is irreducible by OQC3. The five factors are pairwise nonassociate by the proved orbit classification, and each is prime in the original polynomial ring. Consequently

$$\mathcal{J} \cap Q\mathcal{P} = \mathcal{H}_Q\mathcal{J}, \quad \ker(\mathcal{J}_d \rightarrow R_d) = \mathcal{H}_Q\mathcal{J}_{d-10}. \quad (\text{OQX1})$$

Here is the complete divisibility proof. If $f \in \mathcal{J}$ and $f = Qb$, then for every $a = 0, \dots, 4$ the exact equality $g^a f = f$ gives $f = Q_a g^a b$. Thus every one of the five original prime factors divides f . Distinctness of the factors implies that their product divides f : after factoring out Q_0 , primality of Q_1 and the fact that Q_1 does not divide Q_0 force Q_1 to divide the remaining quotient. Repeating this argument with Q_2, Q_3, Q_4 proves $f = \mathcal{H}_Q b_0$. The action permutes the five exact factors cyclically, so $g\mathcal{H}_Q = \mathcal{H}_Q$. Applying g to this last equality and using $gf = f$ gives $\mathcal{H}_Q(gb_0 - b_0) = 0$. The polynomial ring is an integral domain and $\mathcal{H}_Q \neq 0$, whence $gb_0 = b_0$. This proves the first inclusion in OQX1. Conversely an invariant b_0 gives an invariant $\mathcal{H}_Q b_0$ divisible by Q , which proves the reverse inclusion. The product is homogeneous of degree ten. If its product with b_0 is homogeneous of degree d , decomposition of b_0 into homogeneous parts shows that only its part of degree $d - 10$ can be nonzero: each other part has zero product with $\mathcal{H}_Q$ and hence is zero. This proves the graded kernel formula, including $d < 10$.

3.2 The exact invariant quotient and the restriction map

Let $\mathcal{U} = \mathcal{P}/(\mathcal{H}_Q)$ as in OQC16. The following are isomorphisms of graded complex algebras with their specified maps:

$$\mathcal{J}/(\mathcal{H}_Q\mathcal{J}) \xrightarrow{[f] \mapsto [f]_{(\mathcal{H}_Q)}} \mathcal{U}^{(g)} \xrightarrow{\text{res}} J \subseteq R. \quad (\text{OQX2})$$

The action on $\mathcal{U}$ is defined because $\mathcal{H}_Q$ is invariant. If $[f]_{(\mathcal{H}_Q)}$ is invariant, averaging its representatives gives $[Rf]_{(\mathcal{H}_Q)} = [f]_{(\mathcal{H}_Q)}$, since $[g^a f] = [f]$ for every a . Thus the first map is onto. Its kernel before passage to the displayed domain is $\mathcal{J} \cap \mathcal{H}_Q\mathcal{P} = \mathcal{H}_Q\mathcal{J}$: the reverse inclusion is immediate, and for $f = \mathcal{H}_Q b \in \mathcal{J}$, cancellation in $\mathcal{H}_Q(gb - b) = 0$ proves $b \in \mathcal{J}$. This proves that the first map is injective as well.

The full quotient restriction $\text{res} : \mathcal{U} \rightarrow R$ is well defined because $(\mathcal{H}_Q) \subseteq (Q)$; its kernel is exactly $Q\mathcal{P}/\mathcal{H}_Q\mathcal{P}$. For its invariant subspace, choose the invariant representative just constructed. If its restriction vanishes, that representative belongs to $\mathcal{J} \cap Q\mathcal{P}$, which is $\mathcal{H}_Q\mathcal{J}$ by OQX1. Its class in $\mathcal{U}$ is therefore zero. The invariant restriction is injective. Its image is exactly J , since an invariant representative restricts to J by definition and every element of J has such a representative. Both maps are induced by polynomial quotient maps, so they preserve products, sums, complex scalars and degrees. No action of g on R is asserted in the size-five case. In particular the second map retains precisely the invariant image of the original observation, with its complete restriction kernel already computed.

3.3 All original degrees and the exact kernel dimension

The map Θ_k and the original observation in OQC8 identify $\dim J_k$ with $\text{rank } \Lambda_k$ and identify R_k/J_k with $(\ker \Lambda_k)^\vee$. Multiplication by the nonzero polynomial $\mathcal{H}_Q$ is injective in $\mathcal{P}$. Taking dimensions in OQX1 thus gives, for every nonnegative degree d , the exact invariant restriction $\text{rank } \dim J_d = d_{5,d,0} - d_{5,d-10,0}$. At the original degrees this evaluates to

$$\boxed{\text{rank } \Lambda_k = d_{5,k,0} - d_{5,k-10,0} = k^2 - 6k + 17, \quad \dim \ker \Lambda_k = 8k - 16, \quad k = 4l + 1, \quad l \geq 1,} \quad (\text{OQX3})$$

in the size-five outcome of the actual coefficient test. To verify every constant and degree range, for $k \geq 10$ the complete character formula OQC12 gives

$$d_{5,k,0} - d_{5,k-10,0} = \frac{1}{5} \left\{ \binom{k+3}{3} - \binom{k-7}{3} + 4(\epsilon_k - \epsilon_{k-10}) \right\}.$$

The two ϵ values are equal because their indices differ by ten. Expanding the remaining cubics gives

$$\binom{k+3}{3} - \binom{k-7}{3} = \frac{(k^3 + 6k^2 + 11k + 6) - (k^3 - 24k^2 + 191k - 504)}{6} = 5(k^2 - 6k + 17).$$

The only original degrees below ten are five and nine. Their negative-degree terms in OQX1 vanish, while the same exact character formula gives

$$d_{5,5,0} = (56 + 4)/5 = 12 = 5^2 - 6 \cdot 5 + 17, \quad d_{5,9,0} = 220/5 = 44 = 9^2 - 6 \cdot 9 + 17.$$

This proves the first equality of OQX3 for every original $l \geq 1$. Since $\dim E_k = \dim R_k = (k+1)^2$, rank-nullity gives $(k+1)^2 - (k^2 - 6k + 17) = 8k - 16$, proving the second equality. For the original degrees already tabulated in OCS26, the exact five-orbit values are

k	$\text{rank } \Lambda_k$	$\dim \ker \Lambda_k$
5	12	24
9	44	56
13	108	88
17	204	120
21	332	152
25	492	184
29	684	216

These are exact values for this exhaustively classified orbit outcome, replacing the less precise bounds OQC18 wherever that outcome occurs. The full invariant lift and its action defect OQC15–23 remain valid with all original coefficients. The size-one outcome retains its different exact values OQC11–13. No choice between the two outcomes has been made here without evaluating the original period coefficient test, and no estimate for a metric volume has been inferred from its dimension alone.

4 The actual period quadric and its cyclic continuation

This calculation uses the original simple quartet, $m = 1$, and the literal period matrix, amplitude unit and cyclic matrix of BRU and OPG. Its formulas evaluate the transported quadric from those data. All source orders elsewhere in the programme remain unchanged.

4.1 Original roots, entire unit and centred phase

Order the roots and keep their actual values as

$$\begin{aligned} \alpha_1 &= \frac{1}{2} + \delta + i\gamma, & \alpha_2 &= \frac{1}{2} + \delta - i\gamma, & \alpha_3 &= \frac{1}{2} - \delta + i\gamma, & \alpha_4 &= \frac{1}{2} - \delta - i\gamma, \\ 0 < \delta < \frac{1}{2}, & \gamma > 2, & h(s) &= \prod_{j=1}^4 (s - \alpha_j), & g &= 2\xi, & v_h &= g/h. \end{aligned} \quad (\text{PQO1})$$

The four zero orders of g are complete and equal to one. Thus v_h is entire and has nonzero value at each displayed root. Put $w = s - 1/2$, and define the following original constants:

$$\begin{aligned} z_+ &= \delta + i\gamma, & z_- &= \delta - i\gamma, & \lambda &= z_+^2, & \mu &= z_-^2, \\ A &= 2(\gamma^2 - \delta^2), & B &= (\delta^2 + \gamma^2)^2, & \mathfrak{a} &= \Phi_h(1/2) = \frac{1}{160} + \frac{A}{24} + \frac{B}{2}, \\ h(1/2 + w) &= w^4 + Aw^2 + B, & \Phi_h(1/2 + w) &= \mathfrak{a} + \frac{w^5}{5} + \frac{Aw^3}{3} + Bw. \end{aligned} \quad (\text{PQO2})$$

The product of the two quadratics $((w - \delta)^2 + \gamma^2)((w + \delta)^2 + \gamma^2)$ proves the quartic identity. Integrating it and using $\Phi_h(0) = 0$ proves the phase and constant in PQO2. In particular the translation has not removed the original constant $\mathfrak{a}$.

The theta integral IPM.2 has real coefficients, real theta kernel, and an expression unchanged by $s \mapsto 1 - s$. It therefore gives $g(\bar{s}) = \overline{g(s)}$ and $g(1 - s) = g(s)$ directly; the same two identities hold for h . Taking their entire quotient gives

$$v_h(\alpha_1) = v_h(\alpha_4) = a, \quad v_h(\alpha_2) = v_h(\alpha_3) = \bar{a}, \quad a \neq 0. \quad (\text{PQO3})$$

These are the full actual unit values. Their magnitude and phase are not assigned. Set $t_+ = a^{-2}$ and $t_- = \bar{a}^{-2}$. For $j = 0, 1, 2, 3$ retain

$$\sigma_j = t_+ \lambda^j - t_- \mu^j, \quad \sigma_2 = -A\sigma_1 - B\sigma_0, \quad \sigma_3 = (A^2 - B)\sigma_1 + AB\sigma_0. \quad (\text{PQO4})$$

The recurrence follows because $\lambda + \mu = -A$ and $\lambda\mu = B$. Each σ_j is purely imaginary. Furthermore σ_0 and σ_1 cannot both vanish: the first would give $t_+ = t_- \neq 0$, and the second would then give $t_+(\lambda - \mu) = 0$, whereas $\lambda - \mu = 4i\delta\gamma \neq 0$.

4.2 The original periods as an absolutely convergent series

Fix $u \neq 0$ on its original logarithm branch. Let $\ell_j(r) = r \exp(i(\arg u + \pi + 2\pi j)/5)$, $\Gamma_j = -\ell_0 + \ell_j$, $1 \leq j \leq 4$. Let Π integrate the degree-less-than-four representative against $e^{\Phi_h(s)/u} ds$ on these original oriented contours. Write $F_{ja} = \zeta^{ja} - 1$, $\zeta = e^{2\pi i/5}$, $1 \leq j, a \leq 4$. Direct finite geometric sums prove

$$(F^{-1})_{aj} = \frac{1}{5} \zeta^{-ja}, \quad CF = FD, \quad D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4), \quad (\text{PQO5})$$

where $(Cx)_j = x_{j+1} - x_1$ for $j < 4$ and $(Cx)_4 = -x_1$. For example multiplying the asserted inverse by F gives $\frac{1}{5} \sum_{j=1}^4 \zeta^{-ja} (\zeta^{jb} - 1) = \delta_{ab}$; substitution into each row gives $CF = FD$.

Let T be the exact increasing-coefficient translation matrix $T_{rb} = \binom{b}{r} 2^{-(b-r)}$ for $0 \leq r \leq b \leq 3$, and zero otherwise. Thus it sends coefficients of $p(s)$ to those of $p(1/2 + w)$, with determinant one. Define the centred Fourier-period matrix

$$P_{ar}(u) = \frac{1}{5} \sum_{j=1}^4 \zeta^{-ja} \int_{\Gamma_j} e^{(w^5/5 + Aw^3/3 + Bw)/u} w^r dw, \quad 1 \leq a \leq 4, \quad 0 \leq r \leq 3. \quad (\text{PQO6})$$

The contours in this formula start at $w = 0$. They give precisely the translated original period by

$$F^{-1}\Pi = e^{a/u}P(u)T. \quad (\text{PQO7})$$

To justify this exact contour translation, the substitution $w = s - 1/2$ first gives rays starting at $-1/2$. Join each shifted ray at a large radius to the corresponding unshifted ray. On the joining segment their arguments differ by $O(1/r)$, and the leading real phase is $-r^5/(5|u|) + O(r^4)$; all lower phase terms and polynomial factors are dominated by this negative leading term. Those outer joining integrals tend to zero. The two common-origin joining integrals cancel in $-\ell_0 + \ell_j$ with their opposite orientations. Cauchy's theorem for the entire integrand proves equality in the limit. Substitution of the polynomial gives exactly the matrix T .

For every integer $L \geq 1$ define, on the same branch,

$$G_L(u) = u^{L/5} 5^{L/5-1} e^{\pi i L/5} \Gamma(L/5). \quad (\text{PQO8})$$

Integration of the leading phase on the two rays gives $\int_{\Gamma_j} e^{w^5/(5u)} w^{L-1} dw = G_L(u)(\zeta^{jL} - 1)$ by the real substitution $r^5/(5|u|)$ on each ray, including its phase in dw . Expanding the remaining two exponentials now gives the exact entries

$$P_{ar}(u) = \sum_{\substack{p,q \geq 0 \\ r+3p+q+1 \equiv a \pmod{5}}} \frac{A^p B^q}{3^p p! q! u^{p+q}} G_{r+3p+q+1}(u). \quad (\text{PQO9})$$

Terms with $r+3p+q+1 \equiv 0$ have zero two-ray integral in every column and contribute zero before the Fourier map. To prove every exchange, the sum of the absolute values of the unintegrated series on either ray is bounded by a constant times

$$R^r \exp\left(-\frac{R^5}{5|u|} + \frac{|A|R^3}{3|u|} + \frac{|B|R}{|u|}\right).$$

Here R is the radial coordinate and r is the fixed column degree. This integrable majorant proves absolute termwise integration. It is uniform on compact coefficient and nonzero- u sets on the chosen branch. The finite Fourier sum in PQO5 then selects exactly the stated residue class. Thus PQO9 is an actual period formula, with all coefficients and all terms retained.

For additional finite computation, let I_d denote the vector of the four centred ray moments of w^d . Integration of the derivative of $e^{(w^5/5 + Aw^3/3 + Bw)/u} w^d$, with the same vanishing outer endpoints and cancelling inner endpoints, proves

$$I_{d+4} + AI_{d+2} + BI_d = -u d I_{d-1} \quad (d \geq 0), \quad u \partial_B I_d = I_{d+1}, \quad 3u \partial_A I_d = I_{d+3}. \quad (\text{PQO10})$$

The first right side is zero at $d = 0$. Coefficient derivatives are justified by the same compact domination. These are the original polynomial period recurrences, and they compute every higher moment from the first four and the actual coefficients.

The determinant is especially explicit in these centred coordinates:

$$\det \Pi = -(2\pi)^2 u^2 e^{4a/u}, \quad \det F = -25\sqrt{5}, \quad \det P(u) = \frac{(2\pi)^2}{25\sqrt{5}} u^2. \quad (\text{PQO11})$$

For the first equality use the full PDE1–13 determinant calculation at $n = 4$: the phase is $e^{15\pi i} = -1$ and the critical-value sum is $4a$. This critical sum also follows directly by pairing w with $-w$ in the odd centred phase. For F , subtract row zero from the other rows in the five-by-five Fourier Vandermonde and expand its first column. Its phase is $e^{13\pi i} = -1$ and its magnitude $5^{5/2}$, by the finite Fourier orthogonality identity. The last equality follows by taking determinants in PQO7 and using $\det T = 1$. The Gamma grid-product proof PGCC1–6 retains the full evaluated constant in the first equality. In particular every inverse below exists on the full stated domain.

4.3 The actual transformed quadric and every deciding coefficient

Let V be the centred evaluation matrix with rows $(1, z_j, z_j^2, z_j^3)$, where $(z_1, z_2, z_3, z_4) = (z_+, z_-, -z_-, -z_+)$. Its inverse sends the four primary values to their unique degree-less-than-four polynomial. Thus the actual matrix from primary coordinates to the Fourier-period space is

$$B_{\text{per}} = F^{-1} \Pi U J_{\text{prim}} = e^{a/u} P(u) V^{-1} \text{diag}(a, \bar{a}, \bar{a}, a). \quad (\text{PQO12})$$

This follows either by applying all maps to the four CRT idempotents or by evaluating the full unit at each root. It agrees with OCS and OPG in the given original orientation.

For $Q_0(t) = t_1 t_4 - t_2 t_3$, define $Q_u(x) = Q_0(B_{\text{per}}^{-1} x)$. On the unchanged original period coordinate $\mathbf{y} \in \mathbb{C}^4$ the quadratic is $Q_u^{\text{per}}(\mathbf{y}) = Q_0((\Pi U J_{\text{prim}})^{-1} \mathbf{y}) = Q_u(F^{-1} \mathbf{y})$, because $F B_{\text{per}} = \Pi U J_{\text{prim}}$. Thus the exact isomorphism $\mathbf{y} = Fx$ carries the following quadric to the original period space. If $p(w) = p_0 + p_1 w + p_2 w^2 + p_3 w^3$, multiplication gives

$$p(w)p(-w) = p_0^2 + (2p_0 p_2 - p_1^2)w^2 + (p_2^2 - 2p_1 p_3)w^4 - p_3^2 w^6.$$

Consequently the exact symmetric matrix in the centred coefficient frame is

$$S = \begin{pmatrix} \sigma_0 & 0 & \sigma_1 & 0 \\ 0 & -\sigma_1 & 0 & -\sigma_2 \\ \sigma_1 & 0 & \sigma_2 & 0 \\ 0 & -\sigma_2 & 0 & -\sigma_3 \end{pmatrix}, \quad \boxed{Q_u(x) = x^\top H(u)x, \quad H(u) = e^{-2a/u} P(u)^{-\top} S P(u)^{-1}.} \quad (\text{PQO13})$$

Indeed applying the preceding product formula at z_+ and z_- with multipliers t_+ and $-t_-$ gives exactly S . This retains both actual unit values through every σ_j . The two parity blocks have determinants

$$\sigma_0 \sigma_2 - \sigma_1^2 = -t_+ t_- (\lambda - \mu)^2, \quad \sigma_1 \sigma_3 - \sigma_2^2 = -t_+ t_- B(\lambda - \mu)^2.$$

Thus S and $H(u)$ are nondegenerate, with the exact determinant

$$\det H(u) = e^{-8a/u} \frac{B(t_+ t_-)^2 (\lambda - \mu)^4}{(\det P(u))^2} \neq 0. \quad (\text{PQO14})$$

Write $N(u) = \text{adj } P(u)$, with entries N_{ra} , $0 \leq r \leq 3$, $1 \leq a \leq 4$, and retain $\Delta(u) = \det P(u)$ from PQO11. Then every coefficient is given by the explicit minor formula

$$H_{ab}(u) = \frac{e^{-2a/u}}{\Delta(u)^2} \{ \sigma_0 N_{0a} N_{0b} + \sigma_1 (N_{0a} N_{2b} + N_{2a} N_{0b} - N_{1a} N_{1b}) \\ + \sigma_2 (N_{2a} N_{2b} - N_{1a} N_{3b} - N_{3a} N_{1b}) - \sigma_3 N_{3a} N_{3b} \}. \quad (\text{PQO15})$$

This is matrix multiplication in PQO13 using $P^{-1} = N/\Delta$. Each N_{ra} is the signed original three-by-three period minor. Its entries are given by the absolutely convergent series PQO9 or by the original integrals PQO6. The polynomial coefficient of x_a^2 is H_{aa} ; that of $x_a x_b$, $a < b$, is $2H_{ab}$. Both factors are retained in this specified quadratic form.

The exact eight deciding entries are

$$\mathcal{E}(u) = (H_{11}, H_{22}, H_{33}, H_{44}, H_{12}, H_{13}, H_{24}, H_{34}). \quad (\text{PQO16})$$

The cyclic action on quadrics is the pullback $Q_u \mapsto Q_u(D^{-1}x)$; its symmetric matrix is $D^{-\top} H(u) D^{-1}$. Since $D^5 = I$ and $\det D = 1$, the line spanned by a nondegenerate form can be stable only with character one. In fact if its multiplier is η , then $\eta^5 = 1$ by the group action, while taking determinants gives $\eta^4 = 1$ because $\det H(u) \neq 0$. Thus $\eta = 1$. Entrywise, invariance is precisely $(\zeta^{-(a+b)} - 1)H_{ab} = 0$. The only pairs with $a + b \equiv 0$ are $(1, 4)$ and $(2, 3)$. Therefore

The quadric is invariant exactly on $Z = \{u : \mathcal{E}(u) = 0\}$;
 $u \in Z \implies Q_u(x) = 2H_{14}(u)x_1 x_4 + 2H_{23}(u)x_2 x_3, \quad H_{14}(u)H_{23}(u) \neq 0$;
 at every $u \notin Z$ its orbit consists of five distinct quadrics.

(\text{PQO17})

Every assertion follows from the entry test and the rank-four determinant. The orbit size divides five; it is one exactly when the line is stable. Thus no two-member orbit occurs. Moreover $F^{-1}C^{-1} = D^{-1}F^{-1}$ by PQO5, so this is exactly the orbit of $Q_u^{\text{per}}(C^{-j}\mathbf{y})$ in the original period coordinates, with the same five indexed maps. PQO15–17 determine the exact exceptional locus by original period minors and original amplitude values, without assigning generic values to them.

4.4 The cyclic action is the actual original continuation in u

Continue the original logarithm of u by $2\pi i$. Its rays advance from ℓ_j to ℓ_{j+1} , and their oriented differences advance by the exact original matrix C . For the last row $\ell_5 = \ell_0$, giving $-\Gamma_1$. Equivalently every summand of PQO9 acquires the phase

$$e^{2\pi i((r+3p+q+1)/5-p-q)} = \zeta^{r+3p+q+1} = \zeta^a.$$

The original constant $e^{a/u}$ is single-valued under this continuation. Hence

$$\begin{aligned} P(\log u + 2\pi i) &= DP(\log u), & B_{\text{per}}(\log u + 2\pi i) &= DB_{\text{per}}(\log u), \\ H(\log u + 2\pi i) &= D^{-\top} H(\log u) D^{-1}. \end{aligned} \tag{PQO18}$$

The contour argument and the convergent series prove the same map. Thus PQO17 describes the actual period continuation of this original quadric. It introduces no freely chosen cyclic action on its coefficients.

4.5 The large-parameter geometry and a positive-ray obstruction

Let $\varepsilon = u^{-1/5}$ on the same branch, and set $D_\varepsilon = \text{diag}(\varepsilon^{-1}, \varepsilon^{-2}, \varepsilon^{-3}, \varepsilon^{-4})$. The exact series PQO9 gives

$$P(u) = \mathcal{P}(A\varepsilon^2, B\varepsilon^4)D_\varepsilon, \quad \mathcal{P}(0,0) = \text{diag}(g_1, g_2, g_3, g_4), \quad g_a = 5^{a/5-1}e^{\pi ia/5}\Gamma(a/5). \tag{PQO19}$$

The entries of $\mathcal{P}$ are entire in the two displayed arguments, by the same absolute integrable series majorant. The same determinant proof PQO11 applies to every complex pair of lower odd-phase coefficients, since the centred phase remains odd and its critical values pair to zero with full multiplicities. Thus their determinant is the nonzero constant $(2\pi)^2/(25\sqrt{5})$, by PQO11 and $\det D_\varepsilon = u^2$. Their inverse is therefore holomorphic everywhere, given by its adjugate divided by that precise constant. Near $\varepsilon = 0$ it is $\text{diag}(g_a^{-1}) + O(\varepsilon^2)$. This bound follows from its convergent power series after substituting $(A\varepsilon^2, B\varepsilon^4)$; it holds on a full small complex disc, with the fixed original A, B .

If $\sigma_0 \neq 0$, PQO13 and PQO19 give

$$e^{2a/u}H_{11}(u) = \frac{\sigma_0}{g_1^2}\varepsilon^2 + O(\varepsilon^4).$$

If $\sigma_0 = 0$, then $\sigma_1 \neq 0$ by PQO4, and they give

$$e^{2a/u}H_{22}(u) = -\frac{\sigma_1}{g_2^2}\varepsilon^4 + O(\varepsilon^6). \tag{PQO20}$$

For verification of the orders, insert $P^{-1} = D_\varepsilon^{-1}\mathcal{P}^{-1}$ into PQO13. The middle matrix has entries $S_{rs}\varepsilon^{r+s+2}$. Its first possible degree is two, from S_{00} , and its next possible degree is four, from S_{02}, S_{20}, S_{11} . The constant term of $\mathcal{P}^{-1}$ is diagonal, so the indicated leading entry is exactly the displayed nonzero coefficient. Every further contribution has the stated higher degree. This also proves the expansion when some other σ_j vanish.

Consequently for every fixed original quartet and its full actual amplitude there exists a finite radius outside which the actual orbit has five distinct quadrics. This conclusion holds on every original logarithm branch: the estimate is on the full complex ε disc, and its leading coefficient

is nonzero. The set Z of PQO17 is a discrete subset of the logarithmic u surface. Indeed the decisive entry just proved is holomorphic on that connected surface and is not identically zero; its zeros are isolated by its local power series, and Z is contained in that zero set. The explicit finite-radius bound from the companion period-connection calculation strengthens this argument while retaining the same actual coefficients.

Moreover PQO18 multiplies each deciding entry by its nonzero root-of-unity factor, so the condition $\mathcal{E}(u) = 0$ is unchanged by every full logarithm turn. Therefore Z is the inverse image under $\log u \mapsto u$ of a well-defined set $Z_{\text{abs}} \subset \mathbb{C}^*$. On a sufficiently small disc about any nonzero u , choose one logarithm branch. The preceding isolated-zero proof shows that Z_{abs} is discrete and closed on that disc. The explicit radius in the companion calculation gives $|u| < R_*$ throughout Z_{abs} . Consequently its only possible accumulation point in the finite u -plane is zero. In particular each compact annulus about zero contains only finitely many such points: an infinite subset of that compact set would have a nonzero accumulation point, contradicting the proved local isolated-zero property.

For positive real u retain the actual lift $\log u = \ln u + 2\pi i \ell_{\text{br}}$, $\ell_{\text{br}} \in \mathbb{Z}$. The exact series has an additional real structure. Every surviving index in PQO9 is $a + 5L$, and its phase at the lift $\ell_{\text{br}} = 0$ is $e^{\pi i a/5}(-1)^L$. Its remaining coefficients are real. The actual lift multiplies row a by $\zeta^{\ell_{\text{br}} a}$, by PQO18. Absolute convergence therefore proves

$$P(u) = E_{\ell_{\text{br}}} R(u), \quad E_{\ell_{\text{br}}} = D^{\ell_{\text{br}}} \text{diag}(e^{\pi i/5}, e^{2\pi i/5}, e^{3\pi i/5}, e^{4\pi i/5}), \quad R(u) \text{ real and invertible.} \quad (\text{PQO21})$$

Let M be multiplication by w in the increasing centred polynomial frame, so

$$M = \begin{pmatrix} 0 & 0 & 0 & -B \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -A \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad c_3 = \frac{t_+ - t_-}{4i\delta\gamma}, \quad c_1 = \frac{t_+ + t_-}{2} - (\delta^2 - \gamma^2)c_3. \quad (\text{PQO22})$$

Both c_1, c_3 are real by PQO3, and the polynomial $c_1 + c_3 w^2$ takes values t_+, t_-, t_-, t_+ at the four original roots. Hence $c_1 M + c_3 M^3$ is precisely multiplication by $w(j_h v_h)^{-2}$, with its entire unit retained.

The companion period-connection proof establishes the exact identity between cyclic invariance of Q_u and

$$[P(u)(c_1 M + c_3 M^3)P(u)^{-1}, D] = 0. \quad (\text{PQO23})$$

It proves this by the flat residue/reflection form on the original polynomial periods and retains all its constants. Using that proved identity, PQO21 gives a further precise obstruction on the positive ray. Since D has distinct diagonal entries, a matrix commuting with it is diagonal. The matrix in PQO23 is $E_{\ell_{\text{br}}}$ times a real matrix times $E_{\ell_{\text{br}}}^{-1}$; if diagonal, its diagonal entries must therefore be real. Its eigenvalues are exactly

$$\frac{\delta + i\gamma}{a^2}, \quad \frac{\delta - i\gamma}{\bar{a}^2}, \quad -\frac{\delta - i\gamma}{\bar{a}^2}, \quad -\frac{\delta + i\gamma}{a^2}, \quad (\text{PQO24})$$

as follows by evaluation at the four distinct roots of h . Thus every positive-real- u point in Z satisfies the explicit original-unit equation

$$\text{Im} \left(\frac{\delta + i\gamma}{a^2} \right) = 0. \quad (\text{PQO25})$$

If that imaginary part is nonzero for the original unit, the positive ray contains no point of Z . When it is zero, the four displayed eigenvalues are the two nonzero real numbers $r, -r$, each twice, and the complete period-minor tests PQO15–17 still determine Z . The equation is not assumed to hold or fail. Together the exact minor tests, the actual period continuation, the proved large-parameter result and this real-structure obstruction give control of the original quadric family with every amplitude value and phase present.

5 The original period quadric as a residue commutator

Retain $0 < \delta < 1/2$, $\gamma > 2$, the ordered simple quartet $\alpha_1 = 1/2 + \delta + i\gamma$, $\alpha_2 = 1/2 + \delta - i\gamma$, $\alpha_3 = 1/2 - \delta + i\gamma$, $\alpha_4 = 1/2 - \delta - i\gamma$, and $h(s) = \prod_a (s - \alpha_a)$. Put

$$w = s - \frac{1}{2}, \quad r_a = \alpha_a - \frac{1}{2}, \quad \beta = 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad H(w) = w^4 + \beta w^2 + \eta. \quad (\text{RPC1})$$

Thus $r_4 = -r_1$, $r_3 = -r_2$, $r_2 = \overline{r_1}$ and $r_1^2 - r_2^2 = 4i\delta\gamma$. No root or original coordinate is discarded by this translation. Its coefficient isomorphism is $J_c : \mathbb{C}[w]_{<4} \rightarrow E_h$, $J_c p = [p(s - 1/2)]$; its inverse is substitution $s = w + 1/2$ in the unique original representative. The primitive with its original constant is

$$\Phi_h(s) = A + \phi(w), \quad \phi(w) = \frac{w^5}{5} + \frac{\beta w^3}{3} + \eta w, \quad A = \Phi_h(1/2) = \frac{1}{160} + \frac{\beta}{24} + \frac{\eta}{2}. \quad (\text{RPC2})$$

The last equality is forced by $\Phi_h(0) = 0$ and follows by evaluating the displayed odd polynomial at $w = -1/2$.

Let $u \neq 0$ carry the specified original branch of its logarithm. Let Π be the original period map on the rays and orientations of PDE2, and let $F_{ja} = \zeta^{ja} - 1$, $\zeta = e^{2\pi i/5}$, $1 \leq j, a \leq 4$. Its inverse has entries $(F^{-1})_{aj} = \zeta^{-ja}/5$: multiplying the two matrices and using $\sum_{j=1}^4 \zeta^{jb} = 4$ for $5 \mid b$ and -1 otherwise proves this entry by entry. The literal centered coefficient-to-Fourier map is

$$P = F^{-1} \Pi J_c, \quad P_0 = e^{-A/u} P. \quad (\text{RPC3})$$

These are invertible by the complete original period determinant PDE3–13. The contours for P_0 may be taken to be the rays based at $w = 0$ with the original five angles. Indeed translating the old rays gives rays based at $-1/2$. Join corresponding finite endpoints and close each contour at radius R . The integrand is entire. On the bounded-length outer joining curves its exponent is $-R^5/(5|u|) + O(R^4)$, uniformly along the joining curves, so the outer integrals tend to zero. The common finite joining path occurs with opposite coefficients in $-\ell_0 + \ell_j$ and cancels exactly. Cauchy's integral theorem therefore proves this contour identity, including its signs and the factor $e^{A/u}$.

5.1 A flat alternating pairing with all constants

On $\mathbb{C}[w]/(H)$ define the bilinear, rather than sesquilinear, pairing

$$\mathcal{J}(p, q) = \sum_{H(r)=0} \frac{p(r)q(-r)}{H'(r)}. \quad (\text{RPC4})$$

The roots are simple at the original packet. The equivalent polynomial formula is the coefficient of w^3 in the remainder of $p(w)q(-w)$ modulo H . To prove it, use Lagrange interpolation: the coefficient of w^3 in the idempotent at r is $1/H'(r)$. This polynomial formula extends the pairing to every complex value of β, η , including repeated roots, without a division by a repeated-root value. In the increasing frame $1, w, w^2, w^3$, multiplication by w and this pairing are the exact matrices

$$M = \begin{pmatrix} 0 & 0 & 0 & -\eta \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\beta \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & \beta \\ 1 & 0 & -\beta & 0 \end{pmatrix}. \quad (\text{RPC5})$$

For the last matrix, the coefficient functional is zero through degree two, one in degree three, zero in degree four and six, and $-\beta$ in degree five. Inserting the sign $(-1)^j$ from the second argument gives every displayed entry. Thus $J^T = -J$, $\det J = 1$, and direct multiplication gives $M^T J + J M = 0$.

Allow the two original odd-phase coefficients to vary complexly, solely to calculate the existing period map. The ray domination $e^{-r^5/(5|u|)+O(r^3+r)}$ on compact parameter sets justifies all coefficient derivatives and integrations by parts below. Ordinary division of w^{j+3} by H , followed by differential reduction, gives

$$\partial_\eta P_0 = \frac{1}{u} P_0 M, \quad \partial_\beta P_0 = \frac{1}{3u} P_0 (M^3 - uN), \quad N = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{RPC6})$$

For clarity, the quotient derivatives for columns $j = 0, 1, 2, 3$ are $0, 0, 1, 2w$, respectively; hence this is the full correction matrix. The two outer boundary terms vanish, and the two values at the common origin cancel. No differential remainder was replaced by an arithmetic remainder in RPC6.

There is also a direct nonvanishing proof on this entire two-parameter family. The displayed matrices satisfy $\text{Tr } M = 0$, $\text{Tr } M^3 = 0$ and $\text{Tr } N = 0$. The first and third follow from their diagonals; the second follows by multiplying RPC5 three times, or from the vanishing third Newton sum of the even monic polynomial H . Multilinearity of the determinant in its columns then gives $\partial_\eta \det P_0 = \partial_\beta \det P_0 = 0$ from RPC6, without first assuming an inverse exists. At the origin the literal Gamma ray integrals give the diagonal matrix G_u defined in RPC8, whose determinant is nonzero. Hence P_0 is invertible at every point of $\mathbb{C}^2$, justifying the inverses throughout this family.

The matrix identity $M^\top J + JM = 0$ also gives $(M^3)^\top J + JM^3 = 0$. Direct multiplication of RPC5–6 gives

$$\partial_\beta J + \frac{1}{3}(N^\top J + JN) = 0. \quad (\text{RPC7})$$

In particular $\partial_\beta J$ has entries $(2, 3) = 1$, $(3, 2) = -1$ when indices begin at zero, while $N^\top J + JN$ has the opposite threefold entries. Differentiate the inverse matrices in $P_0^{-\top} J P_0^{-1}$. RPC6–7 show that both its β and its η derivatives vanish. This matrix is consequently constant on the connected coefficient space $\mathbb{C}^2$.

Define the unchanged positive-Gamma ray constants

$$g_a = 5^{a/5-1} e^{\pi i a/5} \Gamma(a/5), \quad G = \text{diag}(g_1, g_2, g_3, g_4), \quad G_u = \text{diag}(u^{a/5} g_a)_{a=1}^4. \quad (\text{RPC8})$$

At $\beta = \eta = 0$, literal substitution in each original ray integral gives $P_0 = G_u$. Every g_a is nonzero, since its Gamma factor is the integral of a strictly positive function. We have therefore proved the complete flat-pairing identity

$$\begin{aligned} \Omega_0 &= P_0^{-\top} J P_0^{-1} = G_u^{-\top} J|_{\beta=0} G_u^{-1}, \\ (\Omega_0)_{14} &= -\frac{1}{u g_1 g_4}, \quad (\Omega_0)_{23} = \frac{1}{u g_2 g_3}, \quad (\Omega_0)_{ba} = -(\Omega_0)_{ab}, \\ \Omega &= P^{-\top} J P^{-1} = e^{-2A/u} \Omega_0. \end{aligned} \quad (\text{RPC9})$$

All entries not listed are zero; the indices in RPC9 begin at one. This formulation retains the Gamma constants directly and requires no unevaluated normalization of a period or source mass. For $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$, the two paired sums of weights are five. It follows entry by entry that

$$D^\top \Omega D = \Omega, \quad D^\top \Omega_0 D = \Omega_0. \quad (\text{RPC10})$$

5.2 The full arithmetic unit and the exact quadratic form

Retain the original $v_h = 2\xi/h$ and its complete simple-root unit values. Reflection and conjugation of this original function give

$$v_h(\alpha_1) = v_h(\alpha_4) = a, \quad v_h(\alpha_2) = v_h(\alpha_3) = \bar{a}, \quad a \neq 0. \quad (\text{RPC11})$$

Indeed both numerator and denominator are invariant under $s \mapsto 1-s$ and conjugate under $s \mapsto \bar{s}$, first away from the roots and then by the removable values. The complete zero orders give nonzero values at these simple roots. Write

$$c_3 = \frac{a^{-2} - \bar{a}^{-2}}{4i\delta\gamma}, \quad c_1 = \frac{a^{-2} + \bar{a}^{-2}}{2} - (\delta^2 - \gamma^2)c_3, \quad \mathbf{r}(w) = c_1w + c_3w^3. \quad (\text{RPC12})$$

Both coefficients are real, and $(c_1, c_3) \neq (0, 0)$. Directly, $c_1 + c_3r_1^2 = a^{-2}$ and $c_1 + c_3r_2^2 = \bar{a}^{-2}$; the same equalities hold at the respective opposite roots. Thus $\mathbf{r}(M) = M v_h(M + 1/2)^{-2}$ in the original finite algebra, with the entire function evaluated by its complete original arithmetic unit. RPC12 is an exact interpolation of this specific multiplier.

For $Q_0(t) = t_1t_4 - t_2t_3$, let J_{prim} be the original Lagrange coordinate isomorphism and set

$$B = F^{-1}\Pi U J_{\text{prim}}, \quad Q(x) = Q_0(B^{-1}x), \quad \mathcal{H} = P\mathbf{r}(M)P^{-1}. \quad (\text{RPC13})$$

If $p = P^{-1}x$, the primary components of $B^{-1}x$ are exactly $p(r_1)/a, p(r_2)/\bar{a}, p(r_3)/\bar{a}, p(r_4)/a$. Since $H'(r) = 4r(r^2 + \gamma^2 - \delta^2)$, its values give $r/H'(r) = 1/(8i\delta\gamma)$ on roots one and four, and $r/H'(r) = -1/(8i\delta\gamma)$ on roots two and three. Inserting these four values into RPC4 proves the exact identity

$$Q(x) = 4i\delta\gamma p^\top \mathbf{r}(M)^\top J p = 4i\delta\gamma x^\top \mathcal{H}^\top \Omega x. \quad (\text{RPC14})$$

There is no conjugate transpose in this quadratic identity. Because $\mathbf{r}$ is odd, $\mathbf{r}(M)^\top J + J\mathbf{r}(M) = 0$; hence the matrices in RPC14 are symmetric. This also proves $\mathcal{H}^\top \Omega + \Omega \mathcal{H} = 0$. The four eigenvalues of $\mathbf{r}(M)$ are exactly $r_1/a^2, \bar{r}_1/a^2, -\bar{r}_1/a^2, -r_1/a^2$. They are nonzero, although they need not be distinct. Both B and Q_0 are invertible as linear and quadratic matrices, respectively, so Q always has rank four, including the possible eigenvalue collisions.

By RPC10, substituting Dx into RPC14 transports $\mathcal{H}$ to $D^{-1}\mathcal{H}D$. The resulting symmetric quadratic matrices are equal if and only if the corresponding $\mathcal{H}$ matrices are equal, because Ω is invertible. Consequently

$$Q(Dx) = Q(x) \iff [\mathcal{H}, D] = 0 \iff \mathcal{H}_{ab} = 0 \text{ for every } a \neq b. \quad (\text{RPC15})$$

This is an exact test in the actual original period matrix and unit. The final equivalence uses the four distinct diagonal entries of D . It makes no assumption that their off-diagonal tests are independent.

For completeness, a rank-four quadratic semi-invariant of cyclic weight $j \neq 0$ is impossible: its coefficient in row j could pair only with the missing weight zero, so that row is zero. Thus any semi-invariant rank-four quadratic must have weight zero. Since five is prime, the projective orbit of Q under D has size one or five, and size one is exactly RPC15. Every failure of any displayed off-diagonal test proves size five.

5.3 An explicit original-parameter radius forcing orbit five

Here we prove that RPC15 fails for all sufficiently large $|u|$, with a completely specified finite radius for the given original quartet and unit. The argument evaluates the original periods; it does not choose or insert a replacement amplitude.

Set $b = \beta u^{-2/5}$, $c = \eta u^{-4/5}$, and let $T(b, c)$ be the Fourier period matrix for $z^5/5 + bz^3/3 + cz$ at $u = 1$, in columns $1, z, z^2, z^3$ and on the five rays of angles $(\pi + 2\pi j)/5$. Substitution $w = u^{1/5}z$, with exactly the original branch, gives

$$P_0 = T(b, c)S_u, \quad S_u = \text{diag}(u^{a/5})_{a=1}^4, \quad S_u M S_u^{-1} = u^{1/5}M(b, c), \quad T(0, 0) = G. \quad (\text{RPC16})$$

Every column includes its original differential factor. The matrix $M(b, c)$ is RPC5 with β, η replaced by b, c ; in particular $N_0 := M(0, 0)$ has its three subdiagonal entries equal to one.

The following positive constants are explicit finite real integrals or finite expressions in the original Gamma constants:

$$\begin{aligned}
E_j &= \frac{8}{5} \int_0^\infty r^j (r^3/3 + r) e^{-r^{5/5} + r^{3/3} + r} dr \quad (0 \leq j \leq 3), \\
C_T &= 2 \left(\sum_{j=0}^3 E_j^2 \right)^{1/2}, \quad g_+ = \max_a |g_a|, \quad g_-^{-1} = \max_a |g_a|^{-1}, \\
\kappa &= \max_{1 \leq a < 4} |g_{a+1}/g_a|, \quad K_0 = GN_0 G^{-1}, \\
C_K &= 4C_T g_-^{-1} + 2g_+ g_-^{-1} + 2g_+ g_-^{-2} C_T, \\
C_{K,3} &= C_K(3\kappa^2 + 3\kappa C_K + C_K^2), \quad \epsilon_0 = \min\{1, (2C_T g_-^{-1})^{-1}\}.
\end{aligned} \tag{RPC17}$$

Here g_-^{-2} means $(\max_a |g_a|^{-1})^2$. The negative fifth degree exponent proves convergence of every integral. All norms in the following estimates are the Euclidean operator norm, with the Frobenius norm used explicitly as an upper bound where stated.

Write $\epsilon = |b| + |c|$. When $|b|, |c| \leq 1$, on each ray

$$|e^{bz^3/3+cz} - 1| \leq \epsilon(r^3/3 + r)e^{r^{3/3}+r}.$$

Each contour has two rays, and each entry of F^{-1} has absolute value $1/5$ with four summands. The entry in column j of $T - G$ is therefore at most $E_j \epsilon$. Summing the squares over all four rows proves

$$\|T - G\| \leq \|T - G\|_F \leq C_T \epsilon. \tag{RPC18}$$

For $\epsilon \leq \epsilon_0$, the convergent inverse series for $I + G^{-1}(T - G)$ gives $\|T^{-1}\| \leq 2g_-^{-1}$ and $\|T^{-1} - G^{-1}\| \leq 2g_-^{-2} C_T \epsilon$. Also $\|M(b, c) - N_0\| \leq \epsilon$, $\|M(b, c)\| \leq 2$, and $\|N_0\| = 1$. Subtract the products $K := TM(b, c)T^{-1}$ and K_0 in three terms, changing the left factor, then the middle factor, then the inverse factor. The preceding inequalities give, with the complete RPC17 constants,

$$\|K - K_0\| \leq C_K \epsilon, \quad \|K\| \leq \kappa + C_K \epsilon, \quad \|K^3 - K_0^3\| \leq C_{K,3} \epsilon. \tag{RPC19}$$

For the last assertion use the exact noncommutative identity $K^3 - K_0^3 = (K - K_0)K^2 + K_0(K - K_0)K + K_0^2(K - K_0)$. Its norm is at most $C_K \epsilon$ times $(\kappa + C_K)^2 + \kappa(\kappa + C_K) + \kappa^2$, as claimed.

The original multiplication and full unit in RPC13 now give exactly

$$\mathcal{H} = c_1 u^{1/5} K + c_3 u^{3/5} K^3, \quad (K_0^3)_{41} = g_4/g_1, \quad (K_0)_{21} = g_2/g_1. \tag{RPC20}$$

The scalar $e^{A/u}$ cancels from this conjugation, while it remains in the quadratic form through RPC9. Put $B_* = \beta + \eta > 0$. For $|u| \geq 1$, $\epsilon \leq B_* |u|^{-2/5}$.

If $c_3 \neq 0$, define the completely specified radius

$$T_* = \max \left\{ 1, \frac{B_*}{\epsilon_0}, \frac{2(|c_3| C_{K,3} B_* + |c_1|(\kappa + C_K))}{|c_3 g_4/g_1|} \right\}, \quad R_* = T_*^{5/2}. \tag{RPC21}$$

For every u on its original branch with $|u| \geq R_*$, RPC19–20 imply

$$\left| u^{-3/5} \mathcal{H}_{41} - c_3 g_4/g_1 \right| \leq |u|^{-2/5} (|c_3| C_{K,3} B_* + |c_1|(\kappa + C_K)) \leq \frac{1}{2} |c_3 g_4/g_1|. \tag{RPC22}$$

In particular $\mathcal{H}_{41} \neq 0$. In the quadratic matrix RPC14 this is the nonzero $(1, 1)$ entry $4i\delta\gamma \mathcal{H}_{41} \Omega_{41}$, whose cyclic weight is two.

If $c_3 = 0$, then $c_1 \neq 0$. In this case define

$$T_* = \max \left\{ 1, \frac{B_*}{\epsilon_0}, \frac{2C_K B_*}{|g_2/g_1|} \right\}, \quad R_* = T_*^{5/2}. \tag{RPC23}$$

For every $|u| \geq R_*$, the same estimates give

$$\left| u^{-1/5} \mathcal{H}_{21} - c_1 g_2 / g_1 \right| \leq |c_1| C_K B_* |u|^{-2/5} \leq \frac{1}{2} |c_1 g_2 / g_1|. \quad (\text{RPC24})$$

Thus $\mathcal{H}_{21} \neq 0$. The $(1, 3)$ entry of the quadratic matrix is then $4i\delta\gamma\mathcal{H}_{21}\Omega_{23}$, which has cyclic weight four; symmetry retains its $(3, 1)$ partner. Both cases prove the unconditional conclusion for the actual original parameters and the explicit radius belonging to those parameters:

$$\boxed{|u| \geq R_* \implies \#\{[Q(D^j x)] : 0 \leq j < 5\} = 5.} \quad (\text{RPC25})$$

The bracket denotes the projective class of the quadratic polynomial. RPC21 or RPC23 supplies its radius for every nonzero original unit; neither case assumes a condition on an uncomputed period coefficient.

Finally, the original periods are holomorphic on the connected logarithm cover of $\mathbb{C}^*$, and on each branch domain: locally the contours may be held in common decay sectors and differentiated under the dominated integral, and contour deformation identifies these local definitions. Their determinant never vanishes. Hence all entries of $\mathcal{H}$ are holomorphic there. RPC22 or RPC24 proves that its respective decisive entry is not identically zero on the logarithm cover and, by the identity theorem on that connected cover, on any open branch domain. The zeros of that one entry are isolated, by its convergent local power series. The exact exceptional set

$$\mathcal{E} = \{u : \mathcal{H}_{ab}(u) = 0 \text{ for all } a \neq b\} \quad (\text{RPC26})$$

is a subset of those isolated zeros and satisfies $|u| < R_*$. Thus it has no accumulation inside the branch domain; accumulation at $u = 0$ or a branch boundary has not been excluded. RPC15 is the full calculation specifying this set. No assertion that it is empty at arbitrary finite complex u is made or used.

6 A finite bound for the original kernel volume from its actual dimension

The original packet, sources, and exact coordinate map. Throughout this calculation the original multiplicity is $m = 1$ and

$$k = 4l + 1, \quad l \geq 1, \quad q = (k + 1)^2 \geq 36, \quad c = k/2, \quad 0 < \delta < 1/2, \quad \gamma > 2. \quad (\text{OKV1})$$

Retain every original root and its two coordinates:

$$\begin{aligned} \chi(S) &= \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma], \\ \omega_{ab} &= (2b - k)\gamma - i(2a - k)\delta, \\ P(y) &= i^{-q}\chi(c + iy) = \prod_{a,b=0}^k (y - \omega_{ab}). \end{aligned} \quad (\text{OKV2})$$

The identity follows factor by factor from $iy - (2a - k)\delta - i(2b - k)\gamma = i(y - \omega_{ab})$; it retains the full phase i^q . The algebra map

$$\Phi : \mathbb{C}[y]/(P) \longrightarrow E_\chi := \mathbb{C}[S]/(\chi), \quad [r(y)] \longmapsto [r((S - c)/i)] \quad (\text{OKV3})$$

is well defined because $P((S - c)/i) = i^{-q}\chi(S)$. Its inverse is $[f(S)] \mapsto [f(c + iy)]$, as verified by substitution. Both maps preserve degrees and are inverse on each polynomial ring before taking the quotients. In particular division by P is the original remainder operation through this exact coordinate map.

Use only the two original source orders $s \in \{1, k\}$ and write

$$b_s = s/2, \quad M_s = (2\pi)^{s/2}, \quad dm_{0,s}(y) = \frac{M_s 2^{b_s-2}}{\pi \Gamma(b_s)} \left| \Gamma\left(\frac{b_s}{2} + \frac{iy}{2}\right) \right|^2 dy. \quad (\text{OKV4})$$

The exact source calculation IFB2–5 supplies mass M_s , the transform $\int e^{ty} dm_{0,s} = M_s (\cos t)^{-b_s}$ for $|t| < \pi/2$, and the original monic orthogonal polynomials and norms

$$\begin{aligned} p_0 &= 1, \quad p_1 = y, \quad p_{d+1} = yp_d - d(b_s + d - 1)p_{d-1}, \\ h_d^{(s)} &= \int |p_d|^2 dm_{0,s} = M_s d!(b_s)_d, \quad \varphi_d(y) = p_d(y)/\sqrt{h_d^{(s)}}, \quad u_d(S) = \varphi_d((S - c)/i). \end{aligned} \quad (\text{OKV5})$$

All these integrals are on the same real line, with the original source mass. For example the stated orthogonality and norms also follow directly from its transform and the exact generating function

$$\sum_{d \geq 0} p_d(y) \frac{z^d}{d!} = (1 + z^2)^{-b_s/2} e^{y \arctan z}.$$

Indeed the source integral of the product at sufficiently small real z, w is $M_s(1 - zw)^{-b_s}$. The cosine addition identity and the displayed transform give this equality; the source exponential bound justifies each finite coefficient derivative under the integral. The coefficients of $z^d w^e$ then give zero for $d \neq e$ and $M_s d!(b_s)_d$ for $d = e$. The initial coefficients are $1, y$, and differentiating the generating function gives precisely the recurrence in OKV5. Thus the source norms used below retain their original constants and all finite-degree factors.

Let $C_s : \mathbb{C}^q \rightarrow E_\chi$ have columns $[u_0], \dots, [u_{q-1}]$ in the original frame $(1, S, \dots, S^{q-1})$. Its exact leading coefficients give

$$\det C_s = i^{-q(q-1)/2} \prod_{d=0}^{q-1} (h_d^{(s)})^{-1/2} \neq 0.$$

Retain the original GMB vectors and matrices, including the last row:

$$v_j = C_s^{-1}[u_{q+j}], \quad K_j = I_q + \sum_{a=0}^{j-1} v_a v_a^*, \quad 0 \leq j \leq q+1. \quad (\text{OKV6})$$

The vector v_q has original polynomial degree $2q$.

Exact remainder coefficients in all original roots. Index the full list in OKV2 by $\omega_1, \dots, \omega_q$, with each pair (a, b) occurring exactly once. Define the elementary and complete homogeneous polynomials by their finite sums

$$e_h(\omega) = \sum_{1 \leq a_1 < \dots < a_h \leq q} \omega_{a_1} \cdots \omega_{a_h}, \quad h_t(\omega) = \sum_{n_1 + \dots + n_q = t} \prod_{a=1}^q \omega_a^{n_a}, \quad (\text{OKV7})$$

with $e_0 = h_0 = 1$, $e_h = 0$ outside $0 \leq h \leq q$, and nonnegative integer n_a in the second sum. The notation $h_t(\omega)$ in this paragraph is a polynomial in the roots; the original squared norm continues to be $h_d^{(s)}$ of OKV5. Multiplication of formal power series gives the exact identity

$$\left(\sum_{h=0}^q (-1)^h e_h(\omega) z^h \right) \left(\sum_{t \geq 0} h_t(\omega) z^t \right) = 1,$$

because each geometric series $(1 - \omega_a z)^{-1}$ multiplies its own factor $1 - \omega_a z$. Consequently, for $D = q + j$ with $0 \leq j \leq q$, the quotient and remainder of y^D upon division by P are exactly

$$\begin{aligned} Q_D(y) &= \sum_{t=0}^j h_t(\omega) y^{j-t}, \quad \text{rem}_P(y^D) = y^D - P(y)Q_D(y), \\ [y^a] \text{rem}_P(y^D) &= - \sum_{t=0}^j h_t(\omega) (-1)^{D-t-a} e_{D-t-a}(\omega), \quad 0 \leq a < q. \end{aligned} \quad (\text{OKV8})$$

To verify the quotient assertion, the power-series identity makes the coefficients of $y^D, y^{D-1}, \dots, y^q$ in PQ_D equal to those of y^D . The remaining coefficients are exactly the last line of OKV8. Their degree is less than q , so uniqueness of monic polynomial division proves both assertions. For $0 \leq D < q$ the remainder is y^D itself.

The actual maximal root modulus is

$$R = k\sqrt{\delta^2 + \gamma^2}, \quad D_* := \max\{3, \sqrt{\delta^2 + \gamma^2}\}, \quad T_* := D_* q. \quad (\text{OKV9})$$

Every $|\omega_{ab}| \leq R$, and equality holds at each corner, so this expression for R is exact. In particular $R \leq T_*$. Counting the summands of OKV7 proves

$$|e_h(\omega)| \leq \binom{q}{h} R^h, \quad |h_t(\omega)| \leq \binom{q+t-1}{t} R^t.$$

The second count follows by placing $q-1$ separators among $t+q-1$ positions to encode $(n_1, \dots, n_q)$. For fixed j , the binomial $\binom{q+t-1}{t}$ is increasing in $0 \leq t \leq j$, because its successive ratio is $(q+t)/(t+1) \geq 1$. In each nonzero summand of OKV8 the total root exponent is $(D-t-a)+t = D-a$. As t varies the indices $D-t-a$ are distinct, so the sum of their $\binom{q}{D-t-a}$ is at most 2^q . It follows that

$$|[y^a] \text{rem}_P(y^{q+j})| \leq 2^q \binom{q+j-1}{j} R^{q+j-a} \leq 2^{3q-1} R^{q+j-a}, \quad 0 \leq a < q, \quad 0 \leq j \leq q. \quad (\text{OKV10})$$

For the last inequality use $\binom{q+j-1}{j} \leq 2^{q+j-1} \leq 2^{2q-1}$, by the binomial expansion of $(1+1)^{q+j-1}$. These estimates retain the exact coefficient formula OKV8, including every signed cancellation.

Original Gamma norms of the low monomials. The same recurrence and norms in OKV5 give, without changing the measure,

$$y\varphi_a = \sqrt{(a+1)(b_s+a)} \varphi_{a+1} + \sqrt{a(b_s+a-1)} \varphi_{a-1}, \quad \varphi_{-1} = 0. \quad (\text{OKV11})$$

For a polynomial of degree at most $q-2$, its orthonormal coefficient vector is therefore sent by multiplication by y to the sum of one upward and one downward weighted shift, with output degree at most $q-1$. Each shift has operator norm at most $\sqrt{q(b_s+q)}$: the squared norm of its coefficient vector is bounded by the largest squared weight times the input squared norm. Since $s \leq k < q$ gives $b_s \leq q/2$, the triangle inequality yields

$$\|yf\|_{L^2(m_{0,s})} \leq 2\sqrt{q(b_s+q)} \|f\|_{L^2(m_{0,s})} \leq 3q \|f\|_{L^2(m_{0,s})} \leq T_* \|f\|_{L^2(m_{0,s})}.$$

Here $2\sqrt{3/2} = \sqrt{6} < 3$. Starting from $\|1\|_{L^2(m_{0,s})} = \sqrt{M_s}$ and applying this inequality to $1, y, \dots, y^{a-1}$ proves

$$\boxed{\|y^a\|_{L^2(m_{0,s})} \leq \sqrt{M_s} T_*^a, \quad 0 \leq a < q.} \quad (\text{OKV12})$$

All intermediate polynomial degrees satisfy the stated operator domain.

For any polynomial $f = \sum_a f_a y^a$, write $\mathbf{c}_{T_*}(f) = \sum_a |f_a| T_*^a$. This is a finite coefficient sum, not a change of source or quotient norm. The exact original recurrence gives

$$\mathbf{c}_{T_*}(p_{d+1}) \leq T_* \mathbf{c}_{T_*}(p_d) + d(b_s + d - 1) \mathbf{c}_{T_*}(p_{d-1}).$$

For $1 \leq d < 2q$ its actual positive coefficient is bounded by $d(b_s + d - 1) \leq 2q(b_s + 2q) \leq 5q^2 \leq T_*^2$. The initial values are $1, T_*$. Induction now proves

$$\boxed{\mathbf{c}_{T_*}(p_d) \leq (2T_*)^d, \quad 0 \leq d \leq 2q.} \quad (\text{OKV13})$$

Indeed substituting the two preceding bounds in the recurrence gives $T_*(2T_*)^d + T_*^2(2T_*)^{d-1} = \frac{3}{4}(2T_*)^{d+1} \leq (2T_*)^{d+1}$.

Combining all q coefficients of OKV10 with OKV12, by the triangle inequality in the original L^2 space, gives

$$\|\text{rem}_P(y^D)\|_{L^2(m_{0,s})} \leq \sqrt{M_s} q 2^{3q} T_*^D \quad (0 \leq D \leq 2q).$$

For $D \geq q$, each summand is bounded by $\sqrt{M_s} 2^{3q-1} R^{D-a} T_*^a \leq \sqrt{M_s} 2^{3q-1} T_*^D$ and there are q summands. For $D < q$, the assertion follows from OKV12 and $q 2^{3q} \geq 1$. Linearity of the exact remainder map, followed by OKV13, proves

$$\boxed{\|\text{rem}_P(p_d)\|_{L^2(m_{0,s})} \leq \sqrt{M_s} q 2^{3q} (2T_*)^d, \quad 0 \leq d \leq 2q.} \quad (\text{OKV14})$$

The original remainder vectors and the full matrix norm. The polynomial $\text{rem}_P(p_d)$ has degree less than q . Its expansion in the orthonormal basis $\varphi_0, \dots, \varphi_{q-1}$ has squared coefficient norm equal to its squared original $L^2(m_{0,s})$ norm, by expanding the finite inner product and using OKV5. The exact quotient map OKV3 and definition OKV6 therefore give, for $d = q + j$, $0 \leq j \leq q$,

$$\|v_j\|_2^2 = \frac{\|\text{rem}_P(p_{q+j})\|_{L^2(m_{0,s})}^2}{M_s (q+j)! (b_s)_{q+j}} \leq \frac{q^2 2^{6q} (2T_*)^{2(q+j)}}{(q+j)! (b_s)_{q+j}}. \quad (\text{OKV15})$$

This equality is the actual map of the original quotient coordinates to the remainder polynomial. The full source factor M_s occurs in both the numerator bound and its original norm denominator before the displayed cancellation. No estimate on the conditioning of C_s or a companion matrix is used.

Since $b_s \geq 1/2$, for each integer $d \geq 1$ the first factor of $(b_s)_d$ is at least $1/2$ and every subsequent factor is at least one. Thus $d!(b_s)_d \geq 1/2$. Also $T_* \geq 1$ and $q + j \leq 2q$. Consequently

$$\|v_j\|_2^2 \leq 2q^2 2^{6q} (2D_* q)^{4q}, \quad 0 \leq j \leq q,$$

and the original complete matrix obeys

$$\boxed{1 \leq \|K_{q+1}\|_2 \leq 1 + 2(q+1)q^2 2^{6q} (2D_*q)^{4q}}. \quad (\text{OKV16})$$

To check its norm bound, $\|v_j v_j^*\|_2 = \|v_j\|_2^2$ follows directly from Cauchy–Schwarz, with equality on a nonzero v_j ; it is zero for a zero vector. Add all $q+1$ summands of OKV6 and the norm of I_q . Positivity also proves its lower bound.

Here is an explicit bound of the requested scale, with constants depending only on the original two root coordinates:

$$\boxed{\log \|K_{q+1}\|_2 \leq C(\delta, \gamma) q \log(q+1), \quad C(\delta, \gamma) = 12 + 4 \log(2D_*)}. \quad (\text{OKV17})$$

For a direct verification, put $U = 2(q+1)q^2 2^{6q} (2D_*q)^{4q} \geq 1$ and use $\log(1+U) \leq \log(2U)$. This upper bound is exactly

$$\log 4 + \log(q+1) + 2 \log q + 6q \log 2 + 4q \log(2D_*) + 4q \log q.$$

Write $L = \log(q+1)$. Since $q \geq 36$, one has $\log 4 \leq L$, $\log q \leq L$, $6 \log 2 \leq 2L$, and $L \geq 1$. The last inequality follows already from $\log 4 = \int_1^4 dt/t > 1$, by bounding the integrand below by $1/2$ on $[1, 2]$ and $1/4$ on $[2, 4]$. The preceding display is therefore at most $4L + 6qL + 4 \log(2D_*)qL$, which is bounded by the right side of OKV17. The completely finite and sharper bound OKV16 remains available before this logarithmic estimate.

The four original kernel Grams with their full coefficient frame. Retain the original surjection $\Lambda : E_\chi \rightarrow B_{\text{obs}}$, the actual subspace $\mathcal{K}_\Lambda = \ker \Lambda$, and its original inclusion $I : \mathcal{K}_\Lambda \hookrightarrow E_\chi$. Put $r_K = \dim \mathcal{K}_\Lambda$; no separate formula or upper bound for this actual dimension is needed here. Use its fixed original coefficient frame and write

$$T_s = C_s^{-1} I, \quad G_{q+j-1}^{(s)} = C_s^{-*} K_j^{-1} C_s^{-1}, \quad H_j^{(s)} = I^* G_{q+j-1}^{(s)} I = T_s^* K_j^{-1} T_s, \quad 0 \leq j \leq q+1. \quad (\text{OKV18})$$

For completeness, the quotient metric in this identity follows from the original orthonormal source-coordinate map $A_j = [I_q, v_0, \dots, v_{j-1}]$. It obeys $A_j A_j^* = K_j$. The minimum norm solution of $A_j z = x$ is $z = A_j^* K_j^{-1} x$: it has the required image and is orthogonal to $\ker A_j$. Every other solution differs by such a kernel vector, so the Pythagorean identity gives minimum squared norm $x^* K_j^{-1} x$. The actual remainder is $C_s x$. Substitution proves the original quotient Gram and then its full restriction in OKV18.

Let $\lambda_s = \|K_{q+1}\|_2$. The original update chain gives $I_q = K_0 \preceq K_j \preceq K_{q+1} \preceq \lambda_s I_q$. For positive definite matrices, inversion reverses the order: if $A \preceq B$, conjugate by $A^{-1/2}$ to obtain a matrix with eigenvalues at least one, invert those eigenvalues and conjugate back. Thus the full original kernel forms satisfy

$$\lambda_s^{-1} H_0^{(s)} \preceq H_j^{(s)} \preceq H_0^{(s)}, \quad H_0^{(s)} = T_s^* T_s, \quad H_{j+1}^{(s)} \preceq H_j^{(s)}. \quad (\text{OKV19})$$

For $r_K > 0$, the matrix T_s is injective, so $H_0^{(s)}$ is positive definite. Conjugation in OKV19 by its inverse square root gives eigenvalues in $[\lambda_s^{-1}, 1]$. The determinant identity for that exact congruence is

$$\det[(H_0^{(s)})^{-1/2} H_j^{(s)} (H_0^{(s)})^{-1/2}] = \frac{\det H_j^{(s)}}{\det(T_s^* T_s)}.$$

Therefore

$$-r_K \log \lambda_s \leq \log \det H_j^{(s)} - \log \det(T_s^* T_s) \leq 0. \quad (\text{OKV20})$$

In particular the determinant and conditioning of the actual $T_s^* T_s$ are retained in this identity; no new kernel basis is substituted for its original frame.

Define the original four-endpoint kernel volume, with its literal degrees $q-1, q, 2q-1, 2q$, by

$$F_K^{(s)} = \log \det H_0^{(s)} + \log \det H_1^{(s)} - \log \det H_q^{(s)} - \log \det H_{q+1}^{(s)}. \quad (\text{OKV21})$$

Monotonicity in OKV19 makes each of $\log \det H_0^{(s)} - \log \det H_q^{(s)}$ and $\log \det H_1^{(s)} - \log \det H_{q+1}^{(s)}$ nonnegative. Each is at most $r_K \log \lambda_s$ by OKV20. Consequently the actual kernel volume satisfies the complete finite estimates

$$\begin{aligned} 0 \leq F_K^{(s)} &\leq 2r_K \log \|K_{q+1}\|_2 \\ &\leq 2r_K \log [1 + 2(q+1)q^2 2^{6q} (2D_* q)^{4q}] \\ &\leq 2r_K C(\delta, \gamma) q \log(q+1), \quad s \in \{1, k\}. \end{aligned} \quad (\text{OKV22})$$

For $r_K = 0$, every kernel determinant has its original empty value one, and $F_K^{(s)} = 0$. Hence OKV22 includes this case. The four copies of $\log \det(T_s^* T_s)$ in OKV21 have coefficients $+1, +1, -1, -1$ and cancel exactly only after their presence in OKV20 has been recorded.

The original boundary and arithmetic transitions. At each original degree set

$$Q_j^{(s)} = \left(\Lambda(G_{q+j-1}^{(s)})^{-1} \Lambda^* \right)^{-1}, \quad S_j^{(s)} = (G_{q+j-1}^{(s)})^{-1} \Lambda^* Q_j^{(s)}. \quad (\text{OKV23})$$

Their zero dimensional cases have their empty meaning. Surjectivity of Λ and positivity give the displayed inverses. Direct multiplication proves $\Lambda S_j^{(s)} = I$ and $I^* G_{q+j-1}^{(s)} S_j^{(s)} = 0$. For any fixed original section S_* of Λ , the map $[I, S_*]$ is bijective: its inverse records Λz and the unique kernel coordinate of $z - S_* \Lambda z$. The minimum section differs from S_* by a kernel column. The two frames therefore differ by a block triangular matrix with identity diagonal and determinant one. Taking the determinant of the full metric in the orthogonal frame $[I, S_j^{(s)}]$ gives

$$\det H_j^{(s)} \det Q_j^{(s)} = |\det[I, S_*]|^2 \det G_{q+j-1}^{(s)}. \quad (\text{OKV24})$$

This proves the exact relation with all kernel and boundary directions retained. The same frame factor occurs at all four endpoints. Their signed sum is consequently

$$F_K^{(s)} + F_B^{(s)} = \mathcal{B}_k^{(s)}, \quad F_B^{(s)} = \log \det Q_0^{(s)} + \log \det Q_1^{(s)} - \log \det Q_q^{(s)} - \log \det Q_{q+1}^{(s)}. \quad (\text{OKV25})$$

The original matrices $G_{q+j-1}^{(s)}$ decrease with j , so their inverse forms increase. Compression by Λ and inversion then show that $Q_j^{(s)}$ decreases as well. Thus $F_B^{(s)} \geq 0$, and the finite companion bound is

$$\max\{0, \mathcal{B}_k^{(s)} - 2r_K C(\delta, \gamma) q \log(q+1)\} \leq F_B^{(s)} \leq \mathcal{B}_k^{(s)}. \quad (\text{OKV26})$$

Here the finer middle bound in OKV22 can be used in place of its last line, with the same proof.

Finally the original source identifications are $\mathcal{B}_k^{(1)} = \mathcal{B}_k^\sigma$ and $\mathcal{B}_k^{(k)} = \mathcal{B}_k^0$. Their already proved full source transition is $\mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0$, and the original arithmetic transition is $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma$, with its original finite signed allowance for δ_k^σ . Substitution of the proved exact splitting OKV25 gives, with every source and arithmetic term still present,

$$\mathcal{B}_k^{\text{ar}} = F_K^{(k)} + F_B^{(k)} + \mathcal{H}_k + \delta_k^\sigma = F_K^{(1)} + F_B^{(1)} + \delta_k^\sigma. \quad (\text{OKV27})$$

Thus OKV22 controls the original Gamma kernel summands in this same arithmetic identity. It keeps the separate source transition and arithmetic scalar with their established signs and bounds. The result is finite for the actual $r_K = \dim \ker \Lambda$ and uses no additional asserted dimension estimate or replacement source.

7 The actual simple-quartet kernel and boundary at the original volume scale

This section combines the complete original period calculation, the exact cyclic-orbit rank calculation, and the finite original Gamma kernel estimate. The hypotheses are the actual simple-quartet data already used in the programme, and the period range below has an explicit bound proved from those data. No orbit condition or kernel-rank premise is added to the statement.

7.1 The original domain and the proved period range

Fix the original simple quartet $m = 1$, its coordinates $0 < \delta < 1/2$, $\gamma > 2$, and its actual nonzero amplitude value $a = v_h(1/2 + \delta + i\gamma)$, where $v_h = 2\xi/h$ and every zero order in this quartet is complete. Reflection and conjugation give the other three unit values exactly as in PQO3 and RPC11. For every integer $l \geq 1$ retain

$$k = 4l + 1, \quad q = (k + 1)^2, \quad c = k/2, \quad s \in \{1, k\}. \quad (\text{OMV1})$$

The polynomial χ , the source measures $m_{0,s}$, their masses $(2\pi)^{s/2}$, the original observation $\Lambda_k : E_\chi \rightarrow B_k$, its kernel, and all four endpoint degrees are unchanged.

Let R_* be the finite positive real number defined by RPC17 and RPC21 when the actual c_3 of RPC12 is nonzero, and by RPC17 and RPC23 when it is zero. Those two explicitly evaluated alternatives exhaust the original nonzero unit, because c_1, c_3 cannot both vanish. In particular R_* depends on the fixed quartet and its actual unit and is independent of k . Its constants are finite positive Gamma values and the convergent positive integrals RPC17. The full period proof RPC18–25 establishes

$$|u| \geq R_* \implies \#\{[Q_u(D^j x)] : 0 \leq j < 5\} = 5 \quad (\text{OMV2})$$

on every original logarithm branch. To recall the actual nonvanishing calculation, RPC22 bounds the error in $u^{-3/5}\mathcal{H}_{41}$ by half the nonzero value $|c_3g_4/g_1|$ in the first alternative. RPC24 bounds the error in $u^{-1/5}\mathcal{H}_{21}$ by half $|c_1g_2/g_1|$ in the second. Each is a nonzero off-diagonal entry of the exact $\mathcal{H} = P(c_1M + c_3M^3)P^{-1}$; the flat residue identity RPC14–15 makes it a nonzero noninvariant coefficient of the actual quadric. The degree-five cyclic action and its nondegenerate quadratic matrix then give OMV2. All powers of u use its original branch, and the absolute error estimates hold for every such branch.

The exact invariant-ideal proof OQX1–3 now applies at every point of this proved domain. It gives, for all the original degrees OMV1,

$$\dim B_k = k^2 - 6k + 17, \quad r_K := \dim \ker \Lambda_k = 8k - 16. \quad (\text{OMV3})$$

Specifically, an invariant polynomial vanishing on the original quadric is divisible by all five distinct original orbit factors; its quotient by their invariant product is invariant. Thus the actual observed rank is $d_{5,k,0} - d_{5,k-10,0}$, with negative degrees zero. The full character calculation OQX3 evaluates this difference to $k^2 - 6k + 17$, including the two original low degrees 5, 9. Subtraction from $q = (k + 1)^2$ proves the second equality. This is the dimension of the kernel of the literal period observation, not an assigned ambient dimension or a presumed maximal rank.

7.2 A uniform finite bound on the actual kernel volume

Define, from the original root coordinates only,

$$D_* = \max\{3, \sqrt{\delta^2 + \gamma^2}\}, \quad C_* = 12 + 4 \log(2D_*), \quad \mathfrak{M}_k = (16k - 32)C_*q \log(q + 1). \quad (\text{OMV4})$$

Retain the exact original source-coordinate vectors $v_j = C_s^{-1}[u_{q+j}]$, $K_j = I + \sum_{a < j} v_a v_a^*$, and the full original kernel frame $T_s = C_s^{-1}I$ of OKV6,18. The original kernel metrics at degrees $q + j - 1$

are $H_j^{(s)} = T_s^* K_j^{-1} T_s$. Their actual four-endpoint volume satisfies

$$\boxed{0 \leq F_K^{(s)}(u) \leq (16k - 32) \log[1 + 2(q + 1)q^2 2^{6q} (2D_* q)^{4q}] \leq \mathfrak{M}_k, \quad s \in \{1, k\}, \quad |u| \geq R_*} \quad (\text{OMV5})$$

Here the source index and the actual period dependence of the kernel are retained. The proof is direct substitution of the now proved $r_K = 8k - 16$ into OKV22. Its finite coefficient proof retains every original root in the signed remainder formula, all Gamma recurrence coefficients through degree $2q$, and the full source mass before its exact norm-ratio cancellation. Its metric proof retains $\det(T_s^* T_s)$ at each of the four degrees, and cancels only the four copies with coefficients $+1, +1, -1, -1$. In particular neither the conditioning nor the changing position of the original period kernel has been suppressed. The bound is uniform in the entire period domain OMV2 because its right side depends only on k, δ, γ , and the dimension in OMV3 is the same at every point of that domain.

Write the four original total quotient baselines as $\mathcal{B}_k^{(1)} = \mathcal{B}_k^\sigma$ and $\mathcal{B}_k^{(k)} = \mathcal{B}_k^0$. The full determinant identity OKV24–25, with the original fixed coefficient section, gives

$$F_B^{(s)}(u) = \mathcal{B}_k^{(s)} - F_K^{(s)}(u), \quad \max\{0, \mathcal{B}_k^{(s)} - \mathfrak{M}_k\} \leq F_B^{(s)}(u) \leq \mathcal{B}_k^{(s)}. \quad (\text{OMV6})$$

Both factors in the exact splitting are nonnegative by monotonicity of their original minimum metrics. The literal boundary term is the actual period-vector expression

$$F_B^{(s)}(u) = \log \frac{\det \mathcal{M}_{2q}^{(s)}(u) \det \mathcal{M}_{2q+1}^{(s)}(u)}{\det \mathcal{M}_q^{(s)}(u) \det \mathcal{M}_{q+1}^{(s)}(u)}, \quad \mathcal{M}_D^{(s)}(u) = \sum_{d=0}^{D-1} \lambda_d^{(s)}(u) (\lambda_d^{(s)}(u))^*. \quad (\text{OMV7})$$

The vectors are the complete original primary-period coefficients OPG14 and OCS21 on the actual image of Λ_k . Formula OPG17 proves that each consecutive determinant ratio is $1 + \xi_j$; the unchanged weights $1, 2, \dots, 2, 1$ give exactly OMV7. The companion kernel vectors retain the minimum-section correction OPR2 and the action defect OPG22/OQC22. No invariance of that kernel under the original sum action is needed for OMV5–7.

7.3 The two individual leading coefficients

The complete original baseline proofs BSL16–19 and HBL establish, for fixed original δ, γ and $m = 1$, both source limits

$$\frac{\mathcal{B}_k^{(s)}}{q^2} \longrightarrow C_B = 9 - 8 \log 2 + F(2, \pi) > \frac{769}{17010}, \quad s = 1 \text{ and } s = k. \quad (\text{OMV8})$$

Here $F(2, \pi)$ is exactly the energy functional evaluated in those complete proofs; its original end-point convention and coefficients are retained. No value of it is redefined by the present comparison. Since $q = (k + 1)^2$,

$$\frac{\mathfrak{M}_k}{q^2} = C_* \frac{(16k - 32) \log((k + 1)^2 + 1)}{(k + 1)^2} \longrightarrow 0.$$

For an elementary verification, $((k + 1)^2 + 1) < (k + 2)^2$, and the last fraction is at most $32 \log(k + 2)/k$. The inequality $\log x = \int_1^x dt/t \leq 2(\sqrt{x} - 1)$ for $x \geq 1$ proves that this upper bound tends to zero. The finite estimate OMV5 and the exact identity OMV6 therefore prove

$$\boxed{\frac{F_K^{(1)}(u)}{q^2}, \frac{F_K^{(k)}(u)}{q^2} \longrightarrow 0, \quad \frac{F_B^{(1)}(u)}{q^2}, \frac{F_B^{(k)}(u)}{q^2} \longrightarrow C_B.} \quad (\text{OMV9})$$

These limits are uniform for all $|u| \geq R_*$ and all original branches. Indeed the kernel error is bounded by $\mathfrak{M}_k/q^2$ uniformly, while the boundary error is at most $|\mathcal{B}_k^{(s)}/q^2 - C_B| + \mathfrak{M}_k/q^2$. The baseline is independent of u . This proves the asserted uniformity, rather than only a conclusion for each separately fixed observation. The original combined coefficient has now been resolved into its two individual contributions on the proved simple-quartet period domain.

7.4 Exact propagation to the arithmetic identity and the finer scale

Keep the original signed transitions $\mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0$ and $\delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma$. Their complete finite allowances, including the QGQ simple- multiplicity refinement, remain those of the full current receiver. The exact arithmetic receiving equality becomes

$$\begin{aligned}\mathcal{B}_k^{\text{ar}} - F_B^{(1)}(u) &= F_K^{(1)}(u) + \delta_k^\sigma, \\ \mathcal{B}_k^{\text{ar}} - F_B^{(k)}(u) &= F_K^{(k)}(u) + \mathcal{H}_k + \delta_k^\sigma.\end{aligned}\tag{OMV10}$$

This is substitution in OKV27, so every source and sign is the original one. In particular the finite upper and lower allowances are obtained by adding $[0, \mathfrak{M}_k]$ to the actual interval for δ_k^σ , or to the actual interval for $\mathcal{H}_k + \delta_k^\sigma$, respectively. The original proved relations $\delta_k^\sigma/(kq) \rightarrow 0$ and $\mathcal{H}_k/(kq) \rightarrow -\mathcal{C}_\Gamma/4$, together with $k/q \rightarrow 0$ and OMV5, prove that both differences in OMV10, divided by q^2 , tend to zero uniformly on the stated period domain. This is an exact comparison with the full arithmetic baseline. The literal arithmetic kernel and boundary metrics retain their own source until their corresponding metric comparison is applied.

The finer original Gamma return is still carried by an exact mixed identity:

$$\begin{aligned}F_B^{(1)}(u) - F_B^{(k)}(u) &= \mathcal{H}_k - (F_K^{(1)}(u) - F_K^{(k)}(u)), \\ -\mathfrak{M}_k &\leq F_K^{(1)}(u) - F_K^{(k)}(u) \leq \mathfrak{M}_k, \\ \mathcal{H}_k - \mathfrak{M}_k &\leq F_B^{(1)}(u) - F_B^{(k)}(u) \leq \mathcal{H}_k + \mathfrak{M}_k.\end{aligned}\tag{OMV11}$$

Subtract OMV6 for the two actual source orders to obtain the first line. Each kernel term lies in $[0, \mathfrak{M}_k]$, so their difference lies in the stated interval, proving the other two lines. The error bound here is of order $kq \log(q+1)$, whereas the already proved signed return $\mathcal{H}_k$ has order kq . No sign or limiting coefficient for either finer mixed difference is assigned by OMV11. Its explicit remaining term is the difference of the two original kernel metrics, on the exactly calculated $8k-16$ dimensional period kernel. The full source multiplier and its action on this actual kernel remain the input to that next calculation, rather than a freely selected source order or kernel.

8 The original arithmetic kernel and boundary receive the full source comparison

This calculation uses the proved full-polynomial comparison TW12/TWA3, the original observation OPG1–5, and the complete original Gamma kernel estimate OKV1–27. It gives the exact maps between their source, quotient, kernel and boundary metrics, before taking any determinant. The subsequent simple-quartet conclusion uses the evaluated original period test RPC1–26 and its exact rank calculation OQX1–3. Every source order below is one of the already present orders $1, k$.

8.1 The complete original source and the same observation

Retain the original full-order quartet, with $0 < \delta < 1/2$, $\gamma > 2$, and positive integer complete zero order m . Thus

$$\begin{aligned} \mathcal{R} &= \{\tfrac{1}{2} + \delta + i\gamma, \tfrac{1}{2} + \delta - i\gamma, \tfrac{1}{2} - \delta + i\gamma, \tfrac{1}{2} - \delta - i\gamma\}, \quad h(z) = \prod_{\alpha \in \mathcal{R}} (z - \alpha)^m, \quad v_h = 2\xi/h, \\ k &\geq 3, \quad c = k/2, \quad e = 1 + k(m-1), \quad q = e(k+1)^2, \\ \lambda_{a,b} &= c + (2a-k)\delta + i(2b-k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^e, \quad E = \mathbb{C}[S]/(\chi), \\ w_h(y) &= |v_h(1/2 + iy)|^2/(2\pi), \quad m_{h,k} = w_h^{*k}, \quad d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy/(2\pi). \end{aligned} \quad (\text{AMT1})$$

The original source inner products, conjugate-linear in their first argument, are

$$\langle P, Q \rangle_{\text{ar}} = \int_{\mathbb{R}} \overline{P(c + iy)} Q(c + iy) m_{h,k}(y) dy, \quad \langle P, Q \rangle_{\sigma} = \int_{\mathbb{R}} \overline{P(c + iy)} Q(c + iy) d\sigma(y). \quad (\text{AMT2})$$

No measure mass or complex-coordinate phase is changed. In particular $\int d\sigma = \sqrt{2\pi}$. Write $\mathcal{P}_N$ for polynomials in the original S of degree at most N . In its coefficient frame $(1, S, \dots, S^N)$ let the two Grams be $H_N^{\text{ar}}, H_N^{\sigma}$. For $N \geq q-1$, the literal remainder map $J_N : \mathcal{P}_N \rightarrow E$ is onto and has kernel $\chi \mathcal{P}_{N-q}$; at $N = q-1$ this is the zero vector space. The frame on E is $(1, S, \dots, S^{q-1})$.

Here is the original observation being retained. Put $n = 4m$, $E_h = \mathbb{C}[z]/(h)$, $U = M_{j_h v_h}$, and $\iota[P] = [P(z_1 + \dots + z_k)] \in E_h^{\otimes k}$. The map j_h records every divided Taylor coefficient through order $m-1$ at every original root. Its unit $j_h v_h$ is invertible because the stated order is complete. Fix the original $u \neq 0$ and its logarithm branch. With $\Phi'_h = h$ and $\Phi_h(0) = 0$, let ℓ_j be the outward ray of angle $(\arg u + \pi + 2\pi j)/(n+1)$ and $\Gamma_j = -\ell_0 + \ell_j$. The period map and projector are exactly

$$\begin{aligned} (\Pi[P])_j &= \int_{\Gamma_j} e^{\Phi_h(z)/u} \text{rem}_h P(z) dz, \quad 1 \leq j \leq n, \\ C : \Gamma_j &\mapsto \Gamma_{j+1} - \Gamma_1 \quad (j < n), \quad \Gamma_n \mapsto -\Gamma_1, \quad P_k = \frac{1}{n+1} \sum_{a=0}^n (C^{-a})^{\otimes k}, \end{aligned} \quad (\text{AMT3})$$

$$L = P_k \Pi^{\otimes k} U^{\otimes k} \iota, \quad B = \text{im } L, \quad j : B \hookrightarrow (\mathbb{C}^n)^{\otimes k}, \quad j\Lambda = L, \quad \Lambda : E \twoheadrightarrow B, \quad K = \ker \Lambda.$$

These are the convergent original representative integrals; their existence and full determinant are proved in PDE and OPG. Fix the original coefficient inclusion $I : K \hookrightarrow E$ and section $S_* : B \rightarrow E$ with $\Lambda S_* = I_B$. Let $r = \dim K$ and $b = \dim B = q - r$. All matrices below use these same frames and this same full unit and period observation. Empty Grams have determinant one, and maps to or from a zero space have their unique empty values.

8.2 All constants of the actual full-polynomial inequality

Retain exactly the constants of TW and TWA:

$$\begin{aligned} \alpha &= \pi/2, \quad p = 8m - \frac{9}{2}, \quad B_h = 42 + 8m, \quad M = 21 + 4m, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy > 0, \quad M_\alpha = \int_{\mathbb{R}} e^{\alpha y} w_h(y) dy \in (0, \infty), \\ c_\Gamma &= \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M_\alpha^{k-1}, \quad D_N = [1 + 4(N+M)^2]^M. \end{aligned} \tag{AMT4}$$

Here c_h is the proved original convolution lower-envelope constant TW7 and C_h^{tilt} is the original compact/tail maximum BT12. Their full constructions occur in the preserved arithmetic endpoint and BT providers; they depend on this fixed h and not on k, N, u . Their defining inequalities, proved there with the original numerator, are

$$\begin{aligned} m_{h,k}(y) &\geq c_h \vartheta_h^{k-3} e^{-\alpha(|y|+k-3)} (k-2+|y|)^{-B_h}, \\ m_{h,k}(y) &\leq k C_h^{\text{tilt}} M_\alpha^{k-1} e^{-\alpha|y|} (1+|y|/k)^{-p}, \\ c_\Gamma e^{-\alpha|y|} (1+|y|)^{-1/2} &\leq d\sigma/dy \leq C_\Gamma e^{-\alpha|y|} (1+|y|)^{-1/2}. \end{aligned} \tag{AMT5}$$

For clarity the full polynomial step is given here. The first and third bounds, $k-2+|y| \leq (k-2)(1+|y|)$, and $(1+|y|)^{B_h} \leq 2^{B_h/2}(1+y^2)^M$, give $m_{h,k} \geq a_k(1+y^2)^{-M} d\sigma/dy$. The second and third, using $1+|y|/k \geq (1+|y|)/k$ and $p > 1/2$, give $m_{h,k} \leq A_k d\sigma/dy$. The fixed Gamma orthonormal polynomials obey

$$y e_j = \sqrt{(j+1)(j+1/2)} e_{j+1} + \sqrt{j(j-1/2)} e_{j-1}.$$

The two shifts on degree at most N have norms at most $N+1$ and N , respectively. Thus $\|yP\|_\sigma \leq 2(N+1)\|P\|_\sigma$. Iterating at the successively increased degrees and expanding $(1+y^2)^M$ proves $\int |P|^2 (1+y^2)^M d\sigma \leq D_N \int |P|^2 d\sigma$. Cauchy–Schwarz applied to $|P|(1+y^2)^{-M/2}$ and $|P|(1+y^2)^{M/2}$ then gives the reciprocal lower bound. The coordinate map $P(S) \mapsto P(c+iy)$ has inverse $f(y) \mapsto f((S-c)/i)$, preserves the degree and retains every coefficient through this explicitly invertible substitution. It therefore proves on the complete original source

$$\boxed{\frac{a_k}{D_N} H_N^\sigma \preceq H_N^{\text{ar}} \preceq A_k H_N^\sigma.} \tag{AMT6}$$

All coefficients can be complex in this proof. In particular AMT6 holds on every affine remainder fibre, not only on monic vectors. The two Grams are positive: the Gamma density is positive and the lower bound in AMT5 is positive everywhere. Their finite moments exist by the exponential upper bound.

8.3 Exact remainder, minimum-section, kernel and boundary maps

For $a \in \{\sigma, \text{ar}\}$ put

$$\begin{aligned} G_N^a &= (J_N (H_N^a)^{-1} J_N^*)^{-1}, \quad V_N^a = (H_N^a)^{-1} J_N^* G_N^a : E \longrightarrow \mathcal{P}_N, \\ H_{K,N}^a &= I^* G_N^a I, \quad Q_N^a = (\Lambda (G_N^a)^{-1} \Lambda^*)^{-1}, \quad S_N^a = (G_N^a)^{-1} \Lambda^* Q_N^a : B \longrightarrow E. \end{aligned} \tag{AMT7}$$

Inverses are on the indicated nonzero spaces. For an onto matrix J and positive H , direct multiplication gives $JV = I$ for $V = H^{-1} J^* (JH^{-1} J^*)^{-1}$. Moreover $Z^* H V = Z^* J^* G = 0$ for $JZ = 0$. Therefore every lift of z is $Vz + Z$, with squared norm $z^* G z + Z^* H Z$, proving its unique minimum and its Gram. Applying this proof first to J_N , then to Λ , proves every formula AMT7 and $\Lambda S_N^a = I_B$, $I^* G_N^a S_N^a = 0$. It also proves that $V_N^a S_N^a$ is the minimum lift for the composite ΛJ_N .

The exact comparison map on $\mathcal{P}_N$ is the identity on original coefficients; it commutes with J_N and ΛJ_N . It induces the identity on E, K, B . Between the two spaces of minimum lifts the exact bijection and its inverse are

$$\text{im } V_N^\sigma \longrightarrow \text{im } V_N^{\text{ar}}, \quad V_N^\sigma z \longmapsto V_N^{\text{ar}} z, \quad T_N = V_N^{\text{ar}} J_N|_{\text{im } V_N^\sigma}, \quad T_N^{-1} = V_N^\sigma J_N|_{\text{im } V_N^{\text{ar}}}. \quad (\text{AMT8})$$

Their difference $V_N^{\text{ar}} z - V_N^\sigma z$ lies in $\chi \mathcal{P}_{N-q}$, with its unique coefficient obtained by division by the unchanged monic χ . This also gives the minimum-lift bijection on the preimage of K and the analogous bijection $V_N^\sigma S_N^\sigma y \mapsto V_N^{\text{ar}} S_N^{\text{ar}} y$ for boundary lifts. The latter differences lie in $\ker(\Lambda J_N)$, exactly as multiplication by ΛJ_N verifies.

Minimize both sides of AMT6 over $J_N P = z$. Taking infima preserves each inequality because these are the same nonempty fibres. Restrict the resulting inequality to $z = Ix$, and independently minimize it over $\Lambda z = y$. This proves the three full-vector comparisons

$$\begin{aligned} (a_k/D_N) G_N^\sigma &\preceq G_N^{\text{ar}} \preceq A_k G_N^\sigma, \\ (a_k/D_N) H_{K,N}^\sigma &\preceq H_{K,N}^{\text{ar}} \preceq A_k H_{K,N}^\sigma, \\ (a_k/D_N) Q_N^\sigma &\preceq Q_N^{\text{ar}} \preceq A_k Q_N^\sigma. \end{aligned} \quad (\text{AMT9})$$

The source identity is an exact bounded bijection with the two constants in AMT6. No invariance of K under an additional multiplier was used to obtain its induced identities.

The matrix $[I, S_*]$ is invertible: its inverse sends z to the unique kernel coordinate of $z - S_* \Lambda z$ and the boundary coordinate Λz . Each $S_N^a - S_*$ lies in K , so the change from this frame to $[I, S_N^a]$ is triangular with identity diagonal and determinant one. The latter frame is orthogonal in G_N^a by AMT7. Consequently

$$\det H_{K,N}^a \det Q_N^a = |\det[I, S_*]|^2 \det G_N^a, \quad a \in \{\sigma, \text{ar}\}. \quad (\text{AMT10})$$

This includes $r = 0$ and $b = 0$ and retains the original frame factor.

8.4 The exact nested deficits and their rank factors

Define the actual finite source determinant deficit and its width

$$\Xi_{k,N} = (N+1) \log A_k - \log \frac{\det H_N^{\text{ar}}}{\det H_N^\sigma}, \quad L_N = \log(A_k D_N / a_k) \geq 0. \quad (\text{AMT11})$$

The last sign follows by applying AMT6 to any nonzero vector. Put

$$\begin{aligned} d_{E,N} &= q \log A_k - \log \frac{\det G_N^{\text{ar}}}{\det G_N^\sigma}, \\ d_{K,N} &= r \log A_k - \log \frac{\det H_{K,N}^{\text{ar}}}{\det H_{K,N}^\sigma}, \\ d_{B,N} &= b \log A_k - \log \frac{\det Q_N^{\text{ar}}}{\det Q_N^\sigma}. \end{aligned} \quad (\text{AMT12})$$

AMT9, by congruence with the inverse square root of each comparison Gram, places each eigenvalue of its relative Gram in $[a_k/D_N, A_k]$. Multiplication of all eigenvalues gives $0 \leq d_{K,N} \leq r L_N$, $0 \leq d_{B,N} \leq b L_N$ and $0 \leq d_{E,N} \leq q L_N$.

Here is the full source-to-quotient determinant calculation, including all relation directions. Choose an H_N^σ -orthonormal coefficient frame R_N of $\ker J_N$ solely for this computation, and put $W_N = [R_N, V_N^\sigma (G_N^\sigma)^{-1/2}]$. AMT7 gives $W_N^* H_N^{\text{ar}} W_N = I_{N+1}$. In this frame the positive contraction

$$\mathcal{D}_N = A_k^{-1} W_N^* H_N^{\text{ar}} W_N = \begin{pmatrix} F_N & C'_N \\ (C'_N)^* & A'_N \end{pmatrix} \preceq I_{N+1}$$

has Schur complement

$$T'_N = A'_N - (C'_N)^* F_N^{-1} C'_N = A_k^{-1} (G_N^\sigma)^{-1/2} G_N^{\text{ar}} (G_N^\sigma)^{-1/2}.$$

Indeed minimization in its first block is exactly the original remainder minimum at $z = (G_N^\sigma)^{-1/2} w$. Both F_N and T'_N are positive contractions: positivity follows from positive block elimination, and the upper bound for the Schur complement follows by testing its minimum at zero first coordinate. Since $|\det W_N|^2 = (\det H_N^\sigma)^{-1}$, elimination gives

$$\Xi_{k,N} = d_{R,N} + d_{E,N}, \quad d_{R,N} := -\log \det F_N \geq 0, \quad d_{E,N} = -\log \det T'_N \geq 0. \quad (\text{AMT13})$$

When $N = q - 1$, F_N is empty, $d_{R,N} = 0$, and the same equalities hold without a relation inverse. The determinant identities are independent of the auxiliary orthonormal frame. Dividing the two instances of AMT10 proves $d_{K,N} + d_{B,N} = d_{E,N}$. Thus the complete nested statement is

$$\boxed{\begin{aligned} \Xi_{k,N} &= d_{R,N} + d_{K,N} + d_{B,N}, & d_{R,N}, d_{K,N}, d_{B,N} &\geq 0, \\ 0 &\leq d_{K,N} \leq \min\{rL_N, d_{E,N}\} \leq \min\{rL_N, \Xi_{k,N}\}, \\ 0 &\leq d_{B,N} \leq \min\{bL_N, d_{E,N}\} \leq \min\{bL_N, \Xi_{k,N}\}. \end{aligned}} \quad (\text{AMT14})$$

In particular each metric keeps its own rank factor, and the two deficits retain their exact common sum. Any proved finite upper bound for the original $\Xi_{k,N}$, including the degree-specific HMR44/TWA15 bound, can be substituted in these displayed minima. No new density or source determinant is introduced by this substitution.

8.5 All four original endpoints and the correlated mixed return

Let

$$(N_0, N_1, N_2, N_3) = (q - 1, q, 2q - 1, 2q), \quad (\epsilon_0, \epsilon_1, \epsilon_2, \epsilon_3) = (1, 1, -1, -1),$$

and define, in their original signs,

$$\begin{aligned} F_K^a &= \sum_{j=0}^3 \epsilon_j \log \det H_{K,N_j}^a, & F_B^a &= \sum_{j=0}^3 \epsilon_j \log \det Q_{N_j}^a, \\ \mathcal{B}_k^a &= \sum_{j=0}^3 \epsilon_j \log \det G_{N_j}^a, & \Delta_K^\sigma &= F_K^{\text{ar}} - F_K^\sigma, \quad \Delta_B^\sigma = F_B^{\text{ar}} - F_B^\sigma, \quad \delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma. \end{aligned} \quad (\text{AMT15})$$

AMT10 cancels its four identical frame factors only after these signs are applied. It gives $F_K^a + F_B^a = \mathcal{B}_k^a$ and $\Delta_K^\sigma + \Delta_B^\sigma = \delta_k^\sigma$ exactly. The spaces $\mathcal{P}_N$ increase with the same source norm and remainder map, so their quotient minima decrease. Restriction to K and minimization onto B preserve that order. Pairing N_0 with N_2 and N_1 with N_3 consequently proves $F_K^a, F_B^a \geq 0$.

For $X = K, B$, put $r_K = r$, $r_B = b$, and retain the exact caps

$$c_{X,N} = \min\{r_X L_N, d_{E,N}\}, \quad C_X^- = c_{X,q-1} + c_{X,q}, \quad C_X^+ = c_{X,2q-1} + c_{X,2q}. \quad (\text{AMT16})$$

The entirely source-evaluated caps $\hat{c}_{X,N} = \min\{r_X L_N, \Xi_{k,N}\}$ and the corresponding $\hat{C}_X^\pm$ are at least these values; all conclusions below also hold with the hats. Substituting AMT12 in AMT15 records each endpoint and its sign:

$$\boxed{\begin{aligned} \Delta_K^\sigma &= -d_{K,q-1} - d_{K,q} + d_{K,2q-1} + d_{K,2q}, \\ \Delta_B^\sigma &= -d_{B,q-1} - d_{B,q} + d_{B,2q-1} + d_{B,2q}, \\ \delta_k^\sigma &= -d_{E,q-1} - d_{E,q} + d_{E,2q-1} + d_{E,2q}, \\ -C_X^- &\leq \Delta_X^\sigma \leq C_X^+, \quad X = K, B. \end{aligned}} \quad (\text{AMT17})$$

The four copies of $r_X \log A_k$ cancel here because their signs sum to zero. Since the original total correction is retained, the joint interval is stronger than two independent intervals:

$$\boxed{\begin{aligned} \max\{-C_K^-, \delta_k^\sigma - C_B^+\} &\leq \Delta_K^\sigma \leq \min\{C_K^+, \delta_k^\sigma + C_B^-\}, \\ \max\{-C_B^-, \delta_k^\sigma - C_K^+\} &\leq \Delta_B^\sigma \leq \min\{C_B^+, \delta_k^\sigma + C_K^-\}. \end{aligned}} \quad (\text{AMT18})$$

These follow by subtracting the interval for either component from the exact common sum; they require no independence assertion.

Here are the explicit original four endpoint widths:

$$\begin{aligned} E_k^+ &= L_{q-1} + L_q = 2 \log(A_k/a_k) + M \log[1 + 4(q-1+M)^2] + M \log[1 + 4(q+M)^2], \\ E_k^- &= L_{2q-1} + L_{2q} = 2 \log(A_k/a_k) + M \log[1 + 4(2q-1+M)^2] + M \log[1 + 4(2q+M)^2]. \end{aligned} \quad (\text{AMT19})$$

They are the unchanged TWA5 constants, both nonnegative. AMT17 therefore gives in particular

$$\boxed{-rE_k^+ \leq F_K^{\text{ar}} - F_K^\sigma \leq rE_k^-, \quad -bE_k^+ \leq F_B^{\text{ar}} - F_B^\sigma \leq bE_k^-}. \quad (\text{AMT20})$$

The total correction retains TWA15 with all four individual source deficits:

$$\begin{aligned} \delta_k^\sigma &= \max\{-qE_k^+, -\Xi_{k,q-1} - \Xi_{k,q}\}, \\ \bar{\delta}_k^\sigma &= \min\{qE_k^-, \Xi_{k,2q-1} + \Xi_{k,2q}\}, \quad \underline{\delta}_k^\sigma \leq \delta_k^\sigma \leq \bar{\delta}_k^\sigma. \end{aligned} \quad (\text{AMT21})$$

This also follows immediately from AMT14 and AMT17. It can be inserted into AMT18 when an interval, rather than the actual finite total determinant, is wanted. Thus the lower and upper endpoints are kept in their distinct places throughout the mixed transfer.

For later use, subtraction of the exact logarithms in AMT4 gives

$$\begin{aligned} \mathfrak{s}_h &= \alpha + \log(M_\alpha/\vartheta_h), \quad \mathfrak{c}_h = -\log M_\alpha + 3 \log \vartheta_h - 3\alpha + \log \frac{C_h^{\text{tilt}} C_\Gamma 2^{B_h/2}}{c_h c_\Gamma}, \\ \log(A_k/a_k) &= k\mathfrak{s}_h + (p+1) \log k + B_h \log(k-2) + \mathfrak{c}_h, \quad E_k^\pm = O_h(k + \log q). \end{aligned} \quad (\text{AMT22})$$

The last statement follows from the displayed exact expressions, since M, p, B_h and the two integral constants are fixed and $\log D_N = O_h(1 + \log(N+1))$. It is a consequence of the original constants and contains no change of measure.

8.6 The full original multiplier and its transported observation

On the existing tensor class $k = 4l + 1$, retain

$$\begin{aligned} t_j &= 2j + \frac{1}{2}, \quad \xi_j = c + t_j, \quad f_l(S) = \prod_{j=0}^{l-1} (S - c - t_j), \\ \beta_k &= \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}, \quad dm_{0,k}(y) = \beta_k |f_l(c + iy)|^2 d\sigma(y). \end{aligned} \quad (\text{AMT23})$$

This is the exact Gamma recurrence identity TWA26 with masses $(2\pi)^{k/2}$ and $\sqrt{2\pi}$. For $l = 0$ both f_l and β_k are one. Write $E_\chi : E \rightarrow \mathbb{C}^q$ for the complete divided Taylor map $[P] \mapsto (P^{(r)}(\lambda_{a,b})/r!)_{a,b, 0 \leq r < e}$. It is invertible: a vector in its kernel is divisible by the coprime factors of χ , and dimensions agree. The full multiplication matrix U_f has, in each primary block, entries

$$\begin{aligned} (U_f)_{r,a} &= \begin{cases} f_l^{(r-a)}(\lambda)/(r-a)!, & 0 \leq a \leq r < e, \\ 0, & a > r. \end{cases} \\ \det U_f &= \prod_{a,b=0}^k \prod_{j=0}^{l-1} [(2a-k)\delta - t_j + i(2b-k)\gamma]^e \neq 0. \end{aligned} \quad (\text{AMT24})$$

Indeed k is odd, so every $\lambda_{a,b}$ has nonzero imaginary part, while all ξ_j are real and distinct. The inverse in each block is the finite Taylor inverse of its nonzero constant term. The exact polynomial map and its inverse on its stated range are

$$\mathcal{P}_N \longrightarrow f_l \mathcal{P}_N \subseteq \mathcal{P}_{N+l}, \quad P \longmapsto f_l P, \quad Q \longmapsto Q/f_l. \quad (\text{AMT25})$$

It sends the original fibre $J_N P = z$ bijectively to the product jet fibre $(\mathbf{E}_f Q, \mathbf{E}_\chi Q) = (0, U_f \mathbf{E}_\chi z)$. Vanishing of the l distinct f -values proves divisibility by f_l , and the inverse of U_f proves the converse fibre statement. The original relation space maps to $f_l \chi \mathcal{P}_{N-q}$. On the free χ jet coordinate Y of this constrained product fibre, the exact observed map, its kernel, and a section are

$$\boxed{\begin{aligned} \tilde{\Lambda} &= \Lambda \mathbf{E}_\chi^{-1} U_f^{-1}, & \tilde{\Lambda} U_f \mathbf{E}_\chi &= \Lambda, \\ \ker \tilde{\Lambda} &= U_f \mathbf{E}_\chi K, & \tilde{I} &= U_f \mathbf{E}_\chi I, & \tilde{S}_* &= U_f \mathbf{E}_\chi S_*, & \tilde{\Lambda} \tilde{S}_* &= I_B. \end{aligned}} \quad (\text{AMT26})$$

Both inclusions in the kernel equality follow by applying the displayed invertible map and its inverse. This retains the full original amplitude in Λ and every derivative in U_f . The resulting kernel is proved by transport; it has not been assigned the coordinates of $\mathbf{E}_\chi K$.

For an explicit metric on this transported space, let $b_n(y)$ be the fixed monic Gamma polynomials and put $\pi_n(S) = i^n b_n((S - c)/i)$, whose leading coefficient is one and squared norm is $\gamma_n = \sqrt{2\pi}(2n)!/4^n$. The evaluation kernels are the full finite sums

$$\mathcal{K}_{\chi,L} = \sum_{n=0}^L \frac{\mathbf{E}_\chi \pi_n (\mathbf{E}_\chi \pi_n)^*}{\gamma_n}, \quad \mathcal{K}_{f,L} = \sum_{n=0}^L \frac{\mathbf{E}_f \pi_n (\mathbf{E}_f \pi_n)^*}{\gamma_n}, \quad \mathcal{K}_{\chi,f;L} = \sum_{n=0}^L \frac{\mathbf{E}_\chi \pi_n (\mathbf{E}_f \pi_n)^*}{\gamma_n}.$$

The product evaluation is onto for $L = N + l \geq q + l - 1$, because a polynomial of degree below $q + l$ vanishing at all its divided jets is divisible by the degree- $q + l$ product $f_l \chi$ and is zero. Thus its positive kernel has positive Schur complement

$$\begin{aligned} C_N &= \mathcal{K}_{\chi,N+l} - \mathcal{K}_{\chi,f;N+l} \mathcal{K}_{f,N+l}^{-1} \mathcal{K}_{f,\chi;N+l} > 0, \\ U_N &= (\mathbf{E}_\chi \pi_{N+j} / \sqrt{\gamma_{N+j}})_{j=1}^l, \quad V_N = \mathcal{K}_{\chi,f;N+l} \mathcal{K}_{f,N+l}^{-1/2}, \quad C_N = \mathcal{K}_{\chi,N} + U_N U_N^* - V_N V_N^*. \end{aligned} \quad (\text{AMT27})$$

Empty f blocks have the identity convention. The inverse of the joint positive kernel, evaluated on $(0, Y)$, gives the minimum $Y^* C_N^{-1} Y$: block elimination of the f -block proves this identity with every mixed entry present. Combining it with AMT25 and the coefficient β_k proves the exact tensor-source metrics

$$\boxed{\begin{aligned} G_N^0 &= \beta_k \mathbf{E}_\chi^* U_f^* C_N^{-1} U_f \mathbf{E}_\chi, & H_{K,N}^0 &= \beta_k \tilde{I}^* C_N^{-1} \tilde{I}, \\ Q_N^0 &= \beta_k (\tilde{\Lambda} C_N \tilde{\Lambda}^*)^{-1}, & S_N^0 &= \mathbf{E}_\chi^{-1} U_f^{-1} C_N \tilde{\Lambda}^* (\tilde{\Lambda} C_N \tilde{\Lambda}^*)^{-1}. \end{aligned}} \quad (\text{AMT28})$$

Direct multiplication verifies the section and its orthogonality. The formulas hold on the original E, K, B , with the transported observation exactly as in AMT26. No block or mixed product was dropped from C_N .

The same coefficient identity also supplies a quantitative comparison directly with this original tensor source. Define the positive exact finite factors

$$\Pi_l = \prod_{j=0}^{l-1} t_j^2, \quad U_{l,N}^{\text{poly}} = \prod_{j=0}^{l-1} [4(N+j+1)^2 + t_j^2], \quad \tilde{L}_N = L_N + \log(U_{l,N}^{\text{poly}} / \Pi_l). \quad (\text{AMT29})$$

Pointwise $|f_l(c + iy)|^2 \geq \Pi_l$. Also $\|(iy - t)P\|_\sigma^2 = \|yP\|_\sigma^2 + t^2 \|P\|_\sigma^2$, so the degree-specific multiplication estimate in AMT6, applied successively, gives $\|f_l P\|_\sigma^2 \leq U_{l,N}^{\text{poly}} \|P\|_\sigma^2$. Writing H_N^0 for the unchanged $m_{0,k}$ source Gram yields

$$\beta_k \Pi_l H_N^\sigma \preceq H_N^0 \preceq \beta_k U_{l,N}^{\text{poly}} H_N^\sigma, \quad \frac{a_k}{D_N \beta_k U_{l,N}^{\text{poly}}} H_N^0 \preceq H_N^{\text{ar}} \preceq \frac{A_k}{\beta_k \Pi_l} H_N^0. \quad (\text{AMT30})$$

The second pair follows by using the first upper bound for its lower inequality and the first lower bound for its upper inequality. Minimizing and restricting with the same J_N, I, Λ gives AMT30 for all three metrics in AMT28. Its common upper scalar is $A_k/(\beta_k \Pi_l)$; it is retained before its four signed copies cancel. The same determinant proof as AMT12–17 therefore gives

$$\begin{aligned} \tilde{E}_k^+ &= E_k^+ + \log(U_{l,q-1}^{\text{poly}} U_{l,q}^{\text{poly}} / \Pi_l^2), & \tilde{E}_k^- &= E_k^- + \log(U_{l,2q-1}^{\text{poly}} U_{l,2q}^{\text{poly}} / \Pi_l^2), \\ \boxed{-r\tilde{E}_k^+ \leq F_K^{\text{ar}} - F_K^0 \leq r\tilde{E}_k^-, \quad -b\tilde{E}_k^+ \leq F_B^{\text{ar}} - F_B^0 \leq b\tilde{E}_k^-} \end{aligned} \quad (\text{AMT31})$$

For an explicit bound on the extra widths, $t_j \geq 1/2$ gives

$$0 \leq \log(U_{l,N}^{\text{poly}} / \Pi_l) = \sum_{j=0}^{l-1} \log \left(1 + \frac{4(N+j+1)^2}{t_j^2} \right) \leq l \log[1 + 16(N+l)^2]. \quad (\text{AMT32})$$

Thus $\tilde{E}_k^\pm = O_h(k \log(q+k))$ on the stated endpoints. This is a proved polynomial-source estimate for the exact original f_l , with no inference that its multiplication preserves K .

8.7 The evaluated original simple-quartet arithmetic kernel

Now take the original simple-quartet class $m = 1, k = 4l+1, l \geq 1, q = (k+1)^2$. The parameter u is precisely the period parameter in AMT3, not the real integration coordinate y . RPC21 and RPC23 give an explicit finite radius R_* for the unchanged quartet and arithmetic unit. For reference its exact constants and cases are recorded here. Set

$$\begin{aligned} \beta &= 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad a = v_h(\tfrac{1}{2} + \delta + i\gamma) \neq 0, \\ c_3 &= \frac{a^{-2} - \bar{a}^{-2}}{4i\delta\gamma}, \quad c_1 = \frac{a^{-2} + \bar{a}^{-2}}{2} - (\delta^2 - \gamma^2)c_3, \\ g_j &= 5^{j/5-1} e^{\pi i j/5} \Gamma(j/5) \quad (1 \leq j \leq 4), \quad g_+ = \max_j |g_j|, \quad g_-^{-1} = \max_j |g_j|^{-1}, \quad \kappa = \max_{1 \leq j < 4} |g_{j+1}/g_j|, \\ E_j^{\text{ray}} &= \frac{8}{5} \int_0^\infty r^j (r^3/3 + r) e^{-r^5/5 + r^3/3 + r} dr \quad (0 \leq j \leq 3), \quad C_T = 2 \left(\sum_{j=0}^3 (E_j^{\text{ray}})^2 \right)^{1/2}, \\ C_K &= 4C_T g_-^{-1} + 2g_+ g_-^{-1} + 2g_+ g_-^{-2} C_T, \quad C_{K,3} = C_K(3\kappa^2 + 3\kappa C_K + C_K^2), \\ \epsilon_* &= \min\{1, (2C_T g_-^{-1})^{-1}\}, \quad B_* = \beta + \eta. \end{aligned} \quad (\text{AMT33})$$

Here $g_-^{-2} = (g_-^{-1})^2$. All integrals converge by their negative fifth-degree exponent, and $(c_1, c_3) \neq (0, 0)$ by the original unit values. The exact radius is

$$R_* = \begin{cases} \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2(|c_3|C_{K,3}B_* + |c_1|(\kappa + C_K))}{|c_3 g_4/g_1|} \right\} \right]^{5/2}, & c_3 \neq 0, \\ \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2C_K B_*}{|g_2/g_1|} \right\} \right]^{5/2}, & c_3 = 0. \end{cases} \quad (\text{AMT34})$$

The complete period calculation RPC16–25 proves, for every original branch and every $|u| \geq R_*$, a nonzero original commutator entry: its $(4, 1)$ entry in the first case and its $(2, 1)$ entry in the second. RPC14–15 identifies those entries with the exact test for the original rank-four quadric's cyclic invariance. Thus its actual projective orbit has five elements throughout this domain. OQX1–3, using the product of precisely these five original quadrics, proves

$$\boxed{r = 8k - 16, \quad b = k^2 - 6k + 17, \quad m = 1, \quad k = 4l + 1, \quad l \geq 1, \quad |u| \geq R_*} \quad (\text{AMT35})$$

This is the evaluated original period outcome; no value of the amplitude or of a period coefficient is chosen. The radius depends on the fixed one-factor quartet and unit and is independent of k .

For the same original packet, OKV22 proves for both existing source orders $s = 1, k$, and for the actual K of any rank,

$$\begin{aligned} D_* &= \max\{3, \sqrt{\delta^2 + \gamma^2}\}, \quad C(\delta, \gamma) = 12 + 4 \log(2D_*), \\ Z_q &= 1 + 2(q+1)q^2 2^{6q} (2D_* q)^{4q}, \\ 0 \leq F_K^{(s)} &\leq 2r \log Z_q \leq 2rC(\delta, \gamma)q \log(q+1), \\ F_K^{(1)} &= F_K^\sigma, \quad F_K^{(k)} = F_K^0. \end{aligned} \tag{AMT36}$$

Its proof includes the exact source coefficient map and the unchanged kernel inclusion. Combining it with the already proved AMT17–20 gives the finite arithmetic conclusion

$$\begin{aligned} \max\{0, F_K^\sigma - C_K^-\} &\leq F_K^{\text{ar}} \leq F_K^\sigma + C_K^+, \\ 0 \leq F_K^{\text{ar}} &\leq U_K := 2r \log Z_q + \widehat{C}_K^+ \\ &\leq 2rC(\delta, \gamma)q \log(q+1) + r\widetilde{E}_k^-. \end{aligned} \tag{AMT37}$$

The cap symbols $C_K^\pm, \widehat{C}_K^\pm$ in this formula mean the endpoint sums AMT16; the constant C_K without a superscript in AMT33 belongs only to the explicit period radius. They have different stated indices and are not identified. The alternative finite tensor-source bound $F_K^{\text{ar}} \leq 2r \log Z_q + r\widetilde{E}_k^-$ also follows from AMT31. Formula AMT37 retains the sharper fixed-source widths.

On AMT35 the exact ranks give

$$F_K^{\text{ar}} - F_K^\sigma = O_h(k^2), \quad F_K^{\text{ar}} = O_h(kq \log(q+1)), \quad \frac{F_K^{\text{ar}}}{q^2} \longrightarrow 0. \tag{AMT38}$$

For the first estimate use $r = 8k - 16$ in AMT20 and AMT22, with $\log q = 2 \log(k+1)$. The second follows from AMT37. For the limit, divide its last line by q^2 : its first term is at most $16C(\delta, \gamma)k \log(q+1)/q \rightarrow 0$, and its second is $O_h(k^2/q^2) \rightarrow 0$. Nonnegativity gives the limit by the two bounds. These estimates are uniform over the entire original domain $|u| \geq R_*$, since none of their right sides depends on u . The tensor comparison likewise gives $F_K^{\text{ar}} - F_K^0 = O_h(k^2 \log q) = o(q^2)$ from AMT31–32.

8.8 The original boundary receives the full baseline coefficient

Before using a limit, the exact finite arithmetic identity and its consequence are

$$\begin{aligned} F_B^{\text{ar}} &= \mathcal{B}_k^\sigma + \delta_k^\sigma - F_K^{\text{ar}}, \\ \max\{0, \mathcal{B}_k^\sigma + \delta_k^\sigma - U_K\} &\leq F_B^{\text{ar}} \leq \mathcal{B}_k^\sigma + \delta_k^\sigma, \\ \max\{0, \mathcal{B}_k^\sigma + \underline{\delta}_k^\sigma - U_K\} &\leq F_B^{\text{ar}} \leq \mathcal{B}_k^\sigma + \bar{\delta}_k^\sigma. \end{aligned} \tag{AMT39}$$

The first identity follows from AMT10 and AMT15; AMT37 and $F_K^{\text{ar}}, F_B^{\text{ar}} \geq 0$ prove the first interval, and AMT21 proves the second. The correlated boundary return also has the sharper exact interval AMT18, with every original signed total correction present.

BSL13–19 supplies the complete original finite baseline and its proved coefficient

$$\mathcal{B}_k^\sigma = V_q + \eta_q + \eta_{q+1} - \eta_1, \quad \frac{\mathcal{B}_k^{\text{ar}}}{q^2} \longrightarrow C_B^{\text{base}} := 9 - 8 \log 2 + \mathcal{F}(2, \pi) > \frac{769}{17010} > 0. \tag{AMT40}$$

Here V_q , the three distinct original relation errors η_r , and the energy $\mathcal{F}$ are exactly the BSL/HBL/GEL quantities, whose complete definitions and proofs are in the retained closure. The superscript on C_B^{base} only distinguishes that existing baseline coefficient from the endpoint caps $C_B^\pm$ in AMT16. Its finite BSL14 enclosure can be substituted directly for $\mathcal{B}_k^\sigma$ in AMT39,

keeping the signs of all three η terms; no new evaluation is needed for that substitution. Subtract AMT38 from the exact original identity $F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}} - F_K^{\text{ar}}$. It proves

$$\boxed{\frac{F_B^{\text{ar}}}{q^2} \longrightarrow C_B^{\text{base}} > 0, \quad F_K^{\text{ar}} + F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}}, \quad m = 1, \quad k = 4l + 1, \quad l \rightarrow \infty, \quad |u| \geq R_*} \quad (\text{AMT41})$$

The convergence is uniform over that original period domain by AMT38 and because the full-source baseline is independent of u . Thus the proved original kernel estimate now passes to the actual arithmetic kernel, and the original arithmetic boundary carries the entire positive q^2 coefficient. All earlier full-sum identities remain exact, with their individual mixed contributions now receiving AMT37–41 and their signed finite corrections AMT17–21.

8.9 The same comparison evaluates the mixed Gamma return on the kq scale

The first inequality pair AMT30 directly compares the two existing Gamma sources on the same original coefficient space. Apply its proved minimum and restriction maps AMT7–9 to $X = E, K, B$, writing $H_{E,N}^a = G_N^a$, $H_{B,N}^a = Q_N^a$, and $r_E = q, r_K = r, r_B = b$. At every original endpoint it gives

$$\beta_k \Pi_l H_{X,N}^\sigma \preceq H_{X,N}^0 \preceq \beta_k U_{l,N}^{\text{poly}} H_{X,N}^\sigma. \quad (\text{AMT42})$$

Define the exact relative determinant and the positive degree width

$$\begin{aligned} z_{X,N} &= \log \frac{\det H_{X,N}^0}{\det H_{X,N}^\sigma} - r_X \log(\beta_k \Pi_l), & \omega_N^\Gamma &= \log(U_{l,N}^{\text{poly}} / \Pi_l), \\ 0 &\leq z_{X,N} \leq r_X \omega_N^\Gamma, & z_{K,N} + z_{B,N} &= z_{E,N}, \\ \Omega_k^{\text{low}} &= \omega_{q-1}^\Gamma + \omega_q^\Gamma, & \Omega_k^{\text{high}} &= \omega_{2q-1}^\Gamma + \omega_{2q}^\Gamma. \end{aligned} \quad (\text{AMT43})$$

Congruence and multiplication of the eigenvalues in AMT42 prove the two inequalities. The determinant identity AMT10, applied to the tensor metric AMT28 as well, proves the common sum. These facts preserve the complete original kernel and boundary rather than assigning separate arbitrary errors to their determinants.

Use the precise return orientation

$$\Delta_X^\Gamma = F_X^\sigma - F_X^0 \quad (X = K, B), \quad \mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0 = -\mathfrak{T}_k.$$

The last equality is the original TWA30/BSL18 sign. Substituting all four z terms, and canceling $\log(\beta_k \Pi_l)$ only after the signed sum, proves

$$\boxed{\begin{aligned} \Delta_X^\Gamma &= -z_{X,q-1} - z_{X,q} + z_{X,2q-1} + z_{X,2q}, \\ -r_X \Omega_k^{\text{low}} &\leq \Delta_X^\Gamma \leq r_X \Omega_k^{\text{high}}, \\ \Delta_K^\Gamma + \Delta_B^\Gamma &= \mathcal{H}_k, \\ \mathcal{H}_k - r \Omega_k^{\text{high}} &\leq \Delta_B^\Gamma \leq \mathcal{H}_k + r \Omega_k^{\text{low}}. \end{aligned}} \quad (\text{AMT44})$$

AMT29 displays every factor of both Ω values explicitly; AMT32 gives the finite bounds

$$\begin{aligned} 0 &\leq \Omega_k^{\text{low}} \leq l \log[1 + 16(q-1+l)^2] + l \log[1 + 16(q+l)^2], \\ 0 &\leq \Omega_k^{\text{high}} \leq l \log[1 + 16(2q-1+l)^2] + l \log[1 + 16(2q+l)^2]. \end{aligned} \quad (\text{AMT45})$$

In particular both are $O(k \log(q+1))$ on the original $m = 1$ class. Its actual rank $r = 8k - 16$ therefore proves directly

$$\boxed{\Delta_K^\Gamma = O_h(k^2 \log(q+1)) = o(kq), \quad \frac{F_K^{\text{ar}} - F_K^\sigma}{kq} \longrightarrow 0, \quad \frac{F_K^{\text{ar}} - F_K^0}{kq} \longrightarrow 0.} \quad (\text{AMT46})$$

For the first equality divide the AMT44 bounds by kq and use $q = (k+1)^2$, so $k \log(q+1)/q \rightarrow 0$. The second limit uses AMT38, since $k^2/(kq) = k/q \rightarrow 0$. The last uses the exact identity $F_K^{\text{ar}} - F_K^0 = \Delta_K^\sigma + \Delta_K^\Gamma$. All bounds are uniform for $|u| \geq R_*$.

For completeness the entire finite arithmetic boundary return is also evaluated by those same quantities:

$$\boxed{\begin{aligned} F_B^{\text{ar}} - F_B^0 &= \mathcal{H}_k + \delta_k^\sigma - (F_K^{\text{ar}} - F_K^0), \\ \mathcal{H}_k + \delta_k^\sigma - r\tilde{E}_k^- &\leq F_B^{\text{ar}} - F_B^0 \leq \mathcal{H}_k + \delta_k^\sigma + r\tilde{E}_k^+. \end{aligned}} \quad (\text{AMT47})$$

This uses AMT31 with its exact endpoint constants $\tilde{E}_k^+ = E_k^+ + \Omega_k^{\text{low}}$ and $\tilde{E}_k^- = E_k^- + \Omega_k^{\text{high}}$, rather than a new source comparison. Both tensor and arithmetic source masses and the full original resultant have already been retained in AMT23–30 before these signed cancellations.

The proved GEL18–19 constant is

$$\mathcal{C}_\Gamma = 2 \int_{a_*^2}^{b_*^2} \log x \rho_{2,\pi}(x) dx - 4(2 \log 2 - 1) > 0, \quad \frac{\mathfrak{T}_k}{lq} \rightarrow \mathcal{C}_\Gamma. \quad (\text{AMT48})$$

Here a_*, b_* and $\rho_{2,\pi}$ are exactly the proved GEL8/EIQ equilibrium endpoints and density; their complete definitions and integral proof remain in that provider. The endpoint letters here avoid the already fixed original period parameter u . GEL18a gives $4 \log(27/(8\pi)) \leq \mathcal{C}_\Gamma \leq 6$ and proves the strict positivity. TWA15 also proves the actual source return $\delta_k^\sigma/(kq) \rightarrow 0$. As $l/k \rightarrow 1/4$, AMT44, AMT46 and AMT47 now prove

$$\boxed{\begin{aligned} \frac{F_B^\sigma - F_B^0}{kq} &\rightarrow -\frac{\mathcal{C}_\Gamma}{4}, & \frac{F_B^{\text{ar}} - F_B^0}{kq} &\rightarrow -\frac{\mathcal{C}_\Gamma}{4}, & \frac{F_B^{\text{ar}} - F_B^\sigma}{kq} &\rightarrow 0, \\ \frac{F_B^\sigma - F_B^0}{\mathcal{H}_k} &\rightarrow 1, & \frac{F_B^{\text{ar}} - F_B^0}{\mathcal{H}_k} &\rightarrow 1, & m=1, k=4l+1, l \rightarrow \infty, |u| \geq R_*. \end{aligned}} \quad (\text{AMT49})$$

Indeed the first two limits subtract an $o(kq)$ kernel return from $\mathcal{H}_k = -\mathfrak{T}_k$, whose quotient by kq tends to $-\mathcal{C}_\Gamma/4$. The third follows from $\Delta_B^\sigma = \delta_k^\sigma - \Delta_K^\sigma$. The nonzero limit of $\mathcal{H}_k/(kq)$ makes its denominator nonzero for all sufficiently large original k and proves the last two ratios by division. The uniformity in u follows from the uniform kernel bounds and the period-independent total returns. Thus the full existing source comparison evaluates the mixed Gamma return also at its kq scale, with its finite signed remainders AMT44 and AMT47 still present.

9 Factorial control of the original common kernel at the kq scale

The exact denominator in OKV15 contains the information needed to improve its absolute volume estimate. This proof keeps that denominator and also allows every original primary multiplicity. The original Gamma sources, root coordinates, coefficient maps and observation are retained. The simple-quartet conclusion uses the actual period and rank calculations PQO/RPC and OQC/OQX, with their complete proofs supplied as dependencies.

9.1 Original coordinates, all primary multiplicities, and source norms

Fix the original full-order quartet with $0 < \delta < 1/2$, $\gamma > 2$, and a positive integer primary order m . On the original tensor class set

$$k = 4l + 1, \quad l \geq 1, \quad e_k = 1 + k(m - 1), \quad q = e_k(k + 1)^2, \quad c = k/2, \quad s \in \{1, k\}, \quad b_s = s/2. \quad (\text{KAF1})$$

The original polynomial and its exact real-coordinate representative are

$$\begin{aligned} \chi(S) &= \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^{e_k}, \\ \omega_{ab} &= (2b - k)\gamma - i(2a - k)\delta, \\ P(y) &= i^{-q}\chi(c + iy) = \prod_{a,b=0}^k (y - \omega_{ab})^{e_k}. \end{aligned} \quad (\text{KAF2})$$

In the full list $\omega_1, \dots, \omega_q$, each root ω_{ab} occurs e_k times. The exact maximal modulus remains $R = k\sqrt{\delta^2 + \gamma^2}$. Substitution gives inverse algebra maps

$$\Phi : \mathbb{C}[y]/(P) \longrightarrow E = \mathbb{C}[S]/(\chi), \quad [r] \longmapsto [r((S - c)/i)], \quad \Phi^{-1}[f] = [f(c + iy)]. \quad (\text{KAF3})$$

Indeed $P((S - c)/i) = i^{-q}\chi(S)$, and the two substitutions are inverse before quotienting. Thus every root, multiplicity and factor of i in the original remainder operation is present.

Retain the complete measures and their monic orthogonal polynomials:

$$\begin{aligned} M_s &= (2\pi)^{s/2}, \quad dm_{0,s}(y) = \frac{M_s 2^{b_s-2}}{\pi \Gamma(b_s)} \left| \Gamma\left(\frac{b_s}{2} + \frac{iy}{2}\right) \right|^2 dy, \\ p_0^{(s)} &= 1, \quad p_1^{(s)} = y, \quad p_{d+1}^{(s)} = yp_d^{(s)} - d(b_s + d - 1)p_{d-1}^{(s)}, \\ h_d^{(s)} &= \int |p_d^{(s)}|^2 dm_{0,s} = M_s d!(b_s)_d, \quad \varphi_d^{(s)} = p_d^{(s)} / \sqrt{h_d^{(s)}}, \quad u_d^{(s)}(S) = \varphi_d^{(s)}((S - c)/i). \end{aligned} \quad (\text{KAF4})$$

These are precisely the existing orders $1, k$. Their complete source transform and norm calculation are IFB2–5 and OKV4–5. In particular $\int e^{ty} dm_{0,s} = M_s (\cos t)^{-b_s}$ for $|t| < \pi/2$. The generating series $(1 + z^2)^{-b_s/2} \exp(y \arctan z)$ has coefficients $p_d^{(s)} z^d / d!$. Integrating its product at two sufficiently small real variables gives $M_s (1 - zw)^{-b_s}$, proving the displayed norms and orthogonality by coefficient comparison. The original exponential bound justifies those finite coefficient differentiations under the integral. None of these source statements depends on m .

In the original polynomial frame of E let C_s have columns $[u_0^{(s)}], \dots, [u_{q-1}^{(s)}]$. Its determinant is exactly

$$\begin{aligned} \det C_s &= i^{-q(q-1)/2} \prod_{d=0}^{q-1} (h_d^{(s)})^{-1/2} \neq 0, \\ v_j^{(s)} &= C_s^{-1}[u_{q+j}^{(s)}], \quad K_j^{(s)} = I_q + \sum_{a=0}^{j-1} v_a^{(s)} (v_a^{(s)})^*, \quad 0 \leq j \leq q + 1. \end{aligned} \quad (\text{KAF5})$$

The last included vector has original degree $2q$.

9.2 A degree-dependent constant in the unchanged remainder estimate

Define the actual finite constants

$$A_{k,m} = \max \left\{ 2\sqrt{1 + \frac{k}{2q}}, \frac{R}{q} \right\}, \quad T = qA_{k,m}, \quad \mathcal{A}_{k,m} = 6 \log 2 + 4 + 4 \log A_{k,m}. \quad (\text{KAF6})$$

The letter $A_{k,m}$ in this section is a coefficient-estimate constant; it is not the arithmetic upper source constant A_k of TWA/AMT. In particular $A_{k,m} \geq 2$ and $R \leq T$. For fixed original quartet and fixed m , $q \geq (k+1)^2$ implies

$$A_{k,m} \longrightarrow 2, \quad \mathcal{A}_{k,m} \longrightarrow 4 + 10 \log 2. \quad (\text{KAF7})$$

Indeed $k/q \rightarrow 0$ and $R/q = \sqrt{\delta^2 + \gamma^2} k/q \rightarrow 0$. The root-dependent term in the maximum is retained at every finite k .

Here is the complete remainder estimate needed below. For the full root list define elementary and complete homogeneous polynomials $e_h(\omega)$ and $h_t(\omega)$ by their finite sums, with $e_0 = h_0 = 1$ and $e_h = 0$ outside $0 \leq h \leq q$. The formal identity

$$\left(\sum_{h=0}^q (-1)^h e_h(\omega) z^h \right) \left(\sum_{t \geq 0} h_t(\omega) z^t \right) = 1 \quad (\text{KAF8})$$

follows by multiplying each of the q factors $1 - \omega_a z$ by its geometric series. Repeated roots do not affect this identity. It proves, for $D = q + j$ and $0 \leq j \leq q$, the exact signed formula

$$[y^a] \text{rem}_P(y^{q+j}) = - \sum_{t=0}^j h_t(\omega) (-1)^{q+j-t-a} e_{q+j-t-a}(\omega), \quad 0 \leq a < q. \quad (\text{KAF9})$$

In detail the quotient is $\sum_{t=0}^j h_t(\omega) y^{j-t}$; KAF8 cancels all coefficients of degree at least q in its product with P , and the remaining coefficients are KAF9. This proves the formula by uniqueness of monic division, including all multiplicities.

The defining sums give $|e_h| \leq \binom{q}{h} R^h$ and $|h_t| \leq \binom{q+t-1}{t} R^t$. In every nonzero summand of KAF9 the total root exponent is $q + j - a$. The binomial coefficients of h_t increase with t , and the distinct indices in the e_h sum contribute at most 2^q . Hence

$$|[y^a] \text{rem}_P(y^{q+j})| \leq 2^q \binom{q+j-1}{j} R^{q+j-a} \leq 2^{3q-1} R^{q+j-a}. \quad (\text{KAF10})$$

The last bound is the binomial expansion at one, since $j \leq q$.

The exact orthonormal recurrence from KAF4 has upward and downward weights $\sqrt{(a+1)(b_s+a)}$ and $\sqrt{a(b_s+a-1)}$. On degree at most $q-2$ each shift has norm at most $\sqrt{q(b_s+q)}$. Because $b_s \leq k/2$, their sum has norm at most $2\sqrt{q(k/2+q)} \leq T$. Iterating from the constant, with all intermediate degrees in that range, proves

$$\|y^a\|_{L^2(m_{0,s})} \leq \sqrt{M_s} T^a, \quad 0 \leq a < q. \quad (\text{KAF11})$$

For the finite coefficient sum $c_T(f) = \sum_a [y^a]f |T^a$ the original recurrence gives $c_T(p_{d+1}^{(s)}) \leq T c_T(p_d^{(s)}) + d(b_s + d - 1) c_T(p_{d-1}^{(s)})$. For $1 \leq d < 2q$,

$$d(b_s + d - 1) \leq 2q(k/2 + 2q) = 4q^2 + kq \leq 4q^2 + 2kq \leq T^2.$$

Starting with $c_T(p_0) = 1$, $c_T(p_1) = T$, induction therefore proves $c_T(p_d) \leq (2T)^d$ for $0 \leq d \leq 2q$: the recurrence bound is $(3/4)(2T)^{d+1} \leq (2T)^{d+1}$. Combining KAF10 with all q monomial norms KAF11, then using the linearity of the original remainder map, gives

$$\|\text{rem}_P(p_d^{(s)})\|_{L^2(m_{0,s})} \leq \sqrt{M_s} q 2^{3q} (2T)^d, \quad 0 \leq d \leq 2q. \quad (\text{KAF12})$$

For a monomial of degree $D \geq q$, each coefficient term is at most $\sqrt{M_s} 2^{3q-1} R^{D-a} T^a \leq \sqrt{M_s} 2^{3q-1} T^D$. For $D < q$ use KAF11 directly. These two cases and the proved coefficient sum establish KAF12 without deleting a remainder coefficient or changing the source norm.

9.3 Retaining the factorial denominator

Expansion of the degree-below- q remainder in the orthonormal polynomials of KAF4, followed by the exact map KAF3, proves

$$\|v_j^{(s)}\|_2^2 = \frac{\|\text{rem}_P(p_{q+j}^{(s)})\|_{L^2(m_{0,s})}^2}{M_s(q+j)!(b_s)_{q+j}} \leq \frac{q^2 2^{6q} (2T)^{2(q+j)}}{(q+j)!(b_s)_{q+j}}. \quad (\text{KAF13})$$

The factor M_s occurs in the original numerator and denominator before this cancellation. For every $d \geq 0$,

$$d!(b_s)_d \geq d!(1/2)_d = \frac{(2d)!}{4^d}, \quad \|v_j^{(s)}\|_2^2 \leq q^2 2^{6q} \frac{(4T)^{2(q+j)}}{(2q+2j)!}. \quad (\text{KAF14})$$

The inequality is factor by factor because $b_s \geq 1/2$. The equality follows by separating even and odd factors of $(2d)!$. This is precisely the denominator whose further retention sharpens OKV16–17.

Define the positive exact quantities

$$\eta_{k,m} = \frac{1 - 1/(4q)}{A_{k,m}^2} < \frac{1}{4}, \quad Z_{k,m} = 1 + q^2 2^{6q} \frac{(4q A_{k,m})^{4q}}{(4q)!} \frac{1 - \eta_{k,m}^{q+1}}{1 - \eta_{k,m}}. \quad (\text{KAF15})$$

For $t_d = (4T)^{2d}/(2d)!$, $q+1 \leq d \leq 2q$ gives $t_{d-1}/t_d = (2d)(2d-1)/(16T^2) \leq \eta_{k,m}$. Consequently $\sum_{d=q}^{2q} t_d \leq t_{2q} \sum_{a=0}^q \eta_{k,m}^a$. Adding the $q+1$ actual rank-one updates in KAF5 and using KAF14 proves the finite operator bound

$$\boxed{1 \leq \|K_{q+1}^{(s)}\|_2 \leq Z_{k,m}, \quad s \in \{1, k\}.} \quad (\text{KAF16})$$

Here $\|vv^*\|_2 = \|v\|_2^2$ follows by Cauchy–Schwarz with equality on $v \neq 0$; a zero vector contributes zero.

For an explicit linear bound, monotonicity of $\log x$ on each interval $[j-1, j]$ gives

$$\log n! = \sum_{j=1}^n \log j \geq \int_1^n \log x \, dx = n \log n - n + 1.$$

Thus $(4q A_{k,m})^{4q}/(4q)! \leq e^{-1} (e A_{k,m})^{4q}$, where e is the exponential constant and differs from the primary integer e_k . The geometric sum in KAF15 is at most $4/3$. Therefore $Z_{k,m} \leq 1 + [4/(3e)]X$, with $X = q^2 \exp(q A_{k,m})$. The positive exponential series gives $e > 1 + 1 + 1/2 + 1/6 = 8/3$, so $4/(3e) < 1/2$. Since $A_{k,m} \geq 2$ and $q \geq 36$, $X > 2$, and hence $1 + [4/(3e)]X \leq X$. We have proved

$$\boxed{\log \|K_{q+1}^{(s)}\|_2 \leq \log Z_{k,m} \leq q A_{k,m} + 2 \log q.} \quad (\text{KAF17})$$

The full factorial and finite geometric sum in KAF15 remain available as the sharper finite bound.

9.4 All four actual kernel and boundary Grams

Let $\Lambda : E \twoheadrightarrow B$ be the original period and full-unit observation, let $I : \mathcal{K} \hookrightarrow E$ be its actual kernel inclusion, and put $r = \dim \mathcal{K}$. No rank formula is needed in this subsection, so it applies to every original m . The original quotient and restricted Grams are

$$G_{q+j-1}^{(s)} = C_s^{*-}(K_j^{(s)})^{-1} C_s^{-1}, \quad T_s = C_s^{-1} I, \quad H_j^{(s)} = T_s^* (K_j^{(s)})^{-1} T_s. \quad (\text{KAF18})$$

Indeed the original orthonormal source-coordinate remainder matrix is $[I_q, v_0, \dots, v_{j-1}]$. Multiplication by its adjoint gives K_j ; its unique minimum lift of x is its adjoint applied to $K_j^{-1}x$, with

squared norm $x^* K_j^{-1} x$. This proves both matrices KAF18 in the original coefficient frame. The positive update chain implies

$$Z_{k,m}^{-1} T_s^* T_s \preceq H_j^{(s)} \preceq T_s^* T_s, \quad H_{j+1}^{(s)} \preceq H_j^{(s)}. \quad (\text{KAF19})$$

Inversion reverses positive order by congruence and inversion of positive eigenvalues. Restricting this order with the same injective T_s proves KAF19. After congruence with $(T_s^* T_s)^{-1/2}$, every eigenvalue lies between $Z_{k,m}^{-1}$ and one. Thus $-r \log Z_{k,m} \leq \log \det H_j^{(s)} - \log \det (T_s^* T_s) \leq 0$.

The exact four original degrees are $q-1, q, 2q-1, 2q$. With their unchanged signs define

$$F_K^{(s)} = \log \det H_0^{(s)} + \log \det H_1^{(s)} - \log \det H_q^{(s)} - \log \det H_{q+1}^{(s)}.$$

Pairing degrees $q-1$ with $2q-1$, and q with $2q$, proves nonnegativity by KAF19. Each difference is at most $r \log Z_{k,m}$. The four original frame factors cancel only after these four signs have been recorded. It follows that

$$0 \leq F_K^{(s)} \leq 2r \log Z_{k,m} \leq 2r [q \mathcal{A}_{k,m} + 2 \log q], \quad s \in \{1, k\}. \quad (\text{KAF20})$$

If $r = 0$, every kernel determinant is one and the statement has its same empty-space meaning.

At each original degree the boundary Gram is $Q_j^{(s)} = (\Lambda(G_{q+j-1}^{(s)})^{-1} \Lambda^*)^{-1}$. For any fixed section S_* of Λ , its minimum section $S_j^{(s)} = (G_{q+j-1}^{(s)})^{-1} \Lambda^* Q_j^{(s)}$ differs from S_* by a kernel column and is orthogonal to I in that metric. The change of frame has determinant one. Taking determinants in this orthogonal frame proves, with its original coefficient factor,

$$\det H_j^{(s)} \det Q_j^{(s)} = |\det[I, S_*]|^2 \det G_{q+j-1}^{(s)}.$$

The same four signs now prove the exact splitting $F_K^{(s)} + F_B^{(s)} = \mathcal{B}_k^{(s)}$. Boundary minimum norms decrease with degree, so $F_B^{(s)} \geq 0$. In particular

$$\max\{0, \mathcal{B}_k^{(s)} - 2r \log Z_{k,m}\} \leq F_B^{(s)} \leq \mathcal{B}_k^{(s)}. \quad (\text{KAF21})$$

These are statements about the original kernel and boundary, with all multiplicities and the same observation throughout.

9.5 The actual simple quartet and the arithmetic source

For $m = 1$, the complete RPC calculation supplies an explicit finite R_* from the original unit and quartet. On every logarithm branch and for $|u| \geq R_*$ its original period quadric has five distinct cyclic orbit members. The full invariant restriction ideal calculation OQX then gives $r = 8k - 16$. Substitution in the finite KAF20 gives

$$0 \leq \frac{F_K^{(s)}}{kq} \leq \left(16 - \frac{32}{k}\right) \left(\mathcal{A}_{k,1} + \frac{2 \log q}{q}\right), \quad s \in \{1, k\}, \quad |u| \geq R_*. \quad (\text{KAF22})$$

The original arithmetic-source comparison AMT17–22 proves $-r E_k^+ \leq F_K^{\text{ar}} - F_K^{(1)} \leq r E_k^-$, where the unchanged full constants are

$$\begin{aligned} E_k^+ &= 2 \log(A_k/a_k) + \log D_{q-1} + \log D_q, \\ E_k^- &= 2 \log(A_k/a_k) + \log D_{2q-1} + \log D_{2q}, \\ D_N &= [1 + 4(N + 21 + 4m)^2]^{21+4m}. \end{aligned}$$

Here a_k, A_k are exactly the original arithmetic constants of AMT4, whose full source inequality and minimum-section proof are included there. They retain the actual numerator $2\xi/h$ and all

source masses; AMT22 proves $E_k^\pm = O_h(k + \log q)$. Thus the actual arithmetic metric satisfies the finite bound

$$0 \leq F_K^{\text{ar}} \leq (16k - 32) \log Z_{k,1} + (8k - 16)E_k^-. \quad (\text{KAF23})$$

Combining KAF7, KAF22–23, and $q = (k + 1)^2$ proves uniformly on the entire stated original period domain, for $a \in \{\sigma, 0, \text{ar}\}$,

$$0 \leq \liminf_{k \rightarrow \infty} \frac{F_K^a}{kq} \leq \limsup_{k \rightarrow \infty} \frac{F_K^a}{kq} \leq 16(4 + 10 \log 2). \quad (\text{KAF24})$$

For precision, uniformity means the same finite upper bound holds for every such u and branch at each original k ; therefore it also bounds every sequence of those parameters. The arithmetic extra term divided by kq is at most $(8 - 16/k)E_k^-/q \rightarrow 0$. AMT46 further proves that the three ratios differ by quantities tending uniformly to zero. Thus their limit points coincide for any fixed allowed sequence of period parameters; KAF24 does not assert that this bounded sequence already has a unique limit.

Finally the exact arithmetic identity gives

$$\max\{0, \mathcal{B}_k^{\text{ar}} - (16k - 32) \log Z_{k,1} - (8k - 16)E_k^-\} \leq F_B^{\text{ar}} \leq \mathcal{B}_k^{\text{ar}}. \quad (\text{KAF25})$$

Its full-source value remains $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma = \mathcal{B}_k^0 + \mathcal{H}_k + \delta_k^\sigma$. All exact Gamma and arithmetic signed returns of AMT44–49 are retained. The new calculation improves the absolute common kernel bound to the kq scale, while its exact four-endpoint determinant and actual period dependence remain the objects for the next leading term calculation.

10 An exact conductor frame and a fixed invariant-generator recursion

Retain the original simple-quartet data, $m = 1$, $k = 4l + 1$, $l \geq 1$, the original root order $(++)$, $(+-)$, $(-+)$, $(--)$, and the complete original period and amplitude matrix of OQC1:

$$\mathbf{H} = F^{-1} \Pi U \mathcal{E}_{\text{prim}}.$$

Thus $\det \mathbf{H} \neq 0$ and $Q(x) = Q_0(\mathbf{H}^{-1}x)$, where $Q_0(t) = t_{++}t_{--} - t_{+-}t_{-+}$. Use $gf(x) = f(D^{-1}x)$, $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$, $\zeta = e^{2\pi i/5}$, and the original invariant algebra $\mathcal{J} = \mathbb{C}[x_1, x_2, x_3, x_4]^{(g)}$. This calculation applies on the actual five-orbit period locus, which includes every original logarithm branch with $|u| \geq R_*$ by PQO and RPC. On that locus the five prime polynomials $Q_a = g^a Q$, $0 \leq a < 5$, are pairwise nonassociate. Keep their actual products

$$\mathcal{A} = \prod_{a=1}^4 Q_a, \quad \mathcal{H}_Q = Q\mathcal{A} = \prod_{a=0}^4 Q_a, \quad \deg \mathcal{A} = 8, \quad \deg \mathcal{H}_Q = 10. \quad (\text{OCF1})$$

No coefficient in these polynomials is replaced. The exact orbit decision and the period domain are those of the cited original calculations. The construction below treats every possible nonzero leading coefficient of their actual restriction.

10.1 The original Segre coefficient isomorphism

For $d \geq 0$ let $\mathcal{B}_d$ be the space of biforms of bidegree (d, d) in (z_0, z_1) and (w_0, w_1) . Its ordered basis is

$$e_{ab}^{(d)} = z_0^a z_1^{d-a} w_0^b w_1^{d-b}, \quad 0 \leq a, b \leq d. \quad (\text{OCF2})$$

Pairs are ordered lexicographically, first by a , then by b . The order is used only for the exact coefficient elimination. Set $\mathcal{B}_d = 0$ for $d < 0$. Multiplication satisfies $e_{ab}^{(d)} e_{ij}^{(e)} = e_{a+i, b+j}^{(d+e)}$. The original substitution is

$$t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^T, \quad x = \mathbf{H}t, \quad (\text{OCF3})$$

$$\mathcal{P}_d = \mathbb{C}[x_1, x_2, x_3, x_4]_d, \quad \psi_d : \mathcal{P}_d \longrightarrow \mathcal{B}_d, \quad f \longmapsto f(\mathbf{H}t).$$

The induced map $R_d = \mathcal{P}_d / (Q\mathcal{P}_{d-2}) \longrightarrow \mathcal{B}_d$ is an isomorphism. Here is a coefficient proof with the given orientation. For each (a, b) , choose an integer h between $\max(0, a + b - d)$ and $\min(a, b)$. The monomial $t_{++}^h t_{+-}^{a-h} t_{-+}^{b-h} t_{--}^{d-a-b+h}$ maps to $e_{ab}^{(d)}$, proving surjectivity. Two monomials with the same pair differ in their exponent vectors by an integer multiple of $(1, -1, -1, 1)$. After removing their common monomial, their difference is divisible by $t_{++}t_{--} - t_{+-}t_{-+}$, using $A^n - B^n = (A - B) \sum_{j=0}^{n-1} A^{n-1-j} B^j$. A polynomial in the kernel has coefficient sum zero within every pair, so it is a sum of such differences. The reverse inclusion in the kernel follows by direct substitution. The invertible substitution $x = \mathbf{H}t$ then gives exactly the asserted kernel. Thus these maps preserve multiplication as well as every grading.

Write the full conductor restriction as

$$A = \psi_8(\mathcal{A}) = \sum_{r=0}^8 \sum_{s=0}^8 a_{rs} e_{rs}^{(8)} \neq 0. \quad (\text{OCF4})$$

To verify its nonvanishing, Q is a rank-four irreducible quadric, hence prime in the polynomial ring, and none of $Q_1, \dots, Q_4$ is its associate. Therefore Q does not divide their product. The kernel calculation OCF3 proves OCF4. The ring $\mathcal{B} = \bigoplus_{d \geq 0} \mathcal{B}_d$, with the indicated diagonal grading, embeds in the four-variable polynomial ring; it is an integral domain. Multiplication by A is consequently injective on every $\mathcal{B}_d$.

10.2 Every leading-coefficient case gives an exact strip frame

Let (r_*, s_*) be the lexicographically largest pair for which $a_{r_*s_*} \neq 0$, and put $a_* = a_{r_*s_*}$. This is an actual nonzero coefficient of OCF4. For $d \geq 8$ define

$$\begin{aligned} P_d &= \{(r_* + i, s_* + j) : 0 \leq i, j \leq d - 8\}, \\ E_d^{\text{strip}} &= \{0, \dots, d\}^2 \setminus P_d, \quad W_d = \mathbb{C}^{E_d^{\text{strip}}}. \end{aligned} \tag{OCF5}$$

For $0 \leq d < 8$ put $P_d = \emptyset$, $E_d^{\text{strip}} = \{0, \dots, d\}^2$ and use the same definition of W_d ; for $d < 0$ put $W_d = 0$. Let $S_d : W_d \hookrightarrow \mathbb{C}^{(d+1)^2}$ insert zeros in the positions P_d . All coefficient columns below use OCF2.

The following finite elimination defines $N_d : \mathbb{C}^{(d+1)^2} \rightarrow W_d$ and, for $d \geq 8$, $V_d : \mathbb{C}^{(d+1)^2} \rightarrow \mathbb{C}^{(d-7)^2}$. Starting with the complete input biform f , traverse all $(i, j) \in \{0, \dots, d-8\}^2$ in decreasing lexicographic order. At the step (i, j) , let c be the current coefficient at $(r_* + i, s_* + j)$, record $b_{ij} = c/a_*$, and replace the current biform by

$$f_{\text{current}} - b_{ij} A e_{ij}^{(d-8)}. \tag{OCF6}$$

At completion take the coefficients on E_d^{strip} as $N_d f$ and take (b_{ij}) as $V_d f$. If $d < 8$, N_d is the identity and V_d is the zero map into the zero space. The exact multiplication matrix $M_{A,d}$ gives

$$S_d N_d + M_{A,d} V_d = I, \quad N_d S_d = I_{W_d}, \quad \ker N_d = A \mathcal{B}_{d-8}. \tag{OCF7}$$

To prove these identities, the largest term of $A e_{ij}^{(d-8)}$ is $a_* e_{r_*+i, s_*+j}^{(d)}$; every other term is strictly smaller in the same lexicographic order. Therefore each step zeros its designated position without altering an earlier zero. Summing all subtractions gives the first identity. Input supported on the strip produces zero coefficients at every elimination step, giving the second. Finally, the leading term of Ab , for a nonzero homogeneous biform b , is the product of the two leading terms and belongs to P_d . Indeed every other product is smaller in the additive lexicographic order. Thus $A \mathcal{B}_{d-8} \cap S_d W_d = 0$. The first identity now proves the kernel formula, the uniqueness of the decomposition, and all three identities in OCF7. All operations are linear in the input; their coefficients are explicit rational expressions in the original a_{rs} with powers of the displayed a_* as denominators. The polynomial A itself remains the full polynomial OCF4.

In particular the quotient map induced by N_d is the exact isomorphism

$$\mathcal{B}_d / (A \mathcal{B}_{d-8}) \xrightarrow{\sim} W_d, \quad n_d := \dim W_d = \begin{cases} (d+1)^2, & 0 \leq d < 8, \\ 16d - 48, & d \geq 8. \end{cases} \tag{OCF8}$$

The second formula is $(d+1)^2 - (d-7)^2 = 16d - 48$. The possible choices $(r_*, s_*) \in \{0, \dots, 8\}^2$ form an exhaustive finite partition: the chart for a pair specifies that its coefficient is nonzero and every larger coefficient is zero. Hence all actual coefficient cases have been included. At the first original degree $k = 5$ this construction retains all 36 coordinates; at every original $k \geq 9$ it retains $16k - 48$ coordinates.

10.3 Twenty-five fixed invariant monomials generate all degrees

Define the following fixed finite sets of exponent vectors:

$$G_e = \{\nu \in \mathbb{N}^4 : |\nu| = e, \nu_1 + 2\nu_2 + 3\nu_3 + 4\nu_4 \equiv 0 \pmod{5}\}, \quad 1 \leq e \leq 5, \quad G = \bigcup_{e=1}^5 G_e. \tag{OCF9}$$

Here G_1 is empty. The complete remaining lists, in the original x_1, x_2, x_3, x_4 order, are

e	G_e	
2	$(0, 1, 1, 0), (1, 0, 0, 1)$	
3	$(0, 0, 2, 1), (0, 1, 0, 2), (1, 2, 0, 0), (2, 0, 1, 0)$	
4	$(0, 0, 1, 3), (0, 2, 2, 0), (0, 3, 0, 1), (1, 0, 3, 0),$ $(1, 1, 1, 1), (2, 0, 0, 2), (3, 1, 0, 0)$	(OCF10)
5	$(0, 0, 0, 5), (0, 0, 5, 0), (0, 1, 3, 1), (0, 2, 1, 2),$ $(0, 5, 0, 0), (1, 0, 2, 2), (1, 1, 0, 3), (1, 3, 1, 0),$ $(2, 1, 2, 0), (2, 2, 0, 1), (3, 0, 1, 1), (5, 0, 0, 0).$	

Thus $(|G_1|, \dots, |G_5|) = (0, 2, 4, 7, 12)$ and $|G| = 25$. These lists can also be obtained directly by choosing $0 \leq \nu_1, \nu_2, \nu_3 \leq e$ with their sum at most e , putting $\nu_4 = e - \nu_1 - \nu_2 - \nu_3$, and applying the single displayed congruence, which verifies completeness of each list.

Every positive-degree invariant monomial is a product of members x^ν , $\nu \in G$. To prove this, list the weights of its individual factors in any order. If its degree is at most five, the entire list gives a member of G . If its degree exceeds five, consider the six partial sums of the first five weights, including the empty partial sum, in $\mathbb{Z}/5\mathbb{Z}$. Two coincide; the nonempty block between them has length at most five and total weight zero. Its factors give a divisor x^ν with $\nu \in G$. The complementary monomial again has total weight zero and smaller degree. Induction completes the factorization. Because the action is diagonal, a polynomial is invariant exactly when all its nonzero monomials have weight zero: comparison of each monomial coefficient in $gf = f$ proves this statement. Thus

$$\mathcal{J} = \mathbb{C}[x^\nu : \nu \in G]. \quad (\text{OCF11})$$

All intermediate degrees here are degrees in the original coefficient algebra. This construction introduces no new source distribution and leaves the original source orders 1 and k intact.

For each $\nu \in G_e$ the complete generator biform is

$$f_\nu = \psi_e(x^\nu) = \prod_{j=1}^4 (H_{j1}z_0w_0 + H_{j2}z_0w_1 + H_{j3}z_1w_0 + H_{j4}z_1w_1)^{\nu_j} = \sum_{a,b=0}^e c_{\nu,ab} e_{ab}^{(e)}. \quad (\text{OCF12})$$

For an entirely explicit coefficient formula, sum over all nonnegative 4 by 4 integer arrays (n_{jt}) satisfying $\sum_t n_{jt} = \nu_j$, $\sum_j (n_{j1} + n_{j2}) = a$ and $\sum_j (n_{j1} + n_{j3}) = b$. Then

$$c_{\nu,ab} = \sum_{(n_{jt})} \prod_{j=1}^4 \left(\frac{\nu_j!}{\prod_{t=1}^4 n_{jt}!} \prod_{t=1}^4 H_{jt}^{n_{jt}} \right). \quad (\text{OCF13})$$

The multinomial theorem proves the formula, including every original period and amplitude entry. Every factor has degree at most five and at most 36 possible biform coefficients.

10.4 Exact multiplication on the strip and the invariant recursion

Let $\mathcal{L} = \mathcal{B}/(AB)$, a graded algebra. For $\nu \in G_e$ and $d \geq e$ define

$$m_{\nu,d} : W_{d-e} \longrightarrow W_d, \quad m_{\nu,d} = N_d M_{f_\nu} S_{d-e}. \quad (\text{OCF14})$$

Here M_{f_ν} is the exact coefficient convolution:

$$(M_{f_\nu} S_{d-e} v)_{ab} = \sum_{i,j} c_{\nu,a-i,b-j} (S_{d-e} v)_{ij},$$

with out-of-range indices giving zero. Therefore every entry of OCF14 is supplied by the finite formulas OCF6 and OCF13. For a representative $h \in \mathcal{B}_{d-e}$, OCF7 writes $h - S_{d-e} N_{d-e} h = Ab$. Multiplication by f_ν preserves this ideal. Applying N_d yields the exact identity

$$N_d(f_\nu h) = m_{\nu,d} N_{d-e} h. \quad (\text{OCF15})$$

This proves that OCF14 is multiplication by the specified element of $\mathcal{L}$, including its independence of representatives.

The original invariant image is $J_d = \psi_d(\mathcal{J}_d) \subseteq \mathcal{B}_d$. Set $\bar{J}_d = N_d J_d \subseteq W_d$, and set $\bar{J}_d = 0$ for $d < 0$. The exact recursion is

$$\bar{J}_0 = \mathbb{C} \cdot 1, \quad \bar{J}_d = \sum_{\substack{\nu \in G_e \\ 2 \leq e \leq 5, e \leq d}} m_{\nu,d} \bar{J}_{d-e} \quad (d \geq 1). \quad (\text{OCF16})$$

Indeed OCF11 factors every invariant monomial of degree $d > 0$ as x^ν times an invariant monomial of degree $d - e$. Applying ψ , then OCF15, gives the inclusion from left to right. Conversely every summand is the image of a product of two invariant polynomials, so it belongs to the left side. Linearity proves equality for the complete invariant space. In particular the recursion gives $\bar{J}_1 = 0$ and includes all small degrees.

The conductor ideal is contained in this invariant image with its original trace coefficient:

$$A\mathcal{B}_{d-8} \subseteq J_d, \quad [5R(\mathcal{A}h)]_{(Q)} = [\mathcal{A}h]_{(Q)}, \quad R = \frac{1}{5} \sum_{a=0}^4 g^a. \quad (\text{OCF17})$$

To prove the identity, $g^a \mathcal{A}$ is the product of all five orbit equations except Q_a ; for $a \neq 0$ it contains Q . Thus all four corresponding terms vanish modulo Q , and the $a = 0$ term is the displayed product. Every biform in $\mathcal{B}_{d-8}$ has a polynomial lift by OCF3, so the identity proves the inclusion. It also shows exactly why the quotient used in OCF16 retains the original invariant image.

10.5 Exact ranks, a finite pivot atlas, and the original quotient map

For $d \geq 0$ put

$$d_0(d) = \frac{\binom{d+3}{3} + 4\epsilon_d}{5}, \quad \epsilon_d = \mathbf{1}_{d \equiv 0 \pmod{5}} - \mathbf{1}_{d \equiv 1 \pmod{5}}, \quad d_0(d) = 0 \quad (d < 0). \quad (\text{OCF18})$$

This is the number of degree- d weight-zero monomials: the identity character has generating function $(1 - z)^{-4}$, each of the four nonidentity characters has generating function $\prod_{j=1}^4 (1 - z\zeta^j)^{-1} = (1 - z)/(1 - z^5)$, and averaging their coefficients gives OCF18. Every invariant polynomial divisible by Q is divisible by all five Q_a , hence by $\mathcal{H}_Q$, since they are pairwise nonassociate primes. Its quotient by this product is invariant: apply g and cancel the nonzero invariant product in the integral domain. Conversely each $\mathcal{H}_Q$ times an invariant polynomial is invariant and vanishes modulo Q . Consequently $\ker(\mathcal{J}_d \rightarrow J_d) = \mathcal{H}_Q \mathcal{J}_{d-10}$, and injection of multiplication by the nonzero product gives $\dim J_d = d_0(d) - d_0(d - 10)$. Injection of multiplication by A , OCF7 and OCF17 then prove

$$j_d := \dim \bar{J}_d = d_0(d) - d_0(d - 10) - \max(d - 7, 0)^2. \quad (\text{OCF19})$$

In particular $j_0 = 1$, $j_1 = 0$, $j_5 = 12$, and $j_d = 8d - 32$ for $d \geq 8$. For the last formula at $d \geq 10$, $\epsilon_d = \epsilon_{d-10}$ and direct expansion gives $\binom{d+3}{3} - \binom{d-7}{3} = 5(d^2 - 6d + 17)$; subtracting $(d - 7)^2$ gives $8d - 32$. At $d = 8, 9$ the ranks before this subtraction are 33, 44, respectively, which gives 32, 40 and verifies both remaining cases.

Here is a complete finite construction of all the columns used by OCF16. Start with $Z_0 = (1)$ and the invariant polynomial certificate $F_{0,1} = 1$. Suppose all earlier matrices Z_b have independent columns spanning $\bar{J}_b$, and their columns are $N_b \psi_b(F_{b,h})$ for invariant polynomials $F_{b,h}$. Form the candidate matrix

$$C_d = [m_{\nu,d} Z_{d-e}]_{\nu \in G_e, 2 \leq e \leq 5, e \leq d}. \quad (\text{OCF20})$$

Each candidate column has the explicit invariant certificate $x^\nu F_{d-e,h}$. OCF16 proves that C_d has column image $\bar{J}_d$ and rank j_d . Select j_d columns and j_d rows giving a nonzero determinant. Let Z_d be those columns, let I_d be those row positions, and let T_d be the complementary row positions

in E_d^{strip} . Retain the corresponding invariant certificates. This completes the induction. When $j_d = 0$, use the empty matrix with zero columns, empty I_d , and $T_d = E_d^{\text{strip}}$.

Every selected determinant and every matrix entry is an explicit expression obtained from the original entries of $\mathbf{H}$ and the full a_{rs} . To cover all cases, order the finite choices of column and row subsets and select the first nonzero determinant; the corresponding chart asserts its nonvanishing and the vanishing of all preceding candidates. A nonzero determinant exists because the rank was proved in OCF19. At rank zero no determinant is needed. Together with the exhaustive leading-coefficient charts after OCF8, this gives a finite atlas for each specified k . It requires no generic coefficient premise and supplies no numerical zero test for an unevaluated analytic coefficient: each chart is specified by its exact minors.

Define $\rho_d : W_d \rightarrow \mathbb{C}^{T_d}$ and the full original coefficient map π_d by

$$\rho_d = \text{row}_{T_d} - (Z_d)_{T_d} (Z_d)_{I_d}^{-1} \text{row}_{I_d}, \quad \pi_d = \rho_d N_d. \quad (\text{OCF21})$$

At rank zero the second term is the zero map. If E_{T_d} inserts a column into its T_d positions in W_d , then

$$\begin{aligned} w &= Z_d (Z_d)_{I_d}^{-1} w_{I_d} + E_{T_d} \rho_d w, \\ \rho_d E_{T_d} &= I, \quad \ker \rho_d = \text{im } Z_d = \bar{J}_d, \\ \ker \pi_d &= J_d, \quad \pi_d S_d E_{T_d} = I_{\mathbb{C}^{T_d}}. \end{aligned} \quad (\text{OCF22})$$

For the first line, the subtraction on its right leaves zero I_d rows and exactly $\rho_d w$ on the complementary rows. This proves the first three claims. If $\pi_d f = 0$, then $N_d f \in \bar{J}_d$; choose $j \in J_d$ with $N_d j = N_d f$. Now $f - j \in \ker N_d = \mathcal{A}\mathcal{B}_{d-8} \subseteq J_d$ by OCF17, so $f \in J_d$. The reverse inclusion follows immediately from $N_d J_d = \bar{J}_d$. The last equality uses $N_d S_d = I$. This proves every map and kernel in OCF22.

At the original degrees, these are the exact dimensions

k	n_k	j_k	$ T_k = n_k - j_k$
5	36	12	24
$k \geq 9, k = 4l + 1$	$16k - 48$	$8k - 32$	$8k - 16$.

(OCF23)

The original map $\Theta_k : R_k \xrightarrow{\sim} E_k^\vee$ in OQC7 sends the coefficient basis OCF2 to the original dual primary basis in the same order. Indeed its evaluation on $e^{wY_k}[1]$ is $\exp(((2b-k)\gamma - i(2a-k)\delta)w)$. The corresponding eigenvalue of A_k is $\beta_{ab} = k/2 + (2a-k)\delta + i(2b-k)\gamma$; the eigenvalue of $Y_k = (A_k - kI/2)/i$ is $\omega_{ab} = (\beta_{ab} - k/2)/i = (2b-k)\gamma - i(2a-k)\delta$. These eigenvalues are distinct because $\delta > 0$ and $\gamma > 0$. Their exponentials are independent by the Vandermonde determinant, so the evaluation identifies the covectors exactly. By the original observation identity OQC8, $\Theta_k J_k = \text{im } \Lambda_k^\vee$. Therefore OCF22 supplies the specified isomorphism

$$E_k^\vee / \text{im } \Lambda_k^\vee \xrightarrow{\pi_k} \mathbb{C}^{T_k} \cong (\ker \Lambda_k)^\vee. \quad (\text{OCF24})$$

The displayed π_k uses exactly the original primary coefficient coordinates under Θ_k . Every dualization in this statement is complex-linear. No conjugation is inserted in this coefficient map; the original metric maps can subsequently be composed with it.

Finally, this construction is finite with explicitly bounded compressed matrices. OCF20 has n_d rows and $\sum_{e=2}^5 |G_e| j_{d-e}$ columns, with negative subscripts zero. Both dimensions grow at most linearly in d : for example $n_d \leq 16d + 1$ and the column number is at most $25(16d + 1)$. For $d \geq 13$ the exact column number is $200d - 1632$, obtained from $\sum_e |G_e| = 25$, $\sum_e e|G_e| = 104$, and $j_{d-e} = 8(d-e) - 32$. Only the preceding five graded bases are needed to form the next matrix. Each column is obtained from the at-most-36-coefficient formula OCF12, one exact convolution, and OCF6. A full original biform coefficient column, if used as temporary storage, has $(d+1)^2$ entries; every invariant relation matrix in OCF20 has the stated linear row and column counts. The invariant certificates can be retained as their products and parent-column references, so their expanded high-degree monomials need never be enumerated. The proof of their completeness is OCF11–16, and the exact original quotient and its right inverse are OCF21–24.

10.6 The same recursion with fourteen proved generators

The complete fixed set OCF9 can be reduced to the following fourteen specific members while preserving every map and image just proved:

e	$\mathcal{G}_e$	
2	x_1x_4, x_2x_3	
3	$x_1^2x_3, x_1x_2^2, x_2x_4^2, x_3^2x_4$	(OCF25)
4	$x_1^3x_2, x_1x_3^2, x_2^3x_4, x_3x_4^3$	
5	$x_1^5, x_2^5, x_3^5, x_4^5$	

Here is a complete proof. Call a positive-degree invariant monomial indecomposable when it is not the product of two positive-degree invariant monomials. The divisor proof preceding OCF11 shows that each indecomposable has degree at most five. If it contains both x_1, x_4 , it is their product, since otherwise division by this invariant quadratic leaves a nonconstant invariant monomial. The same assertion holds for x_2, x_3 . After these cases are removed, at most two variables occur: every three indices contain one of the two opposite pairs. One variable gives precisely its fifth power. The remaining two-element supports are $\{1, 2\}, \{1, 3\}, \{2, 4\}, \{3, 4\}$. For positive exponents (a, b) with $a + b \leq 5$, the equation $a + 2b \equiv 0 \pmod{5}$ has exactly $(a, b) = (3, 1), (1, 2)$, and $a + 3b \equiv 0 \pmod{5}$ has exactly $(a, b) = (2, 1), (1, 3)$. For $\{2, 4\}$ divide the weights by 2 in the field $\mathbb{Z}/5\mathbb{Z}$, giving the first list; for $\{3, 4\}$ divide by 3, giving the second. These are exactly the eight mixed monomials of degrees three and four in OCF25. Every listed monomial has no proper invariant divisor, as is also seen from these lists and the excluded opposite pairs. This proves the classification. Factoring any invariant monomial repeatedly into indecomposables terminates because positive total degree strictly decreases. Hence OCF25 generates the entire invariant algebra.

Use $\mathcal{G} = \bigcup_{e=2}^5 \mathcal{G}_e$ in place of $\{x^\nu : \nu \in G\}$ in OCF12–16 and OCF20. The same factorization argument proves the same exact invariant image; all coefficient formulas, invariant certificates, pivot charts, quotient maps and ranks remain valid. The new recursion has 2, 4, 4, 4 generators in degrees 2, 3, 4, 5, respectively. Thus for $d \geq 13$ its precise matrix dimensions and rank are

$$C_d^{(14)} \in \text{Mat}_{(16d-48) \times (112d-864)}(\mathbb{C}), \quad \text{rank } C_d^{(14)} = 8d - 32. \quad (\text{OCF26})$$

Indeed the number of generators is 14 and the sum of their degrees is $2 \cdot 2 + 4 \cdot 3 + 4 \cdot 4 + 4 \cdot 5 = 52$; substituting the already proved $j_{d-e} = 8(d-e) - 32$ gives $14(8d-32) - 8 \cdot 52 = 112d - 864$. At smaller degrees the exact number of columns is the same sum of the appropriate j_{d-e} from OCF19, with negative subscripts zero. All coefficients in this reduced recursion are still the original coefficients of OCF12–14.

11 The exact common kernel frame and its original Gamma metrics

Retain the original simple quartet $m = 1$, $k = 4l + 1$, $l \geq 1$, $q = (k + 1)^2$, $c = k/2$, $0 < \delta < 1/2$ and $\gamma > 2$. The original period parameter and its branch are fixed, on the proved original OCF rank domain $|u| \geq R_*$, with R_* the retained explicit period-orbit radius. The kernel below is the kernel of that one fixed original observation $\Lambda : E \rightarrow B$. It is common to the two source metrics and to their four degree endpoints. No intersection over different period parameters or different observations is made.

Order all pairs (a, b) , $0 \leq a, b \leq k$, lexicographically once and retain that order. Define

$$\begin{aligned} \beta_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, & \omega_{ab} &= \frac{\beta_{ab} - c}{i} = (2b - k)\gamma - i(2a - k)\delta, \\ \chi(S) &= \prod_{a,b=0}^k (S - \beta_{ab}), & E &= \mathbb{C}[S]/(\chi), & \varepsilon_{ab} &= \left[\prod_{(a',b') \neq (a,b)} \frac{S - \beta_{a'b'}}{\beta_{ab} - \beta_{a'b'}} \right], & (\text{OKM1}) \\ V_\beta &= ((\beta_{ab})^d)_{(a,b), 0 \leq d < q}. \end{aligned}$$

The points are distinct: equality of their real and imaginary parts forces both indices equal. Hence every denominator displayed above is nonzero. Evaluation at all points gives $\varepsilon_{ab}(\beta_{a'b'}) = \mathbf{1}_{(a,b)=(a',b')}$. Polynomial interpolation proves that these are a basis of E , and the evaluation map from increasing polynomial coordinates to this primary basis is exactly V_β . Its determinant is the ordered Vandermonde product, so it is invertible.

11.1 The exact dual and primal maps of the strip quotient

Use the coefficient presentation of OQC7–8 in Segre coordinates. A biform of bidegree (k, k) has the full coefficient expression

$$F = \sum_{a,b=0}^k f_{ab} X_1^a X_0^{k-a} Y_1^b Y_0^{k-b}. \quad (\text{OKM2})$$

The literal variable dictionary to the OCF biform coefficients is $X_1 = z_0$, $X_0 = z_1$, $Y_1 = w_0$, $Y_0 = w_1$. The Segre substitution has $t_{++} = X_1 Y_1$, $t_{+-} = X_1 Y_0$, $t_{-+} = X_0 Y_1$, $t_{--} = X_0 Y_0$. At the original exponential vector take $X_1 = e^{-i\delta w}$, $X_0 = e^{i\delta w}$, $Y_1 = e^{\gamma w}$, $Y_0 = e^{-\gamma w}$. Then the monomial with coefficient f_{ab} is $e^{\omega_{ab} w}$. The original coefficient identification includes the entire invertible matrix H of OQC1 before this Segre presentation; its unit and period entries remain in the matrices produced by the strip calculation. OQC7 is characterized by evaluation on $e^{wY}[1]$. Since $e^{wY}[1] = \sum_{a,b} e^{\omega_{ab} w} \varepsilon_{ab}$, the independent exponentials in OQC4 therefore give the precise complex-linear dual pairing

$$\Theta(F)(\varepsilon_{ab}) = f_{ab}, \quad \Theta(F)(x) = f^\top x_{\text{prim}}. \quad (\text{OKM3})$$

There is no binomial coefficient or complex conjugation in this complex-linear dual pairing. The statement follows by equating all coefficients of the distinct exponentials, whose independence is proved by their Vandermonde derivative matrix in OQC4.

Let J be exactly the coefficient subspace of the original observed dual image, as identified in OQC8. The original strip and invariant elimination provider OCF constructs its literal surjection

$$\pi : \mathbb{C}^q \twoheadrightarrow \mathbb{C}^{r_K}, \quad \ker \pi = J, \quad r_K = 8k - 16. \quad (\text{OKM4})$$

This input is the actual OCF matrix, including its stated period coefficients and its valid pivot choice. The coefficient pairing OKM3 and OQC8 identify $K := \ker \Lambda = J^\perp$ in primary coordinates, where $J^\perp = \{x : f^\top x = 0 \text{ for every } f \in J\}$. Consequently its exact primal frame and polynomial frame are

$$I_{\text{prim}} = \pi^\top : \mathbb{C}^{r_K} \hookrightarrow E_{\text{prim}}, \quad I_{\text{poly}} = V_\beta^{-1} \pi^\top : \mathbb{C}^{r_K} \hookrightarrow E. \quad (\text{OKM5})$$

Here is a direct proof, including inverse coordinates. Choose any of the explicit right sections L_π of the surjective matrix π , so $\pi L_\pi = I_{r_K}$. For $x = \pi^\top v$ and $f \in J$, $f^\top x = (\pi f)^\top v = 0$. The transpose is injective because $L_\pi^\top \pi^\top = I_{r_K}$. Conversely, if $x \in J^\perp$, then $(I - L_\pi \pi)f \in J$ for every f , so $x^\top (I - L_\pi \pi)f = 0$ for every f . Thus $x = \pi^\top L_\pi^\top x$. This proves equality of the image and K , and its inverse on that image is $L_\pi^\top$. In polynomial coordinates the inverse is $L_\pi^\top V_\beta$. This proves the full typed primal arrow, rather than selecting a Hermitian transpose in place of it.

11.2 The complete original source and quotient metric

For either original source $s \in \{1, k\}$ write $b_s = s/2$ and $M_s = (2\pi)^{s/2}$. The unchanged real-coordinate measure is

$$dm_{0,s}(y) = M_s \frac{2^{b_s}}{4\pi\Gamma(b_s)} |\Gamma(b_s/2 + iy/2)|^2 dy, \quad S = c + iy. \quad (\text{OKM6})$$

The complete canonical OGN1–3 proof gives the monic real orthogonal polynomials and all their squared norms:

$$\begin{aligned} p_0(y) = 1, \quad p_1(y) = y, \quad p_{d+1}(y) = yp_d(y) - d(b_s + d - 1)p_{d-1}(y), \\ h_d = M_s d! (b_s)_d, \quad u_d(S) = \frac{p_d((S - c)/i)}{\sqrt{h_d}}. \end{aligned} \quad (\text{OKM7})$$

The square root is positive. Thus u_d has the original leading coefficient $i^{-d}/\sqrt{h_d}$ in S and is orthonormal on the line in OKM6. No source mass or phase has been removed.

For a polynomial cutoff $D \geq q - 1$ let

$$\begin{aligned} T_D &= \left(\frac{p_d(\omega_{ab})}{\sqrt{h_d}} \right)_{(a,b), 0 \leq d \leq D}, \quad R_D = T_D T_D^*, \\ G_D^{\text{prim}} &= R_D^{-1}, \quad G_D^{\text{poly}} = V_\beta^* R_D^{-1} V_\beta, \quad H_D^K = \bar{\pi} R_D^{-1} \pi^\top. \end{aligned} \quad (\text{OKM8})$$

The star denotes conjugate transpose, while the superscript $\top$ is ordinary transpose. In particular $(\pi^\top)^* = \bar{\pi}$, with precisely the indicated matrix dimensions.

These are the actual minimum quotient metrics, as follows directly from the original remainder map. In the orthonormal coefficient frame $u_0, \dots, u_D$, the remainder map followed by primary evaluation is exactly T_D , column by column. Its first q columns are invertible: the monic polynomials $p_0, \dots, p_{q-1}$ evaluated at the distinct ω_{ab} have a nonzero Vandermonde determinant, and all $\sqrt{h_d}$ are nonzero. Thus $R_D > 0$. For a prescribed primary vector x , the vector $T_D^* R_D^{-1} x$ maps to x . It is orthogonal to $\ker T_D$, because its inner product with a kernel vector contains T_D applied to that vector. Every other lift differs by such a vector, so its squared norm is $x^* R_D^{-1} x$ plus the squared norm of that difference. This proves the first quotient metric in OKM8. The identity $x_{\text{prim}} = V_\beta x_{\text{poly}}$ proves the second by direct pairing. Restriction to the exact frame OKM5 then gives

$$I_{\text{poly}}^* G_D^{\text{poly}} I_{\text{poly}} = (\pi^\top)^* R_D^{-1} \pi^\top = H_D^K > 0. \quad (\text{OKM9})$$

The last positivity follows from injectivity of $\pi^\top$. The map $K \hookrightarrow E$ is the fixed original coefficient map. No invariance of that kernel under multiplication is used.

11.3 Christoffel–Darboux with the exact source mass

For arbitrary $z, w \in \mathbb{C}$ and $D \geq 0$, the complete polynomial identity is

$$\sum_{d=0}^D \frac{p_d(z) p_d(w)}{h_d} = \begin{cases} \frac{p_{D+1}(z) p_D(w) - p_D(z) p_{D+1}(w)}{h_D(z - w)}, & z \neq w, \\ \frac{p'_{D+1}(z) p_D(z) - p'_D(z) p_{D+1}(z)}{h_D}, & z = w. \end{cases} \quad (\text{OKM10})$$

To prove it, put $a_d = d(b_s + d - 1)$, with $a_0 = 0$, and set $p_{-1} = 0$ for this recurrence expression. The recurrence in OKM7 gives

$$(z - w)p_d(z)p_d(w) = p_{d+1}(z)p_d(w) - p_d(z)p_{d+1}(w) - a_d(p_d(z)p_{d-1}(w) - p_{d-1}(z)p_d(w)). \quad (\text{OKM11})$$

For $d \geq 1$, the full norm identity is $a_d/h_d = 1/h_{d-1}$. Divide OKM11 by h_d and sum. Every adjacent term cancels, leaving just the degree- D numerator divided by h_D ; the $d = 0$ lower term is zero. This proves the first line of OKM10. Both sides of the resulting equality after multiplication by $z - w$ are polynomials. Differentiating with respect to z and then setting $z = w$ proves the second line with the displayed order and sign. This also proves the value at every confluent point without an assertion about a limiting singular quotient. The denominator is the full $h_D = M_s D!(b_s)_D$, including M_s .

Since all coefficients of the recurrence are real, the matrix entries in OKM8 are therefore exactly the values of OKM10 at

$$z = \omega_{ab}, \quad w = \overline{\omega_{a'b'}}. \quad (\text{OKM12})$$

Their indexed conjugation and confluent locus are

$$\begin{aligned} \overline{\omega_{a'b'}} &= \omega_{k-a',b'}, & \omega_{ab} - \overline{\omega_{a'b'}} &= 2(b - b')\gamma - 2i(a + a' - k)\delta, \\ \omega_{ab} &= \overline{\omega_{a'b'}} & \iff & b = b' \text{ and } a + a' = k. \end{aligned} \quad (\text{OKM13})$$

These follow by substituting OKM1; positivity of δ, γ proves both implications in the last line. Thus the conjugation map on the point indices is precisely $(a, b) \mapsto (k - a, b)$. The confluent formula of OKM10 applies at these indexed entries. For the retained odd k , its matrix diagonal has $a' = a, b' = b$, so $2a = k$ cannot occur; the literal Hermitian diagonal is given by the first line of OKM10. The conjugate-index entries use its second line. All entries, including the positive diagonal, are simultaneously defined by the original polynomial sum on the left of OKM10.

11.4 Every pivot change and the four original endpoints

Let π' be another valid strip/invariant elimination matrix for the same original J , with the same surjective codomain dimension. Choose right sections $L_\pi, L_{\pi'}$ and put

$$A_{\pi',\pi} = \pi' L_\pi, \quad A_{\pi,\pi'} = \pi L_{\pi'}. \quad (\text{OKM14})$$

Because π' vanishes on $J = \ker \pi$, one has $\pi' L_\pi \pi = \pi'$. Thus $\pi' = A_{\pi',\pi} \pi$. Reversing the roles gives the opposite identity, and surjectivity of each matrix then proves $A_{\pi',\pi} A_{\pi,\pi'} = I$ and $A_{\pi,\pi'} A_{\pi',\pi} = I$. This proves invertibility and also independence from the chosen right sections. Writing $A = A_{\pi',\pi}$ for these identities gives the exact frame and Gram changes

$$\begin{aligned} I'_{\text{prim}} &= I_{\text{prim}} A^\top, & I'_{\text{poly}} &= I_{\text{poly}} A^\top, \\ (H_D^K)' &= \overline{A} H_D^K A^\top, & \det(H_D^K)' &= |\det A|^2 \det H_D^K. \end{aligned} \quad (\text{OKM15})$$

The first two formulas follow by transposing $\pi' = A\pi$. Substitution into OKM9 proves the congruence and determinant. This retains all actual pivot determinants. They have not been assigned modulus one.

The kernel is the same fixed $K = \ker \Lambda$ used in MRI1–5. If its original BRU inclusion is I_0 and its fixed-section kernel coordinate map is κ_0 , let

$$C_K = \kappa_0 I_{\text{poly}}, \quad C_K^{-1} = L_\pi^\top V_\beta I_0. \quad (\text{OKM16})$$

The equalities $I_0 \kappa_0 x = x$ for $x \in K$ and $L_\pi^\top V_\beta I_{\text{poly}} = I_{r_K}$ prove that these are inverse matrices and $I_0 C_K = I_{\text{poly}}$. Consequently $H_D^K = C_K^* (I_0^* G_D^{\text{poly}} I_0) C_K$. The matrix C_K is independent of D and

of the choice of source order s . The source metrics change; the two specified coefficient frames do not.

It follows that the complete original four-endpoint kernel scalar is

$$\boxed{F_K^{(s)} = \log \det H_{q-1}^K + \log \det H_q^K - \log \det H_{2q-1}^K - \log \det H_{2q}^K.} \quad (\text{OKM17})$$

Indeed MRI3 gives the actual consecutive kernel determinant ratio

$$\frac{\det(I_0^* G_{q+j-1}^{\text{poly}} I_0)}{\det(I_0^* G_{q+j}^{\text{poly}} I_0)} = \frac{1 + t_j}{1 + \xi_j}.$$

Each factor $|\det C_K|^2$ cancels in that ratio by OKM16. The weighted sum with endpoint weights $1, 2, \dots, 2, 1$ therefore has precisely the four signs in OKM17, as in MRI4. This verifies the original degree cutoffs, rather than shifting the column count in T_D . Under any valid pivot change OKM15, the added scalar $2 \log |\det A|$ has total coefficient $1 + 1 - 1 - 1 = 0$, proving invariance of OKM17 with its original frames.

For emphasis, the full mass in OKM7 appears in every column of T_D and in the denominator of OKM10. If that common mass is factored explicitly, write $T_D = M_s^{-1/2} \hat{T}_D$, where $\hat{T}_D$ has exactly the same numerator polynomials and the denominator $\sqrt{d!(b_s)_d}$. Then $H_D^K = M_s \bar{\pi} (\hat{T}_D \hat{T}_D^*)^{-1} \pi^\top$, so its determinant has factor $M_s^{r_K}$. It cancels only in the displayed signed four-endpoint scalar, whose four ranks are equal. The individual metrics in OKM8–9 retain it. No inserted Gamma order, discarded action defect, or kernel-invariance assertion enters any of these calculations.

12 The original kernel graph, boundary basis and complete action

This section uses the complete conductor coefficient construction OCF and its exact original dual identification in OKM. Fix the original simple quartet, its complete amplitude unit and periods, the original parameter on any logarithm branch with $|u| \geq R_*$, and $k = 4l + 1$, $l \geq 1$. Put $q = (k + 1)^2$, $r = 8k - 16$, and $\mathcal{I}_k = \{(a, b) : 0 \leq a, b \leq k\}$. The roots, centre and centred eigenvalues remain

$$c = k/2, \quad \beta_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \omega_{ab} = (\beta_{ab} - c)/i = (2b - k)\gamma - i(2a - k)\delta. \quad (\text{OKA1})$$

The original algebra is $E = \mathbb{C}[S]/(\chi)$, with $\chi(S) = \prod_{(a,b) \in \mathcal{I}_k} (S - \beta_{ab})$. Let ε_{ab} be its original primary idempotents and $\mathcal{V} : E \rightarrow \mathbb{C}^{\mathcal{I}_k}$ the evaluation isomorphism $[p] \mapsto (p(\beta_{ab}))_{ab}$. Its inverse sends e_{ab} to

$$\varepsilon_{ab}(S) = \frac{\chi(S)}{(S - \beta_{ab})\chi'(\beta_{ab})}. \quad (\text{OKA2})$$

These denominators are nonzero: equality of two roots in OKA1, on comparing real and imaginary parts and using $\delta > 0$, $\gamma > 2$, implies equality of both indices. Evaluation proves that the polynomials in OKA2 are the coordinate idempotents, so their columns give the exact inverse, with no changed root or polynomial coefficient convention.

12.1 The coefficient quotient gives a graph in the original primary basis

Let $\pi : \mathbb{C}^{\mathcal{I}_k} \rightarrow \mathbb{C}^r$ be the surjective coefficient quotient of OCF, whose kernel is precisely the image J_k of the invariant polynomials under the original restriction. The domain here consists of the coefficients of the biforms $z_0^a z_1^{k-a} w_0^b w_1^{k-b}$, and is the complex-linear dual of the displayed primary coordinate space. Let $T \subset \mathcal{I}_k$ be the final OCF complement of its invariant-image pivot rows, viewed as a subset of the full biform coefficient grid. Its ordered coordinate inclusion is $E_T : \mathbb{C}^r \rightarrow \mathbb{C}^{\mathcal{I}_k}$. The explicit OCF formula gives

$$\pi E_T = I_r, \quad N = \pi^T : \mathbb{C}^r \longrightarrow \mathbb{C}^{\mathcal{I}_k}, \quad E_T^T N = I_r. \quad (\text{OKA3})$$

Indeed a retained strip monomial in T already has zero conductor remainder outside its own coordinate, zero coordinates on the invariant pivot rows, and its indicated coordinate equal to one. Both reductions in OCF therefore leave that quotient coordinate equal to its standard unit vector. Transposing gives the last identity. This uses ordinary transpose, because the restriction quotient and its dual are complex-linear maps.

Write $J = \mathcal{I}_k \setminus T$ only for the complementary index set in this section; the invariant image retains the name J_k . Denote coordinate restrictions by x_T, x_J , and put $N_J = E_J^T N$. The original kernel inclusion is consequently

$$I_K = \mathcal{V}^{-1} N : \mathbb{C}^r \hookrightarrow E, \quad \text{im } I_K = K := \ker \Lambda, \quad N = \begin{pmatrix} I_r \\ N_J \end{pmatrix}_{T,J}. \quad (\text{OKA4})$$

For completeness, a primary vector v pairs with a biform coefficient vector f by $f^T v$. The annihilator of $\ker \pi = J_k$ is $\text{im } \pi^T$: containment follows by multiplication, while surjectivity of π gives both subspaces dimension r . OQC8 identifies precisely this annihilator with the primary coordinates of the fixed original $\ker \Lambda$. This proves both the original image assertion and its direction in OKA4. In polynomial coordinates the same inclusion is $V_\beta^{-1} N$, where $(V_\beta)_{(a,b),d} = \beta_{ab}^d$ for $0 \leq d < q$; this is the full coefficient matrix of OKA2.

12.2 An explicit basis of the actual boundary

For $j \in J$ define the actual boundary vector $b_j = \Lambda \varepsilon_j$. These vectors form a basis of the original image $B = \text{im } \Lambda$. To prove independence, if $\sum_{j \in J} c_j \varepsilon_j$ lies in K , its primary coordinates equal $N \zeta$.

Its T coordinates vanish, so OKA3 gives $\zeta = 0$ and all $c_j = 0$. To prove spanning, use the literal decomposition

$$x = Nx_T + E_J(x_J - N_Jx_T), \quad \Lambda\mathcal{V}^{-1}x = \sum_{j \in J} b_j(x_J - N_Jx_T)_j. \quad (\text{OKA5})$$

The first equality follows separately on its T and J coordinates. Apply Λ , which kills the first term by OKA4, to obtain the second. Thus no observation has been substituted: in this basis the specified original observation has the exact coefficient matrix $(-N_J, I_{q-r})$.

Here are the complete original period and tensor coordinates of these boundary vectors. Write the quartet roots as $\alpha_{\epsilon, \eta} = 1/2 + \epsilon\delta + i\eta\gamma$, where $\epsilon, \eta \in \{+1, -1\}$ and the signs $+, -$ in subscripts denote these same two numbers. In a subscript α below means that complex root, indexed by its sign pair. Keep the original one-factor primary idempotents $\varepsilon_\alpha^h \in E_h$. Set

$$v_\alpha = \Pi U \varepsilon_\alpha^h = v_h(\alpha) \Pi \varepsilon_\alpha^h, \quad Z_+ = z_0, \quad Z_- = z_1, \quad W_+ = w_0, \quad W_- = w_1, \quad \mathbf{v}(z, w) = \sum_{\epsilon, \eta} v_{\epsilon\eta} Z_\epsilon W_\eta. \quad (\text{OKA6})$$

The vectors belong to the original one-factor period space; all entries of Π and all four actual nonzero amplitude values remain present. For every $(a, b) \in \mathcal{I}_k$,

$$j\Lambda\varepsilon_{ab} = [z_0^a z_1^{k-a} w_0^b w_1^{k-b}] P_k(\mathbf{v}(z, w)^{\otimes k}), \quad P_k = \frac{1}{5} \sum_{h=0}^4 (C^{-h})^{\otimes k}. \quad (\text{OKA7})$$

The original sum inclusion $\iota : E \hookrightarrow E_h^{\otimes k}$ sends ε_{ab} to the sum of the primary tensor idempotents whose word has a positive real signs and b positive imaginary signs. This assertion follows by evaluation: on a primary word, the sum variable takes exactly the value β_{ab} of its two counts, and OKA2 evaluates to one for those counts and zero otherwise. Applying $(\Pi U)^{\otimes k}$ gives precisely the corresponding sum of the tensors of the v_α in OKA6. Expanding the tensor power and taking the indicated biform coefficient selects exactly the same words, each once. The original definition $j\Lambda = P_k(\Pi U)^{\otimes k} \iota$ proves OKA7, including the literal inverse powers and $1/5$. Its specialization to $j \in J$ therefore supplies the actual basis b_j of OKA5. Neither a different target nor a numerical period matrix has been introduced.

The fixed, source-independent kernel coordinate and section are now explicitly

$$\kappa = E_T^T \mathcal{V} : E \rightarrow \mathbb{C}^r, \quad S_* : B \rightarrow E, \quad S_* b_j = \varepsilon_j, \quad (\text{OKA8})$$

$$I_K \kappa + S_* \Lambda = I_E, \quad \kappa I_K = I_r, \quad \kappa S_* = 0, \quad \Lambda S_* = I_B.$$

Every equality follows by applying OKA5 to an arbitrary primary vector. In particular this is an actual splitting of the fixed original short exact sequence, used unchanged for both source orders and for all polynomial degrees.

12.3 The whole original action and its retained defect

The original centred sum action is $Y = (S - c)/i$ on E . Under $\mathcal{V}$ it is the diagonal matrix of the ω_{ab} in OKA1. Write its two diagonal submatrices as Ω_T and Ω_J . In the exact coordinates $e = I_K \zeta + S_* b$ of OKA8, it has the complete matrix

$$Y \longleftrightarrow \begin{pmatrix} \Omega_T & 0 \\ \Omega_J N_J - N_J \Omega_T & \Omega_J \end{pmatrix}, \quad (\Omega_J N_J - N_J \Omega_T)_{jt} = (\omega_j - \omega_t)(N_J)_{jt}. \quad (\text{OKA9})$$

Indeed the primary vector of e is $(\zeta, N_J \zeta + b)$. Applying the original diagonal matrix gives $(\Omega_T \zeta, \Omega_J N_J \zeta + \Omega_J b)$. Applying the coordinates in OKA5 subtracts $N_J \Omega_T \zeta$ from its second component and proves every block in OKA9. This proves, with its precise codomain,

$$Y I_K - I_K \Omega_T = S_* (\Omega_J N_J - N_J \Omega_T) : \mathbb{C}^r \rightarrow E, \quad \Lambda Y I_K = \Omega_J N_J - N_J \Omega_T : \mathbb{C}^r \rightarrow B, \quad (\text{OKA10})$$

where the matrix on the right uses the basis b_j . The chosen complementary span of the original primary idempotents is stable under Y ; no stability of K has been used. Its entire failure is the displayed coefficient map.

For each original source $s \in \{1, k\}$ retain the monic Gamma polynomials $p_d^{(s)}$, the complete norms $h_d^{(s)} = (2\pi)^{s/2} d!(s/2)_d$, and the original $e_d^{(s)} = [p_d^{(s)}(Y)]/\sqrt{h_d^{(s)}}$. Then the fixed-section kernel and boundary coordinates of every actual remainder, including all degrees $d \geq q$, are

$$\begin{aligned} (\kappa e_d^{(s)})_t &= \frac{p_d^{(s)}(\omega_t)}{\sqrt{h_d^{(s)}}}, \\ (\Lambda e_d^{(s)})_j &= \frac{p_d^{(s)}(\omega_j) - \sum_{t \in T} (N_J)_{jt} p_d^{(s)}(\omega_t)}{\sqrt{h_d^{(s)}}}. \end{aligned} \tag{OKA11}$$

Polynomial evaluation in the original primary algebra gives the full vector $p_d^{(s)}(\omega)/\sqrt{h_d^{(s)}}$ even when its polynomial representative must be reduced modulo χ . Applying OKA5 and OKA8 proves OKA11. Thus the formula includes the original finite remainder operation and every phase and source mass. Composing its second line with the actual columns OKA7 gives exactly the period-vector increments of OPG. Its first line is the fixed-section kernel vector used by OPG and OPR; the degree-dependent minimum-section correction in OPR remains required for the mixed norm. The metric computation in OKM applies to the common inclusion OKA4 and does not replace that correction by the fixed-section vector alone.

13 The complete homogeneous geometry of the original multiplicity curve

13.1 The actual primary, unit, period and Fourier coefficient map

Retain the original full-order quartet and every original period parameter. Write

$$\begin{aligned} \alpha_{\epsilon,\eta} &= \frac{1}{2} + \epsilon\delta + i\eta\gamma, \quad \epsilon, \eta \in \{-1, 1\}, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \\ m &\geq 1, \quad d_m = m - 1, \quad n = 4m, \quad N = 4m + 1, \quad h(z) = \prod_{\epsilon,\eta} (z - \alpha_{\epsilon,\eta})^m, \\ E_h &= \mathbb{C}[z]/(h), \quad A_h = M_z, \quad v_h = 2\xi/h, \quad U = M_{j_h v_h}, \quad v_h(\alpha_{\epsilon,\eta}) \neq 0. \end{aligned} \quad (\text{OMG1})$$

The last inequalities follow from the stipulated complete original zero order m . The map j_h records all divided Taylor coefficients of orders $0, \dots, m-1$ at all four roots. Let $\mathcal{E}_{\text{prim}} : \mathbb{C}^{4m} \rightarrow E_h$ send its coordinate (α, d) to the original primary element $\varepsilon_\alpha(z - \alpha)^d$, $0 \leq d < m$. Explicitly its representative is

$$E_{\alpha,d}(z) = \text{rem}_h \left[h_\alpha(z) \sum_{j=0}^{m-1} \frac{(h_\alpha^{-1})^{(j)}(\alpha)}{j!} (z - \alpha)^{j+d} \right], \quad h_\alpha = h/(z - \alpha)^m.$$

The Taylor inverse is defined since $h_\alpha(\alpha) \neq 0$. It gives the idempotent one on its own primary factor and zero on the others. The four coprime factors therefore prove that $\mathcal{E}_{\text{prim}}$ is an isomorphism and include every nilpotent degree. In this basis the full matrix of U is

$$U_{\text{prim};\alpha,a;\alpha,d} = \begin{cases} v_h^{(a-d)}(\alpha)/(a-d)!, & a \geq d, \\ 0, & a < d, \end{cases} \quad \det U = \prod_{\alpha} v_h(\alpha)^m \neq 0. \quad (\text{OMG2})$$

Other primary blocks are zero. Inverting its constant coefficient plus its nilpotent part by the finite geometric series proves invertibility with all derivatives retained.

Keep the original $u \neq 0$ and its logarithm branch, the primitive $\Phi'_h = h$, $\Phi_h(0) = 0$, the outward rays ℓ_j of angles $(\arg u + \pi + 2\pi j)/N$, and $\Gamma_j = -\ell_0 + \ell_j$. The exact period and Fourier maps are

$$\begin{aligned} (\Pi[P])_j &= \int_{\Gamma_j} e^{\Phi_h(z)/u} \text{rem}_h P(z) dz, \quad 1 \leq j \leq n, \\ \zeta &= e^{2\pi i/N}, \quad F_{jt} = \zeta^{jt} - 1, \quad (F^{-1})_{tj} = \zeta^{-tj}/N \quad (1 \leq j, t \leq n), \quad C = F \text{diag}(\zeta, \dots, \zeta^n) F^{-1}. \end{aligned} \quad (\text{OMG3})$$

The full original period theorem PDE/OPG gives the convergence and invertibility of Π on these representatives. The Fourier inverse follows directly from $\sum_{j=1}^{N-1} \zeta^{-tj} (\zeta^{sj} - 1) = N \mathbf{1}_{t=s}$. It is the same literal cyclic matrix as OCS4, including its inverse powers in the original average. Its inherited Gram is $F^*F = N(I_n + \mathbf{1}\mathbf{1}^*)$, as expansion of its entries gives.

Define the exact diagonal map and coefficient matrix

$$\begin{aligned} D_m^{\text{jet}} &= \text{diag} \left(i^{-d}/d! \right)_{\alpha, 0 \leq d < m}, \quad \mathbf{H}_m = F^{-1} \Pi U \mathcal{E}_{\text{prim}} D_m^{\text{jet}}, \\ (\mathbf{H}_m)_{t;\alpha,d} &= \frac{i^{-d}}{d!} \sum_{a=d}^{m-1} \left[\frac{1}{N} \sum_{j=1}^n \zeta^{-tj} \int_{\Gamma_j} e^{\Phi_h(z)/u} E_{\alpha,a}(z) dz \right] \frac{v_h^{(a-d)}(\alpha)}{(a-d)!}. \end{aligned} \quad (\text{OMG4})$$

This is exactly OCS13, now regarded as one square matrix. All factors in its first line are invertible. In particular

$$\det \mathbf{H}_m = (\det F)^{-1} \det \Pi \left(\prod_{\alpha} v_h(\alpha)^m \right) \det \mathcal{E}_{\text{prim}} \prod_{\alpha} \prod_{d=0}^{m-1} \frac{i^{-d}}{d!} \neq 0,$$

with the determinant of $\mathcal{E}_{\text{prim}}$ taken in the fixed original coefficient frame. None of these factors is prescribed.

Set $\omega_{\epsilon,\eta} = \eta\gamma - i\epsilon\delta$ and let

$$z_{\epsilon,\eta,d}(w) = w^d e^{\omega_{\epsilon,\eta} w}, \quad x(w) = F^{-1} \Pi U e^{w(A_h - 1/2)/i} [1]. \quad (\text{OMG5})$$

The terminating exponential on each primary factor has coefficient $i^{-d} w^d / d!$ on $\varepsilon_\alpha(z - \alpha)^d$. Multiplication by OMG2 and application of the literal period maps therefore prove the exact identity and inverse coordinate map

$$\boxed{x(w) = H_m z(w), \quad z(w) = H_m^{-1} x(w).} \quad (\text{OMG6})$$

For the original Euclidean period metric, its Gram on these primary curve coordinates is $H_m^* N(I_n + \mathbf{11}^*) H_m$. Thus the coefficient change also retains its full metric and all off-diagonal entries, without claiming orthogonality.

13.2 The complete homogeneous ideal, with quadratic generators

Use binary indices $a, b \in \{0, 1\}$ for $\epsilon = 2a - 1$, $\eta = 2b - 1$. Let $\mathcal{P}_Z = \mathbb{C}[Z_{a,b,d} : a, b \in \{0, 1\}, 0 \leq d \leq d_m]$ with each variable of degree one. Define the algebra map

$$\Phi_m : \mathcal{P}_Z \longrightarrow \mathbb{C}[X_0, X_1, Y_0, Y_1, T_0, T_1], \quad Z_{a,b,d} \longmapsto X_a Y_b T_0^{d_m - d} T_1^d. \quad (\text{OMG7})$$

Its image is the complete diagonal graded algebra

$$R_m = \bigoplus_{k \geq 0} \mathbb{C}[X_0, X_1]_k \otimes \mathbb{C}[Y_0, Y_1]_k \otimes \mathbb{C}[T_0, T_1]_{k d_m}. \quad (\text{OMG8})$$

Indeed a basis monomial of its degree- k component is

$$X_0^{k-a} X_1^a Y_0^{k-b} Y_1^b T_0^{k d_m - D} T_1^D, \quad 0 \leq a, b \leq k, \quad 0 \leq D \leq k d_m. \quad (\text{OMG9})$$

It is a product of k images in OMG7: choose k binary first indices with sum a , independently k binary second indices with sum b , and k integers in $[0, d_m]$ with sum D ; the last choice follows by filling successive slots up to d_m . Combining the three lists into rows gives the required preimage monomial. Distinct displayed target monomials are linearly independent.

Let $\mathfrak{t}_m$ be generated by every quadratic binomial

$$Z_{a,b,d} Z_{a',b',d'} - Z_{c,e,f} Z_{c',e',f'} \quad \text{with} \quad a + a' = c + c', \quad b + b' = e + e', \quad d + d' = f + f', \quad (\text{OMG10})$$

where every index belongs to its original range. Repeated variables are allowed. Then

$$\boxed{\ker \Phi_m = \mathfrak{t}_m, \quad \mathcal{P}_Z / \mathfrak{t}_m \simeq R_m.} \quad (\text{OMG11})$$

Here is a complete ideal proof. Each displayed quadratic has zero image. Two degree- k monomials have the same image exactly when the sums of their first indices, second indices and third indices agree. Regard their k factors temporarily as labelled rows. Align the first indices with the target list by pairwise swaps of first indices, keeping each row's other indices fixed. Each swap is an OMG10 relation. Next align the second indices by the same operation, keeping the aligned first indices fixed. Finally, if the third indices differ from their target list, choose a row below its target and a row above its target. Increase the former third index by one and decrease the latter by one. These entries stay in $[0, d_m]$ because their target entries do; the total sum is unchanged and the sum of absolute deviations decreases by two. This terminates at the target list. Each transfer is another OMG10 relation. Thus the difference between any two monomials with the same image is a sum of multiples of the quadrics, by telescoping the successive monomial differences. Labels only specify the sequence of polynomial maps; the products are the original commutative monomials

throughout. For any polynomial in $\ker \Phi_m$, collect its coefficients by their target monomial. Each such coefficient sum is zero by target monomial independence. Subtracting one chosen monomial in each group expresses that whole group as a linear combination of the proved monomial differences. This proves the reverse inclusion for all degrees and hence the entire ideal equality.

Every degree- k original curve monomial has the function

$$w^D \exp\{[(2b - k)\gamma - i(2a - k)\delta]w\} \quad (0 \leq a, b \leq k, \ 0 \leq D \leq kd_m). \quad (\text{OMG12})$$

The frequencies are distinct: equality of real parts gives equality of b and equality of imaginary parts gives equality of a . To prove independence with all polynomial powers, suppose $\sum_{a,b} P_{a,b}(w)e^{\omega_{a,b}w} = 0$, with $\deg P_{a,b} \leq kd_m$. Fix a pair and apply $\prod_{(a',b') \neq (a,b)} (\partial_w - \omega_{a',b'})^{kd_m+1}$. This kills every other summand. On its surviving polynomial each factor is $\partial_w + (\omega_{a,b} - \omega_{a',b'})$, invertible on the space of degree at most kd_m because differentiation is nilpotent and the constant is nonzero. The surviving polynomial is therefore zero. Repeating this for every pair proves independence.

Thus the full homogeneous ideal of the original primary curve is exactly $\mathfrak{t}_m$: a homogeneous polynomial restricts to zero if and only if its coefficient sums at every target monomial OMG9 are zero. In particular

$$\boxed{\dim R_{m,k} = (k+1)^2[1 + k(m-1)] = q_k, \quad \ker\{\mathcal{P}_{Z,k} \rightarrow \text{Hol}(\mathbb{C}), f \mapsto f(z(w))\} = \mathfrak{t}_{m,k}.} \quad (\text{OMG13})$$

This statement concerns the homogeneous ideal, equivalently the projective closure of $[z(w)]$. Every one of its degrees has been computed, including degree zero. No assertion about a different nonhomogeneous affine ideal is used.

13.3 The projective model and its complete chart inverses

For $m \geq 2$, the morphism

$$(\mathbb{P}^1)^3 \longrightarrow \mathbb{P}^{4m-1}, \quad ([X_0 : X_1], [Y_0 : Y_1], [T_0 : T_1]) \longmapsto [X_a Y_b T_0^{d_m-d} T_1^d]_{a,b,d} \quad (\text{OMG14})$$

has multidegree $(1, 1, m-1)$. At least one first, second and third coordinate is nonzero; their corresponding extreme product is nonzero, so the map is everywhere defined. Its image and scheme structure are exactly $\text{Proj } R_m$, with complete closed ideal OMG10. Here are explicit inverses proving this claim. The eight charts $Z_{a,b,0} \neq 0$ and $Z_{a,b,d_m} \neq 0$ cover $\text{Proj } R_m$. Indeed OMG11 gives

$$Z_{a,b,d}^{d_m} = Z_{a,b,0}^{d_m-d} Z_{a,b,d_m}^d \quad \text{in } R_m.$$

A homogeneous prime containing every extreme variable contains every variable by this equality, and hence gives no point of Proj . On $Z_{a,b,0} \neq 0$ the three inverse affine coordinates are

$$x = Z_{1-a,b,0}/Z_{a,b,0}, \quad y = Z_{a,1-b,0}/Z_{a,b,0}, \quad t = Z_{a,b,1}/Z_{a,b,0}.$$

For every variable its ratio to $Z_{a,b,0}$ is $x^{1_{a' \neq a}} y^{1_{b' \neq b}} t^d$, by OMG7 and its proved injective quotient. Consequently the degree-zero localized ring is $\mathbb{C}[x, y, t]$. These three coordinates are algebraically independent, since their images in the parameter fraction field are $X_{1-a}/X_a, Y_{1-b}/Y_b, T_1/T_0$. For the chart at Z_{a,b,d_m} replace the last ratio by $t = Z_{a,b,d_m-1}/Z_{a,b,d_m}$; the general ratio has exponent $d_m - d$ in t . Its coordinate ring is again $\mathbb{C}[x, y, t]$. On chart overlaps these formulas invert the same parameter ratios, so they glue to the inverse of OMG14. This proves the isomorphism of schemes and its full defining homogeneous ideal. For $m = 1$ the third factor has degree zero and the same argument uses the four first/second charts: the model is the Segre $(\mathbb{P}^1)^2$ with ideal $Z_{0,0,0}Z_{1,1,0} - Z_{0,1,0}Z_{1,0,0}$.

In the actual Fourier coordinates let $\mathcal{P}_x = \mathbb{C}[x_1, \dots, x_n]$ and define

$$\mathfrak{a} = \{f(x) : f(H_m Z) \in \mathfrak{t}_m\}, \quad R = \mathcal{P}_x / \mathfrak{a}. \quad (\text{OMG15})$$

The invertible substitution OMG6 proves that $\mathfrak{a}$ is the complete homogeneous ideal of $x(w)$ and that $R \simeq R_m$. Its explicit generators are all OMG10 quadrics with every Z replaced by its component of $H_m^{-1}x$. These are the actual period and unit coefficients of OMG4, and $\mathfrak{a}$ is prime because the target ring in OMG7 is a domain. Thus the original curve's projective closure is the precise linear image of OMG14 under H_m , with no prescribed substitute matrix.

13.4 The exact graded source-dual algebra and every factorial

For each $k \geq 0$ retain the original sum algebra

$$e_k = 1 + kd_m, \quad c_k = k/2, \quad \beta_{k,a,b} = c_k + (2a - k)\delta + i(2b - k)\gamma, \\ \chi_k(S) = \prod_{a,b=0}^k (S - \beta_{k,a,b})^{e_k}, \quad E_k = \mathbb{C}[S]/(\chi_k), \quad Y_k = (M_S - c_k I)/i. \quad (\text{OMG16})$$

For $k = 0$ this is $E_0 = \mathbb{C}[S]/(S) = \mathbb{C}$. Let $H_{k,a,b,D} = \varepsilon_{k,a,b}(S - \beta_{k,a,b})^D$, $0 \leq D < e_k$, be the complete original primary basis obtained by the same Taylor inverse as in OMG2. It satisfies

$$e^{wY_k}[1] = \sum_{a,b=0}^k e^{\omega_{k,a,b}w} \sum_{D=0}^{e_k-1} \frac{i^{-D}w^D}{D!} H_{k,a,b,D}, \quad \omega_{k,a,b} = (2b - k)\gamma - i(2a - k)\delta. \quad (\text{OMG17})$$

For $f \in \mathcal{P}_{x,k}$ expand its original function uniquely as $f(x(w)) = \sum c_{a,b,D}(f)w^D e^{\omega_{k,a,b}w}$ using OMG12. Define the complex linear dual map, without a conjugation,

$$\Theta_k(f)(H_{k,a,b,D}) = i^D D! c_{a,b,D}(f). \quad (\text{OMG18})$$

Then OMG17 proves exactly $\Theta_k(f)(e^{wY_k}[1]) = f(x(w))$. The independence and surjectivity in OMG9–13 prove that this map is onto $E_k^\vee$, with kernel $\mathfrak{a}_k$. Consequently it induces a specified vector-space isomorphism

$$\boxed{\Theta_k : R_k \xrightarrow{\sim} E_k^\vee}. \quad (\text{OMG19})$$

All phases and factorials are those of OCS17; for a monomial x^ν its coefficients are precisely the finite multinomial OCS15 rule with the entries OMG4.

The collection of these vector spaces has an exact graded algebra structure coming from the original sum maps. For $k, j \geq 0$ put

$$\Delta_{k,j} : E_{k+j} \longrightarrow E_k \otimes E_j, \quad [P(S)] \longmapsto [P(S_1 + S_2)]. \quad (\text{OMG20})$$

On a primary pair the eigenvalue is the displayed sum $\beta_{k,a,b} + \beta_{j,a',b'} = \beta_{k+j,a+a',b+b'}$. The sum of the two nilpotents has index $e_k + e_j - 1 = e_{k+j}$: its power $e_k + e_j - 2$ has nonzero top coefficient $\binom{e_k+e_j-2}{e_k-1}$ and its next power is zero. Every pair of total sign counts occurs. Thus the full minimal polynomial of $S_1 + S_2$ is exactly χ_{k+j} , proving both well-definedness and injectivity of OMG20. It also gives the complete primary formula

$$\Delta_{k,j} H_{k+j,A,B,D} = \sum_{a+a'=A, b+b'=B} \sum_{\substack{d+d'=D \\ 0 \leq d < e_k, 0 \leq d' < e_j}} \binom{D}{d} H_{k,a,b,d} \otimes H_{j,a',b',d'}. \quad (\text{OMG21})$$

Indeed the total primary idempotent is one precisely on the matching eigenvalue pairs, and the nilpotent power expands by the ordinary binomial theorem there. All other components vanish. Substitution of the sum of three coordinates proves coassociativity; interchange of the two coordinates proves cocommutativity. The $k = 0$ maps give the count. Hence

$$(\ell \star \ell')(v) = (\ell \otimes \ell')(\Delta_{k,j}v) \quad (\ell \in E_k^\vee, \ell' \in E_j^\vee) \quad (\text{OMG22})$$

is an associative commutative graded product with unit in $E_0^\vee$. For the exact dual basis $\lambda_{k,a,b,D} = i^D D! H_{k,a,b,D}^\vee$, OMG21 gives

$$\lambda_{k,a,b,D} \star \lambda_{j,a',b',D'} = \lambda_{k+j,a+a',b+b',D+D'}.$$

The coefficient is exactly $\binom{D+D'}{D} i^D D! i^{D'} D'! = i^{D+D'} (D+D')!$; this records the cancellation before stating the basis product. Equivalently, OMG20 sends $e^{wY_{k+j}}[1]$ to $e^{wY_k}[1] \otimes e^{wY_j}[1]$, with $c_{k+j} = c_k + c_j$. Evaluating OMG18 on this identity proves

$$\Theta : R \xrightarrow{\sim} \bigoplus_{k \geq 0} (E_k^\vee, \star) \text{ is an isomorphism of graded algebras.} \quad (\text{OMG23})$$

Its source multiplication is the actual polynomial multiplication; its target multiplication has been constructed from the original sum coalgebra rather than assigning a multiplication between unrelated finite-dual vector spaces.

13.5 The full semisimple and nilpotent action in this geometry

On $\mathcal{P}_Z$ define the degree-preserving derivation

$$\partial_Z Z_{a,b,d} = \omega_{2a-1,2b-1} Z_{a,b,d} + \begin{cases} dZ_{a,b,d-1}, & d > 0, \\ 0, & d = 0. \end{cases} \quad (\text{OMG24})$$

On the parameter ring OMG7 it is induced by $\partial X_a = -i(2a-1)\delta X_a$, $\partial Y_b = (2b-1)\gamma Y_b$, $\partial T_0 = 0$, $\partial T_1 = T_0$. The product rule verifies OMG24 and commutation with Φ_m ; therefore the complete ideal $\mathfrak{t}_m$ is stable. The corresponding exact flow on the parameters is

$$(X_a, Y_b, T_0, T_1) \mapsto (e^{-i(2a-1)\delta w} X_a, e^{(2b-1)\gamma w} Y_b, T_0, T_1 + wT_0).$$

At $(X_0, X_1, Y_0, Y_1, T_0, T_1) = (1, 1, 1, 1, 0)$ it gives precisely the original functions OMG5. This is the entire stated analytic flow on the explicit algebraic variety, with all polynomial degrees.

If B_Z is the matrix in OMG24 on the column of variables, its rows have diagonal ω_α and entry d in column $(\alpha, d-1)$ of row (α, d) . In actual coordinates the derivation is

$$\partial_x x = \mathcal{A}x, \quad \mathcal{A} = H_m B_Z H_m^{-1}. \quad (\text{OMG25})$$

The inverse substitution proves $\partial_x \mathfrak{a} \subseteq \mathfrak{a}$, and its flow differentiates the original $x(w)$. On the exact degree- k basis of OMG9/OMG18 the induced action is

$$\partial_x \lambda_{k,a,b,D} = \omega_{k,a,b} \lambda_{k,a,b,D} + \begin{cases} D \lambda_{k,a,b,D-1}, & D > 0, \\ 0, & D = 0. \end{cases} \quad \Theta_k \partial_x = Y_k^\vee \Theta_k. \quad (\text{OMG26})$$

The displayed action on the basis uses its identification OMG23. For a direct verification, on the original primal primary basis $Y_k H_D = \omega H_D + i^{-1} H_{D+1}$ with the last term zero at $D = e_k - 1$. Applying $i^D D! H_D^\vee$ proves the exact lowering coefficient D in its dual action. Each frequency block has all e_k terms, with nonzero top iterated derivative $(e_k - 1)!$. Thus every original primary nilpotent has exactly the same index e_k in this model. The full annihilating polynomial is retained with its leading phase:

$$\chi_k(c_k + i\partial_x) = 0 \quad \text{on } R_k, \quad \chi_k(c_k + iT) = i^{q_k} \prod_{a,b=0}^k (T - \omega_{k,a,b})^{e_k}. \quad (\text{OMG27})$$

Distinct frequencies and the nonzero top chain prove that this degree q_k is minimal. The original S action corresponds to $c_k + i\partial_x$ on degree k , or to $\frac{1}{2}\mathbf{E} + i\partial_x$ on the graded algebra, where $\mathbf{E}$ is the degree derivation. These equalities specify every constant, sign and nilpotent action under OMG23.

13.6 The exact original cyclic observation as invariant restriction

Let $D = \text{diag}(\zeta, \dots, \zeta^n)$ and let $G = \mathbb{Z}/N$ act on $\mathcal{P}_x$ by $(gf)(x) = f(D^{-1}x)$. Thus a degree- k monomial x^ν has character $\zeta^{-\sum t\nu_t}$. Write

$$\mathcal{J} = \mathcal{P}_x^G, \quad \mathcal{J} = \text{im}(\mathcal{J} \longrightarrow R). \quad (\text{OMG28})$$

These are graded algebras. The coefficient image of the original observation is exactly their restriction at every degree, as follows. Let f_t be column t of F and let S_ν be the sum of all distinct ordered words in the f_t having multiplicity vector ν . Disjoint word supports make the S_ν a basis of the permutation-invariant tensor space. The expansion and projector are

$$(Fx)^{\otimes k} = \sum_{|\nu|=k} x^\nu S_\nu, \quad P_k(Fx)^{\otimes k} = \sum_{\substack{|\nu|=k \\ \sum t\nu_t=0 \pmod{N}}} x^\nu S_\nu, \quad P_k = \frac{1}{N} \sum_{a=0}^{N-1} (C^{-a})^{\otimes k}.$$

Every ordered word occurs once; the factor $k!/\prod \nu_t!$ is the number of terms in S_ν , not a missing coefficient. The original sum injection is the iterated map OMG20 into $E_h^{\otimes k}$. Hence the original observation $L_k = P_k \Pi^{\otimes k} U^{\otimes k} \iota_k = j\Lambda_k$ sends $e^{wY_k}[1]$ to the displayed projected curve with $x = x(w)$. The coefficient covectors of its surviving S_ν pull back to exactly $\Theta_k(x^\nu)$. They span the full dual of the image, since every covector on a subspace of a finite dimensional space extends to its ambient space. Consequently

$$\boxed{\begin{array}{l} \Theta_k(\mathcal{J}_k) = \text{im } \Lambda_k^\vee = (\ker \Lambda_k)^\perp, \quad \text{rank } \Lambda_k = \dim \mathcal{J}_k, \\ R_k/\mathcal{J}_k \xrightarrow{\Theta_k|_{K_k}} K_k^\vee \text{ is an isomorphism of vector spaces,} \quad K_k = \ker \Lambda_k. \end{array}} \quad (\text{OMG29})$$

For the last statement, a covector on E_k vanishes on K_k exactly when it factors through $E_k/K_k \simeq \text{im } \Lambda_k$. Restriction to K_k is onto by extension of a basis. This proves the claimed quotient and its precise type: the degree- k quotient is a vector-space quotient by $\mathcal{J}_k$.

13.7 The actual orbit ideal, including every stabilizer

The whole ideal $\mathfrak{a}$ is generated by the explicit quadratic space in OMG15. Let B_2 be its coefficient matrix in the ordered degree-two monomial frame of $\mathcal{P}_x$. Repeated or dependent generator rows are harmless. For $a \in \mathbb{Z}/N$ let $D_{2,a}$ be diagonal with entry $\zeta^{-a \sum t\nu_t}$ at monomial x^ν . The actual stabilizer and its exact finite test are

$$H = \{a : g^a \mathfrak{a} = \mathfrak{a}\} = \left\{ a : \text{rank} \begin{bmatrix} B_2 \\ B_2 D_{2,a} \end{bmatrix} = \text{rank } B_2 \right\}. \quad (\text{OMG30})$$

Equality of these row spaces is equivalent to equality of their quadratically generated ideals; their dimensions agree under the invertible diagonal map. This proves the finite test using the actual entries OMG4. It prescribes no generic coefficient value. The set H is a subgroup by composition and inverse of ideal equalities. Put $s = |H|$ and $d = N/s$. Since G is cyclic, its cosets have representatives $0, \dots, d-1$ and $H = \{0, d, 2d, \dots, (s-1)d\}$. Define all distinct component ideals and the union algebra by

$$\mathfrak{a}_a = g^a \mathfrak{a} \quad (0 \leq a < d), \quad \mathfrak{u} = \bigcap_{a=0}^{d-1} \mathfrak{a}_a, \quad A = \mathcal{P}_x/\mathfrak{u}. \quad (\text{OMG31})$$

These are distinct homogeneous prime ideals with the same Hilbert function OMG13. Their intersection is a homogeneous radical ideal stable under the original cyclic action. It is the complete ideal of their scheme-theoretic union by the definition of the union's ideal. No product formula for this intersection is substituted.

For any invariant polynomial, membership in one component ideal implies membership in all its translates, by applying every g^a . Thus the exact invariant restriction and its inverse description are

$$\boxed{\begin{array}{c} \mathcal{J} \cap \mathfrak{a} = \mathcal{J} \cap \mathfrak{u}, \\ \mathcal{J}/(\mathcal{J} \cap \mathfrak{u}) \xrightarrow{[f] \mapsto [f]_{\mathfrak{u}}} A^G \xrightarrow{\text{res}_0} \mathcal{J} \subseteq R \end{array}} \text{ are isomorphisms of graded algebras.} \quad (\text{OMG32})$$

To prove surjectivity of the first map, average any representative of an invariant class by $N^{-1} \sum g^a$; its class is unchanged. Its kernel is exactly the displayed intersection. For the second, use an invariant representative: a zero restriction lies in $\mathcal{J} \cap \mathfrak{a} = \mathcal{J} \cap \mathfrak{u}$ and hence is already zero in A^G . Its image is $\mathcal{J}$ by definition. This gives the invariant image for every actual H , including nontrivial stabilizers of a composite group order $4m + 1$.

13.8 The complete component conductor and an explicit finite witness

Set $R_a = \mathcal{P}_x/\mathfrak{a}_a$. The simultaneous restriction

$$A \hookrightarrow \tilde{A} := \prod_{a=0}^{d-1} R_a, \quad [f]_{\mathfrak{u}} \mapsto ([f]_{\mathfrak{a}_a})_a \quad (\text{OMG33})$$

is injective precisely because its kernel before quotienting is the intersection OMG31. Its component conductor ideals are

$$\mathfrak{c}_a = \frac{(\bigcap_{b \neq a} \mathfrak{a}_b) + \mathfrak{a}_a}{\mathfrak{a}_a} \subseteq R_a, \quad \bigcap_{b \neq a} \mathfrak{a}_b = \mathcal{P}_x \quad (d = 1). \quad (\text{OMG34})$$

The conductor of this specified inclusion is exactly

$$\boxed{\{t \in \tilde{A} : t\tilde{A} \subseteq A\} = \prod_{a=0}^{d-1} \mathfrak{c}_a \subseteq A.} \quad (\text{OMG35})$$

Indeed a tuple supported solely on component a lies in the image of A if and only if its value belongs to $\mathfrak{c}_a$, by lifting its other zero values to the intersection of the other ideals. If t is in the conductor, multiplication by each component idempotent of $\tilde{A}$ gives such a supported tuple; thus each component of t lies in $\mathfrak{c}_a$. Conversely these are ideals of R_a . Multiplying a tuple with all its components in the indicated ideals by any element of $\tilde{A}$ gives a sum of such supported tuples, each lying in A . This proves both inclusions and the full formula. In particular this conductor is also the annihilator of $\tilde{A}/A$ in A .

The action of G permutes these components with its original coefficient substitutions. Projection of an invariant tuple to component zero gives an isomorphism

$$\tilde{A}^G \xrightarrow{\sim} R^H, \quad r \in R^H \mapsto (g^a r)_{0 \leq a < d} \text{ for its inverse.} \quad (\text{OMG36})$$

The inverse is independent of coset representatives because r is fixed by H , and invariance follows by reindexing the components. The same argument applies to the conductor, whose ideals are permuted. Equations OMG32–36 consequently prove the exact inclusions of graded subalgebras and conductor ideals

$$\boxed{\mathfrak{c}_0^H \subseteq \mathcal{J} \subseteq R^H, \quad \mathfrak{c}_0^H R^H \subseteq \mathcal{J}.} \quad (\text{OMG37})$$

Here $\mathfrak{c}_0^H$ denotes invariant elements of the specified component ideal and is an ideal of R^H . An invariant tuple in the conductor already lies in A , so its projection belongs to $\mathcal{J}$ by OMG32. This proves the first inclusion; the ideal property gives the second.

There is a nonzero homogeneous element of this conductor with a completely bounded degree. For every $a \neq 0$ choose from the explicit transformed quadratic list of $\mathfrak{a}_a$ a quadratic $f_a \notin \mathfrak{a}_0$. Such a choice exists: containment of the entire quadratic space in that of $\mathfrak{a}_0$, which has the same dimension, would make the spaces equal and their generated ideals equal, contrary to distinctness. Membership is the finite row-space test in the same degree-two monomial frame. The product $f = \prod_{a=1}^{d-1} f_a$ lies in all other component ideals. It is nonzero modulo $\mathfrak{a}_0$ because that ideal is prime and none of its factors belongs to it. Thus $\bar{f} \in \mathfrak{c}_0$ is nonzero of degree $2(d-1)$. The group H preserves $\mathfrak{c}_0$ and acts on the domain R . Its exact norm

$$c_H = \prod_{h \in H} g^h \bar{f} \in \mathfrak{c}_0^H, \quad \deg c_H = D_H = 2s(d-1) = 2(N-s), \quad c_H \neq 0 \quad (\text{OMG38})$$

is invariant by permutation of its factors and nonzero in the domain. For $d=1$ take the empty product $c_H = 1$, $D_H = 0$; all statements remain exact. Multiplication by this actual conductor element is injective and gives the finite dimension bounds, with negative graded pieces set to zero,

$$\begin{aligned} (R^H)_{k-D_H} &\xrightarrow{\times c_H} \mathcal{J}_k \hookrightarrow (R^H)_k, \\ \dim(R^H)_{k-D_H} &\leq \text{rank } \Lambda_k \leq \dim(R^H)_k, \\ q_k - \dim(R^H)_k &\leq \dim K_k \leq q_k - \dim(R^H)_{k-D_H}. \end{aligned} \quad (\text{OMG39})$$

Every ideal, group and matrix in these expressions is defined by the actual original coefficient map. In particular the stabilizer has its finite equality test OMG30 throughout; no particular value of it has been inserted into the dimensions. To compute the displayed invariant dimensions directly, let the actual H substitutions act on the basis OMG9 through $Z \mapsto H_m^{-1} D^{-h} H_m Z$ and reduce with OMG10. This is an explicit finite matrix on q_k coordinates. The idempotent average $s^{-1} \sum_{h \in H} g^h$ has trace equal to its rank, proving $\dim(R^H)_k = s^{-1} \sum_{h \in H} \text{Tr}(g^h | R_k)$. Thus the dimensions in OMG39 also have complete original finite coefficient formulas for every actual stabilizer.

13.9 The exact action defect and all subsequently observed directions

The derivation ∂_x of OMG25 acts on R with all primary directions OMG26. The possible failure of its restriction to the observed image is the specified finite linear map

$$\mathcal{J}_k \longrightarrow R_k / \mathcal{J}_k, \quad j \longmapsto [\partial_x j]. \quad (\text{OMG40})$$

Via OMG29 it is exactly the complex linear dual of the original map $\Lambda_k Y_k I_K : K_k \rightarrow B_k$. Indeed a j corresponds to the covector $\lambda \Lambda_k$ on E_k for a unique boundary covector λ ; evaluating OMG26 on $I_K v$ gives $\lambda \Lambda_k Y_k I_K v$, proving the identification and all domains and codomains. The action on ambient polynomials also has the exact conjugation formula

$$g^a \partial_x g^{-a} = \partial_{D^a A D^{-a}}, \quad g^a \partial_x g^{-a} - \partial_x = \partial_{D^a A D^{-a} - A}. \quad (\text{OMG41})$$

For a linear vector field $\partial_M f = (Mx) \cdot \nabla f$, the chain rule proves this by substituting $D^{-a}x$ after differentiating $f(D^a x)$. This formula retains every original period/unit coefficient in $\mathcal{A}$ and does not require the original ideal or the observed image to be preserved by extra ambient actions.

Finally the subspace invisible to all original iterates has the complete model

$$\begin{aligned} \mathcal{K}_{\infty, k} &= \bigcap_{r=0}^{q_k-1} \ker(\Lambda_k Y_k^r), \\ R_k / \sum_{r=0}^{q_k-1} \partial_x^r \mathcal{J}_k &\xrightarrow{\Theta_k | \mathcal{K}_{\infty, k}} \mathcal{K}_{\infty, k}^\vee \quad \text{is a vector-space isomorphism.} \end{aligned} \quad (\text{OMG42})$$

In finite dimensional duality the annihilator of an intersection of kernels is the sum of the images of their dual maps: this follows by taking the dual of the single stacked map with all those components, whose kernel is precisely the intersection. Equations OMG26 and OMG29 identify these images with $\partial_x^r \mathcal{J}_k$, proving OMG42. The exact original polynomial OMG27 expresses every higher iterate as a linear combination of the first q_k , so the same intersection is taken over all nonnegative iterates and is Y_k -invariant. This proves the full nilpotent and cyclic-action receiver in the new homogeneous model for every original multiplicity, with its actual observation, periods and amplitude jets throughout.

14 The absolute common kernel: the new finite result

This complete incoming calculation sharpens the current absolute common kernel estimate. Its original comparison was against the preceding logarithmic bound; the intervening KAF chapter already established the $O(kq)$ scale. The new finite envelope reduces its uniform limsup constant from $16(4 + 10 \log 2)$ to $32 \log(25 \log 3 / (4\pi))$. AKJ below takes the pointwise minimum of both finite bounds. No limiting kernel coefficient is inferred from these upper bounds.

Retain the actual hypothetical complete quartet and its entire quotient

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad h(z) = \prod_{\alpha \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (z - \alpha)^m, \quad v_h = 2\xi/h.$$

The full primary unit is $U = M_{j_h v_h}$. The original tensor order, centre, polynomial, and coefficient quotient are

$$\begin{aligned} k &= 4l + 1, \quad l \geq 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad c = k/2, \\ \lambda_{a,b} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^e, \quad E = \mathbb{C}[S]/(\chi). \end{aligned} \quad (\text{AKS1})$$

For the principal conclusion $m = 1$. The source-operator estimates below also hold with the stated repeated-root list for $m > 1$, but no simple-quartet rank is assigned to those multiplicities.

Let $\Lambda_k(u) : E \rightarrow B_k(u)$ be the original observation, $K(u) = \ker \Lambda_k(u)$, and $I : K(u) \hookrightarrow E$ its original coefficient inclusion. For a source $s \in \{1, k\}$, let $G_N^{(s)}$ be the original minimum-remainder Gram, in the frame $(1, S, \dots, S^{q-1})$. Define $H_{K,N}^{(s)} = I^* G_N^{(s)} I$ and retain exactly

$$F_K^{(s)} = \log \det H_{K,q-1}^{(s)} + \log \det H_{K,q}^{(s)} - \log \det H_{K,2q-1}^{(s)} - \log \det H_{K,2q}^{(s)}. \quad (\text{AKS2})$$

Every empty determinant is one. All conjugate transposes refer to the specified coordinate matrices, not to a bilinear residue pairing.

Here are entirely explicit positive numbers used in the bound:

$$\begin{aligned} R &= k\sqrt{\delta^2 + \gamma^2}, \quad b_s = s/2, \quad t_q = \frac{\pi}{2} - \frac{1}{q}, \quad \mathbf{a}_q = \frac{2 \log 3}{t_q}, \quad \mathbf{s}_q = \frac{\mathbf{a}_q^q - 1}{\mathbf{a}_q - 1}, \\ \mathbf{b}_{s,q} &= \left[\frac{9}{25} \sin \frac{1}{q} \right]^{-b_s} \exp \left(2R \left(\log 3 + \frac{t_q}{2} \right) \right), \\ \mathcal{Z}_{s,j} &= 1 + \mathbf{b}_{s,q} \mathbf{s}_q^2 \sum_{d=q}^{q+j-1} \frac{d!}{(b_s)_d} \left(\frac{25}{16} \right)^d, \quad \mathcal{L}_{s,j} = \log \mathcal{Z}_{s,j} \quad (0 \leq j \leq q + 1). \end{aligned} \quad (\text{AKS3})$$

The empty sum gives $\mathcal{Z}_{s,0} = 1$. In the admitted family $q \geq 36$, so $t_q \in (0, \pi/2)$ and $\mathbf{a}_q > 1$. These parameters are radii in the existing Laplace and generating functions; neither is a new source order.

Main theorem (absolute finite bound). For the original simple quartet, every original logarithm branch, and every $|u| \geq R_*$, where R_* is precisely RPC21 or RPC23 (recorded in Section 20), put $r = 8k - 16$. Then

$$0 \leq F_K^{(s)}(k, u) \leq r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}), \quad s \in \{1, k\}. \quad (\text{AKS4})$$

There is a stronger, fully period-dependent interval in AKS28 below. In particular, with

$$\kappa_* = 32 \log \left(\frac{25 \log 3}{4\pi} \right), \quad 25.02078097649 < \kappa_* < 25.02078097650, \quad (\text{AKS5})$$

one has the uniform statement

$$\boxed{0 \leq \limsup_{\substack{k \rightarrow \infty \\ k \equiv 1 \pmod{4}}} \sup_{\substack{|u| \geq R_* \\ \text{original branches}}} \frac{F_K^{(s)}(k, u)}{kq} \leq \kappa_*}. \quad (\text{AKS6})$$

Here the source may be $s = 1$ for every k , or $s = k$. The same bound holds for the actual arithmetic kernel. Its exact finite allowances are in AKS34–36. Thus the absolute kernel is $O_h(kq)$ on this domain, not only $O_h(kq \log q)$. Neither AKS5 nor AKS6 calls κ_* the actual leading coefficient.

15 An exact coefficient realization of the actual lost directions

15.1 The period matrix and invariant restriction, with their full coefficients

For $m = 1$, order the one-factor roots as $(++)$, $(+-)$, $(-+)$, $(--)$, and let $\mathcal{E}_{\text{prim}}$ have their original Lagrange idempotents as columns. Set $\zeta = \exp(2\pi i/5)$ and $F_{jt} = \zeta^{jt} - 1$, $1 \leq j, t \leq 4$. The exact inverse is $(F^{-1})_{tj} = \zeta^{-tj}/5$. Retain

$$\begin{aligned} \Pi[P]_j &= \int_{\Gamma_j} e^{\Phi_h(z)/u} \text{rem}_h P(z) dz, & \Phi'_h &= h, & \Phi_h(0) &= 0, \\ \Gamma_j &= -\ell_0 + \ell_j, & \arg \ell_j &= (\arg u + \pi + 2\pi j)/5, \\ \mathbf{H} &= F^{-1} \Pi U \mathcal{E}_{\text{prim}}, & D &= \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4), \\ Q(x) &= Q_0(\mathbf{H}^{-1}x), & Q_0(t) &= t_1 t_4 - t_2 t_3, & Q_a(x) &= Q(D^{-a}x). \end{aligned} \quad (\text{AKS7})$$

The integral always uses the degree-less-than-four representative. Invertibility of Π , all contour phases, and its full determinant are the PDE/PQO results in the supplied proofs. The four unit values are $(a, \bar{a}, \bar{a}, a)$, with the actual $a = v_h(\rho) \neq 0$; no value of a is selected. Hence $\mathbf{H}$ is the original invertible matrix.

The following formulas make the coefficient realization directly computable. Write

$$\mathcal{I}_{k,0} = \{\nu \in \mathbb{Z}_{\geq 0}^4 : |\nu| = k, \nu_1 + 2\nu_2 + 3\nu_3 + 4\nu_4 \equiv 0 \pmod{5}\}.$$

For $\nu \in \mathcal{I}_{k,0}$ define $B_{\nu;(a,b)}$ by summing over 4×4 arrays $n_{t,\alpha} \geq 0$ with row sums ν_t , total plus-real count a , and total plus-imaginary count b :

$$B_{\nu;(a,b)} = \sum_{\substack{\sum_{\alpha} n_{t,\alpha} = \nu_t \\ \sum_{t,\alpha: \epsilon_{\alpha} = +} n_{t,\alpha} = a \\ \sum_{t,\alpha: \eta_{\alpha} = +} n_{t,\alpha} = b}} \left(\prod_{t=1}^4 \frac{\nu_t!}{\prod_{\alpha} n_{t,\alpha}!} \right) \prod_{t,\alpha} \mathbf{H}_{t,\alpha}^{n_{t,\alpha}}. \quad (\text{AKS8})$$

Let $\mathbf{V}_{(a,b),d} = \lambda_{a,b}^d$ for $0 \leq d < q$ and set $\mathbf{A} = \mathbf{B}\mathbf{V}$. All entries of $\mathbf{A}$ are now finite expressions in the actual periods and unit. There is no generic period matrix in this definition.

For completeness, put $t_{\alpha}(w) = \exp((\eta_{\alpha}\gamma - i\epsilon_{\alpha}\delta)w)$ and $x(w) = \mathbf{H}t(w)$. Expansion gives

$$x(w)^{\nu} = \sum_{a,b} B_{\nu;(a,b)} e^{\omega_{a,b} w}, \quad \omega_{a,b} = (2b - k)\gamma - i(2a - k)\delta.$$

In the original primary algebra, $e^{wY}[1] = \sum_{a,b} e^{\omega_{a,b} w} \varepsilon_{a,b}$, where $Y = (M_S - cI)/i$. The exponentials are independent by the Vandermonde of the distinct $\omega_{a,b}$. OCS9's unscaled symmetric word basis therefore identifies the original observation with

$$j\Lambda_k = \mathcal{F}_0 \mathbf{A}, \quad \ker \mathbf{A} = K(u) \quad \text{in the original coefficient frame.} \quad (\text{AKS9})$$

Every multinomial in AKS8 belongs to its actual ordered words; the word multiplicity is not inserted a second time. The exact inherited row Gram of $\mathcal{F}_0$ is the restriction to $\mathcal{I}_{k,0}$ of

$$\mathcal{G}_{\nu,\mu} = \sum_{\substack{n_{tu} \geq 0 \\ \sum_u n_{tu} = \nu_t \\ \sum_t n_{tu} = \mu_u}} \frac{k!}{\prod_{t,u} n_{tu}!} \prod_{t,u} [5(\delta_{tu} + 1)]^{n_{tu}}. \quad (\text{AKS10})$$

This follows by taking the coefficient of $\bar{x}^\nu y^\mu$ in $(\bar{x}^T F^* F y)^k$, since $F^* F = 5(I + \mathbf{1}\mathbf{1}^*)$. It is positive definite and includes every row cross term.

Here is the precise role of the orbit ideal. The map sending a homogeneous polynomial f to the covector specified by $\Theta_k(f)(e^{wY}[1]) = f(x(w))$ has kernel $Q\mathbb{C}[x]_{k-2}$ and is onto $E^\vee$. Indeed equal exponential indices correspond exactly to monomial differences divisible by $t_1 t_4 - t_2 t_3$; the displayed Vandermonde proves that there are no other relations. Its invariant image is $\text{im } \Lambda_k^\vee$. On the five-member domain, the five Q_a are distinct prime quadrics. An invariant polynomial divisible by Q is divisible by each Q_a , and therefore by their full product; its quotient is invariant by cancellation in the polynomial ring. Consequently

$$\mathbb{C}[x]^{\langle D \rangle} \cap (Q) = \left(\prod_{a=0}^4 Q_a \right) \mathbb{C}[x]^{\langle D \rangle}, \quad (\text{AKS11})$$

$$K(u)^\vee \simeq \mathbb{C}[x]_k / (\mathbb{C}[x]_k^{\langle D \rangle} + Q\mathbb{C}[x]_{k-2}),$$

$$b := \text{rank } \mathbf{A} = d_{5,k,0} - d_{5,k-10,0} = k^2 - 6k + 17, \quad r = q - b = 8k - 16.$$

Negative degrees are zero. The count is $d_{5,d,0} = ((\binom{d+3}{3} + 4\epsilon_d)/5$, $\epsilon_d = \mathbf{1}_{d \equiv 0(5)} - \mathbf{1}_{d \equiv 1(5)}$; subtracting the cubics gives the stated expression for $k \geq 10$, and $k = 5, 9$ give 12, 44 directly. This is the inherited OQX calculation, repeated to specify which kernel AKS9 realizes. The source metric is still to be applied to this kernel, not to the invariant ring itself.

15.2 A determinant formula requiring no assumed nonzero pivot

Let $\mathcal{G}_0$ be the full positive row Gram in AKS10 and set

$$W = \mathbf{A}^* \mathcal{G}_0 \mathbf{A}, \quad \sigma_j = [z^j] \det(I_q + zW), \quad P_K = \frac{1}{\sigma_b} \sum_{j=0}^b (-1)^j \sigma_{b-j} W^j. \quad (\text{AKS12})$$

Since W has exactly b positive eigenvalues, $\sigma_b > 0$. The scalar polynomial in AKS12 is one at zero and zero at every nonzero eigenvalue: it is the normalized product of the factors $\lambda_i - z$ for the b positive eigenvalues. Hence $P_K = P_K^* = P_K^2$ and $\text{im } P_K = K(u)$. This is the projector derived from the literal observation, not a freely chosen projection. Replacing $\mathcal{G}_0$ by any other positive coordinate Gram leaves this same coefficient-orthogonal kernel projector, since its image and orthogonal complement are unchanged. No such replacement is needed here.

At every original endpoint define

$$D_{K,N}^{(s)} = [z^r] \det(I_q + z P_K G_N^{(s)} P_K). \quad (\text{AKS13})$$

These numbers are strictly positive for $r > 0$ and are one for $r = 0$. To prove their exact metric meaning, choose an orthonormal coefficient basis U_K of $\text{im } P_K$ just for the proof. In a unitary extension, $P_K G_N P_K$ has its positive block $U_K^* G_N U_K$ and a zero block. Thus $D_{K,N} = \det(U_K^* G_N U_K)$. Write the original inclusion as $I = U_K T$; then $|\det T|^2 = \det(I^* I)$ and

$$\det H_{K,N}^{(s)} = \det(I^* I) D_{K,N}^{(s)}, \quad \boxed{F_K^{(s)} = \log \frac{D_{K,q-1}^{(s)} D_{K,q}^{(s)}}{D_{K,2q-1}^{(s)} D_{K,2q}^{(s)}}}. \quad (\text{AKS14})$$

This supplies an equivalent exact coefficient determinant formula for all the $8k - 16$ lost directions, including the entire period dependence. The original frame factor is present at each endpoint and cancels only with its four actual signs. It is unnecessary to suppose any particular minor nonzero. Alternatively, selecting the first independent r columns of P_K gives an explicit coefficient frame; the determinant formula avoids even that chart choice.

For $m > 1$ replace AKS8–9 by OCS12–18’s full confluent matrix $\mathfrak{A}_{\nu;(a,b,D)} = i^D D! c_{\nu;a,b,D}$ and its primary isomorphism. AKS12–14 are then identical with the actual rank, not with the rank in AKS11. All primary nilpotents and all unit derivatives remain in that matrix.

16 A bounded exact remainder operator on the original source

16.1 The complete source norms and coefficient map

For either existing source set $M_s = (2\pi)^{s/2}$ and retain

$$dm_{0,s}(y) = \frac{M_s 2^{b_s}}{4\pi\Gamma(b_s)} |\Gamma(b_s/2 + iy/2)|^2 dy, \quad \int e^{ty} dm_{0,s} = M_s (\cos t)^{-b_s}, \quad |t| < \pi/2. \quad (\text{AKS15})$$

The supplied IGO1–3 proof derives this transform by the beta integral, Fourier inversion and a contour shift within $|\operatorname{Im} x| < \pi/2$. In particular it includes the exact masses $M_1 = \sqrt{2\pi}$ and $M_k = (2\pi)^{k/2}$ and exponential domination on every smaller strip. The original real monic polynomials satisfy

$$\begin{aligned} p_0 &= 1, \quad p_1 = y, \quad p_{d+1} = yp_d - d(b_s + d - 1)p_{d-1}, \\ \sum_{d \geq 0} \frac{p_d(y)}{d!} z^d &= (1 + z^2)^{-b_s/2} e^{y \arctan z}, \quad h_d^{(s)} = \|p_d\|^2 = M_s d! (b_s)_d. \end{aligned} \quad (\text{AKS16})$$

The generating formula follows by differentiating in z . To check the normalization directly, integrate the product of two generating functions at small real z, w . AKS15 and the cosine addition identity give $M_s(1 - zw)^{-b_s}$. Taking any finite pair of derivatives at zero, justified by the stated exponential domination, gives orthogonality and the norms in AKS16. Thus the leading phase in $u_d(S) = p_d((S - c)/i) / \sqrt{h_d^{(s)}}$ is exactly i^{-d} . The same generating function agrees with NIST DLMF 18.23.7 after the original parameter rescaling; it is derived here from the supplied recurrence, not substituted for it.

List the original roots $\omega_\nu = (\lambda_{a,b} - c)/i$ in lexicographic order, repeating each exactly e times, and put

$$P(y) = i^{-q} \chi(c + iy) = \prod_{\nu=1}^q (y - \omega_\nu), \quad \Phi : [f(y)] \mapsto [f((S - c)/i)]. \quad (\text{AKS17})$$

The inverse is substitution $S = c + iy$, and $P((S - c)/i) = i^{-q} \chi(S)$. This is an exact algebra isomorphism, including the complete confluent orders; there is no monomial comparison divisor. Let C_s have columns $[u_0], \dots, [u_{q-1}]$. Its nonzero determinant is $i^{-q(q-1)/2} \prod_{d < q} (h_d^{(s)})^{-1/2}$. Define

$$v_j = C_s^{-1} [u_{q+j}], \quad \mathbf{K}_j = I_q + \sum_{a=0}^{j-1} v_a v_a^*, \quad 0 \leq j \leq q + 1. \quad (\text{AKS18})$$

Parseval in the low-degree original orthonormal polynomial basis proves

$$\|v_j\|_2^2 = \frac{\|\operatorname{rem}_P p_{q+j}\|_{m_{0,s}}^2}{M_s (q+j)! (b_s)_{q+j}}. \quad (\text{AKS19})$$

This is the full norm of the actual remainder, not a diagonal entry of an approximated quotient Gram.

16.2 Newton interpolation with all roots and their orders

For $0 \leq a < q$ define $N_a(y) = \prod_{\nu=1}^a (y - \omega_\nu)$, with $N_0 = 1$. For every complex v , the unique entire-function remainder is

$$\text{rem}_P(e^{vy}) = \sum_{a=0}^{q-1} [\omega_1, \dots, \omega_{a+1}] e^{v \cdot} N_a(y), \quad |[\omega_1, \dots, \omega_{a+1}] e^{v \cdot}| \leq \frac{|v|^a}{a!} e^{R|v|}. \quad (\text{AKS20})$$

Here divided differences at repeated nodes mean Hermite divided differences, not evaluation with a zero denominator.

A proof covering the confluent case is useful. On the standard a -simplex with barycentric coordinates $\theta_1, \dots, \theta_{a+1}$,

$$[\omega_1, \dots, \omega_{a+1}] e^{v \cdot} = v^a \int_{\Delta_a} e^{v \sum \theta_\nu \omega_\nu} d\theta.$$

The integral of $\prod \theta_\nu^{n_\nu}$ is $\prod n_\nu! / (a + \sum n_\nu)!$, as follows by iterated integration of integer beta integrals. Expanding the exponential therefore gives $\sum_{n \geq 0} v^{a+n} h_n(\omega_1, \dots, \omega_{a+1}) / (a+n)!$. For monomials, the divided difference of y^{a+n} is exactly h_n : this follows by multiplying the geometric series $\prod (1 - \omega_\nu z)^{-1}$ and the Newton division recurrence. It is a polynomial identity in the nodes and so also holds at repeated nodes. This proves the integral formula. Iterated Newton division for polynomials, followed by the convergent exponential series, proves the first equality in AKS20, including every required derivative. Finally the simplex has volume $1/a!$ and $|\sum \theta_\nu \omega_\nu| \leq R$, proving its stated bound.

For any $0 < t < \pi/2$, put $\tau = 2/t$. Positivity of every term of the cosh series in AKS15 gives

$$\|y^a\|_{m_{0,s}} \leq \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a!.$$

Indeed its square is at most $M_s (\cos t)^{-b_s} (2a)! t^{-2a}$ and $(2a)! \leq 4^a (a!)^2$. Expanding the *whole* Newton polynomial, then using $|e_j(\omega_1, \dots, \omega_a)| \leq \binom{a}{j} R^j$, proves

$$\begin{aligned} \|N_a\|_{m_{0,s}} &\leq \sqrt{M_s} (\cos t)^{-b_s/2} \sum_{j=0}^a \binom{a}{j} R^{a-j} \tau^j j! \\ &= \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a! \sum_{n=0}^a \frac{(R/\tau)^n}{n!} \\ &\leq \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a! e^{R/\tau}. \end{aligned} \quad (\text{AKS21})$$

There are no omitted polynomial coefficients. Combining AKS20–21 gives for every complex v

$$\|\text{rem}_P(e^{vy})\|_{m_{0,s}} \leq \sqrt{M_s} (\cos t)^{-b_s/2} e^{R(|v|+t/2)} \sum_{a=0}^{q-1} (2|v|/t)^a. \quad (\text{AKS22})$$

16.3 Cauchy extraction in the actual Gamma generating function

Fix $0 < z < 1$ and write $a_z = \text{arctanh } z$. For $|w| = z$ the defining power series gives $|\arctan w| \leq a_z$ and $|1 + w^2| \geq 1 - z^2$. Apply the literal finite-dimensional remainder map to AKS16 and use Cauchy's formula for its d th coefficient. The norm triangle inequality on that circle and AKS22 give

$$\begin{aligned} \frac{\|\text{rem}_P p_d\|_{m_{0,s}}}{d!} &\leq z^{-d} (1 - z^2)^{-b_s/2} \sqrt{M_s} (\cos t)^{-b_s/2} \\ &\quad \times e^{R(a_z+t/2)} \sum_{a=0}^{q-1} (2a_z/t)^a. \end{aligned} \quad (\text{AKS23})$$

Cauchy's formula is legitimate for the finite remainder vector: its entries are holomorphic for $|w| < 1$ by the actual matrix generating identity OPG7. The source norm is taken only after this finite-vector identity; there is no unjustified integral of a nonintegrable generating function on the large circle.

Substitution of AKS23 into AKS19 retains the factorial denominator and cancels the two occurrences of the original mass, giving

$$\|v_j\|_2^2 \leq \frac{d!}{(b_s)_d} z^{-2d} [\cos t(1 - z^2)]^{-b_s} e^{2R(az+t/2)} \left[\sum_{a=0}^{q-1} (2a_z/t)^a \right]^2, \quad d = q + j. \quad (\text{AKS24})$$

Adding the *complete* rank-one positive matrices in AKS18 now proves $K_j \preceq \mathcal{Z}_{s,j} I_q$ after setting $z = 4/5$ and $t = t_q$. Indeed $a_{4/5} = \log 3$, $1 - z^2 = 9/25$, and $\cos t_q = \sin(1/q)$, so the exact scalar obtained is AKS3. The high index $j = q + 1$ includes the original polynomial of degree $2q$.

There is also a root-by-root finite refinement. Put $\mu_{s,d} = \int y^{2d} dm_{0,s}$, $R_a = \max_{\nu \leq a+1} |\omega_\nu|$, and

$$B_{s,a}^{\text{root}} = \sum_{d=0}^a |e_{a-d}(\omega_1, \dots, \omega_a)| \sqrt{\mu_{s,d}}, \quad (\text{AKS25})$$

$$J_s^{\text{root}}(z) = \frac{(1 - z^2)^{-b_s}}{M_s} \left[\sum_{a=0}^{q-1} \frac{a_z^a}{a!} e^{a_z R_a} B_{s,a}^{\text{root}} \right]^2.$$

Replace the scalar multiplying $d!z^{-2d}/(b_s)_d$ in AKS24 by $J_s^{\text{root}}(z)$ to get a valid sharper bound, by the same proof before AKS21 is relaxed. Every moment is exactly $M_s(2d)! [w^{2d}] (\cos w)^{-b_s}$. Thus this refinement keeps every actual root coefficient and a finite original moment formula. AKS21 bounds it by the explicit scalar in AKS24. The proof never replaces the source determinant by a product of diagonal moments: AKS24 is a full-vector operator bound, and the next step restricts the inverse of the full covariance.

17 The four original kernel Grams and the evaluated scale

The original orthonormal source map at degree $q + j - 1$ is $C_s[I_q, v_0, \dots, v_{j-1}]$. Its minimum lift of $C_s x$ is $[I_q, v_0, \dots, v_{j-1}]^* K_j^{-1} x$. It has that image and is orthogonal to the full relation kernel, so Pythagoras proves

$$G_{q+j-1}^{(s)} = C_s^{-*} K_j^{-1} C_s^{-1}, \quad H_j := H_{K,q+j-1}^{(s)} = T_s^* K_j^{-1} T_s, \quad T_s = C_s^{-1} I. \quad (\text{AKS26})$$

This verifies the source-minimum step in the coefficient determinant AKS13–14. Neither a cross block nor any original relation is removed.

Since $I_q \preceq K_j \preceq \mathcal{Z}_{s,j} I_q$, inversion and compression give $\mathcal{Z}_{s,j}^{-1} H_0 \preceq H_j \preceq H_0$, where $H_0 = T_s^* T_s$. Define $d_j = -\log \det(H_0^{-1/2} H_j H_0^{-1/2})$. Congruence and all r eigenvalues give

$$0 = d_0 \leq d_1 \leq \dots \leq d_{q+1}, \quad 0 \leq d_j \leq r \mathcal{L}_{s,j}, \quad F_K^{(s)} = d_q + d_{q+1} - d_1. \quad (\text{AKS27})$$

The identity uses precisely the four endpoints in AKS2. In particular it proves AKS4, as well as the sharper interval

$$d_1 \leq F_K^{(s)} \leq r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1. \quad (\text{AKS28})$$

For a directly calculable expression for its lower endpoint set $V = \|v_0\|^2$ and $\theta = v_0^* T_s (T_s^* T_s)^{-1} T_s^* v_0$. Then

$$0 \leq \theta \leq V, \quad d_1 = \log \frac{1 + V}{1 + V - \theta}. \quad (\text{AKS29})$$

Indeed the inverse of $I + v_0 v_0^*$ is $I - v_0 v_0^*/(1 + V)$; take its full compressed determinant. For $r = 0$ set $\theta = d_1 = 0$. In GMB/OPR notation $V = t_0$ and $V - \theta = \xi_0$, so this is exactly the first original section-corrected kernel increment, not an independently chosen kernel observation.

Here is an evaluated finite remainder for AKS6. Put

$$\begin{aligned} a_\infty &= \frac{4 \log 3}{\pi}, & c_0 &= \log \frac{25 \log 3}{4\pi} > 0, \\ E_{s,q}(R) &= \log 2 + \frac{4}{\pi - 2/q} + \frac{s}{2} \log \frac{25\pi q}{18} + 2R \left(\log 3 + \frac{\pi}{4} \right) \\ &\quad + \log((q+1)(4q+1)) - 2 \log(a_\infty - 1). \end{aligned} \tag{AKS30}$$

Then for both source orders, every $q \geq 36$, and the same actual kernel,

$$\boxed{\mathcal{L}_{s,q}, \mathcal{L}_{s,q+1} \leq 2c_0 q + E_{s,q}(R), \quad 0 \leq F_K^{(s)} \leq 4rc_0 q + 2rE_{s,q}(R).} \tag{AKS31}$$

To verify every factor, $b_s \geq 1/2$ gives

$$\frac{d!}{(b_s)_d} \leq \frac{4^d}{\binom{2d}{d}} \leq 2d + 1.$$

The last inequality uses that the central binomial is the largest of the $2d + 1$ positive coefficients whose sum is 4^d . Thus the sum in AKS3 is at most $(q+1)(4q+1)(5/4)^{4q}$. Also $\mathfrak{s}_q \leq \mathfrak{a}_q^q/(a_\infty - 1)$ and

$$2q \log \mathfrak{a}_q \leq 2q \log a_\infty + \frac{4}{\pi - 2/q}, \quad \sin(1/q) \geq \frac{2}{\pi q}.$$

The first follows from $-\log(1-x) \leq x/(1-x)$ at $x = 2/(\pi q)$; the second is the chord bound for sine on $[0, \pi/2]$. Use $t_q \leq \pi/2$ in the remaining exponential. The order- q terms combine to $4q \log(5/4) + 2q \log a_\infty = 2c_0 q$. The upper bound for the logarithm of the added positive scalar is positive, so $\log(1 + e^x) \leq \log 2 + \max(0, x)$ proves the first bound in AKS31. AKS28 proves the second.

On the original simple family $q = (k+1)^2$, $R = O_h(k)$ and $r = 8k - 16$. Consequently $E_{1,q}(R) = O_h(k + \log q)$ and $E_{k,q}(R) = O_h(k \log q)$. Dividing AKS31 by kq proves AKS6, with $32c_0 = \kappa_*$. In fact the fixed-source upper bound is

$$F_K^{(1)} \leq (32k - 64)c_0 q + (16k - 32)E_{1,q}(R), \tag{AKS32}$$

whose explicit second term is $O_h(k^2)$. The tensor-source version has an $O_h(k^2 \log q)$ upper remainder. These are remainders of upper bounds, not assertions about the actual signed error or leading value. The proof is uniform in all the period parameters in the stated domain: no norm or condition number of a period matrix occurs in AKS30.

The precise earlier loss. OKV15 had already retained $\|v_j\|^2 \leq q^2 2^{6q} (2D_* q)^{2d} / (d!(b_s)_d)$, $d = q + j$. Replacing its denominator by $1/2$ caused the old $q \log q$ exponent. Even without AKS20–24 one obtains the valid repair

$$\|K_{q+1}\| \leq 1 + (q+1)q^2 2^{6q} \frac{(4D_* q)^{4q}}{(4q)!}, \quad D_* \geq 3. \tag{AKS33}$$

Here $d!(b_s)_d \geq (2d)!/4^d$ and the remaining factorial ratio increases up to $d = 2q$ for $D_* \geq 3$. The elementary inequality $\log n! \geq n \log n - n + 1$ already makes the logarithm of AKS33 $O_h(q)$. The Newton calculation strengthens this repair to the packet-independent limsup constant AKS5, while retaining R in its finite remainder.

Exact enclosure of the displayed numerical ceiling. For $x > 1$ and $z = (x - 1)/(x + 1)$ the integrated geometric series gives

$$0 \leq \log x - 2 \sum_{j=0}^{39} \frac{z^{2j+1}}{2j+1} \leq \frac{2z^{81}}{81(1-z^2)}.$$

For $0 < x < 1$, the alternating sum through $j = 39$ for $\arctan x$ has error between zero and $x^{81}/81$. The tangent addition formula, with the angle in $(0, \pi/2)$, proves $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$: the tangent of $4 \arctan(1/5) - \arctan(1/239)$ is exactly one. Use the resulting rational lower and upper bounds for π and for $\log 3$, then apply the first displayed bound to the two endpoints of $25 \log 3/(4\pi)$. Multiplication by 32 gives, with outward decimal rounding, the interval

$$\kappa_* \in [25.0207809764994871, 25.0207809764994872].$$

The accompanying rational-arithmetic implementation checks these inequalities by integer cross multiplication, rather than a floating-point logarithm. This proves the less finely rounded strict interval AKS5.

18 The complete arithmetic and mixed receivers

18.1 The unchanged finite arithmetic allowances

Use exactly AMT4's constants. To prevent a loss hidden in order notation, write them here:

$$\begin{aligned} \alpha &= \pi/2, \quad p = 8m - 9/2, \quad B_h = 42 + 8m, \quad M = 21 + 4m, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy, \quad M_\alpha = \int e^{\alpha y} w_h(y) dy, \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \\ c_\Gamma &= \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M_\alpha^{k-1}, \\ D_N &= [1 + 4(N + M)^2]^M, \quad L_N^{\text{ar}} = \log(A_k D_N / a_k), \\ E_k^+ &= L_{q-1}^{\text{ar}} + L_q^{\text{ar}}, \quad E_k^- = L_{2q-1}^{\text{ar}} + L_{2q}^{\text{ar}}. \end{aligned} \tag{AKS34}$$

The constants c_h, C_h^{tilt} are the original TW/BT envelope constants, not new choices. Their actual inequalities and complete source construction are preserved in AMT4–6 and the current source closure. Minimize AMT6 on the same remainder fibre and then restrict to $I(K)$. AMT9–20 gives, with the original arithmetic source w_h^{*k} ,

$$-rE_k^+ \leq F_K^{\text{ar}} - F_K^{(1)} \leq rE_k^-.$$

Let $\underline{U}_s = d_1^{(s)}$ and $\overline{U}_s = r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1^{(s)}$. The new finite conclusion is

$$\boxed{\max\{0, \underline{U}_1 - rE_k^+\} \leq F_K^{\text{ar}} \leq \overline{U}_1 + rE_k^-} \tag{AKS35}$$

This improves AMT37's absolute upper bound. For the sharper correlated version retain $d_{E,N} = q \log A_k - \log(\det G_N^{\text{ar}} / \det G_N^{(1)})$ and $c_{X,N} = \min\{r_X L_N^{\text{ar}}, d_{E,N}\}$, where $r_K = r$, $r_B = q - r$. Put $C_X^- = c_{X,q-1} + c_{X,q}$, $C_X^+ = c_{X,2q-1} + c_{X,2q}$. AMT18 then gives

$$\begin{aligned} \max\{0, \underline{U}_1 - C_K^-, \underline{U}_1 + \delta_k^\sigma - C_B^+\} &\leq F_K^{\text{ar}} \\ &\leq \overline{U}_1 + \min\{C_K^+, \delta_k^\sigma + C_B^-\}. \end{aligned} \tag{AKS36}$$

One may replace $d_{E,N}$ in the caps by the original full-source $\Xi_{k,N}$ or its proved finite majorant. The scalar $\delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma$ retains exactly AMT21:

$$\begin{aligned} \max\{-qE_k^+, -\Xi_{k,q-1} - \Xi_{k,q}\} &\leq \delta_k^\sigma \\ &\leq \min\{qE_k^-, \Xi_{k,2q-1} + \Xi_{k,2q}\}. \end{aligned} \quad (\text{AKS37})$$

Since $E_k^\pm = O_h(k + \log q)$ by the exact AMT22 expression, AKS35 and AKS32 prove the arithmetic version of AKS6, with an explicit $O_h(k^2)$ upper remainder. Nonnegativity follows from the actual nested source minima, as in AMT15, not from comparison with a possibly negative lower endpoint.

18.2 The original Gamma multiplier, not a replacement higher source

Retain $t_j = 2j + 1/2$, $f_l(S) = \prod_{j=0}^{l-1} (S - c - t_j)$ and

$$dm_{0,k} = \beta_k |f_l(c + iy)|^2 d\sigma, \quad \beta_k = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}, \quad T_l = \prod_{j=0}^{l-1} t_j^2, \quad U_{l,N} = \prod_{j=0}^{l-1} [4(N + j + 1)^2 + t_j^2]. \quad (\text{AKS38})$$

The exact source map is $\mathcal{P}_N \rightarrow f_l \mathcal{P}_N \subset \mathcal{P}_{N+l}$, $P \mapsto f_l P$, with inverse division on this specified subspace. Let $\mathbf{E}_\chi$ be the complete divided-jet map and U_f its Leibniz multiplication matrix, $(U_f)_{r,a} = f_l^{(r-a)}(\lambda)/(r-a)!$ for $r \geq a$. AMT25–28 gives the precise transported observation and kernel

$$\begin{aligned} \tilde{\Lambda} &= \Lambda \mathbf{E}_\chi^{-1} U_f^{-1}, \quad \tilde{I} = U_f \mathbf{E}_\chi I, \quad \ker \tilde{\Lambda} = U_f \mathbf{E}_\chi K, \\ C_N &= \mathcal{K}_{\chi,N+l} - \mathcal{K}_{\chi,f;N+l} \mathcal{K}_{f,N+l}^{-1} \mathcal{K}_{f,\chi;N+l}, \\ H_{K,N}^{(k)} &= \beta_k \tilde{I}^* C_N^{-1} \tilde{I}. \end{aligned} \quad (\text{AKS39})$$

Each covariance is the original finite sum of full polynomial jet outer products divided by its original monic norm. These formulas retain the extra f constraints and their full cross covariance. In particular $M_f K = K$ is not assumed. AKS26 calculates the same tensor metric directly in its own orthonormal source; the two representations agree by their identical source norm and identical constrained minimum.

Define

$$w_N = \log(U_{l,N}/T_l), \quad \Omega^{\text{low}} = w_{q-1} + w_q, \quad \Omega^{\text{high}} = w_{2q-1} + w_{2q}.$$

The already proved same-coefficient comparison AMT42–44 retains all mass factors before cancellation and gives

$$-r\Omega^{\text{low}} \leq \Delta_K^\Gamma := F_K^{(1)} - F_K^{(k)} \leq r\Omega^{\text{high}}. \quad (\text{AKS40})$$

It therefore sharpens the two independent absolute bounds by

$$\begin{aligned} \max\{U_k, \underline{U}_1 - r\Omega^{\text{high}}, 0\} &\leq F_K^{(k)} \\ &\leq \min\{\bar{U}_k, \bar{U}_1 + r\Omega^{\text{low}}\}. \end{aligned} \quad (\text{AKS41})$$

The original finite estimate $0 \leq w_N \leq l \log[1 + 16(N + l)^2]$ proves that AKS40 is $o(kq)$ on the actual rank- r domain. Along with AKS35 this preserves the completed fact that all three source kernels differ by $o(kq)$. AKS4, not AKS40, is the new absolute-scale calculation.

18.3 Boundary, QGQ and arithmetic receiving identities

At each endpoint retain the original minimum section $S_N^{(s)} = (G_N^{(s)})^{-1} \Lambda^* (\Lambda(G_N^{(s)})^{-1} \Lambda^*)^{-1}$. The full Schur identity is

$$\det(I^* G_N^{(s)} I) \det Q_N^{(s)} = |\det[I, S_*]|^2 \det G_N^{(s)}, \quad Q_N^{(s)} = (\Lambda(G_N^{(s)})^{-1} \Lambda^*)^{-1}. \quad (\text{AKS42})$$

Thus the new finite absolute boundary interval is

$$\max\{0, \mathcal{B}_k^{(s)} - \overline{U}_s\} \leq F_B^{(s)} \leq \mathcal{B}_k^{(s)} - \underline{U}_s. \quad (\text{AKS43})$$

The baselines in this formula are the *complete* original BSL/HBL objects, with their full finite relation errors. No q^2 coefficient is assigned to K ; its proved coefficient there remains zero.

Write $\mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0 = -\mathfrak{T}_k$. The exact full identities still read

$$\begin{aligned} F_B^{(1)} - F_B^{(k)} &= \mathcal{H}_k - \Delta_K^\Gamma, \\ \boxed{\mathcal{B}_k^{\text{ar}} = F_K^{(k)} + F_B^{(k)} + \mathcal{H}_k + \delta_k^\sigma = F_K^{(1)} + F_B^{(1)} + \delta_k^\sigma.} \end{aligned} \quad (\text{AKS44})$$

The Gamma interval is not re-estimated in this argument. For either complete QRI1 or QRI2 interval $[\omega^-, \omega^+]$ for its already calculated centre W_k , retain its exact source errors $e_0 \in [-a_0, b_0]$ and $e_q + e_{q+1} \in [-b_k^+, b_k^-]$. Then

$$\underline{\mathcal{H}}_k = -\omega^+ - b_k^- - a_0, \quad \overline{\mathcal{H}}_k = -\omega^- + b_k^+ + b_0, \quad \mathcal{H}_k \in [\underline{\mathcal{H}}_k, \overline{\mathcal{H}}_k]. \quad (\text{AKS45})$$

These are exactly QRI3, not new unspecified error choices. The complete QRI1–9, QGT and QGQ definitions remain verbatim in each updated receiver; the new interval is intersected with all their retained tighter finite intervals. In particular the simple-quartet nonzero error radius stays

$$\limsup \left| \frac{\mathcal{H}_k + lq\mathcal{C}_\Gamma}{q} + \frac{\partial_\alpha \mathcal{L}(2, \pi)}{8} - \frac{\log 2}{4} \right| \leq \frac{A_{\text{QGT}}}{\sqrt{2}}. \quad (\text{AKS46})$$

No zero-radius or $o(q)$ arithmetic assertion is introduced.

For example the complete finite absolute defect relative to the tensor boundary, with the actual Gamma and arithmetic scalars retained, is

$$\underline{U}_k + \mathcal{H}_k + \delta_k^\sigma \leq \mathcal{B}_k^{\text{ar}} - F_B^{(k)} \leq \overline{U}_k + \mathcal{H}_k + \delta_k^\sigma. \quad (\text{AKS47})$$

Inserting AKS41 makes this sharper. Replacing $\mathcal{H}_k$ by the appropriate endpoints of AKS45 and δ_k^σ by AKS37 gives its fully inherited finite scalar enclosure with the same signs. Also $F_B^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma - F_K^{\text{ar}}$; subtract the interval AKS35 or AKS36 to get its new absolute boundary interval. This retains the exact correlations rather than taking independent hypothetical source errors.

The existing limit is $\mathcal{H}_k/(kq) \rightarrow -\mathcal{C}_\Gamma/4$ and $\delta_k^\sigma/(kq) \rightarrow 0$. Therefore the new interval implies

$$\begin{aligned} \liminf_k \inf_{|u| \geq R_*} \frac{\mathcal{B}_k^{\text{ar}} - F_B^{(k)}}{kq} &\geq -\mathcal{C}_\Gamma/4, \\ \limsup_k \sup_{|u| \geq R_*} \frac{\mathcal{B}_k^{\text{ar}} - F_B^{(k)}}{kq} &\leq \kappa_* - \mathcal{C}_\Gamma/4. \end{aligned} \quad (\text{AKS48})$$

All original branches are included in the infima and suprema. This does not decide its sign. The already proved boundary *difference* limits of AMT49 are unchanged. Every subsequential limit of the absolute normalized kernel lies in $[0, \kappa_*]$ and is shared by its three source versions when they are evaluated on the same sequence of period parameters. Existence of a single limit is not proved by boundedness.

Finally the HC residual remains exactly $\mathcal{R}_k = \mathcal{B}_k^{\text{ar}} - 4q \log(D_h k)$, with its original D_h and original nonnegative contraction, phase and trace terms. AKS44 changes none of them. The supplied positive q^2 baseline and its complete Gamma correction are not re-presented as new results here.

19 Actions, support faces, and the actual arithmetic class

The coefficient formula AKS12 supplies all four blocks of the same sum action: $P_K Y P_K$, $P_K Y (I - P_K)$, $(I - P_K) Y P_K$ and $(I - P_K) Y (I - P_K)$. There is no claim that either off-diagonal block vanishes. In the actual minimum-section coordinates the corresponding formula remains

$$[I, S_N]^{-1} Y [I, S_N] = \begin{pmatrix} H_{K,N}^{-1} I^* G_N Y I & H_{K,N}^{-1} I^* G_N Y S_N \\ \Lambda Y I & \Lambda Y S_N \end{pmatrix}. \quad (\text{AKS49})$$

Its proof is multiplication by the inverse coordinate maps $H_{K,N}^{-1} I^* G_N$ and Λ . In the invariant-ring model, the same action is the derivation $\mathcal{D}f = (H\Omega H^{-1}x) \cdot \nabla f$, $\Omega = \text{diag}(\gamma - i\delta, -\gamma - i\delta, \gamma + i\delta, -\gamma + i\delta)$, with $\mathcal{D}Q = 0$. For the actual conductor $\mathcal{A}_Q = \prod_{a=1}^4 Q_a$, its full defect is $\mathcal{D}\mathcal{A}_Q = \sum_{a=1}^4 (\mathcal{D}Q_a) \prod_{b \neq a, 1 \leq b \leq 4} Q_b$, exactly OQC22. Thus this coefficient realization preserves the original action leaving the observed dual.

In particular the section-corrected vector is still OPR2's

$$C_s r_j^{\ker} = e_{q+j} - S_{q+j-1}^{(s)} \Lambda e_{q+j} = I(\zeta_{q+j} - \kappa S_{q+j-1}^{(s)} \Lambda e_{q+j}),$$

with its full denominator $1 + \xi_j$ in the kernel increment. The operator proof estimates the complete covariance and its actual restriction, so none of these coupled terms is replaced by a raw fixed-section vector. The all-iterate kernel stays $\bigcap_{a=0}^{q-1} \ker(\Lambda Y^a)$; nothing in AKS4 declares it zero.

On the existing arithmetic coefficient realization retain $\mathcal{V}_{h,k} P = P(D_1 + \dots + D_k) F_h^{\otimes k}$ and $J^{(k)} \mathcal{V}_{h,k} = \eta \text{rem}_\chi$, where $\eta = U^{\otimes k} \iota$ is the original injection. AKS17 and the Newton coefficients compute precisely this same remainder. Any two polynomial lifts differ by χQ in the original relation space, so their original arithmetic cohomology classes agree. The specified primitive of the difference is EP.48, with the full differential and cochain signs proved in AKP3–5 below; it depends on the lift difference. The source comparison to w_h^{*k} is the identity on original polynomial coefficients, followed by the actual minima as in AMT8; it is not a replacement arithmetic source.

The scalar object stays $G(R) = \{\tau\} \sqcup R^\bullet$, with supported zero $0_R^\bullet$, injective amplitude/support pair, and infinite integer target. Every coefficient map above is applied on each original present support label and outer leg. It commutes with the specified coefficient inclusions because both compositions apply the same linear map to the same coefficient; a present zero is not converted into absence. No new base object, purity condition, or identification of the sum action with a Frobenius action is used. The actual distinguished class and its higher-jet directions remain in the full fibre, even when the observation kills them.

20 Exact period domain and the interior alternative

For reference the inherited radius used above is completely specified as follows; these are AMT33–34/RPC17–23, with no new constants. Put

$$\begin{aligned} \beta &= 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad a = v_h(\rho) \neq 0, \\ c_3 &= (a^{-2} - \bar{a}^{-2})/(4i\delta\gamma), \quad c_1 = (a^{-2} + \bar{a}^{-2})/2 - (\delta^2 - \gamma^2)c_3, \\ g_j &= 5^{j/5-1} e^{\pi i j/5} \Gamma(j/5), \quad g_+ = \max_j |g_j|, \quad g_-^{-1} = \max_j |g_j|^{-1}, \quad \kappa = \max_{j \leq 4} |g_{j+1}/g_j|, \\ E_j^{\text{ray}} &= \frac{8}{5} \int_0^\infty r^j (r^3/3 + r) e^{-r^5/5 + r^3/3 + r} dr, \quad C_T = 2 \left(\sum_{j=0}^3 (E_j^{\text{ray}})^2 \right)^{1/2}, \\ C_K &= 4C_T g_-^{-1} + 2g_+ g_-^{-1} + 2g_+ g_-^{-2} C_T, \\ C_{K,3} &= C_K(3\kappa^2 + 3\kappa C_K + C_K^2), \quad \epsilon_* = \min\{1, (2C_T g_-^{-1})^{-1}\}, \quad B_* = \beta + \eta. \end{aligned} \quad (\text{AKS50})$$

Then

$$R_* = \begin{cases} \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2(|c_3|C_{K,3}B_* + |c_1|(\kappa + C_K))}{|c_3g_4/g_1|} \right\} \right]^{5/2}, & c_3 \neq 0, \\ \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2C_KB_*}{|g_2/g_1|} \right\} \right]^{5/2}, & c_3 = 0. \end{cases} \quad (\text{AKS51})$$

The actual unit implies $(c_1, c_3) \neq (0, 0)$ and all integrals converge. RPC's full period comparison proves a nonzero $(4, 1)$ or $(2, 1)$ commutator entry, respectively, for $|u| \geq R_*$ on every branch. It therefore proves the five-member orbit needed in AKS11.

Inside this radius the same coefficient determinant AKS12–14 and the operator bound AKS24 remain valid. The orbit is decided by the original noninvariant quadratic coefficients, not by the radius alone. Since Q is nondegenerate, its projective orbit has either one or five members. A one-member orbit must have character one: its character has both fourth and fifth powers equal to one, by the quadratic determinant and $D^5 = I$. In that case $Q = a'x_1x_4 + b'x_2x_3$ with $a'b' \neq 0$, and

$$b = d_{5,k,0} - d_{5,k-2,0} = \left\lfloor \frac{(k+1)^2}{5} \right\rfloor + \mathbf{1}_{k \equiv 0 \text{ or } 3 \pmod{5}}, \quad r = q - b. \quad (\text{AKS52})$$

If a noninvariant coefficient is nonzero, AKS11 applies instead. These are the complete OQC alternatives. The bound AKS31 always uses that actual r ; it does not extend the $8k - 16$ specialization to an unevaluated exceptional point. This calculation does not assert that the interior exceptional set is empty.

21 Verification scope and the remaining absolute coefficient

The new mathematical content is AKS20–32 and its application to the actual coefficient determinant AKS12–14, followed by the finite receiving improvements AKS35–48. The complete source and orbit identifications used in that application are the supplied OCS/OQC/OQX, RPC, OPG/OPR and AMT results; their exact predecessors are preserved. AKS33 pinpoints a repair obtainable immediately from the earlier proof, whereas the Newton bound supplies the sharper explicit constant.

The bound *at* scale kq is now finite and uniform. What is not proved is a matching lower coefficient, a unique limit for $F_K^{(1)}/(kq)$, or a sign for the interval AKS48. Those questions involve the orientation of the original period kernel within the inverse remainder covariance; replacing that compression by a scalar bound loses precisely that information. This is the finite-bounds outcome allowed by the current request, not a claim to have evaluated an asymptotic coefficient. The source differences are already completed inputs and are not renamed as the remaining problem.

The accompanying exact checks distinguish finite algebraic diagnostics from the analytic proof for the full family. They do not assert that a diagnostic root is a zero of ξ , that a diagnostic matrix equals an actual period matrix, or that a new Lean theorem has been checked. No RH conclusion is drawn from this kernel bound.

22 The new ceiling in the original coefficient frame

The complete incoming AKS proof sharpens the previously proved KAF bound. Both estimates apply to the same original remainder and the same kernel. We retain all their finite constants. In particular, for $m = 1$, $k = 4l + 1$, $l \geq 1$, $q = (k + 1)^2$, $c = k/2$, $0 < \delta < 1/2$, $\gamma > 2$, and every original branch with $|u| \geq R_*$, put $r = 8k - 16$. The AKS constant is

$$\kappa_* = 32 \log \frac{25 \log 3}{4\pi} \in (25.02078097649, 25.02078097650). \quad (\text{AKJ1})$$

AKS20–32 proves the full finite operator inequality from confluent Newton division and coefficient extraction. Its parameters t and z are extraction radii in the existing source, whose orders remain $1, k$. For $s \in \{1, k\}$ define, using the exact KAF and AKS quantities,

$$\begin{aligned} L_s &= d_1^{(s)}, \\ U_s &= \min\{2r \log Z_{k,1}, r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1^{(s)}\}. \end{aligned} \quad (\text{AKJ2})$$

The first entry of the minimum is KAF’s factorial estimate; the second is AKS28, with its original first compressed increment retained. Thus $L_s \leq F_K^{(s)} \leq U_s$. This pointwise minimum retains whichever finite estimate is sharper at the actual original parameters. It does not assume that the asymptotically smaller ceiling wins at every degree.

The two exact original coordinate constructions are related as follows. Use OKM’s invertible ordered Vandermonde V_β and its actual quotient matrix π . Put $I_\pi = V_\beta^{-1} \pi^\top$. AKS12 constructs the coefficient-orthogonal projection P_K from the actual observation and its complete ordered-word Gram. Directly,

$$P_K = I_\pi (I_\pi^* I_\pi)^{-1} I_\pi^*, \quad D_{K,N}^{(s)} = \frac{\det(\bar{\pi}(R_N^{(s)})^{-1} \pi^\top)}{\det(I_\pi^* I_\pi)}. \quad (\text{AKJ3})$$

Indeed I_π is injective with image K by OKM5. The first displayed matrix is Hermitian and idempotent, fixes this image, and kills its coefficient-orthogonal complement, so it is exactly AKS12’s projection. Apply AKS14 with inclusion I_π and then OKM8–9 to obtain the second identity. The positive denominator is independent of endpoint and source. Its exponent is $1 + 1 - 1 - 1 = 0$ in the four-endpoint quotient. This proves the exact morphism between the incoming projector determinant and the existing conductor-frame determinant, retaining the original coefficient metric and every frame factor.

With the original AMT correlated caps $C_X^\pm$ and scalar δ_k^σ , the following joint intervals hold:

$$\begin{aligned} \max\{0, L_1 - C_K^-, L_1 + \delta_k^\sigma - C_B^+\} &\leq F_K^{\text{ar}} \\ &\leq U_1 + \min\{C_K^+, \delta_k^\sigma + C_B^-\}, \\ \max\{L_k, L_1 - r\Omega^{\text{high}}, 0\} &\leq F_K^{(k)} \leq \min\{U_k, U_1 + r\Omega^{\text{low}}\}. \end{aligned} \quad (\text{AKJ4})$$

For proof, add each retained AMT lower or upper source-difference bound to the corresponding endpoint of AKJ2, and intersect the resulting intervals. In the first line the upper and lower source differences are the same AMT18 correlated caps used in AKS36; no independent errors have replaced their actual values. Every boundary interval is obtained by subtracting these kernel intervals from the exact original total $\mathcal{B}_k^a = F_K^a + F_B^a$. Consequently all three original kernels have uniform limsup at most κ_* after division by kq . The source differences remain $o(kq)$. A matching leading value is still to be calculated from these same matrices.

23 Residue detection expressed in every original primary coefficient

Retain now every original multiplicity $m \geq 1$, $e = 1 + k(m - 1)$, $q = e(k + 1)^2$, and the full polynomial

$$\chi(S) = \prod_{a,b=0}^k (S - \beta_{ab})^e, \quad \beta_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad E = \mathbb{C}[S]/(\chi). \quad (\text{AKJ5})$$

All distinct pairs give distinct roots. For a root β write $z = S - \beta$, $h_\beta(S) = \chi(S)/(S - \beta)^e$, and let ε_β be its original CRT idempotent. Define exactly $\ell([f]) = [S^{q-1}] \text{rem}_\chi f$. For every polynomial class the primary expansion gives

$$\ell(f) = \sum_{\beta} [z^{e-1}] \frac{f(\beta + z)}{h_\beta(\beta + z)}. \quad (\text{AKJ6})$$

The quotient on the right is a truncated power series through degree $e - 1$. To prove the identity, partial-fraction division of f/χ gives at each finite pole the coefficient of $(S - \beta)^{-1}$ shown on the right. Expanding those principal parts at infinity, only their first-order terms contribute to S^{-1} . Polynomial terms contribute nothing. For the unique representative of degree below q , the same S^{-1} coefficient is its coefficient of S^{q-1} . This proves AKJ6 without deleting any repeated-root term. Since $h_\beta(\beta) \neq 0$, the bilinear pairing $(f, g) \mapsto \ell(fg)$ is nondegenerate: if the first nonzero local coefficient of f is $f_t z^t$, take $g = \varepsilon_\beta z^{e-1-t} h_\beta$ to obtain $\ell(fg) = f_t \neq 0$.

Let λ_ν be the specified zero-charge coefficient rows of the original OCS observation, so the injection $\mathcal{F}_0$ gives $j\Lambda = \mathcal{F}_0 \mathfrak{A} \mathcal{H}^{-1}$. Retain the exact entries

$$a_{\nu,\beta,D} := \lambda_\nu(\varepsilon_\beta z^D) = i^D D! c_{\nu;abD}, \quad 0 \leq D < e. \quad (\text{AKJ7})$$

Here $c_{\nu;abD}$ is the full OCS multinomial in the original period and unit jets; $\beta = \beta_{ab}$. It is not an assigned coefficient. The residue representative of each row has the explicit formula

$$A_\nu = \left[\sum_{\beta} \varepsilon_\beta h_\beta(S) \sum_{D=0}^{e-1} a_{\nu,\beta,D} (S - \beta)^{e-1-D} \right], \quad \lambda_\nu(x) = \ell(x A_\nu). \quad (\text{AKJ8})$$

Indeed, restrict to one primary factor and substitute AKJ8 into AKJ6. For $x = \varepsilon_\beta z^D$, multiplication cancels h_β and extracts exactly $a_{\nu,\beta,D}$. These vectors are a basis, proving the asserted equality for every x . This gives the concrete inverse of the residue equivalence used in the incoming formalization, on the actual OCS rows, with their factorials and phases.

Define the integer t_β from these actual coefficients by

$$t_\beta = \begin{cases} -1, & a_{\nu,\beta,D} = 0 \text{ for every } \nu, D, \\ \max\{D : a_{\nu,\beta,D} \neq 0 \text{ for some } \nu\}, & \text{otherwise,} \end{cases} \quad n_\beta = e - 1 - t_\beta. \quad (\text{AKJ9})$$

Define $\mathcal{O}_q : E \rightarrow B^q$ by $\mathcal{O}_q(x) = (\Lambda M_S^j x)_{0 \leq j < q}$, where $B = \text{im } \Lambda$ is the original observation image. Thus $0 \leq n_\beta \leq e$. The complete original permanently invisible submodule and its exact dimension are

$$\begin{aligned} K_\infty &= \bigcap_{j=0}^{q-1} \ker(\Lambda M_S^j) = \bigoplus_{\beta} \varepsilon_\beta (z^{e-n_\beta}) \subset E, \\ \dim K_\infty &= \sum_{\beta} n_\beta, \quad \text{rank } \mathcal{O}_q = q - \sum_{\beta} n_\beta. \end{aligned} \quad (\text{AKJ10})$$

Here $(z^e) = 0$ and (z^0) is the entire local algebra. To prove this, vanishing of the first q observations says $\ell(S^j x A_\nu) = 0$ for all $j < q, \nu$. Nondegeneracy just proved and the power basis imply $x A_\nu = 0$

for every row. Conversely those products being zero kills every iterate. At the factor β , AKJ8 is a unit h_β times its displayed reverse coefficient polynomial. The minimum valuation of all A_ν is precisely n_β . Its common annihilator is (z^{e-n_β}) : multiplication by a coefficient of minimal valuation is a unit times z^{n_β} , and all other coefficients have at least that valuation. This proves the module formula and its dimension; rank-nullity proves the last equality. Affine substitution $M_S = c + iY$ gives an invertible triangular change between the first q powers of the two operators, so the same kernel is obtained using the original Y -stack of OCS.

In particular all iterates are jointly injective exactly when every original highest-jet column $a_{\nu,\beta,e-1}$ has a nonzero entry. For $m = 1$, AKJ8 reduces to

$$A_\nu(\beta) = \chi'(\beta)B_{\nu,\beta}, \quad K_\infty = \text{span}\{\varepsilon_\beta : B_{\nu,\beta} = 0 \text{ for all } \nu\}. \quad (\text{AKJ11})$$

This is an exact test on the AKS8 matrix, not an assumption that it passes. It constructs the inclusion $K_\infty \hookrightarrow K$: each listed primary ideal is killed by all rows, hence in particular by their original one-step observation. The source-minimum metric on this subspace is its restriction from OKM8, retaining the same original Gram. Full-iterate injectivity alone therefore supplies no estimate of the one-step kernel determinant AKJ3.

24 The actual conductor confines the kernel's polynomial degrees

Return to $m = 1$, $k = 4l + 1 \geq 9$ and the original five-member period domain. Let $\mathcal{A}_Q = \prod_{j=1}^4 Q_j$ and write its full Segre restriction from OCF as

$$A = \sum_{r,s=0}^8 a_{rs} X_1^r X_0^{8-r} Y_1^s Y_0^{8-s} \neq 0. \quad (\text{AKJ12})$$

OCF proves nonvanishing and retains all actual H coefficients. No particular a_{rs} is presumed nonzero. Write $k' = k - 8$, $q' = (k - 7)^2$ and

$$b_{rs} = 4 + (2r - 8)\delta + i(2s - 8)\gamma, \quad (\mathcal{T}_A p)(S') = \sum_{r,s=0}^8 a_{rs} p(S' + b_{rs}), \quad \mu_j = \sum_{r,s} a_{rs} b_{rs}^j. \quad (\text{AKJ13})$$

All 81 numbers b_{rs} are distinct. Their Vandermonde is invertible, so some μ_j with $0 \leq j \leq 80$ is nonzero. Define v to be the first such index. The exact finite differential expression on degree-below- q polynomials is

$$\mathcal{T}_A = \sum_{j=0}^{q-1} \frac{\mu_j}{j!} \partial_S^j = \frac{\mu_v}{v!} \partial_S^v B(\partial_S), \quad B(X) = 1 + \sum_{j=v+1}^{q-1} \frac{v! \mu_j}{j! \mu_v} X^{j-v}. \quad (\text{AKJ14})$$

This is Taylor's finite formula. Hence it kills polynomials of degree below v and lowers every higher degree by exactly v . The operator $B(\partial_S)$ is invertible: with $U = B(\partial_S) - I$, its inverse is the finite sum $\sum_{a=0}^{q-1} (-U)^a$. Terms beyond nilpotency are zero. Thus $\mathcal{T}_A$ maps polynomials of degree at most d onto those of degree at most $d - v$ when $d \geq v$.

Let $\chi_{k'}$ be the original polynomial for tensor degree k' . The conductor's exact annihilator in the original coefficient algebra is

$$V_A = \{p \in \mathbb{C}[S]_{< q} : \chi_{k'} \mid \mathcal{T}_A p\}, \quad K \subset V_A. \quad (\text{AKJ15})$$

To prove this map, pair the biform Ag , of bidegree (k, k) , with the primary values of p . If g has coefficient g_{ab} at bidegree (k', k') , convolution of all original coefficients gives the pairing

$$\sum_{a,b=0}^{k'} g_{ab} \sum_{r,s=0}^8 a_{rs} p(\beta_{ab}^{(k')} + b_{rs}) = \sum_{a,b=0}^{k'} g_{ab} (\mathcal{T}_A p)(\beta_{ab}^{(k')}). \quad (\text{AKJ16})$$

The addition identity for the roots follows directly from their retained centres and both signs. Vanishing for every g is exactly vanishing at all q' distinct roots, hence the divisibility in AKJ15. OCF's proved inclusion $AC[X, Y]_{(k', k')} \subset J_k$ then gives $K \subset V_A$ by its original complex-linear coefficient pairing.

The following is a full degree-adapted basis of V_A . Define $\mathcal{J}_v S^d = d! S^{d+v} / (d+v)!$, so $\partial_S^v \mathcal{J}_v = I$, and set

$$1, S, \dots, S^{v-1}, \quad P_h = \frac{(q' + h + v)!}{(q' + h)!} B(\partial_S)^{-1} \mathcal{J}_v(\chi_{k'} S^h), \quad (AKJ17)$$

$$0 \leq h \leq q - q' - v - 1.$$

The first list is empty when $v = 0$. Since $q - q' = 16k - 48 \geq 96 > v$, the range of the second list is nonnegative. Each P_h is monic of degree $q' + h + v$ and maps under $\mathcal{T}_A$ to $\mu_v(q' + h + v)! / (v!(q' + h)!)$ times $\chi_{k'} S^h$. These statements follow from AKJ14, the inverse's constant term one, and the leading coefficient of $\mathcal{J}_v$. The images span every multiple of $\chi_{k'}$ of degree at most $q - v - 1$, and the low list spans the exact kernel of $\mathcal{T}_A$. This proves spanning and independence. In particular $\dim V_A = q - q' = 16k - 48$.

Choose the original OCF basis of $J_k / (AC[X, Y]_{(k', k')})$, whose exact dimension is $j_k = 8k - 32$. Evaluate each chosen biform row on the primary values of every polynomial in AKJ17. This gives an explicit j_k by $(16k - 48)$ matrix Ξ , with the full original period coefficients. Its kernel maps isomorphically onto K by the displayed basis. Its rank is j_k : the original J_k and conductor annihilator duality, or dimension $\dim V_A - \dim K$, gives this rank. For the submatrix $\Xi_{\leq d}$ containing only basis columns of degree at most d , the exact formula is

$$\dim(K \cap \mathbb{C}[S]_{\leq d}) = \#\{\text{degrees in AKJ17 at most } d\} - \text{rank } \Xi_{\leq d}. \quad (AKJ18)$$

Distinct monic degrees ensure that no omitted higher column can cancel to form a degree-at-most- d polynomial. Restricting the coefficient isomorphism to this prefix and applying rank-nullity proves AKJ18. It uses actual minors and makes no assertion of a generic prefix rank.

Let $d_0 < \dots < d_{r-1}$ be the degree jumps of the actual kernel. The basis degrees of V_A , in increasing order, are i for $i < v$, and $q' + i$ for $i \geq v$. Inclusion and dimension of each degree prefix, together with $K \subset \mathbb{C}[S]_{< q}$, imply

$$i + q' \mathbf{1}_{i \geq v} \leq d_i \leq q - r + i. \quad (AKJ19)$$

For the lower inequality the i th independent kernel vector cannot appear earlier than the i th independent vector of V_A . For the upper inequality, intersection of an r -dimensional subspace of $\mathbb{C}^q$ with its $(q - r + i + 1)$ -dimensional degree prefix has dimension at least $i + 1$. Summing yields the exact exterior-degree bounds

$$\frac{r(r-1)}{2} + \max\{r-v, 0\}q' \leq \sum_{i=0}^{r-1} d_i \leq r(q-r) + \frac{r(r-1)}{2}, \quad (AKJ20)$$

$$0 \leq rq - \sum_i d_i \leq \frac{r(r+1)}{2} + rj_k + \min\{r, v\}q'.$$

The difference between the first line's upper and lower endpoints is $rj_k + \min\{r, v\}q'$, since $q - q' = r + j_k$. Therefore the original kernel's exterior degree is $rq - O(k^2)$, with the exact bound above and $v \leq 80$, uniformly in the proved period domain. The affine map $S = c + iy$ preserves all degree prefixes and multiplies a degree- d leading coefficient by i^d . Thus these are also the degrees in the original Gamma coordinate, with all phases known. They specify the kernel's position without assigning a condition number to its frame.

25 The exact conductor translations in the original source norm

Keep either source $s \in \{1, k\}$, $b = s/2$, $M = (2\pi)^{s/2}$ and the original $p_n, h_n = Mn!(b)_n$. On polynomials of degree at most D , let $\varphi_n = p_n / \sqrt{h_n}$ and $w_n = n! / (b)_n$. Differentiating the original

generating identity AKS16 in y gives

$$\partial_y \varphi_n = \sum_{\substack{1 \leq a \leq n \\ a \text{ odd}}} \frac{(-1)^{(a-1)/2}}{a} \sqrt{\frac{w_n}{w_{n-a}}} \varphi_{n-a}. \quad (\text{AKJ21})$$

Indeed differentiation multiplies the generating series by $\arctan z = \sum_{a \text{ odd}} (-1)^{(a-1)/2} z^a / a$. Equating its finite coefficient and substituting the full norms gives exactly AKJ21; the two copies of M are cancelled only at this point.

Write $H_D = \sum_{j=1}^D 1/j$ and $H_0 = 0$. The original source norm obeys

$$\|\partial_y\| \leq \sqrt{H_D(H_D + 4)} \leq H_D + 2, \quad \|f(y+z)\| \leq e^{|z|(H_D+2)} \|f\|. \quad (\text{AKJ22})$$

Here is a proof retaining both source orders. For $n > m \geq 1$ and $b \geq 1/2$,

$$\frac{w_n}{w_m} \leq \prod_{j=m+1}^n \frac{j}{j-1/2} \leq \sqrt{\frac{n}{m}}, \quad w_n \leq 2\sqrt{n} \quad (n \geq 1). \quad (\text{AKJ23})$$

The factor inequality follows by squaring and using $j(j-1) \leq (j-1/2)^2$; the product telescopes. For $m = 0$, separate the first factor, at most two. Apply the weighted Schur estimate to the absolute matrix in AKJ21 with positive weights $\vartheta_n = 1/\sqrt{w_n}$. Its weighted row sum is at most H_D . Its weighted column sum at n is at most $H_n + 4$: for $1 \leq m < n$ the excess over $1/(n-m)$ is at most

$$\frac{\sqrt{n/m} - 1}{n-m} = \frac{1}{\sqrt{m}(\sqrt{n} + \sqrt{m})} \leq \frac{1}{\sqrt{nm}},$$

whose sum is at most two by comparison with $\int_0^n t^{-1/2} dt$. The $m = 0$ term is at most $2/\sqrt{n} \leq 2$. Enlarging the odd-index sums to all indices preserves these bounds. For completeness, if a nonnegative matrix has weighted row bound A and column bound B , Cauchy–Schwarz gives $|(Tx)_m|^2 \leq A \vartheta_m \sum_n T_{mn} |x_n|^2 / \vartheta_n$; summing and using the column bound gives $\|Tx\|^2 \leq AB \|x\|^2$. This proves the first bound in AKJ22, including $D = 0$ where the derivative is zero. The exact finite Taylor identity $f(y+z) = \exp(z\partial_y)f(y)$ and its norm series prove the second. The inverse translation is obtained by $-z$ with the same bound.

There is consequently an exact original-source upper control for AKJ13. For $f(y) = p(c_k + iy)$, evaluate its output at $c_{k-8} + iy$, retaining $c_k = k/2$ and $c_{k-8} = k/2 - 4$. Then

$$(\mathcal{T}_{Ap})(c_{k-8} + iy) = \sum_{r,s=0}^8 a_{rs} f\left(y + \frac{b_{rs} - 4}{i}\right), \quad (\text{AKJ24})$$

and therefore

$$\|\mathcal{T}_{Ap}\|_{s, c_{k-8}} \leq \left(\sum_{r,s} |a_{rs}| e^{|b_{rs}-4|(H_D+2)} \right) \|p\|_{s, c_k}. \quad (\text{AKJ25})$$

The notation $\|p\|_{s,c} = \|p(c + iy)\|_{m_0,s}$ changes only the explicit centre; the measure, mass and source order s are unchanged. AKJ24 proves the centre correction rather than discarding it. For a fixed original period and quartet the coefficient in AKJ25 grows at most polynomially in D , since $H_D \leq 1 + \log D$ for $D \geq 1$ and $|b_{rs} - 4| \leq 8\sqrt{\delta^2 + \gamma^2}$. The entire actual coefficient sum remains visible. The finite inverse in AKJ17 has the specified moment denominators μ_v ; AKJ25 asserts no bound on that inverse. Together AKJ18–25 provide exact degree information and a source-norm estimate for the actual conductor map, to be used in the remaining four-endpoint compression calculation. No leading kernel value or kernel-invariance assumption has replaced that calculation.

26 The original theta class and the checked metric interface

The absolute base and original coefficient resolution are retained:

$$\mathbf{F}_{1,\tau} = \{\tau, 1\}, \quad A_\tau(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi - \hat{\psi})} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V. \quad (\text{AKP1})$$

The four-point chart has $+, - < \eta < \sigma$ and coefficients $(V, V, \mathcal{B}, 0)$, with the original restrictions Θ and $J\Theta = \Theta\mathcal{F}$. Its tensor resolution identifies the top cohomology with $Q^{\otimes k}$. AT1–2a supplies the complete original construction. In its notation, retaining the full unit $v_h = j_h(2\xi/h)$, the actual cochain and arithmetic class maps are

$$\begin{aligned} \mathcal{V}_{h,k}(P) &= P(D_1 + \cdots + D_k)F_h^{\otimes k}, & E &= \mathbb{C}[S]/(\chi), \\ \eta_{h,k}[P] &= v_h^{\otimes k} P(A_h^{(k)})1, & q^{(k)}\mathcal{V}_{h,k}(P) &= \sigma_h^{\otimes k} \eta_{h,k}[P]. \end{aligned} \quad (\text{AKP2})$$

Here $\sigma_h : E_h \hookrightarrow Q$ is the original section and $\eta_{h,k} : E \hookrightarrow E_h^{\otimes k}$ is the full-jet injection. Both injections are part of the supplied source proof; the cochain map has not been replaced by an arbitrary observation.

For completeness, the exact representative-change calculation is as follows. If $P' = P + \chi\kappa$, perform successive monic divisions of $\chi(s_1 + \cdots + s_k)$ by $h(s_a)$. Its final remainder is zero because χ is the original annihilator of the sum in $\bigotimes_a \mathbb{C}[s_a]/(h)$. Thus the actual divisions produce $Q_a(\mathbf{s})$ satisfying

$$\chi(s_1 + \cdots + s_k) = \sum_{a=1}^k h(s_a) Q_a(\mathbf{s}). \quad (\text{AKP3})$$

The chosen ordered division determines these polynomials; no uniqueness of all such decompositions is asserted. On the original tensor complex set

$$\Psi_\kappa = \sum_{a=1}^k (-1)^{a-1} Q_a(\mathbf{D}) \kappa \left(\sum_b D_b \right) (F_h^{\otimes(a-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-a)}). \quad (\text{AKP4})$$

Its differential is

$$d\Psi_\kappa = \mathcal{V}_{h,k}(\chi\kappa) = \mathcal{V}_{h,k}(P') - \mathcal{V}_{h,k}(P). \quad (\text{AKP5})$$

Indeed the only nonzero differential in each summand acts at its ϕ_0 position. The tensor sign is $(-1)^{a-1}$, cancelling the displayed sign. The identity $\Theta\phi_0 = h(D)F_h$ gives the a th term of AKP3; all operators commute with this differential. Summing proves AKP5 with the original signs. This is EP.48, expressed at the current receiving map. Consequently the quotient classes in AKP2 agree. The chosen primitive for their difference depends on κ . At an existing support label ℓ , this quotient map sends the relation to $(\ell, 0)$ and sends external absence τ to τ . The coordinate change has therefore preserved the arithmetic class and its support through an explicit cochain homotopy.

This calculation specifies the correction to the sentence following AKS49: equality of cohomology classes is retained, with AKP4 the primitive of the change. Membership in an additional period-observation kernel alone supplies no such primitive. Its proved map is the inclusion $K_\infty \hookrightarrow K \hookrightarrow E$ of AKJ10–11, followed by the original arithmetic injection in AKP2. In particular, a nonzero coefficient in this additional kernel remains a nonzero original arithmetic class under that injection. Its zero observed value and its arithmetic class are related by these exact maps.

26.1 The complete minimum-section calculation at each endpoint

Let G be one of the original positive Hermitian coefficient Grams, write $K_G = G^{-1}$, and let $\Lambda : E \rightarrow B$ be the actual observation onto its image. Define

$$Q_G = (\Lambda K_G \Lambda^*)^{-1}, \quad S_G = K_G \Lambda^* Q_G. \quad (\text{AKP6})$$

The inverse exists because Λ^* is injective and K_G is positive definite. Direct multiplication, using Hermitian symmetry, gives

$$\Lambda S_G = I_B, \quad S_G^* G = Q_G \Lambda, \quad S_G^* G S_G = Q_G. \quad (\text{AKP7})$$

For every rectangular column family X put $Z = \Lambda X$ and $R = X - S_G Z$. Then

$$\Lambda R = 0, \quad R^* G S_G = 0, \quad R^* G R = X^* G X - Z^* Q_G Z \succeq 0. \quad (\text{AKP8})$$

The first two equalities follow from AKP7. In the expansion of the last Gram, each cross term is $Z^* Q_G Z$ and the section Gram is the same matrix. The resulting single subtraction proves the identity, including every off-diagonal pairing. Any other lift is $S_G Z + R'$ with $\Lambda R' = 0$. Its Gram is $Z^* Q_G Z + R'^* G R'$, proving the minimum property by the same orthogonality calculation.

Retain a fixed section S_* , the actual inclusion I_K , and original coordinate map κ_K satisfying $I_K \kappa_K = I_E - S_* \Lambda$. Applying this identity to R gives

$$R = I_K (\kappa_K X - \kappa_K S_G \Lambda X). \quad (\text{AKP9})$$

Thus the fixed-section coordinates include their specified minimum correction. For one original new column e_D , define $t_D = e_D^* G e_D$, $\lambda_D = \Lambda e_D$ and $\xi_D = \lambda_D^* Q_G \lambda_D$. AKP8 yields $t_D - \xi_D \geq 0$, so the exact ordered factors satisfy

$$(1 + \xi_D) \left(1 + \frac{t_D - \xi_D}{1 + \xi_D} \right) = 1 + t_D. \quad (\text{AKP10})$$

The denominator is positive and is retained. These are the full OPR minimum-section identities, now proved at the original Grams of AKS26 and OKM8. For an empty observation image, the section is the zero map, the Gram on that image has determinant one, and the same identities reduce to $R = X$. No empty present support is deleted.

The new formalization in draft PR32 checks these finite identities and the original observation-iterate and residue-annihilator maps. Its checked implementation is

8e78bc7c240b04d297ade6afdadfd863e0c6db7b.

The later source and status head is

9cec8482f2ecf978cf8bdb67b4c080b9dd74f5d6.

AKJ6–11 evaluates the residue inverse in every actual primary-jet coefficient, giving the exact map to those checked declarations. The analytic AKS estimate, the conductor metric estimates, and the polynomial-gcd dimension corollary have the full written proofs stated here or in their supplied sources; they are outside that Lean run.

27 The finite conductor inverse in the two original Gamma norms

Current refinement. The finite estimates in this chapter remain valid. RUI5–15 and CEP29–49 below sharpen their exterior consequence: complementary singular values and the exact source quotient give an $o(kq)$ conductor error at all four original endpoints, for a fixed actual period. The full original constants, moments and low-degree graph are retained in those proofs.

Retain AKJ12–25 and the complete OCF conductor coefficient polynomial. Thus $m = 1$, $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $0 < \delta < 1/2$, $\gamma > 2$, and the original period belongs to the proved five-orbit locus. Its fixed 4 by 4 matrix $\mathbf{H}$ determines the four quadrics and all 81 coefficients a_{rs} of their full product. They remain the same coefficients as the tensor degree k varies. Put

$$b_{rs} = 4 + (2r - 8)\delta + i(2s - 8)\gamma, \quad \mu_j = \sum_{r,s=0}^8 a_{rs} b_{rs}^j, \quad v = \min\{j : \mu_j \neq 0\} \leq 80. \quad (\text{AIE1})$$

The existence and bound for v are the full distinct-node Vandermonde argument of AKJ13. Throughout, $0 \leq D \leq q - 1$ denotes the polynomial degree cutoff. Every norm below is the original Gamma norm of source order $s \in \{1, k\}$, with

$$b = s/2, \quad M = (2\pi)^{s/2}, \quad h_n = Mn!(b)_n, \quad (\text{AIE2})$$

$$\varphi_n = p_n/\sqrt{h_n}, \quad w_n = n!/(b)_n, \quad \|p\|_{s,c} = \|p(c + iy)\|_{m_0,s}.$$

The centre c is displayed wherever it changes. No source order, source mass, coefficient, or period is assigned a replacement value.

27.1 A coefficient-controlled disk for the literal finite symbol

The exact symbol used in AKJ14 and AKJ17 is

$$B_q(z) = 1 + \sum_{j=v+1}^{q-1} \frac{v! \mu_j}{j! \mu_v} z^{j-v}. \quad (\text{AIE3})$$

The sum is empty if its lower index exceeds its upper index. Define the positive, original-data constants

$$A_*^{\text{abs}} = \sum_{r,s} |a_{rs}|, \quad B_*^{\text{sh}} = \max_{r,s} |b_{rs}|, \quad C_v = \frac{A_*^{\text{abs}} (B_*^{\text{sh}})^v}{|\mu_v|}, \quad (\text{AIE4})$$

$$R_0 = \frac{1}{B_*^{\text{sh}}} \min \left\{ 1, \frac{v+1}{2eC_v} \right\}, \quad \rho_0 = \tanh R_0.$$

Here $A_*^{\text{abs}} > 0$, $B_*^{\text{sh}} \geq 4$, $C_v \geq 1$, $R_0 > 0$, and $0 < \rho_0 < 1$. The superscripts distinguish these constants from the original conductor polynomial and from every earlier radius or pivot coefficient. The exact moment μ_v , including any cancellation in it, remains in the denominator.

For every $|z| \leq R_0$ one has

$$|B_q(z) - 1| \leq \frac{v!}{|\mu_v|} \sum_{j=v+1}^{q-1} \frac{|\mu_j|}{j!} |z|^{j-v} \leq C_v \frac{B_*^{\text{sh}} |z|}{v+1} \exp(B_*^{\text{sh}} |z|) \leq \frac{1}{2}. \quad (\text{AIE5})$$

To prove the middle inequality, use $|\mu_j| \leq A_*^{\text{abs}} (B_*^{\text{sh}})^j$, write $j = v+1+h$, and apply $v!/(v+1+h)! \leq 1/((v+1)h!)$. Summation of the exponential series gives the displayed bound. Definition AIE4 gives $B_*^{\text{sh}} R_0 \leq 1$ and $C_v B_*^{\text{sh}} R_0/(v+1) \leq 1/(2e)$, proving the last inequality. The first upper bound in AIE5 retains every individual finite moment and may be used directly for a larger verified radius. The explicit R_0 proves a positive disk for every actual coefficient chart; no uniform lower bound for $|\mu_v|$ over varying periods is asserted.

The analytic function

$$F_q(z) = \frac{1}{B_q(-i \arctan z)} = \sum_{a=0}^{\infty} f_{q,a} z^a \quad (\text{AIE6})$$

is well defined on a neighbourhood of the closed disk $|z| \leq \rho_0$, and $|F_q(z)| \leq 2$ there. Indeed the power series of $\arctan$ gives $|\arctan z| \leq \operatorname{arctanh} |z| \leq R_0$; AIE5 gives $|B_q(-i \arctan z)| \geq 1/2$. Compactness gives an open neighbourhood with nonzero denominator. Cauchy's coefficient formula therefore proves

$$f_{q,0} = 1, \quad |f_{q,a}| \leq 2\rho_0^{-a} \quad (a \geq 1). \quad (\text{AIE7})$$

Although the literal symbol B_q changes with q , the disk and its bound depend only on the original fixed conductor and shifts.

27.2 The exact coefficient map and its norm

Let $\mathcal{P}_D$ denote the polynomials in y of degree at most D , with the orthonormal basis in AIE2. Define the lowering map

$$L_s \varphi_0 = 0, \quad L_s \varphi_n = \sqrt{\frac{w_n}{w_{n-1}}} \varphi_{n-1} \quad (1 \leq n \leq D). \quad (\text{AIE8})$$

Its powers have exactly the entries $L_s^a \varphi_n = \sqrt{w_n/w_{n-a}} \varphi_{n-a}$ for $a \leq n$ and vanish for $a > n$. Thus $L_s^{D+1} = 0$. AKJ21 gives the polynomial identity $\partial_y = \arctan L_s$, with every power above degree D zero. For the coordinate pullback $\mathcal{C}_c p(y) = p(c + iy)$, the chain rule is $\mathcal{C}_c \partial_S \mathcal{C}_c^{-1} = -i \partial_y$. Consequently the inverse of the literal finite operator is

$$\begin{aligned} \mathcal{C}_c B_q(\partial_S)^{-1} \mathcal{C}_c^{-1} &= F_q(L_s), \\ F_q(L_s) \varphi_n &= \sum_{a=0}^n f_{q,a} \sqrt{\frac{w_n}{w_{n-a}}} \varphi_{n-a}. \end{aligned} \quad (\text{AIE9})$$

For proof, evaluate the analytic power-series identity $B_q(-i \arctan z) F_q(z) = 1$ in the nilpotent matrix L_s . All relevant coefficients are finite, so multiplication is an exact finite polynomial identity. This also proves that AIE9 is the same inverse as the finite nilpotent Neumann sum in AKJ14. The factor $-i$ is forced by the original coordinate map and has been retained.

Define the exact weight ratio and the resulting bound

$$\begin{aligned} \alpha_{D,s} &= \max_{0 \leq m \leq n \leq D} \sqrt{\frac{w_n}{w_m}}, \\ N_{D,s} &= \alpha_{D,s} \left(1 + 2 \sum_{a=1}^D \rho_0^{-a} \right) \leq \frac{2\alpha_{D,s}}{1 - \rho_0} \rho_0^{-D}. \end{aligned} \quad (\text{AIE10})$$

Every absolute row sum and every absolute column sum of AIE9 is at most $\alpha_{D,s} \sum_{a=0}^D |f_{q,a}|$. Cauchy–Schwarz in each row followed by the column bound proves that the operator norm is at most this number; AIE7 gives $N_{D,s}$. It follows that, at every unchanged centre c ,

$$\|B_q(\partial_S)^{-1} p\|_{s,c} \leq N_{D,s} \|p\|_{s,c}. \quad (\text{AIE11})$$

When $s = k$, $b = k/2 \geq 1$ and $w_n/w_{n-1} = n/(b+n-1) \leq 1$, so $\alpha_{D,k} = 1$. When $s = 1$, the same ratio is greater than one, so $\alpha_{D,1} = \sqrt{D!/(1/2)_D}$; for $D \geq 1$, the exact factor estimate of AKJ23 yields $\alpha_{D,1} \leq \sqrt{2} D^{1/4}$. At $D = 0$, both values are one. In particular

$$\log \|B_q(\partial_S)^{-1}\|_{s,c} \leq D \log(\coth R_0) + \log \frac{2\alpha_{D,s}}{1 - \tanh R_0} = O_{H,\delta,\gamma}(D + \log(D+1)). \quad (\text{AIE12})$$

For the fixed original quartet and period, $q = (k+1)^2$ and $D \leq q-1$ imply that this logarithm divided by kq tends to zero. This estimate holds for both original source orders at once; its constants explicitly retain the actual μ_v .

27.3 The infinite symbol differs by an exact low-degree map

To record precisely the truncation in AIE3, define the entire function

$$B_\infty(z) = \frac{v!}{\mu_v z^v} \sum_{r,s} a_{rs} e^{b_{rs} z}, \quad B_\infty(0) = 1. \quad (\text{AIE13})$$

Its apparent singularity is removable by the definition of v . On polynomials of degree at most $D \leq q-1$, set $E_q = B_\infty(\partial_S) - B_q(\partial_S)$; its exact expression is

$$E_q = \sum_{n=q-v}^D \frac{v! \mu_{n+v}}{(n+v)! \mu_v} \partial_S^n. \quad (\text{AIE14})$$

An empty sum is zero. Its image has degree at most $D - q + v \leq v-1$. Both symbols have constant term one, hence both are invertible in this finite polynomial algebra. Their inverses obey

$$B_q(\partial_S)^{-1} - B_\infty(\partial_S)^{-1} = B_q(\partial_S)^{-1} E_q B_\infty(\partial_S)^{-1}, \quad (\text{AIE15})$$

whose image is contained in $\mathbb{C}[S]_{<v}$ and whose rank is at most v . Equation AIE15 follows by multiplying out its right side. The degree statement holds because every factor is a power series in the degree-lowering derivative. Thus the infinite-symbol formula can be related to the literal AKJ inverse by the displayed exact correction; it cannot replace that inverse without this map.

27.4 An explicit right inverse of the original conductor

Assume now $D \geq v$. Define the raising map, on the degree ranges where its image lies in $\mathcal{P}_D$, by

$$J_s \varphi_n = \sqrt{\frac{w_n}{w_{n+1}}} \varphi_{n+1} = \sqrt{\frac{b+n}{n+1}} \varphi_{n+1}. \quad (\text{AIE16})$$

Then $L_s^v J_s^v = I$ on $\mathcal{P}_{D-v}$ and $\|J_s^v\| \leq \max\{1, b^{v/2}\}$. The latter follows from the displayed coefficient, at each raising step, by $(b+n)/(n+1) \leq \max\{1, b\}$.

The analytic function $G(z) = z/\arctan z$, with $G(0) = 1$, satisfies $|G(z)| \leq 2$ on $|z| < 1$. Indeed

$$\frac{\arctan z}{z} = \int_0^1 \frac{dt}{1+t^2 z^2}, \quad \operatorname{Re} \frac{1}{1+u} > \frac{1}{2} \quad (|u| < 1). \quad (\text{AIE17})$$

The real-part inequality follows from $2 \operatorname{Re}(1+\bar{u}) > |1+u|^2$, which reduces to $1 > |u|^2$. It proves that the integral is nonzero with real part greater than $1/2$, including the removable value at zero. Every Taylor coefficient of $G(z)^v$ is therefore bounded by 2^v : apply Cauchy's formula on radius $t < 1$ and let t increase to one for each fixed coefficient.

Define a map between the specified degree spaces by

$$R_{v,s} = G(L_s)^v J_s^v : \mathcal{P}_{D-v} \longrightarrow \mathcal{P}_D. \quad (\text{AIE18})$$

Since all functions of L_s commute, AIE8 and the power-series identity $(\arctan z)^v G(z)^v = z^v$ give $\partial_y^v R_{v,s} = L_s^v J_s^v = I$. This proves the right-inverse identity even at the top allowed degree. The same row and column argument as for AIE10, now with every coefficient bounded by 2^v , gives

$$\|R_{v,s}\| \leq 2^v (D+1) \alpha_{D,s} \max\{1, b^{v/2}\}. \quad (\text{AIE19})$$

When $v = 0$ the map is exactly the identity; AIE19 remains a valid upper bound. Define the corresponding map on polynomials in the original S coordinate by

$$\mathcal{R}_{v,s,c} = \mathcal{C}_c^{-1} i^v R_{v,s} \mathcal{C}_c. \quad (\text{AIE20})$$

The chain rule in AIE9 proves $\partial_S^v \mathcal{R}_{v,s,c} = I$, with exactly the factor $(-i)^v i^v = 1$. This is a map in the original polynomial algebra; the source norm only specifies its coefficients through the explicitly given original orthonormal basis.

Let $c_k = k/2$ and $c' = c_{k-8} = k/2 - 4$. The full conductor of AKJ13 has the following right inverse on $\mathbb{C}[S']_{\leq D-v}$:

$$\mathcal{R}_{A,s}g = \frac{v!}{\mu_v} B_q(\partial_S)^{-1} \mathcal{R}_{v,s,c'}g, \quad \mathcal{T}_A \mathcal{R}_{A,s}g = g. \quad (\text{AIE21})$$

To check it, use the exact finite factorization $\mathcal{T}_A = (\mu_v/v!) \partial_S^v B_q(\partial_S)$ from AKJ14 on degree at most D . The two inverse factors cancel and AIE20 gives the stated identity. In the identity the output variable is S' ; the differential expression uses the same numerical polynomial coefficients, so the domain and codomain are as displayed.

Set $H_D = \sum_{j=1}^D 1/j$, $H_0 = 0$, and

$$C_{A,D,s} = \frac{v!}{|\mu_v|} e^{4(H_D+2)} N_{D,s} 2^v (D+1) \alpha_{D,s} \max\{1, b^{v/2}\}. \quad (\text{AIE22})$$

Then the original-centre norm estimate is

$$\|\mathcal{R}_{A,s}g\|_{s,c_k} \leq C_{A,D,s} \|g\|_{s,c'}. \quad (\text{AIE23})$$

Indeed AIE11 and AIE19 first prove the bound without the exponential factor, with both norms at c' . For any output polynomial p , $p(c_k + iy) = p(c' + i(y - 4i))$, so AKJ22 applied to the exact shift $-4i$ supplies the remaining factor. Thus the four-unit centre difference is retained in the typed norm map. For the fixed actual quartet and period, $v \leq 80$ and $b \in \{1/2, k/2\}$ give

$$\log C_{A,D,s} \leq D \log(\coth R_0) + O_{H,\delta,\gamma}(\log(D+1) + \log(k+1)), \quad \frac{\log C_{A,D,s}}{kq} \longrightarrow 0. \quad (\text{AIE24})$$

Every factor in the first assertion is written in AIE22; the asymptotic statement follows from $D \leq (k+1)^2 - 1$ and the fixed finite v . No varying-period bound is inferred from this statement.

For comparison with the exact monomial lift $\mathcal{J}_v$ of AKJ17, put $L_g = \mathcal{R}_{v,s,c'}g - \mathcal{J}_v g$. Both terms have v th derivative g , so $L_g \in \mathbb{C}[S]_{<v}$. More explicitly,

$$L_g = \sum_{a=0}^{v-1} [S^a](\mathcal{R}_{v,s,c'}g) S^a, \quad \mathcal{R}_{A,s}g - \frac{v!}{\mu_v} B_q(\partial_S)^{-1} \mathcal{J}_v g = \frac{v!}{\mu_v} B_q(\partial_S)^{-1} L_g. \quad (\text{AIE25})$$

The first identity uses the exact zero coefficients of $\mathcal{J}_v g$ in degrees below v . This proves the complete degree-below- v correction between the two lifts, rather than discarding their different choices of integration constants.

27.5 The exact quotient and exterior transfer supplied by this bound

The kernel of $\mathcal{T}_A$ on degree at most $D \geq v$ is exactly $\mathbb{C}[S]_{<v}$ by AKJ14. Give the quotient $\mathbb{C}[S]_{\leq D}/\mathbb{C}[S]_{<v}$ its Hilbert quotient norm from the original source at c_k , and give its target $\mathbb{C}[S']_{\leq D-v}$ the original source at c' . The induced conductor map $\overline{\mathcal{T}}_A$ is an isomorphism, because AIE21 proves surjectivity and the displayed kernel proves injectivity. Define the full upper norm from AKJ25 by

$$N_{A,D} = \sum_{r,s} |a_{rs}| e^{|b_{rs}-4|(H_D+2)}. \quad (\text{AIE26})$$

For every target g , its minimum-norm preimage class obeys

$$\frac{\|g\|_{s,c'}}{N_{A,D}} \leq \|\overline{\mathcal{T}}_A^{-1} g\|_{\text{quot}} \leq C_{A,D,s} \|g\|_{s,c'}. \quad (\text{AIE27})$$

For the left inequality, apply AKJ25 to every preimage and take the infimum. For the right inequality, use the actual preimage AIE21. The infimum is attained since the omitted subspace is a closed finite-dimensional subspace. These arguments prove the norm statement with the specified quotient map, rather than assigning the coefficient norm to the quotient.

In particular AIE21 gives the exact decomposition

$$V_A = \mathbb{C}[S]_{<v} \oplus \mathcal{R}_{A,s}(\chi_{k-8}\mathbb{C}[S']_{\leq D-v-q'}), \quad q' = (k-7)^2, \quad (\text{AIE28})$$

where a negative degree means the zero space and V_A here denotes the AKJ15 space restricted to degree at most D . Indeed the image under $\mathcal{T}_A$ of a member of V_A is precisely a multiple of χ_{k-8} in the displayed degree range. Subtracting its lift leaves the exact kernel. The directness follows by applying $\mathcal{T}_A$. Equation AIE25 identifies this decomposition with the earlier monic basis, including its low-degree correction.

For any t -dimensional target subspace and any basis column matrix X in its original orthonormal coordinates, AIE27 implies the full Gram determinant inequalities

$$N_{A,D}^{-2t} \det(X^*X) \leq \det(X^*(\overline{\mathcal{T}}_A^{-1})^* \overline{\mathcal{T}}_A^{-1} X) \leq C_{A,D,s}^{2t} \det(X^*X). \quad (\text{AIE29})$$

For proof, AIE27 is a two-sided positive Hermitian matrix inequality after congruence by X . Conjugating by $(X^*X)^{-1/2}$ gives eigenvalues between the two scalar squared bounds, whose product is the displayed determinant ratio. This proves the exact exterior transfer and all powers of the norm constants.

Similarly, for any t -plane $U \subset \mathbb{C}[S]_{\leq D}$, AIE11 gives $\|\bigwedge^t(B_q(\partial_S)^{-1}|_U)\| \leq N_{D,s}^t$. Although $B_q(\partial_S)$ is triangular with every diagonal entry one and hence determinant one on the full polynomial space, that full-space determinant does not cancel its restriction to an arbitrary subspace. The exact restricted determinant is $\det(X^*(B_q^{-1})^* B_q^{-1} X) / \det(X^*X)$ for a basis matrix X of that subspace; this is the map to which AIE11 applies.

For the original kernel rank $r = 8k - 16$ and $D \asymp q$, the upper logarithm in these exterior bounds is $O_{H,\delta,\gamma}(rq) = O_{H,\delta,\gamma}(kq)$. Therefore AIE12 and AIE24 prove a sub- kq logarithmic bound for each operator, while AIE29 retains the rank factor required for volumes. They provide the original conductor inverse, its full low-degree comparison and its quotient metric control. A sub- kq error for the actual rank- r four-endpoint determinant is not a consequence of these scalar bounds; that remaining calculation must use the actual restricted matrices and the existing endpoint correlation.

28 Subleading exterior control from the actual finite conductor

Retain the original conductor of AKJ12–25 and AIE1–29. In particular the quartet, full unit, original period and branch are fixed in the proved five-orbit domain, $m = 1$, $k = 4l + 1 \geq 9$, and $q = (k + 1)^2$. All 81 coefficients a_{rs} are those of the actual product of four period quadrics. They are fixed as k varies. Retain

$$b_{rs} = 4 + (2r - 8)\delta + i(2s - 8)\gamma, \quad \mu_j = \sum_{r,s=0}^8 a_{rs} b_{rs}^j, \quad v = \min\{j : \mu_j \neq 0\} \leq 80, \quad (\text{RUI1})$$

$$A_{\text{abs}} = \sum_{r,s} |a_{rs}|, \quad B_{\text{sh}} = \max_{r,s} |b_{rs}|, \quad C_v = \frac{A_{\text{abs}} B_{\text{sh}}^v}{|\mu_v|} \geq 1.$$

No coefficient or moment denominator is replaced by a generic value. The finite bound for v is AKJ13's distinct-node Vandermonde proof. The inequality $C_v \geq 1$ follows from the triangle inequality applied to the actual v th moment. The source measure at order $s \in \{1, k\}$ is exactly

$$dm_{0,s}(y) = (2\pi)^{s/2} \frac{2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad \|p\|_{s,c} = \|p(c + iy)\|_{L^2(m_{0,s})}. \quad (\text{RUI2})$$

Every centre appearing below is stated, and the whole mass remains in the source norm. Put $H_D = \sum_{j=1}^D 1/j$, $H_0 = 0$.

28.1 The forward polynomial bound and exact determinant

For an integer cutoff $L \geq v + 1$, define the literal finite symbol

$$B_L(z) = 1 + \sum_{a=1}^{L-v-1} \frac{v! \mu_{v+a}}{(v+a)! \mu_v} z^a. \quad (\text{RUI3})$$

The empty sum is zero. This includes the actual coefficient symbol B_q of AKJ14 and the source symbol B_{N+1} needed at a larger source cutoff N . For every displayed coefficient, including the constant coefficient, one has

$$\left| \frac{v! \mu_{v+a}}{(v+a)! \mu_v} \right| \leq C_v \frac{v! B_{\text{sh}}^a}{(v+a)!} \leq C_v \frac{B_{\text{sh}}^a}{a!}. \quad (\text{RUI4})$$

The first inequality is the full moment bound $|\mu_{v+a}| \leq A_{\text{abs}} B_{\text{sh}}^{v+a}$. The second follows from $(v+a)! \geq v! a!$.

On $\mathbb{C}[S]_{\leq D}$ with the norm RUI2, the original coordinate pullback sends ∂_S to $-i\partial_y$. AKJ21–23 proves $\|\partial_S\| \leq H_D + 2$, uniformly in both original source orders. Applying RUI4 term by term to the finite operator polynomial gives

$$\|B_L(\partial_S)\|_{s,c} \leq M_D, \quad M_D = C_v \exp(B_{\text{sh}}(H_D + 2)) \geq 1. \quad (\text{RUI5})$$

Indeed each power has norm at most $(H_D + 2)^a$; extending this nonnegative finite sum to the exponential series proves the displayed bound. The constant term is one and every other term strictly lowers degree. Thus, on this exact $(D + 1)$ -dimensional polynomial space,

$$\det B_L(\partial_S) = 1. \quad (\text{RUI6})$$

This is its determinant in the original increasing-power basis, and therefore in every basis of the same source space, including the original orthonormal Gamma basis. The assertion preserves the source measure; it is not a rescaling of this determinant.

28.2 Complementary singular values retain the full exterior norm

Let $n = D + 1$ and $T = B_L(\partial_S)$ acting on this same Hilbert space. Write its singular values as $\sigma_1 \geq \dots \geq \sigma_n > 0$. RUI6 gives $\prod_i \sigma_i = 1$. For every $0 \leq t \leq n$,

$$\begin{aligned} \left\| \bigwedge^t T^{-1} \right\| &= \prod_{j=n-t+1}^n \sigma_j^{-1} = \prod_{j=1}^{n-t} \sigma_j = \left\| \bigwedge^{n-t} T \right\| \leq M_D^{n-t}, \\ \left\| \bigwedge^t T \right\| &\leq M_D^t. \end{aligned} \tag{RUI7}$$

For a proof of the exterior norm formula, the singular-value decomposition expresses T as two unitary maps and a diagonal map with these positive entries. Exterior powers of the unitary maps preserve the induced norm; the diagonal exterior map has entries equal to the products over all t -subsets. Its largest entry is the product of the largest t singular values. Applying this to T^{-1} and using the full determinant proves every equality in RUI7. Empty products cover $t = 0, n$.

Consequently, for any actual t -dimensional subspace and any full-rank basis column matrix X in the original orthonormal source coordinates,

$$M_D^{-2t} \det(X^* X) \leq \det(X^* (T^{-1})^* T^{-1} X) \leq M_D^{2(n-t)} \det(X^* X). \tag{RUI8}$$

For the upper inequality apply the first exterior norm of RUI7 to the decomposable exterior vector of the columns of X . Its squared norm is their Gram determinant. For the lower inequality, put $Y = T^{-1}X$ and apply the second bound of RUI7 to $TY = X$. This proves the lower bound on this same subspace without requiring that it be invariant. Every original off-diagonal Gram entry is retained by the displayed matrices.

In particular, for fixed original period and quartet, uniformly in $s \in \{1, k\}$, every $0 \leq D \leq 2q$ and every allowed t and X ,

$$\left| \log \frac{\det(X^* (T^{-1})^* T^{-1} X)}{\det(X^* X)} \right| \leq 2(D+1)(\log C_v + B_{\text{sh}}(H_D + 2)) = O_{\text{actual}}(q \log(q+1)) = o(kq). \tag{RUI9}$$

The harmonic estimate $H_D \leq 1 + \log D$ for $D \geq 1$ follows by integrating $1/x$ on consecutive intervals; $D = 0$ is immediate. Since $q = (k+1)^2$, the displayed upper bound divided by kq tends to zero, independently of D, t, X . This is a uniform estimate over the actual subspaces, not a presumed favourable position of the kernel. Its constants are fixed only after fixing the original period and quartet; no uniform bound on $1/|\mu_v|$ over moving periods is asserted. The same inequalities for T follow by exchanging the image subspace and its inverse image.

RUI9 improves the rank-multiplied estimate at the end of AIE29. The earlier estimate is valid, but its factor t on a single inverse operator bound is unnecessary. The complementary exterior identity uses the exact full determinant and supplies the sharper bound on the same original matrices.

28.3 The actual conductor quotient inherits the sharper volume control

Let $D \geq v$ and use the literal source truncation $L = D + 1$. Taylor's finite formula gives, on degree at most D ,

$$\mathcal{T}_A = \sum_{j=v}^D \frac{\mu_j}{j!} \partial_S^j = \frac{\mu_v}{v!} \partial_S^v B_{D+1}(\partial_S), \quad \ker \mathcal{T}_A = \mathbb{C}[S]_{<v}. \tag{RUI10}$$

The first equality is exactly the translation operator in AKJ13; all lower moments vanish by their proved definition. The factor $B_{D+1}(\partial_S)$ is invertible and preserves each degree filtration, so the last equality follows from the derivative kernel.

Use the original source centre $c_k = k/2$ and target centre $c' = c_{k-8} = k/2 - 4$, both with source order s . AIE16–20 gives the full right inverse $\mathcal{R}_{v,s,c'}$ of ∂_S^v with norm at most

$$P_{D,s} = 2^v(D+1)\alpha_{D,s} \max\{1, (s/2)^{v/2}\}, \quad \alpha_{D,s} = \max_{0 \leq m \leq n \leq D} \sqrt{\frac{n!/(s/2)_n}{m!/(s/2)_m}}. \quad (\text{RUI11})$$

In particular $\alpha_{D,k} = 1$, while $\alpha_{D,1} \leq \sqrt{2} \max\{1, D^{1/4}\}$; the full derivation and the phase i^v are the original AIE16–20 construction. The exact conductor right inverse is

$$\mathcal{R}_{A,s,D} = \frac{v!}{\mu_v} B_{D+1}(\partial_S)^{-1} \mathcal{R}_{v,s,c'}. \quad (\text{RUI12})$$

Multiplying by RUI10 proves $\mathcal{T}_A \mathcal{R}_{A,s,D} = I$. The numerical polynomial coefficients use the original source and target variables, as in AIE21. The centre change from c' to c_k is the original shift $y \mapsto y - 4i$, of norm at most $\exp(4(H_D + 2))$ by AKJ22.

Give the quotient $\mathbb{C}[S]_{\leq D}/\mathbb{C}[S]_{<v}$ its minimum norm from the source at c_k . The induced conductor $\overline{\mathcal{T}}_A$ is an isomorphism onto $\mathbb{C}[S']_{\leq D-v}$ with its source norm at c' . Define

$$L_{D,s} = \frac{v!}{|\mu_v|} e^{4(H_D+2)} P_{D,s}, \quad N_{A,D} = \sum_{r,s} |a_{rs}| e^{|b_{rs}-4|(H_D+2)}. \quad (\text{RUI13})$$

For any t -dimensional target subspace, with original basis matrix X , its exact quotient Gram ratio satisfies

$$N_{A,D}^{-2t} \det(X^* X) \leq \det(X^* (\overline{\mathcal{T}}_A^{-1})^* \overline{\mathcal{T}}_A^{-1} X) \leq L_{D,s}^{2t} M_D^{2(D+1-t)} \det(X^* X). \quad (\text{RUI14})$$

To prove the upper bound, use the actual lift RUI12. Apply $\mathcal{R}_{v,s,c'}$ to the t basis columns, retaining its specified image subspace. RUI7 bounds the exterior power of the inverse B_{D+1} on that image by M_D^{D+1-t} . The scalar, right-inverse norm, and centre translation give $L_{D,s}^t$. Projection to the minimum lift in the quotient is an orthogonal projection on the same source Hilbert space, so its exterior norm is at most one. This proves the complete upper Gram inequality. For the lower bound, apply the forward original conductor to the minimum lifts. AKJ24–25 gives its norm $N_{A,D}$; taking its t th exterior power proves the result. This proof retains the coupled Gram and the full quotient metric.

Every factor in RUI13 is explicit in the fixed original data. For the asymptotic assertion restrict now to $v \leq D \leq 2q$. Since $v \leq 80$, $s \in \{1, k\}$ and $t \leq D+1-v \leq 2q+1$, both logarithmic endpoints of RUI14 have absolute value at most

$$2t(|\log L_{D,s}| + |\log N_{A,D}|) + 2(D+1) \log M_D = O_{\text{actual}}(q \log(q+1) + q \log(k+1)) = o(kq). \quad (\text{RUI15})$$

For the logarithm of $N_{A,D}$, use $A_{\text{abs}} \leq N_{A,D} \leq A_{\text{abs}} \exp(\max |b_{rs} - 4|(H_D + 2))$. For $L_{D,s}$, its displayed factors and the positive lower bound $P_{D,s} \geq 1$ give the same logarithmic estimate, retaining $v!/|\mu_v|$. This proves the absolute bounds even if either constant is below one. The harmonic bound and $q = (k+1)^2$ then prove the final limit. No dimension- q scalar comparison has silently acquired an extra factor of q in this argument.

At the original coefficient cutoff $D = q - 1$, RUI14 is the sharpened version of AIE29, on precisely the same conductor quotient and original coefficient subspaces. At a larger source cutoff $D = N$, its literal symbol is B_{N+1} , and RUI10 proves the exact extension. The old symbol B_q cannot be substituted without recording the finite difference. Passing this estimate through the original remainder square requires the actual image of the relation ideal; that square, its image graph, and the endpoint quotient metrics are retained in the succeeding calculation.

29 The original conductor at every source endpoint

Retain the complete simple-quartet conductor of OCF and AKJ, its actual original period matrix, and $k = 4l + 1 \geq 9$ on the proved five-orbit locus. Write

$$\begin{aligned}
 q &= (k+1)^2, & k' &= k-8, & q' &= (k-7)^2, & \Delta &= q - q' = 16k - 48, \\
 c &= k/2, & c' &= k/2 - 4, & \beta_{ij} &= c + (2i - k)\delta + i(2j - k)\gamma, \\
 & & \beta'_{ab} &= c' + (2a - k')\delta + i(2b - k')\gamma, \\
 A &= \sum_{r,s=0}^8 a_{rs} e_{rs}^{(8)} \neq 0, & b_{rs} &= 4 + (2r - 8)\delta + i(2s - 8)\gamma, \\
 \mu_j &= \sum_{r,s} a_{rs} b_{rs}^j, & v &= \min\{j : \mu_j \neq 0\} \leq 80.
 \end{aligned} \tag{CEP1}$$

The factor i before an imaginary term is the imaginary unit; subscript indices are integers in their displayed ranges. The original polynomials are $\chi_k(S) = \prod_{i,j=0}^k (S - \beta_{ij})$ and $\chi_{k'}(S') = \prod_{a,b=0}^{k'} (S' - \beta'_{ab})$. Both have distinct roots. Set $E_k = \mathbb{C}[S]/(\chi_k)$ and $E_{k'} = \mathbb{C}[S']/(\chi_{k'})$. Every source below has the original order $s \in \{1, k\}$ and full mass $M_s = (2\pi)^{s/2}$, with $b = s/2$ and $h_n = M_s n! (b)_n$. In particular the lower grid does not change source order k to $k - 8$.

29.1 The full source, remainder and primary-coordinate square

For every polynomial define the original finite-translation map

$$(\mathcal{T}_A p)(S') = \sum_{r,s=0}^8 a_{rs} p(S' + b_{rs}). \tag{CEP2}$$

For $0 \leq a, b \leq k'$ and $0 \leq r, s \leq 8$, direct substitution gives $\beta'_{ab} + b_{rs} = \beta_{a+r, b+s}$. All indices on the right lie between zero and k . Consequently

$$(\mathcal{T}_A(\chi_k f))(\beta'_{ab}) = 0 \quad \text{for every polynomial } f. \tag{CEP3}$$

Division by the monic $\chi_{k'}$ and evaluation at its distinct roots then show $\chi_{k'} \mid \mathcal{T}_A(\chi_k f)$. Thus CEP2 induces the specified complex-linear map $\bar{\mathcal{T}}_A : E_k \longrightarrow E_{k'}$.

In the original ordered primary coordinates its matrix is

$$C_{(a,b),(i,j)} = \begin{cases} a_{i-a, j-b}, & 0 \leq i-a, j-b \leq 8, \\ 0, & \text{otherwise.} \end{cases} \quad C \in \text{Mat}_{q' \times q}(\mathbb{C}). \tag{CEP4}$$

Evaluation of CEP2 proves this entry formula, including its absence of conjugation or binomial coefficients. Its ordinary transpose $C^\top$ is the exact biform coefficient multiplication by the nonzero polynomial A . That multiplication is injective because the biform ring embeds in a polynomial integral domain. Therefore C has rank q' and is surjective. With the original ordered Vandermonde matrices $V_\beta, V_{\beta'}$, the polynomial coordinate matrix is exactly $V_{\beta'}^{-1} C V_\beta$. The kernel is the AKJ15 space

$$V_A = \{[p] \in E_k : \chi_{k'} \mid \mathcal{T}_A p\}, \quad \dim V_A = \Delta. \tag{CEP5}$$

CEP3 proves independence of the representative, and CEP4 proves the dimension. The original common kernel $K = \ker \Lambda$ is contained in V_A by the full conductor inclusion OCF17.

For every endpoint $N \in \{q-1, q, 2q-1, 2q\}$, put $d = N - v$ and define

$$\begin{aligned}
 \mathcal{P}_N &= \mathbb{C}[S]_{\leq N}, & \mathcal{P}'_d &= \mathbb{C}[S']_{\leq d}, \\
 U_N : \mathcal{P}_N &\longrightarrow E_k, & p &\longmapsto [p], & V_d : \mathcal{P}'_d &\longrightarrow E_{k'}, & g &\longmapsto [g].
 \end{aligned}$$

The top horizontal map is precisely $\mathcal{T}_{A,N} : \mathcal{P}_N \longrightarrow \mathcal{P}'_{N-v}$. Taylor's finite identity and the first nonzero moment show that CEP2 lowers every degree at least v by exactly v and kills exactly $\mathbb{C}[S]_{<v}$. The exact commutative square is

$$\begin{array}{ccc} \mathcal{P}_N & \xrightarrow{\mathcal{T}_{A,N}} & \mathcal{P}'_{N-v} \\ U_N \downarrow & & \downarrow V_{N-v} \\ E_k & \xrightarrow{\bar{\mathcal{T}}_A} & E_{k'}. \end{array} \quad (\text{CEP6})$$

The lower remainder is surjective because $N - v \geq q - 1 - v \geq q' - 1$; the latter inequality is $\Delta \geq v$, and $\Delta \geq 96 > 80 \geq v$.

The map is complex-linear with its following full action defect. If $M_S, M_{S'}$ denote the original multiplication operators, then

$$CM_S - M_{S'}C = C_b, \quad (C_b)_{(a,b),(i,j)} = \begin{cases} a_{i-a,j-b}b_{i-a,j-b}, & 0 \leq i-a, j-b \leq 8, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{CEP7})$$

Indeed each entry is multiplied by $\beta_{ij} - \beta'_{ab} = b_{i-a,j-b}$. In the original centred coordinates $Y = (M_S - cI)/i$, $Y' = (M_{S'} - c'I)/i$, this becomes $CY - Y'C = (C_b - 4C)/i$. Thus the exact relation to multiplication retains its coefficient defect.

29.2 The literal finite inverse at the larger cutoffs

On $\mathcal{P}_N$, define the literal finite symbol

$$B_{N+1}(z) = 1 + \sum_{j=v+1}^N \frac{v! \mu_j}{j! \mu_v} z^{j-v}, \quad \mathcal{T}_{A,N} = \frac{\mu_v}{v!} \partial_S^v B_{N+1}(\partial_S). \quad (\text{CEP8})$$

The expression on the right is evaluated in the output variable S' . Taylor expansion of each term in CEP2 through degree N proves this identity coefficient by coefficient. Its constant term one makes $B_{N+1}(\partial_S)$ invertible: its difference from the identity lowers degree, so the inverse is its finite nilpotent Neumann sum through power N . A right inverse of ∂_S^v on the polynomial spaces exists by integration of each monomial; composition proves surjectivity of the top map in CEP6.

The relation to the earlier literal B_q of AKJ14 is

$$\begin{aligned} D_{N,q} &= B_{N+1}(\partial_S) - B_q(\partial_S) = \sum_{j=q}^N \frac{v! \mu_j}{j! \mu_v} \partial_S^{j-v}, \\ B_{N+1}(\partial_S)^{-1} - B_q(\partial_S)^{-1} &= -B_{N+1}(\partial_S)^{-1} D_{N,q} B_q(\partial_S)^{-1}, \\ \mathcal{T}_{A,N} - \frac{\mu_v}{v!} \partial_S^v B_q(\partial_S) &= \sum_{j=q}^N \frac{\mu_j}{j!} \partial_S^j. \end{aligned} \quad (\text{CEP9})$$

Empty sums are zero. The first two differences have image contained in $\mathbb{C}[S]_{\leq N-q+v}$; the last has image contained in $\mathbb{C}[S']_{\leq N-q}$. The inverse formula follows by multiplication, and its degree assertion follows since both inverse factors preserve every degree prefix. At $N = q - 1$ all differences vanish. At the other three endpoints these are the full retained defects; their ranks are not assigned the cutoff-independent bound v .

For comparison, the entire symbol $B_\infty(z) = v! \sum_{r,s} a_{rs} e^{b_{rs}z} / (\mu_v z^v)$, with removable value one at zero, obeys at each cutoff

$$B_\infty(\partial_S) - B_{N+1}(\partial_S) = \sum_{n=N+1-v}^N \frac{v! \mu_{n+v}}{(n+v)! \mu_v} \partial_S^n. \quad (\text{CEP10})$$

Its image and the image of the corresponding inverse difference lie in $\mathbb{C}[S]_{<v}$. This follows from the first derivative degree $N + 1 - v$, and from the inverse identity with the same order as AIE15. Thus the rank-at-most- v comparison is with the correctly growing finite symbol in CEP8.

For completeness the original-source inverse estimate extends directly to these literal symbols. Retain the exact constants $A_*^{\text{abs}}, B_*^{\text{sh}}, C_v, R_0, \rho_0$ of AIE4. For every $|z| \leq R_0$ the finite coefficient sum in CEP8 is at most the full absolute moment sum:

$$|B_{N+1}(z) - 1| \leq \frac{v!}{|\mu_v|} \sum_{j=v+1}^N \frac{|\mu_j|}{j!} |z|^{j-v} \leq C_v \frac{B_*^{\text{sh}} |z|}{v+1} e^{B_*^{\text{sh}} |z|} \leq \frac{1}{2}. \quad (\text{CEP11})$$

The middle inequality follows by writing $j = v+1+h$ and using $v!/(v+1+h)! \leq 1/((v+1)h!)$. Hence $F_{N+1}(z) = 1/B_{N+1}(-i \arctan z)$ has coefficient absolute values at most $2\rho_0^{-a}$ on degree $a \geq 1$, and constant coefficient one. This is Cauchy's formula on $|z| = \rho_0$, since $|\arctan z| \leq \operatorname{arctanh} \rho_0 = R_0$.

In the original orthonormal Gamma basis, the lowering matrix $L_s \varphi_n = \sqrt{w_n/w_{n-1}} \varphi_{n-1}$, $w_n = n!/(b)_n$, satisfies $\partial_y = \arctan L_s$ and $L_s^{N+1} = 0$. Evaluation of the power-series inverse in this nilpotent matrix therefore gives the exact finite operator $F_{N+1}(L_s)$, with entries $[z^{n-m}]F_{N+1}(z) \sqrt{w_n/w_m}$. The chain rule $S = c + iy$ supplies the retained $-i$. Bounding the absolute row and column sums gives

$$\|B_{N+1}(\partial_S)^{-1}\|_{s,c} \leq \alpha_{N,s} \left(1 + 2 \sum_{a=1}^N \rho_0^{-a}\right), \quad \alpha_{N,s} = \max_{0 \leq m \leq n \leq N} \sqrt{w_n/w_m}. \quad (\text{CEP12})$$

This proves the larger-cutoff assertion for the actual finite symbol rather than substituting N into an identity for B_q . Likewise the explicit AIE18 raising right inverse on these degree spaces, followed by i^v , $B_{N+1}(\partial_S)^{-1}$ and $v!/\mu_v$, is a right inverse of CEP8. Its comparison with the monomial integral remains exactly the degree-below- v correction in AIE25, now with this literal inverse.

29.3 The exact source operator matrix and its full determinant

Use the unchanged original orthonormal polynomials $u_n(S) = p_n((S - c)/i)/\sqrt{h_n}$ in $\mathcal{P}_N$ and $u'_m(S') = p_m((S' - c')/i)/\sqrt{h_m}$ in $\mathcal{P}'_{N-v}$. Let A_N denote the matrix of $\mathcal{T}_{A,N}$ in these two frames. Define the analytic coefficient series at zero

$$t(z) = \sum_{r,s} a_{rs} \exp\left(\frac{b_{rs} - 4}{i} \arctan z\right) = \sum_{\ell \geq 0} t_\ell z^\ell. \quad (\text{CEP13})$$

Then its exact entries are

$$(A_N)_{m,n} = \begin{cases} t_{n-m} \sqrt{w_n/w_m}, & n \geq m, \\ 0, & n < m, \end{cases} \quad 0 \leq m \leq N - v, \quad 0 \leq n \leq N. \quad (\text{CEP14})$$

To prove this, the original generating identity $\sum p_n(y) z^n/n! = (1 + z^2)^{-b/2} e^{y \arctan z}$ shows that translation by η multiplies the series by $e^{\eta \arctan z}$. Extracting its finite coefficients and dividing by the complete $\sqrt{h_n}$ gives $\sqrt{w_n/w_m}$ after the two copies of M_s cancel at this specified step. Evaluation of CEP2 at $c' + iy$ uses $p(c + i(y + (b_{rs} - 4)/i))$, proving the exact centre shift in CEP13.

For $\ell < v$, the shifted moment $\sum a_{rs} (b_{rs} - 4)^\ell$ is zero by the binomial theorem and the definition of v ; at $\ell = v$ it equals μ_v . Since $\arctan z = z + O(z^3)$, this gives

$$t_0 = \cdots = t_{v-1} = 0, \quad t_v = \frac{(-i)^v \mu_v}{v!} \neq 0. \quad (\text{CEP15})$$

The first v columns of A_N vanish. Its remaining square block $\hat{A}_N$, with columns $v, \dots, N$, is upper triangular with diagonal $t_v \sqrt{w_{m+v}/w_m}$. Thus its rank is $N - v + 1$, its kernel in the source is

exactly the first v orthonormal coordinates, and

$$\begin{aligned}\det \hat{A}_N &= \left(\frac{(-i)^v \mu_v}{v!} \right)^{N-v+1} \prod_{m=0}^{N-v} \sqrt{\frac{w_{m+v}}{w_m}}, \\ \det(A_N A_N^*) &= \left| \frac{\mu_v}{v!} \right|^{2(N-v+1)} \prod_{m=0}^{N-v} \frac{w_{m+v}}{w_m}.\end{aligned}\tag{CEP16}$$

This retains the complex determinant phase and the original moment. The operator norm also has the exact original-source upper bound

$$\|A_N\| \leq M_{A,N} := \sum_{r,s} |a_{rs}| e^{|b_{rs}-4|(H_N+2)}, \quad H_N = \sum_{j=1}^N 1/j.\tag{CEP17}$$

Indeed AKJ22 bounds each exact translation in CEP13 by its displayed factor, and the triangle inequality gives the sum.

There is a closed four-endpoint expression for CEP16. For the signs $\sigma = (1, 1, -1, -1)$ at the ordered endpoints, write $\mathcal{D}_A = \sum_N \sigma_N \log \det(A_N A_N^*)$. Cancellation of the full products gives

$$\begin{aligned}\mathcal{D}_A &= -4q \log \left| \frac{\mu_v}{v!} \right| + \mathcal{W}_{v,s}, \\ \mathcal{W}_{v,s} &= \sum_{a=0}^{v-1} (\log w_{q-1-a} + \log w_{q-a} - \log w_{2q-1-a} - \log w_{2q-a}).\end{aligned}\tag{CEP18}$$

For $v = 0$ the second sum is zero. The product identity used here is $\prod_{m=0}^{N-v} w_{m+v}/w_m = \prod_{a=0}^{v-1} w_{N-a}/\prod_{a=0}^{v-1} w_a$, obtained by cancellation of consecutive products; it remains valid when the two index intervals overlap.

Put $c_b^{\log} = \max\{1, b-1\}$. For every $j \geq 1$ and $b \geq 1/2$,

$$\left| \log \frac{j}{b+j-1} \right| \leq \frac{c_b^{\log}}{j}.\tag{CEP19}$$

For $b \geq 1$ use $\log(1 + (b-1)/j) \leq (b-1)/j$. For $1/2 \leq b < 1$, put $a = 1 - b \leq 1/2$ and use $-\log(1 - a/j) \leq a/(j-a) \leq 1/j$; the last inequality is $aj \leq j-a$, which follows from $a \leq 1/2$ and $j \geq 1$. Summing CEP19 over the two intervals of length q in each summand of CEP18 proves the completely explicit estimate

$$|\mathcal{W}_{v,s}| \leq \frac{2v c_b^{\log} q}{q-v}, \quad |\mathcal{D}_A| \leq 4q \left| \log \left| \frac{\mu_v}{v!} \right| \right| + \frac{2v c_b^{\log} q}{q-v}.\tag{CEP20}$$

All denominators are positive since $q \geq 100 > v$. For either original source and the fixed actual quartet and period, the displayed bound is $O_{\text{actual}}(q+k) = o(kq)$.

29.4 The induced original quotient metric and its correction

Let the matrices of the two remainder maps in the orthonormal source frames and original primary target frames be denoted by the same U_N, V_d as their maps. Explicitly

$$(U_N)_{(i,j),n} = \frac{p_n((\beta_{ij} - c)/i)}{\sqrt{h_n}}, \quad (V_d)_{(a,b),m} = \frac{p_m((\beta'_{ab} - c')/i)}{\sqrt{h_m}}.\tag{CEP21}$$

They have sizes q by $N+1$ and q' by $N-v+1$, respectively. Their first q , respectively q' , columns have nonzero evaluation Vandermonde determinant, so both have full row rank. The square CEP6 is the precise matrix identity

$$CU_N = V_d A_N, \quad R_N = U_N U_N^*, \quad R'_d = V_d V_d^*, \quad G_N = R_N^{-1}, \quad G'_d = (R'_d)^{-1}.\tag{CEP22}$$

Here G_N is the original upper primary quotient metric of OKM8, and G'_d is the lower-grid metric in the same source order s . The induced metric on E_k/V_A , in its specified target coordinates $E_{k'}$, is

$$Q_N = (CG_N^{-1}C^*)^{-1} = (V_d A_N A_N^* V_d^*)^{-1}. \quad (\text{CEP23})$$

To prove the first assertion, the minimum of $x^* G_N x$ under $Cx = y$ is attained at $x = G_N^{-1} C^* (CG_N^{-1} C^*)^{-1} y$. Every other lift differs by an element of $\ker C$ orthogonal to this lift in the G_N metric, proving the minimum and its Gram matrix. The second equality is CEP22.

The exact correction to the lower original Gamma quotient metric is

$$\begin{aligned} W_d &= V_d^* (R'_d)^{-1/2}, \quad W_d^* W_d = I_{q'}, \\ K_{A,N} &= W_d^* A_N A_N^* W_d > 0, \quad Q_N = (R'_d)^{-1/2} K_{A,N}^{-1} (R'_d)^{-1/2}, \\ \log \det Q_N &= \log \det G'_d - \log \det K_{A,N}. \end{aligned} \quad (\text{CEP24})$$

All square roots are the unique positive Hermitian square roots. These formulas follow by multiplying out the definition of $K_{A,N}$ and inverting. In particular the passage between the induced metric and the lower original source metric retains this actual compressed covariance.

If Z_d is any orthonormal basis of $\ker V_d$, completing W_d to a unitary matrix, the further exact determinant factorization is

$$\det K_{A,N} = \det(A_N A_N^*) \det(Z_d^* (A_N A_N^*)^{-1} Z_d). \quad (\text{CEP25})$$

For a proof put $D = A_N A_N^* > 0$ and write its block matrix in the unitary frame $[W_d, Z_d]$. The upper left block is $K_{A,N}$. Block Gaussian elimination gives $\det D = \det K_{A,N} \det(D_{22} - D_{21} K_{A,N}^{-1} D_{12})$. The lower right block of D^{-1} is the inverse of that Schur complement, proving CEP25. The empty-complement case has determinant one. The dimension of this retained complement is $N - v + 1 - q'$.

Choose any fixed full frame $I_V : \mathbb{C}^\Delta \hookrightarrow E_k$ of $\ker C$ and a fixed right section L_C of C . Then $F_V = [I_V, L_C]$ is invertible and

$$\det(I_V^* G_N I_V) = |\det F_V|^2 \frac{\det G_N}{\det Q_N} = |\det F_V|^2 \frac{\det G_N}{\det G'_d} \det K_{A,N}. \quad (\text{CEP26})$$

Indeed $CF_V = [0, I]$, so the lower right block of $(F_V^* G_N F_V)^{-1}$ is $CG_N^{-1} C^* = Q_N^{-1}$. The Schur determinant argument just used gives CEP26. The frame factor is independent of source cutoff and cancels only after applying the four signs.

For the actual inclusion $K \subset V_A$, choose its original coefficient frame, and let $\Xi : \mathbb{C}^\Delta \rightarrow \mathbb{C}^{\Delta-r}$ have its coordinate kernel, $r = 8k - 16$, as in AKJ18. With $H_{V,N} = I_V^* G_N I_V$, the quotient metric is $Q_{\Xi,N} = (\Xi H_{V,N}^{-1} \Xi^*)^{-1}$. The same proof with a frame of $\ker \Xi$ and a right section gives, with all constant frame factors cancelled by the endpoint signs,

$$F_K^{(s)} = \sum_N \sigma_N \log \det G_N - \sum_N \sigma_N \log \det G'_{N-v} + \sum_N \sigma_N \log \det K_{A,N} - \sum_N \sigma_N \log \det Q_{\Xi,N}. \quad (\text{CEP27})$$

Thus the conductor quotient and the remaining original observation quotient have both been retained in the determinant relation. The lower source cutoffs in this equation are exactly $q - 1 - v, q - v, 2q - 1 - v, 2q - v$. Relative to the lower grid's own four cutoffs they have the respective shifts $\Delta - v, \Delta - v, 2\Delta - v, 2\Delta - v$.

29.5 An exact graph quotient with controlled exterior metric error

In original orthonormal coordinates define

$$S_N : \mathcal{P}_N \rightarrow \mathbb{C}^v \oplus \mathcal{P}'_{N-v}, \quad S_N p = (P_{<v} p, \mathcal{T}_{A,N} p). \quad (\text{CEP28})$$

Here $P_{<v}$ records the first v original upper orthonormal coefficients, and the target carries the orthogonal direct-sum norm with those coordinates and the unchanged lower Gamma source. The first v source coordinates span the polynomials of degree below v . Thus CEP14–16 give its exact block matrix $S_N = \text{diag}(I_v, \hat{A}_N)$ in the ordered source split at degree v . In particular it is invertible, and

$$\begin{aligned} \|S_N\| &\leq M_N := \max\{1, M_{A,N}\}, \\ |\det S_N| &= \left| \frac{\mu_v}{v!} \right|^{N-v+1} \prod_{m=0}^{N-v} \sqrt{\frac{w_{m+v}}{w_m}}, \\ |\log |\det S_N|| &\leq L_N := (N-v+1) \left| \log \left| \frac{\mu_v}{v!} \right| \right| + v c_b^{\log} H_N. \end{aligned} \quad (\text{CEP29})$$

The last bound follows by the consecutive-product identity preceding CEP19 and $|\log w_j| \leq c_b^{\log} H_j$: the two lists of v logarithms are each bounded by $v c_b^{\log} H_N$, followed by the factor $1/2$.

The upper and lower source ideals at this cutoff are

$$I_N = \chi_k \mathbb{C}[S]_{\leq N-q}, \quad I'_N = \chi_{k'} \mathbb{C}[S']_{\leq N-v-q'}. \quad (\text{CEP30})$$

A negative degree specifies the zero subspace. Let $\mathcal{W}_N = U_N^{-1}(V_A)$. The exact image and ideal graph under CEP28 are

$$\begin{aligned} S_N \mathcal{W}_N &= \mathbb{C}^v \oplus I'_N, \\ S_N I_N &= \{(P_{<v} h, \mathcal{T}_{A,N} h) : h \in I_N\} \subset \mathbb{C}^v \oplus I'_N, \\ V_A &\xrightarrow{\simeq} (\mathbb{C}^v \oplus I'_N) / (S_N I_N), \quad U_N p \mapsto [S_N p]. \end{aligned} \quad (\text{CEP31})$$

The first equality follows because $\mathcal{W}_N$ is the preimage of I'_N under the top map of CEP6, and S_N is invertible. CEP3 proves the ideal inclusion. If representatives differ by I_N , their images differ by $S_N I_N$, so the last map is well defined. Conversely every class in the target has a preimage under S_N in $\mathcal{W}_N$, and the kernel consists exactly of I_N . This proves the isomorphism. Its norm on the left is the restriction of G_N : all lifts of a member of V_A already belong to $\mathcal{W}_N$, so their minimum source norm is the quotient norm of $\mathcal{W}_N / I_N$. The low-degree graph component $P_{<v} h$ is explicitly retained.

For reference the ideal dimensions at the four original endpoints are

N	$\dim I_N$	$\dim I'_N$	$\dim((\mathbb{C}^v \oplus I'_N) / S_N I_N)$
$q-1$	0	$\Delta - v$	Δ
q	1	$\Delta + 1 - v$	Δ
$2q-1$	q	$q + \Delta - v$	Δ
$2q$	$q+1$	$q + \Delta + 1 - v$	Δ .

(CEP32)

These follow by counting monomials in CEP30; S_N preserves dimensions. Moreover $\mathcal{T}_A|_{I_N}$ is injective because a nonzero member has degree at least $q > v$ and its image has degree exactly v less. Hence the second coordinate of the retained ideal graph already has dimension $\dim I_N$.

Here is the complete exterior norm estimate for this exact map. For an invertible complex matrix S of dimension n and $0 \leq t \leq n$, its singular values $\sigma_1 \geq \dots \geq \sigma_n > 0$ give

$$\left\| \bigwedge^t S^{-1} \right\| = \frac{\left\| \bigwedge^{n-t} S \right\|}{|\det S|}. \quad (\text{CEP33})$$

Indeed the numerator is $\sigma_1 \cdots \sigma_{n-t}$ and the denominator is the product of all singular values; their ratio is the product of the t largest singular values of S^{-1} . This includes $t = 0, n$. For $n = N+1$, CEP29 therefore gives the finite bounds

$$\begin{aligned} \log^+ \left\| \bigwedge^t S_N \right\| &\leq t \log M_N, \\ \log^+ \left\| \bigwedge^t S_N^{-1} \right\| &\leq (N+1-t) \log M_N + L_N. \end{aligned} \quad (\text{CEP34})$$

Here $\log^+ x = \max\{0, \log x\}$. The right sides are nonnegative. Thus for all these ranks both bounds are at most

$$E_N = (N + 1) \log M_N + L_N. \quad (\text{CEP35})$$

With $R_{\text{sh}} = \max_{r,s} |b_{rs} - 4|$ and $A_*^{\text{abs}} = \sum |a_{rs}|$, equation CEP17 gives $\log M_N \leq \log^+ A_*^{\text{abs}} + R_{\text{sh}}(H_N + 2)$. For the fixed original quartet and period, v, R_{sh}, μ_v and A_*^{abs} are fixed while $c_b^{\log} = O(k)$. Since $N \leq 2q$ and $H_N \leq 1 + \log N$, all four explicit bounds in CEP35 satisfy

$$E_N = O_{\text{actual}}(q \log(q + 1) + k \log(q + 1)) = o(kq). \quad (\text{CEP36})$$

This uses only the forward norm and the exact determinant of the original conductor map.

We now prove the exact relation to the metric of the original kernel, including the quotient projection. In a finite-dimensional Hilbert space let $I \subset W$ and let S be invertible. The induced map $W/I \rightarrow SW/SI$ sends a minimal lift $x \in W \cap I^\perp$ to the orthogonal projection of Sx off SI . Orthogonal projection is a contraction on every exterior power: in an orthonormal basis its singular values are zeros and ones. Therefore, on any t -dimensional subspace of W/I , the exterior norm of the induced map is at most $\|\bigwedge^t S\|$. Applying the identical argument to S^{-1} gives the inverse bound. The minimal-lift identification is isometric since every other lift differs by a perpendicular element of I , whose squared norm adds. This proves the quotient assertion with no choice of coefficient norm.

Apply this result with $W = \mathcal{W}_N$, $I = I_N$, and the actual rank- r subspace $K \subset V_A$. The target subspace is the explicit image of K under CEP31; denote its Gram matrix in the transported original frame by $\tilde{H}_{K,N}$. Let $I_K : \mathbb{C}^r \hookrightarrow E_k$ be that original primary frame, and write its original minimum source lift as $Y_N = U_N^* G_N I_K$. Let Z_N be the matrix in the original orthonormal source coordinates of the independent columns $\chi_k(S) S^a$, $0 \leq a \leq N - q$. If there are no columns, all following products involving them are zero. The exact target orthogonal projection and Gram matrix are

$$\begin{aligned} P_N^{\text{graph}} &= I - S_N Z_N (Z_N^* S_N^* S_N Z_N)^{-1} Z_N^* S_N^*, \\ \tilde{H}_{K,N} &= Y_N^* S_N^* P_N^{\text{graph}} S_N Y_N, \quad H_{K,N} = Y_N^* Y_N = I_K^* G_N I_K. \end{aligned} \quad (\text{CEP37})$$

The inverted Gram matrix is positive definite because its columns are independent and S_N is invertible. Multiplication shows that P_N^{graph} is Hermitian, idempotent, kills $S_N I_N$ and fixes its orthogonal complement. Thus CEP37 is precisely the minimum quotient Gram of CEP31, and its kernel frame has full rank because CEP31 is injective. Replacing the lift Y_N by any lift of the same primary frame adds columns in I_N , which the projection kills; hence every choice of lift gives this same matrix.

Applying the proved exterior quotient estimate and squaring volumes gives

$$\begin{aligned} -2 \log \left\| \bigwedge^r S_N^{-1} \right\| &\leq \log \det \tilde{H}_{K,N} - \log \det H_{K,N} \\ &\leq 2 \log \left\| \bigwedge^r S_N \right\|, \\ \left| \log \det \tilde{H}_{K,N} - \log \det H_{K,N} \right| &\leq 2E_N. \end{aligned} \quad (\text{CEP38})$$

In detail, divide the chosen frame by its positive Gram square root so it is orthonormal in the source quotient. The squared norm of its exterior image is exactly the determinant ratio. The forward and inverse exterior estimates prove the two inequalities, and CEP34–35 prove the last assertion.

Consequently the following reduction of the actual four-endpoint kernel determinant is proved:

$$\begin{aligned} \tilde{F}_K^{(s)} &= \sum_N \sigma_N \log \det \tilde{H}_{K,N}, \\ |F_K^{(s)} - \tilde{F}_K^{(s)}| &\leq 2 \sum_{N \in \{q-1, q, 2q-1, 2q\}} E_N = o(kq). \end{aligned} \quad (\text{CEP39})$$

The frame is the original fixed frame of K at every endpoint; CEP37 gives all four transported matrices explicitly. Their target spaces retain the original ideal graphs, the different source cutoffs, all conductor coefficients, and the full original Gamma masses. CEP39 proves that the leading kq coefficient is unchanged by this particular graph-quotient transport. It supplies no value for that coefficient: its remaining determinant is the displayed, completely specified four-endpoint graph quotient.

29.6 All ideal graph entries as original polynomial products

Each individual shift in CEP2 induces the unital algebra map $\sigma_{rs} : E_k \rightarrow E_{k'}, [p] \mapsto [p(S' + b_{rs})]$. The divisibility in CEP3, applied to the individual shifted polynomial, proves independence of the representative; substitution preserves multiplication and one. Thus $\bar{\mathcal{T}}_A = \sum a_{rs} \sigma_{rs}$ is its exact linear combination. For any $x, y \in E_k$, the complete multiplicative defect of that linear combination is

$$\bar{\mathcal{T}}_A(xy) - \bar{\mathcal{T}}_A(x)\bar{\mathcal{T}}_A(y) = \sum_{r,s} a_{rs} \sigma_{rs}(x) \sigma_{rs}(y) - \sum_{r,s,t,u} a_{rs} a_{tu} \sigma_{rs}(x) \sigma_{tu}(y). \quad (\text{CEP40})$$

This follows by applying the proved individual multiplicativity and expanding the product of the two complete sums.

There is a complete finite product for the map on the source ideal. For each original shift put

$$d_{rs}(S') = \prod_{\substack{0 \leq i,j \leq k \\ (i,j) \notin \{r,\dots,r+k'\} \times \{s,\dots,s+k'\}}} (S' + b_{rs} - \beta_{ij}). \quad (\text{CEP41})$$

This is monic of degree Δ , and $\chi_k(S' + b_{rs}) = \chi_{k'}(S') d_{rs}(S')$. Indeed CEP3's root addition identifies exactly the q' factors in the indicated rectangle with the factors of $\chi_{k'}$. All remaining factors are precisely the product in CEP41. Consequently for every $h \in \mathbb{C}[S]_{\leq L}$, $L = N - q$, the exact ideal map is

$$\begin{aligned} \mathcal{T}_{A,N}(\chi_k h) &= \chi_{k'} \mathcal{U}_{A,L} h, \\ (\mathcal{U}_{A,L} h)(S') &= \sum_{r,s} a_{rs} d_{rs}(S') h(S' + b_{rs}) \in \mathbb{C}[S']_{\leq L+\Delta-v}. \end{aligned} \quad (\text{CEP42})$$

The equality follows by the product identity just proved. The degree statement follows from CEP8: if h has degree t and leading coefficient $h_t \neq 0$, the leading coefficient of its displayed image is $h_t \binom{q+t}{v} \mu_v$ at degree $t + \Delta - v$. This follows because $\chi_k, \chi_{k'}$ are monic and $\mathcal{T}_A$ lowers degree exactly by v with its stated leading derivative. Thus $\mathcal{U}_{A,L}$ is injective. At $L = -1$ the domain and image are zero.

This gives a further exact sequence, with all maps specified:

$$0 \longrightarrow \mathbb{C}[S]_{<v} \longrightarrow V_A \longrightarrow \mathbb{C}[S']_{\leq L+\Delta-v} / \mathcal{U}_{A,L} \mathbb{C}[S]_{\leq L} \longrightarrow 0. \quad (\text{CEP43})$$

The first map is the original quotient class. For the second, lift a member of V_A to $p \in \mathcal{W}_N$, divide $\mathcal{T}_A p$ by $\chi_{k'}$, and take its class in the last space. A change of lift by $\chi_k h$ adds exactly $\mathcal{U}_{A,L} h$ by CEP42, proving independence. Every target polynomial times $\chi_{k'}$ has a preimage under the onto map $\mathcal{T}_{A,N}$, proving surjectivity. If the image is zero, subtract the corresponding $\chi_k h$; the remaining polynomial is killed by $\mathcal{T}_A$ and hence has degree below v . Its class injects because $v < q$. This proves exactness. The last space has dimension $\Delta - v$ at every original endpoint, as is also obtained by subtracting the two polynomial dimensions.

For a literal matrix version of the ideal graph, equip $\mathbb{C}[S']_{\leq L+\Delta-v}$ with the ordered lower orthonormal Gamma coefficient basis, and write $M_{\chi',N}$ for multiplication by the full original $\chi_{k'}$ into $\mathcal{P}'_{N-v}$. It is injective. The map

$$D_{\chi',N} = \text{diag}(I_v, M_{\chi',N}) : \mathbb{C}^v \oplus \mathbb{C}[S']_{\leq L+\Delta-v} \longrightarrow \mathbb{C}^v \oplus I'_N \quad (\text{CEP44})$$

is an isomorphism onto the indicated subspace. Its pullback metric is exactly $G_{\chi',N} = \text{diag}(I_v, M_{\chi',N}^* M_{\chi',N})$; this follows by taking the squared norm of each component. For the domain polynomial coefficient frame of h , let $Z_{\chi,N}h$ be the original upper orthonormal coefficients of $\chi_k h$. In these coordinates the complete graph relation matrix is

$$\mathcal{Z}_{A,N} = \begin{pmatrix} P_{<v} Z_{\chi,N} \\ [\mathcal{U}_{A,L}] \end{pmatrix}, \quad D_{\chi',N} \mathcal{Z}_{A,N} = S_N Z_{\chi,N}. \quad (\text{CEP45})$$

The first component is the original low-degree projection, and the second consists of the exact finite products and translations in CEP41–42. The equality is CEP28 and CEP42. It proves that the graph quotient in CEP31 is precisely the quotient by $\text{im } \mathcal{Z}_{A,N}$ in the displayed metric $G_{\chi',N}$. All matrix inverses and minimum projections in CEP37 can therefore be computed using these full original polynomial products. No coefficient of the ideal, source metric or low-degree graph component is discarded by this additional coordinate isomorphism.

29.7 The compressed conductor correction is subleading at every endpoint

The exact exterior estimate also controls the large compressed covariance in CEP24 without multiplying a scalar inverse bound by q' . Write $n' = N - v + 1$ and retain $A_N = [0, \hat{A}_N]$ with its exact square block in CEP16. Since $W_d^* W_d = I_{q'}$, the coefficient matrix of its image under $\hat{A}_N^*$ gives

$$K_{A,N} = (\hat{A}_N^* W_d)^* (\hat{A}_N^* W_d). \quad (\text{CEP46})$$

The squared norm of the exterior product of the q' columns on the right is its determinant. The norm of the exterior product of the columns of W_d is one, because those columns are orthonormal. The forward estimate therefore gives $\det K_{A,N} \leq \|\hat{A}_N\|^{2q'} \leq M_N^{2q'}$. For the reverse inequality apply $\bigwedge^{q'} \hat{A}_N^*$ to the same image columns; their image is the orthonormal column wedge of W_d . Thus

$$\det K_{A,N} \geq \left\| \bigwedge^{q'} \hat{A}_N^{-1} \right\|^{-2}. \quad (\text{CEP47})$$

The singular values of an adjoint are the same, so the adjoint inverse in this argument has exactly the displayed exterior norm. The singular complement identity CEP33 in dimension n' gives $\|\bigwedge^{q'} \hat{A}_N^{-1}\| \leq M_N^{n'-q'} / |\det \hat{A}_N|$. CEP29 bounds the absolute logarithm of the last determinant by L_N . Consequently the complete two-sided estimate is

$$\begin{aligned} -2((N - v + 1 - q') \log M_N + L_N) &\leq \log \det K_{A,N} \leq 2q' \log M_N, \\ |\log \det K_{A,N}| &\leq 2E_N = o(kq) \quad \text{at every original endpoint and source order.} \end{aligned} \quad (\text{CEP48})$$

All ranks in this argument are their actual ranks, $q' \leq N - v + 1$ as proved after CEP6. In particular the estimate applies to the complete original lower quotient, whose rank grows quadratically.

Define the fully specified total-source difference and the remaining original observation quotient return by

$$\begin{aligned} \mathcal{T}_k^{(s)} &= \sum_N \sigma_N (\log \det G_N - \log \det G'_{N-v}), \\ F_{\Xi,k}^{(s)} &= \sum_N \sigma_N \log \det Q_{\Xi,N}, \\ \mathcal{E}_{A,k}^{(s)} &= \sum_N \sigma_N \log \det K_{A,N}. \end{aligned}$$

Equations CEP26–27 and CEP48 prove the exact identities and bounds

$$\begin{aligned} F_{V_A}^{(s)} &= \mathcal{T}_k^{(s)} + \mathcal{E}_{A,k}^{(s)}, \quad F_K^{(s)} = \mathcal{T}_k^{(s)} - F_{\Xi,k}^{(s)} + \mathcal{E}_{A,k}^{(s)}, \\ |\mathcal{E}_{A,k}^{(s)}| &\leq 2 \sum_N E_N = o(kq), \\ \frac{F_K^{(s)} - \mathcal{T}_k^{(s)} + F_{\Xi,k}^{(s)}}{kq} &\longrightarrow 0. \end{aligned} \quad (\text{CEP49})$$

Here $F_{V_A}^{(s)}$ uses its fixed original coefficient frame, and $Q_{\Xi,N}$ is the exact quotient metric of CEP27. The frame factors cancel with coefficients $1 + 1 - 1 - 1$ only in these returns, as proved in CEP26. Thus the leading metric effect of the full conductor source transport has now been controlled. The remaining rank- $8k - 32$ observation quotient and the shifted lower-grid source determinants are still their explicit original matrices; CEP49 carries them forward into the next calculation.

30 The total-source difference as an outer-strip quotient

Retain CEP1–49, with $m = 1$, $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $k' = k - 8$, $q' = (k - 7)^2$, $\Delta = q - q' = 16k - 48$, and the original fixed $0 < \delta < 1/2$, $\gamma > 2$. The conductor has its actual first nonzero moment index $0 \leq v \leq 80$. At each of the four original endpoints $N \in \{q - 1, q, 2q - 1, 2q\}$ put $t = N - q + 1$ and $d = N - v$. The source order remains $s \in \{1, k\}$ at both grids, with

$$b = s/2, \quad M_s = (2\pi)^{s/2}, \quad dm_s(y) = M_s \frac{2^b}{4\pi\Gamma(b)} |\Gamma(b/2 + iy/2)|^2 dy, \quad h_n = M_s n!(b)_n. \quad (\text{CTV1})$$

These are the complete original measure and monic norms of OKM6–7. The source centres are $c = k/2$ and $c' = k/2 - 4$. The total-source difference in CEP49 is

$$\mathcal{T}_k^{(s)} = \sum_N \sigma_N (\log \det G_N - \log \det G'_{N-v}), \quad \sigma = (1, 1, -1, -1), \quad (\text{CTV2})$$

where both matrices use their original primary frames. All calculations below retain these four cutoffs and the same two source orders.

30.1 The exact nested root polynomials in the original real coordinate

Define the full monic polynomials

$$\begin{aligned} P_k(y) &= i^{-q} \chi_k(c + iy) = \prod_{a,b=0}^k (y - \omega_{ab}), & \omega_{ab} &= (2b - k)\gamma - i(2a - k)\delta, \\ P_{k'}(y) &= i^{-q'} \chi_{k'}(c' + iy) = \prod_{a,b=0}^{k'} (y - \omega'_{ab}), & \omega'_{ab} &= (2b - k')\gamma - i(2a - k')\delta. \end{aligned} \quad (\text{CTV3})$$

The original phases i^{-q} and $i^{-q'}$ have modulus one; they occur exactly as written in the coefficient maps. For every lower pair one has $\omega'_{ab} = \omega_{a+4,b+4}$. Hence, with the complete outer index set $\mathcal{O}_k = \{0, \dots, k\}^2 \setminus \{4, \dots, k-4\}^2$, the exact factorization is

$$P_k(y) = P_{k'}(y) P_{\partial,k}(y), \quad P_{\partial,k}(y) = \prod_{(a,b) \in \mathcal{O}_k} (y - \omega_{ab}), \quad \deg P_{\partial,k} = \Delta. \quad (\text{CTV4})$$

Direct substitution proves the root equality; counting the retained central rectangle gives q' roots, leaving Δ distinct outer roots. The products are monic, so the root equality fixes the factor without any additional constant.

Since k' is odd, $|2a - k'| \geq 1$ for every lower index. For real y this proves, factor by factor,

$$|y - \omega'_{ab}| \geq |\operatorname{Im} \omega'_{ab}| \geq \delta, \quad |P_{k'}(y)| \geq \delta^{q'}. \quad (\text{CTV5})$$

Also $|\omega'_{ab}| \leq R' = k' \sqrt{\delta^2 + \gamma^2}$. These estimates keep every original factor; no root has been moved.

30.2 The full monic source and relation determinant identities

For a nonzero polynomial P and an integer $a \geq 0$ define the relation moment determinant

$$D_a(P; s) = \det \left[\int_{\mathbb{R}} y^{i+j} |P(y)|^2 dm_s(y) \right]_{0 \leq i, j < a}, \quad D_0(P; s) = 1. \quad (\text{CTV6})$$

Every nonempty matrix is positive definite: the measure is strictly positive on the real line, a nonzero polynomial has only finitely many zeros, and the squared modulus of its product with P is positive on an open interval. All moments exist because the original source has its finite exponential moments in CTV14 below.

In increasing monomial y coordinates, the quotient by a monic polynomial P of degree a from source degree $L \geq a - 1$ has determinant

$$\det G_{P,L}^y = \frac{\prod_{n=0}^L h_n}{D_{L-a+1}(P; s)}. \quad (\text{CTV7})$$

Here the quotient frame is $1, y, \dots, y^{a-1}$. For proof use the source frame

$$(1, y, \dots, y^{a-1}, P, yP, \dots, y^{L-a}P).$$

Its increasing-degree coefficient matrix has diagonal entries one and no entries above the degree of a column, hence determinant one. The last block is exactly the relation Gram in CTV6. Minimizing over those last coordinates gives its Schur complement, whose determinant is the full source determinant divided by the relation determinant. Monic orthogonalization by the original p_n also has determinant one, so that full determinant is $\prod_{n=0}^L h_n$. This proves CTV7 including the empty relation case. It is the full-source analogue of BSL3, with the original Gamma order retained.

Let $V_\beta, V_{\beta'}$ be the original ordered Vandermonde matrices in the S, S' variables. The exact coordinate pullback $S = c + iy$ sends S^j to $(c + iy)^j$ and has diagonal coefficients i^j ; its determinant is $i^{a(a-1)/2}$ in degree below a . Its modulus is one, so its Gram congruence has determinant factor one. The same statement holds with c' . Evaluation in primary coordinates then gives

$$\begin{aligned} \det G_N &= \frac{\prod_{n=0}^N h_n}{D_t(P_k; s) |\det V_\beta|^2}, \\ \det G'_{N-v} &= \frac{\prod_{n=0}^{N-v} h_n}{D_{t+\Delta-v}(P_{k'}; s) |\det V_{\beta'}|^2}. \end{aligned} \quad (\text{CTV8})$$

The relation phase i^q or $i^{q'}$ multiplying each source relation column has squared modulus one, as used in this equality. The Vandermonde factors are nonzero by the original distinctness of all roots. Their logarithmic difference is constant across the four endpoints and has signed coefficient zero. Therefore CTV8 proves the exact finite expression

$$\mathcal{T}_k^{(s)} = \sum_N \sigma_N \left[\sum_{n=N-v+1}^N \log h_n + \log D_{t+\Delta-v}(P_{k'}; s) - \log D_t(P_k; s) \right]. \quad (\text{CTV9})$$

The first sum is empty for $v = 0$. In particular both low relation determinants and both high relation determinants remain present.

30.3 The exact quotient of the outer-strip polynomial

Introduce the positive measure

$$d\zeta_{k',s}(y) = |P_{k'}(y)|^2 dm_s(y). \quad (\text{CTV10})$$

This measure is the exact pullback of the original source metric under the multiplication map $\mathcal{M}_{P_{k'}} : f \mapsto P_{k'}f$. Indeed $\|\mathcal{M}_{P_{k'}}f\|_{m_s}^2 = \int |f|^2 d\zeta_{k',s}$, including the same full factor M_s . On polynomials of degree at most $N - q'$ its image is the specified subspace $P_{k'}\mathbb{C}[y]_{\leq N-q'}$ of the original degree-at-most- N source. The map is injective because $P_{k'}$ is nonzero. Thus CTV10 records a proved metric map to the original source, rather than an assigned new Gamma order.

For $L = N - q' = t + \Delta - 1$, let $Q_{\partial,L}^{(s)}$ be the minimum quotient Gram of

$$\mathbb{C}[y]_{\leq L} \longrightarrow \mathbb{C}[y]/(P_{\partial,k}) \quad (\text{CTV11})$$

in the source norm CTV10 and in the increasing monomial remainder frame of dimension Δ . The source degree is at least $\Delta - 1$ at every endpoint. The same monic-frame Schur proof as in CTV7, now using CTV4, gives

$$\det Q_{\partial,L}^{(s)} = \frac{D_{t+\Delta}(P_{k'}; s)}{D_t(P_k; s)}. \quad (\text{CTV12})$$

In fact the ideal columns are $P_{\partial,k}y^j$, $0 \leq j < t$; their Gram in CTV10 is exactly $D_t(P_k; s)$ by CTV4. The numerator is the Gram of the complete monomial source of dimension $L + 1 = t + \Delta$. This proves every source and relation factor in CTV12.

Write $\lambda_j^{(k',s)}$ for the squared norm of the monic orthogonal polynomial of degree j in CTV10. This polynomial and its norm are completely determined by the positive moment matrices: subtract from y^j its projection onto lower powers. If H_j is that lower Gram and a_j its inner-product column with y^j , the lower coefficients are $-H_j^{-1}a_j$. Completing the square proves minimality among all monic polynomials of degree j ; its squared norm is the Schur complement. Thus $\lambda_j^{(k',s)} = D_{j+1}(P_{k'}; s)/D_j(P_{k'}; s) > 0$. Multiplying these identities and substituting CTV12 into CTV9 gives

$$\begin{aligned} \mathcal{F}_{\partial,k}^{(s)} &= \log \det Q_{\partial,\Delta-1}^{(s)} + \log \det Q_{\partial,\Delta}^{(s)} - \log \det Q_{\partial,q+\Delta-1}^{(s)} - \log \det Q_{\partial,q+\Delta}^{(s)}, \\ \mathcal{T}_k^{(s)} &= \mathcal{F}_{\partial,k}^{(s)} + \sum_N \sigma_N \mathcal{R}_{N,v}^{(s)}, \\ \mathcal{R}_{N,v}^{(s)} &= \sum_{a=0}^{v-1} \left(\log h_{N-a} - \log \lambda_{N-q'-a}^{(k',s)} \right). \end{aligned} \quad (\text{CTV13})$$

Every weighted norm index is nonnegative: $N - q' - (v - 1) \geq \Delta - v \geq 16$. The total difference therefore has an exact rank- Δ quotient presentation with precisely the displayed norm tails.

30.4 Finite bounds for all norm tails in both original sources

The original Gamma generating identity gives the full exponential moment $\int e^{ty} dm_s(y) = M_s(\cos t)^{-b}$ for $|t| < \pi/2$. By the evenness of the measure this also equals $\int \cosh(ty) dm_s(y)$. The nonnegative Taylor terms of $\cosh$ imply

$$\begin{aligned} \int y^{2j} dm_s(y) &\leq M_s(\cos t_0)^{-b} \frac{(2j)!}{t_0^{2j}}, \\ \|y^j\|_{m_s} &\leq \sqrt{M_s}(\cos t_0)^{-b/2} \mathfrak{c}_{\text{mom}}^j j!, \quad t_0 = \frac{\pi}{4}, \quad \mathfrak{c}_{\text{mom}} = \frac{2}{t_0} = \frac{8}{\pi}. \end{aligned} \quad (\text{CTV14})$$

The second inequality uses $(2j)! \leq 4^j (j!)^2$, proved by $\binom{2j}{j} \leq \sum_{a=0}^{2j} \binom{2j}{a} = 4^j$. Thus every constant in this moment estimate includes the original source mass and order.

For a monic polynomial of degree n whose roots have modulus at most R' , its coefficient of y^a has absolute value at most $\binom{n}{a} (R')^{n-a}$: it is the sum of those products of $n - a$ distinct roots. The triangle inequality and CTV14 therefore give the full bound

$$\|P\|_{m_s} \leq \sqrt{M_s}(\cos t_0)^{-b/2} \mathfrak{c}_{\text{mom}}^n n! \sum_{h=0}^n \frac{(R'/\mathfrak{c}_{\text{mom}})^h}{h!} \leq \sqrt{M_s}(\cos t_0)^{-b/2} \mathfrak{c}_{\text{mom}}^n n! e^{R'/\mathfrak{c}_{\text{mom}}}. \quad (\text{CTV15})$$

The monic test polynomial y^j in the minimization defining $\lambda_j^{(k',s)}$ gives the polynomial $P_{k'}(y)y^j$ in CTV15, of degree $n = q' + j$; its extra j roots are exactly zero and also satisfy the stated bound. For the lower bound, CTV5 applies to every competing monic polynomial, whose original Gamma squared norm is at least h_j . Taking the minimum proves

$$\delta^{2q'} h_j \leq \lambda_j^{(k',s)} \leq M_s(\cos t_0)^{-b} \mathfrak{c}_{\text{mom}}^{2(q'+j)} ((q' + j)!)^2 e^{2R'/\mathfrak{c}_{\text{mom}}}. \quad (\text{CTV16})$$

No lower bound on an actual period coefficient is used here. The lower root polynomial itself supplies the strictly positive real-line lower bound.

For each term of CTV13 put $n = N - a$ and $j = n - q'$. Subtracting the logarithms in CTV16 from the exact $\log h_n = \log M_s + \log n! + \log(b)_n$ yields

$$\begin{aligned} b \log(\cos t_0) + \log \frac{(b)_n}{n!} - 2n \log \mathbf{c}_{\text{mom}} - \frac{2R'}{\mathbf{c}_{\text{mom}}} &\leq \log h_n - \log \lambda_j^{(k',s)} \\ &\leq \sum_{r=j+1}^n (\log r + \log(b+r-1)) - 2q' \log \delta. \end{aligned} \quad (\text{CTV17})$$

The two full $\log M_s$ contributions cancel in this displayed comparison because both norms arise from exactly CTV1. They have not been removed from either individual norm.

Let $c_b^{\log} = \max\{1, b-1\}$. As proved directly in CEP19, $|\log(j/(b+j-1))| \leq c_b^{\log}/j$ for $b \geq 1/2$: use $\log(1+x) \leq x$ for $b \geq 1$; for $1/2 \leq b < 1$ use $-\log(1-a/j) \leq a/(j-a) \leq 1/j$, $a = 1-b$. Summation gives $|\log((b)_n/n!)| \leq c_b^{\log} H_n$. With all original endpoints $N \geq 99$, define the explicit nonnegative numbers

$$\begin{aligned} L_N^{\text{tail}} &= c_b^{\log} H_N + \frac{b}{2} \log 2 + 2N \log(8/\pi) + \frac{\pi R'}{4}, \\ U_N^{\text{tail}} &= q'(\log N + \log(b+N-1) + 2|\log \delta|), \\ C_{N,v}^{\text{tail}} &= v \max\{L_N^{\text{tail}}, U_N^{\text{tail}}\}. \end{aligned} \quad (\text{CTV18})$$

Indeed $\cos(\pi/4) = 2^{-1/2}$ and $2R'/\mathbf{c}_{\text{mom}} = \pi R'/4$. In the upper sum of CTV17 there are exactly q' terms, and each is at most $\log N + \log(b+N-1)$. Thus CTV17 proves

$$-v L_N^{\text{tail}} \leq \mathcal{R}_{N,v}^{(s)} \leq v U_N^{\text{tail}}, \quad |\mathcal{R}_{N,v}^{(s)}| \leq C_{N,v}^{\text{tail}}. \quad (\text{CTV19})$$

When $v = 0$ the equality $\mathcal{R}_{N,0}^{(s)} = 0$ and the zero bound follow from the empty sum. These estimates apply for all actual endpoints and for both unchanged source orders simultaneously.

30.5 The proved first-variation reduction and its remaining matrix

Since $N \leq 2q$, $q' \leq q$, $b \leq k/2$ and $R' \leq k\sqrt{\delta^2 + \gamma^2}$, equations CTV18–19 give $C_{N,v}^{\text{tail}} = O_{\delta,\gamma}(q \log(q+1) + k \log(q+1))$ uniformly for every integer $0 \leq v \leq 80$. To see this directly, use $H_N \leq 1 + \log N$ in the first line of CTV18 and the displayed upper bounds in its second; v is bounded by the fixed integer 80. Dividing by kq tends to zero, since $\log(q+1)/k \rightarrow 0$ for $q = (k+1)^2$. Consequently CTV13 and the finite bounds just proved yield

$$\left| \mathcal{T}_k^{(s)} - \mathcal{F}_{\partial,k}^{(s)} \right| \leq \sum_N C_{N,v}^{\text{tail}} = o(kq). \quad (\text{CTV20})$$

The estimate is uniform over the possible actual moment indices v ; it requires no bound on the remaining period coordinates. Every polynomial and measure defining $\mathcal{F}_{\partial,k}^{(s)}$ is independent of that index.

For an explicit matrix target, let H_L^ζ be the moment Gram of $1, y, \dots, y^L$ in the full measure CTV10, and let $\mathbf{R}_{\partial,L}$ be the literal remainder matrix modulo the full polynomial $P_{\partial,k}$ in CTV4. The exact metric is

$$Q_{\partial,L}^{(s)} = (\mathbf{R}_{\partial,L}(H_L^\zeta)^{-1} \mathbf{R}_{\partial,L}^*)^{-1}. \quad (\text{CTV21})$$

The map is onto because its first Δ columns are the identity in the remainder frame. The minimum source lift of a coefficient vector x is $(H_L^\zeta)^{-1} \mathbf{R}_{\partial,L}^* Q_{\partial,L}^{(s)} x$; it maps to x and is perpendicular to the kernel. Adding any other lift adds a perpendicular kernel vector, proving the stated Gram. This also proves direct equality with the quotient used in CTV11–12.

Combining the complete exact CEP49 and CTV13 identities gives

$$\begin{aligned}
F_K^{(s)} &= \mathcal{F}_{\partial,k}^{(s)} - F_{\Xi,k}^{(s)} + \mathcal{E}_{A,k}^{(s)} + \sum_N \sigma_N \mathcal{R}_{N,v}^{(s)}, \\
\left| F_K^{(s)} - \mathcal{F}_{\partial,k}^{(s)} + F_{\Xi,k}^{(s)} \right| &\leq 2 \sum_N E_N + \sum_N C_{N,v}^{\text{tail}} = o(kq).
\end{aligned} \tag{CTV22}$$

Here the first error has its fixed-actual-period scope from CEP49; the new tail bound has the stronger uniform-in- v scope just proved. Thus the remaining leading calculation now consists of the explicit outer-strip quotient of rank $16k - 48$ and the original observation quotient of rank $8k - 32$. Both have their complete original metric maps and four cutoffs specified. No value of either leading volume is assigned by this reduction, and no derivative of the earlier q^2 asymptotic law was used.

31 The actual observation remaining inside the conductor kernel

Retain the original simple quartet, its actual unit and period matrix $\mathbf{H} = F^{-1}\Pi U \mathcal{E}_{\text{prim}}$, and its proved five-member quadric orbit. Put $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = q - q' = 16k - 48$, $r = 8k - 16$, and $j = 8k - 32$. The period parameter and logarithm branch are fixed at a point on this proved locus. All constructions below use the original ordered primary coordinates at

$$\beta_{ab} = k/2 + (2a - k)\delta + i(2b - k)\gamma, \quad \omega_{ab} = (2b - k)\gamma - i(2a - k)\delta, \quad 0 \leq a, b \leq k. \quad (\text{ROQ1})$$

The full conductor is the actual biform $A = \sum_{a,b=0}^8 a_{ab} e_{ab}^{(8)}$ of OCF4. Its matrix $C : \mathbb{C}^q \rightarrow \mathbb{C}^{q'}$ is exactly CEP4: $C_{(a,b),(i,h)} = a_{i-a,h-b}$ when both differences belong to $\{0, \dots, 8\}$, and zero otherwise. Thus $C^\top$ is multiplication by the entire A , and $V_A = \ker C$. The original observation is Λ , and $K = \ker \Lambda \subset V_A$. Ordinary transpose throughout the coefficient duality is denoted by $\top$; a star denotes conjugate transpose in a displayed metric.

31.1 An explicit residual map and its actual boundary basis

Use the complete OCF6 elimination, with its actual largest nonzero coefficient $a_{r_*s_*}$ and every remaining a_{ab} . To avoid confusing its degree- k matrices with the source cutoff, write them as

$$\mathbf{N} : \mathbb{C}^q \rightarrow \mathbb{C}^\Delta, \quad \mathbf{S} : \mathbb{C}^\Delta \hookrightarrow \mathbb{C}^q, \quad \mathbf{V} : \mathbb{C}^q \rightarrow \mathbb{C}^{q'}, \quad \mathbf{S}\mathbf{N} + C^\top \mathbf{V} = I_q, \quad \mathbf{N}\mathbf{S} = I_\Delta. \quad (\text{ROQ2})$$

These are the exact identities OCF7. Specifically, at each decreasing lexicographic conductor position the elimination subtracts its current coefficient divided by $a_{r_*s_*}$ times the entire translated biform $A e_{ab}^{(k-8)}$. This specifies all entries of $\mathbf{N}, \mathbf{V}$; $\mathbf{S}$ inserts the retained strip coordinates. It also proves $\ker \mathbf{N} = \text{im } C^\top$. Taking the annihilator for the bilinear coefficient pairing therefore gives

$$I_V = \mathbf{N}^\top : \mathbb{C}^\Delta \xrightarrow{\sim} V_A, \quad I_V^{-1} = \mathbf{S}^\top|_{V_A}. \quad (\text{ROQ3})$$

For a direct verification, $C\mathbf{N}^\top = 0$ by transposing $\mathbf{N}C^\top = 0$, and $\mathbf{S}^\top \mathbf{N}^\top = I$. If $Cx = 0$, transposing the first identity of ROQ2 gives $x = \mathbf{N}^\top \mathbf{S}^\top x$.

Retain the j independent original invariant certificates $F_{k,h}(x)$ selected by OCF20, and their entire biform coefficient columns f_h :

$$\mathbf{F} = [f_1, \dots, f_j], \quad f_h = \text{coeff}(F_{k,h}(\mathbf{H}t)), \quad \mathbf{Z} = \mathbf{N}\mathbf{F} \in \text{Mat}_{\Delta \times j}(\mathbb{C}), \quad \mathbf{D} = \mathbf{F}^\top. \quad (\text{ROQ4})$$

Here $t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^\top$ and the coefficient order is the full original biform order. Every certificate is supplied by the fourteen-generator recursion OCF25–26, or equivalently the complete twenty-five-generator recursion OCF9–20. At a recursion step its certificate is the actual product $x^\nu F_{k-e,h}$, and its column is $\mathbf{N}_k M_{f_\nu} \mathbf{S}_{k-e} Z_{k-e,h}$. The coefficients of f_ν are the full multinomials OCF13 in the actual $\mathbf{H}$. Thus ROQ4 specifies finite entries without choosing a substitute period or introducing a freely chosen observation. The proved image and rank are $\text{im } \mathbf{Z} = \mathbf{N}J_k$ and $\text{rank } \mathbf{Z} = j$.

Let I be the j retained pivot rows of $\mathbf{Z}$, let T be their complement among the Δ strip rows, and put $\mathbf{Z}_I = E_I^\top \mathbf{Z}$, $\mathbf{Z}_T = E_T^\top \mathbf{Z}$. The original OCF selection guarantees that $\mathbf{Z}_I$ is invertible on its exhaustive exact pivot chart. ROQ2 applied to $\mathbf{F}$ gives the full coefficient identity

$$\mathbf{F} = \mathbf{S}\mathbf{Z} + C^\top \mathbf{V}\mathbf{F}, \quad \mathbf{D} = \mathbf{Z}^\top \mathbf{S}^\top + (\mathbf{V}\mathbf{F})^\top C. \quad (\text{ROQ5})$$

In particular the unreduced invariant rows and the reduced strip rows agree on V_A , with precisely the displayed conductor multiple as their ambient difference. The residual observation has the explicit forms

$$\Xi : V_A \rightarrow \mathbb{C}^j, \quad x \mapsto Dx, \quad \Xi I_V = \mathbf{Z}^\top, \quad L = \mathbf{N}^\top E_I (\mathbf{Z}_I^\top)^{-1}, \quad CL = 0, \quad DL = I_j. \quad (\text{ROQ6})$$

The last three equalities follow by multiplication and ROQ3–5. Moreover $\ker \Xi = K$: the rows C and D together span the coefficient space J_k . Indeed conductor multiplication spans $\ker \mathbf{N}$, its image is contained in J_k by OCF17, and the f_h project to the basis Z of $\mathbf{N}J_k$. Thus a primary vector killed by both C and D is exactly the annihilator of J_k , which is the original K by OKM3–5. This proves rank j , surjectivity, and the complete exact sequence

$$0 \longrightarrow K \longrightarrow V_A \xrightarrow{\Xi} \mathbb{C}^j \longrightarrow 0. \quad (\text{ROQ7})$$

For every $y \in \mathbb{C}^j$ define $\mathbf{b}(y) = \Lambda Ly$. These vectors give the literal isomorphism $\mathbf{b} : \mathbb{C}^j \rightarrow \Lambda(V_A)$. They span because for $x \in V_A$, $x - LDx \in K$ by ROQ6–7. They are independent because $\Lambda Ly = 0$ implies $Ly \in K$, hence $y = DLy = 0$. The inverse on the displayed image is $\Lambda x \mapsto Dx$ for $x \in V_A$: replacing x by another such lift changes it by K , which D kills. In particular the boundary basis vectors are exactly $\mathbf{b}_h = \sum_{a,b} L_{(a,b),h} \Lambda \varepsilon_{ab}$. Each $\Lambda \varepsilon_{ab}$ is the full original projected tensor coefficient in OKA7, including its literal $1/5$ cyclic average.

The whole original observation can be recovered from these rows as well. Take an actual right inverse L_C of C , for example the inverse of the first nonzero square column minor with its coordinate insertion, and define

$$\mathbf{L} = [L_C - LDL_C, L], \quad \mathbf{R} = \begin{pmatrix} C \\ D \end{pmatrix}, \quad \mathbf{R}\mathbf{L} = I_{q'+j}, \quad \Lambda = (\Lambda\mathbf{L})\mathbf{R}. \quad (\text{ROQ8})$$

The first identity follows by multiplying its four blocks. For the last, $\mathbf{R}(x - \mathbf{L}\mathbf{R}x) = 0$, whose kernel is K by ROQ7. Thus the $q' + j = q - r$ columns of $\Lambda\mathbf{L}$ form an actual basis of the original boundary: applying its kernel and then $\mathbf{R}\mathbf{L} = I$ proves independence, and ROQ8 proves spanning. The last j columns are exactly $\mathbf{b}_h$.

For completeness the inherited period metric on this residual basis is also explicit. Let B be the complete matrix AKS8 in primary coordinates, so $j\Lambda = \mathcal{F}_0 B$. If ν, μ range over the original weight-zero degree- k monomials, set

$$(\mathcal{G}_0)_{\nu\mu} = \sum_{\substack{n_{tu} \geq 0 \\ \sum_u n_{tu} = \nu_t \\ \sum_t n_{tu} = \mu_u}} \frac{k!}{\prod_{t,u} n_{tu}!} \prod_{t,u} [5(\delta_{tu} + 1)]^{n_{tu}}, \quad P_{\text{per}} = L^* B^* \mathcal{G}_0 B L > 0. \quad (\text{ROQ9})$$

Expansion of the original word inner product gives $\mathcal{G}_0 = \mathcal{F}_0^* \mathcal{F}_0$ as in AKS10, so the displayed matrix is exactly the Gram of the independent vectors $j\mathbf{b}_h$. Its strict positivity follows from that independence and injectivity of j . All off-diagonal word terms and every actual entry of $\mathbf{H}$ remain in ROQ9.

31.2 All four induced Gamma metrics and their fixed determinant factor

For either original source $s \in \{1, k\}$ put $b_s = s/2$, $M_s = (2\pi)^{s/2}$ and retain

$$\begin{aligned} p_0 &= 1, & p_1 &= y, & p_{n+1} &= yp_n - n(b_s + n - 1)p_{n-1}, \\ h_n &= M_s n! (b_s)_n, & (u_n)_{ab} &= p_n(\omega_{ab}) / \sqrt{h_n}. \end{aligned} \quad (\text{ROQ10})$$

Thus $U_N = [u_0, \dots, u_N]$ is precisely CEP21 and $R_N = U_N U_N^* > 0$, $G_N = R_N^{-1}$ for every $N \geq q - 1$. The positivity follows from its first q columns, whose determinant is the original Vandermonde times their nonzero leading coefficients. Define

$$H_{V,N} = \bar{\mathbf{N}} G_N \mathbf{N}^\top, \quad Q_{\Xi,N} = (Z^\top H_{V,N}^{-1} \bar{Z})^{-1} > 0. \quad (\text{ROQ11})$$

These are matrices of sizes Δ and j , respectively. The minimum of $z^* H_{V,N} z$ under $Z^\top z = y$ is attained at $z = H_{V,N}^{-1} \bar{Z} Q_{\Xi,N} y$. Its difference from any other lift lies in $\ker Z^\top$ and is perpendicular in the $H_{V,N}$ metric. Its squared norm is $y^* Q_{\Xi,N} y$. Thus ROQ11 calculates exactly the induced

metric of ROQ7 in the actual boundary coordinates ROQ6. This proof also identifies it with the quotient metric appearing in CEP27 and CEP49.

The actual kernel frame in the strip is

$$\rho = E_T^\top - Z_T Z_I^{-1} E_I^\top, \quad I_K = \mathbf{N}^\top \rho^\top, \quad F_{\text{str}} = [\rho^\top, E_I (Z_I^\top)^{-1}]. \quad (\text{ROQ12})$$

The first two matrices are precisely the OCF/OKM frame. In the ordered rows (T, I) the last one has blocks $\begin{pmatrix} I_r & 0 \\ -(Z_I^\top)^{-1} Z_T^\top & (Z_I^\top)^{-1} \end{pmatrix}$. Consequently $|\det F_{\text{str}}|^2 = |\det Z_I|^{-2}$, including the absolute value of the row-order permutation. Block elimination in $F_{\text{str}}^* H_{V,N} F_{\text{str}}$ gives the exact identity

$$\det(I_K^* G_N I_K) \det Q_{\Xi,N} = \frac{\det H_{V,N}}{|\det Z_I|^2}. \quad (\text{ROQ13})$$

Indeed its upper block is the kernel Gram and its Schur complement is the minimum quotient metric of the previous paragraph, since $Z^\top F_{\text{str}} = [0, I_j]$. The determinant of a block matrix is the determinant of its upper block times that Schur complement, while congruence contributes $|\det F_{\text{str}}|^2$. This proves ROQ13 without assigning modulus one to the actual pivot.

Define $F_{\Xi,k}^{(s)}$ by the four ordered endpoints $\mathcal{N} = (q-1, q, 2q-1, 2q)$ and signs $\sigma = (1, 1, -1, -1)$:

$$F_{\Xi,k}^{(s)} = \sum_{N \in \mathcal{N}} \sigma_N \log \det Q_{\Xi,N}. \quad (\text{ROQ14})$$

The fixed factor $|\det Z_I|^{-2}$ in ROQ13 cancels with these signs. If a different exact certificate basis is selected, its residual rows on V_A are $D' = MD$ for an invertible $j \times j$ matrix M . This follows because both lists are bases of the dual of V_A/K . For an explicit inverse formula, $M = D'L$, and the old and new right sections give its inverse; composing the two surjections proves both inverse identities. The new metric is $Q'_{\Xi,N} = M^{-*} Q_{\Xi,N} M^{-1}$, directly from their identical minimum problem. Its determinant changes by $|\det M|^{-2}$, which also cancels in ROQ14. No pivot or source mass has been removed at an individual endpoint. In fact ROQ10 gives $Q_{\Xi,N} = M_s \widehat{Q}_{\Xi,N}$ when the common mass is explicitly factored out of every column; its determinant carries M_s^j . The metric relative to the retained period Gram ROQ9 has determinant $\det Q_{\Xi,N} / \det P_{\text{per}}$, whose fixed denominator cancels only in the same four-endpoint sum.

31.3 The exact constrained covariance in the original source

The residual metric can be computed without inverting the strip frame. Define, using the complete original covariance,

$$\begin{aligned} A_N^C &= C R_N C^*, & B_N^D &= D R_N C^*, & T_N^D &= D R_N D^*, \\ W_N &= R_N - R_N C^* (A_N^C)^{-1} C R_N, \\ \Omega_N &= D W_N D^* = T_N^D - B_N^D (A_N^C)^{-1} (B_N^D)^*. \end{aligned} \quad (\text{ROQ15})$$

Since C has full row rank and $R_N > 0$, $A_N^C > 0$. There is the exact identity

$$W_N = \mathbf{N}^\top H_{V,N}^{-1} \overline{\mathbf{N}}, \quad \Omega_N = Z^\top H_{V,N}^{-1} \overline{Z} > 0, \quad Q_{\Xi,N} = \Omega_N^{-1}. \quad (\text{ROQ16})$$

To prove its first equality, $CW_N = 0$, W_N is Hermitian, and for $x \in \ker C$ direct multiplication gives $W_N G_N x = x$. The other displayed candidate is Hermitian, has image in $\ker C$ by ROQ3, and has the same property on $x = \mathbf{N}^\top z$. Such a matrix is unique: for their Hermitian difference H , its image lies in V_A and $HG_N x = 0$ for $x \in V_A$. For arbitrary y and $x \in V_A$, this yields $x^* G_N H y = (H G_N x)^* y = 0$. Putting $x = H y \in V_A$ gives $(H y)^* G_N (H y) = 0$, hence $H y = 0$. This proves equality. Now use $D \mathbf{N}^\top = Z^\top$ from ROQ6 and take the adjoint to prove the other two identities.

Every entry in ROQ15 is the finite sum of original column products: $A_N^C = \sum_{n=0}^N (Cu_n)(Cu_n)^*$, $B_N^D = \sum_{n=0}^N (Du_n)(Cu_n)^*$, and $T_N^D = \sum_{n=0}^N (Du_n)(Du_n)^*$. The conductor column is exactly $Cu_n = V_{N-v} A_N e_n$ by CEP22 when $n \leq N$; its vanishing low columns and centre difference are precisely CEP13–16. The residual column is explicitly $(Du_n)_h = \sum_{a,b} (f_h)_{ab} p_n(\omega_{ab}) / \sqrt{h_n}$. In particular ROQ15 retains all conductor–observation cross terms. Replacing D by $D + TC$ leaves Ω_N exactly unchanged because $CW_N = 0 = W_N C^*$. Applied to ROQ5, this proves that the full and strip row presentations give exactly the same metric, rather than assuming their ambient coefficient matrices coincide.

31.4 Exact increments and a positive residual four-endpoint return

For $N \geq q - 1$ let $u = u_{N+1}$ be the next original column ROQ10 and define

$$\begin{aligned} a_N &= 1 + (Cu)^*(A_N^C)^{-1}(Cu) > 0, \\ w_N &= u - R_N C^*(A_N^C)^{-1}Cu \in V_A, \quad z_N = Dw_N = Du - B_N^D(A_N^C)^{-1}Cu, \\ \eta_N &= \frac{z_N^* Q_{\Xi, N} z_N}{a_N} \geq 0. \end{aligned} \tag{ROQ17}$$

The complete updates are

$$\begin{aligned} W_{N+1} &= W_N + w_N w_N^* / a_N, \quad \Omega_{N+1} = \Omega_N + z_N z_N^* / a_N, \\ Q_{\Xi, N+1} &= Q_{\Xi, N} - \frac{Q_{\Xi, N} z_N z_N^* Q_{\Xi, N}}{a_N + z_N^* Q_{\Xi, N} z_N}, \\ \log \det Q_{\Xi, N} - \log \det Q_{\Xi, N+1} &= \log(1 + \eta_N). \end{aligned} \tag{ROQ18}$$

Here is a direct proof of the covariance update. Put $a = Cu$ and $v = R_N C^*(A_N^C)^{-1}a$. Since $R_{N+1} = R_N + uu^*$, $A_{N+1}^C = A_N^C + aa^*$, whose inverse is $(A_N^C)^{-1} - (A_N^C)^{-1}aa^*(A_N^C)^{-1}/a_N$; multiplying by $A_N^C + aa^*$ verifies this inverse exactly. For the full covariance calculation write $\alpha = a^*(A_N^C)^{-1}a$. The subtracted covariance product expands as

$$R_N C^*(A_N^C)^{-1} C R_N + vu^* + uv^* + \alpha uu^* - \frac{(v + \alpha u)(v + \alpha u)^*}{1 + \alpha}.$$

Subtracting this from $R_N + uu^*$ and the old W_N leaves $(u - v)(u - v)^*/(1 + \alpha)$, proving the first update. Multiplying by D, D^* proves the second. The same inverse identity proves the third, and the determinant identity $\det(H + zz^*/a) = \det H(1 + z^* H^{-1} z/a)$ proves the fourth. For that identity conjugate by $H^{-1/2}$; the rank-one matrix $H^{-1/2} z z^* H^{-1/2}/a$ has one eigenvalue $z^* H^{-1} z/a$ and all its others zero. This completes the update proof.

Telescoping with the four actual endpoints, rather than changing the source degree, gives the exact finite sum

$$F_{\Xi, k}^{(s)} = \log(1 + \eta_{q-1}) + 2 \sum_{N=q}^{2q-2} \log(1 + \eta_N) + \log(1 + \eta_{2q-1}) \geq 0. \tag{ROQ19}$$

Indeed $\log \det Q_{q-1} - \log \det Q_{2q-1}$ sums the differences for $N = q - 1, \dots, 2q - 2$, while $\log \det Q_q - \log \det Q_{2q}$ sums those for $N = q, \dots, 2q - 1$. This proves each endpoint weight. Its first increment $\lambda_s = \log(1 + \eta_{q-1})$ is a completely specified period-dependent nonnegative lower bound for $F_{\Xi, k}^{(s)}$. The formulas also characterize equality: $\lambda_s = 0$ exactly when $Du_q = B_{q-1}^D(A_{q-1}^C)^{-1}Cu_q$; the full return vanishes exactly when this corresponding row residual is zero for every column $u_q, \dots, u_{2q}$. Positivity of $Q_{\Xi, N}$ proves both directions, without asserting a nonzero generic residual.

31.5 Finite scalar enclosures and propagation into CEP49

Retain the exact AKS3 scalars $\mathcal{Z}_{s,t}$ and $\mathcal{L}_{s,t} = \log \mathcal{Z}_{s,t}$, whose full original polynomial covariance bound is proved in AKS20–27. In primary coordinates it says

$$\mathcal{Z}_{s,t}^{-1} G_{q-1} \preceq G_{q+t-1} \preceq G_{q-1}, \quad 0 \leq t \leq q+1. \quad (\text{ROQ20})$$

The unchanged coefficient map from AKS26 to primary coordinates is the ordered V_β , so congruence by that invertible map proves exactly the displayed statement. Restrict it to V_A and take the minimum on the fixed fibre $Dx = y$ in V_A . Inequalities of squared norms hold for every such x , hence also for their attained minima. Thus $\mathcal{Z}_{s,t}^{-1} Q_{\Xi,q-1} \preceq Q_{\Xi,q+t-1} \preceq Q_{\Xi,q-1}$. If $d_t = \log \det Q_{\Xi,q-1} - \log \det Q_{\Xi,q+t-1}$, the eigenvalues after congruence by $Q_{\Xi,q-1}^{-1/2}$ give $0 \leq d_t \leq j\mathcal{L}_{s,t}$, and ROQ18 gives their monotonicity. Since $d_1 = \lambda_s$, the exact four-endpoint identity $F_{\Xi,k}^{(s)} = d_q + d_{q+1} - d_1$ proves

$$\lambda_s \leq F_{\Xi,k}^{(s)} \leq j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - \lambda_s. \quad (\text{ROQ21})$$

The actual rank j enters only through this proved eigenvalue bound; the determinant and increment formulas ROQ11–19 retain its full orientation. No leading volume is assigned from a dimension.

Let $\epsilon_{A,k}^{(s)} = 2 \sum_{N \in \mathcal{N}} E_N$ with the complete original constants of CEP29–36. CEP49 gives $F_K^{(s)} = \mathcal{T}_k^{(s)} - F_{\Xi,k}^{(s)} + \mathcal{E}_{A,k}^{(s)}$ and $|\mathcal{E}_{A,k}^{(s)}| \leq \epsilon_{A,k}^{(s)} = o(kq)$ for a fixed actual period. Substitution of ROQ21 proves

$$\begin{aligned} \mathcal{T}_k^{(s)} - \epsilon_{A,k}^{(s)} - j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) + \lambda_s &\leq F_K^{(s)}, \\ F_K^{(s)} &\leq \mathcal{T}_k^{(s)} + \epsilon_{A,k}^{(s)} - \lambda_s. \end{aligned} \quad (\text{ROQ22})$$

Intersecting this interval with the full AKJ2 or AKJ4 intervals preserves each of their finite constants and source correlations. More precisely, one may use the exact value $\mathcal{E}_{A,k}^{(s)}$ in place of its upper or lower envelope in the appropriate side. Formula ROQ19 supplies every additional original positive subtraction, rather than only the first one. These are finite results on the residual quotient itself; they do not claim that its leading kq coefficient has been evaluated.

31.6 The complete residual quotient in the conductor graph

At a fixed original endpoint N , let $\mathcal{P}_N$, I_N , $\mathcal{W}_N = U_N^{-1}(V_A)$ and the actual source isomorphism $S_N p = (P_{<v} p, \mathcal{T}_{A,N} p)$ be precisely CEP28–31. The source carries its original orthonormal Gamma coordinates of order $s = 1$ or $s = k$. The induced linear isomorphism on the residual observation quotient is

$$\mathcal{W}_N / U_N^{-1}(K) \xrightarrow{[p] \mapsto [S_N p]} S_N \mathcal{W}_N / S_N U_N^{-1}(K), \quad [p] \mapsto D U_N p \in \mathcal{C}^j. \quad (\text{ROQ23})$$

The last map identifies the left quotient with V_A/K and its fixed coordinates ROQ6. Its kernel and surjectivity follow from ROQ7 and surjectivity of U_N ; applying the invertible S_N proves the other isomorphism. Thus all relation directions, including the source ideal and the entire actual K , are retained in the transported denominator.

Here are exact matrices of that denominator and the resulting metric. Let Z_N^χ be the original source coefficient columns $\chi_k S^a$, $0 \leq a \leq N - q$, with no columns when $N = q - 1$. Take the actual primary frames I_K of ROQ12 and L of ROQ6 and set

$$Y_{K,N} = U_N^* G_N I_K, \quad Y_{L,N} = U_N^* G_N L, \quad J_N = [Z_N^\chi, Y_{K,N}]. \quad (\text{ROQ24})$$

The columns of $Y_{K,N}$ are the minimum lifts of I_K by the same orthogonality proof as OKM8. In particular they are perpendicular to $I_N = \ker U_N$, so J_N has independent columns and its image

is exactly $U_N^{-1}(K)$. It has $N + 1 - q + r$ columns, while $\dim \mathcal{W}_N = N + 1 - q + \Delta$, leaving precisely j quotient dimensions. The transported complete relation projector and Gram are

$$\begin{aligned} P_N^{\Xi, \text{graph}} &= I - S_N J_N (J_N^* S_N^* S_N J_N)^{-1} J_N^* S_N^*, \\ \tilde{Q}_{\Xi, N} &= Y_{L, N}^* S_N^* P_N^{\Xi, \text{graph}} S_N Y_{L, N} > 0. \end{aligned} \tag{ROQ25}$$

The inverse exists because S_N is invertible and J_N has independent columns. Multiplication verifies that the projector is Hermitian, idempotent and projects onto the orthogonal complement of the entire $S_N U_N^{-1}(K)$. Thus its Gram is the minimum norm of the transported quotient ROQ23. Replacing $Y_{L, N}$ by another source lift of L adds columns in I_N , which this projector kills. Replacing the section L by another section of ROQ6 adds columns in K ; their lifts lie in the span of J_N and are killed. Therefore ROQ25 computes the specified quotient metric independently of either choice.

Before applying S_N , the same projection gives

$$Q_{\Xi, N} = L^* G_N L - L^* G_N I_K (I_K^* G_N I_K)^{-1} I_K^* G_N L. \tag{ROQ26}$$

To prove it, $Y_{L, N}$ and $Y_{K, N}$ are perpendicular to the ideal, so projection off J_N only subtracts $Y_{K, N} (Y_{K, N}^* Y_{K, N})^{-1} Y_{K, N}^* Y_{L, N}$. Using $U_N U_N^* = R_N$ in their Gram products yields exactly ROQ26. It is also the minimum of the original G_N norm on $Ly + K$, so it agrees with the already proved ROQ11 and ROQ16.

The all-rank exterior estimate CEP33–35 applies to ROQ23: represent a quotient class by its minimum lift in the denominator's orthogonal complement, apply S_N , then project off the target denominator. This projection has every exterior norm at most one. Consequently the induced j th exterior norm is at most $\|\bigwedge^j S_N\|$, and its inverse is at most $\|\bigwedge^j S_N^{-1}\|$ by the identical argument for S_N^{-1} . CEP34 bounds both logarithms from above by E_N . Applying them to the fixed frame ROQ6 and squaring exterior volumes gives

$$\left| \log \det \tilde{Q}_{\Xi, N} - \log \det Q_{\Xi, N} \right| \leq 2E_N, \quad \left| \sum_{N \in \mathcal{N}} \sigma_N \log \det \tilde{Q}_{\Xi, N} - F_{\Xi, k}^{(s)} \right| \leq \epsilon_{A, k}^{(s)} = o(kq). \tag{ROQ27}$$

All cross terms in $J_N^* S_N^* S_N J_N$ remain present in ROQ25, including $P_{<v}(\chi_k S^a)$ and its pairings with the minimum kernel lifts. Its ideal columns can equally be generated from the full products d_{rs} and maps $\mathcal{U}_{A, L}$ in CEP41–45, because those formulas prove exactly $S_N Z_N^\chi$ column by column. This supplies the typed relation between the explicit residual observation and the conductor graph at all four original cutoffs.

32 A quantitative power comparison for the actual outer-strip quotient

Retain the original real parameters and integers

$$\begin{aligned} 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad k = 4l + 1 \geq 9, \\ q = (k+1)^2, \quad k' = k-8, \quad q' = (k-7)^2, \quad \Delta = q - q' = 16k - 48. \end{aligned} \quad (\text{OSP1})$$

The two source orders in every formula below are precisely $s = 1, k$. Set $b = s/2$, $M_s = (2\pi)^{s/2}$, and retain the full original measure

$$dm_s(y) = M_s \frac{2^b}{4\pi\Gamma(b)} |\Gamma(b/2 + iy/2)|^2 dy. \quad (\text{OSP2})$$

In the original coordinate $S = k/2 + iy$, and respectively $S' = k/2 - 4 + iy$, the complete monic root polynomials are

$$\begin{aligned} P_k(y) &= i^{-q} \chi_k(k/2 + iy) = \prod_{a, b_0=0}^k [y - (2b_0 - k)\gamma + i(2a - k)\delta], \\ P_{k'}(y) &= i^{-q'} \chi_{k'}(k/2 - 4 + iy) = \prod_{a, b_0=0}^{k'} [y - (2b_0 - k')\gamma + i(2a - k')\delta]. \end{aligned} \quad (\text{OSP3})$$

The index b_0 in these products is distinct from the source parameter $b = s/2$. Each root occurs with exactly its displayed multiplicity. The map $(a, b_0) \mapsto (a+4, b_0+4)$ identifies the second root list with the central subrectangle of the first, so

$$P_k = P_{k'} P_{\partial, k}, \quad P_{\partial, k}(y) = \prod_{(a, b_0) \in \{0, \dots, k\}^2 \setminus \{4, \dots, k-4\}^2} [y - (2b_0 - k)\gamma + i(2a - k)\delta]. \quad (\text{OSP4})$$

This proves the factorization and its monic constant; the outer degree is Δ . Both complete root lists are invariant under $\omega \mapsto -\omega$. Every root has modulus at most

$$R = k\sqrt{\delta^2 + \gamma^2}. \quad (\text{OSP5})$$

32.1 The two original source masses and a proved interval lower bound

We first record the source facts needed in the comparison, including their constants and proofs. For $a > 0$ and real x , the two Euler integrals and the change of variables in the beta integral give

$$\frac{\Gamma(a+ix)\Gamma(a-ix)}{\Gamma(2a)} = \int_{\mathbb{R}} e^{ixu} (2 \cosh(u/2))^{-2a} du. \quad (\text{OSP6})$$

For example the intermediate beta integral is $\int_0^\infty t^{a+ix-1} (1+t)^{-2a} dt$, and $t = e^u$ gives the displayed expression. Absolute convergence follows from $a > 0$. Rotating Euler's integral through $\text{sgn}(y)\theta$, $0 < \theta < \pi/2$, gives

$$|\Gamma(a + iy/2)| \leq e^{-\theta|y|/2} \Gamma(a) (\cos \theta)^{-a}. \quad (\text{OSP7})$$

The rotation follows from Cauchy's theorem on the intervening sector: the large circular arc tends to zero because its real exponential has positive cosine, and the small one tends to zero because $a > 0$. The remaining radial integral has absolute value at most $\Gamma(a)(\cos \theta)^{-a}$. In particular the Fourier transform in OSP6 is integrable. Fourier inversion, followed by $y = 2x$, now gives

$$\int_{\mathbb{R}} e^{i\xi y} dm_s(y) = M_s (\cosh \xi)^{-b}, \quad \int_{\mathbb{R}} dm_s = M_s. \quad (\text{OSP8})$$

The same identity holds analytically for $|\operatorname{Im} \xi| < \pi/2$: OSP7 with any larger $\theta < \pi/2$ supplies an integrable dominating function on each smaller strip. Substituting $\xi = -it$ proves

$$\int_{\mathbb{R}} e^{ty} dm_s(y) = M_s(\cos t)^{-b}, \quad |t| < \pi/2. \quad (\text{OSP9})$$

This also proves finiteness of all moments used below.

Write

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}, \quad \ell_s = \frac{s-1}{4}, \quad \beta_s = \frac{M_s}{\sqrt{2}\Gamma(s/2)}. \quad (\text{OSP10})$$

The two values of ℓ_s are the integers $0, l$. The reflection identity gives a useful explicit lower bound

$$\frac{d\sigma}{dy}(y) \geq C_L e^{-\pi|y|}. \quad (\text{OSP11})$$

Here is its proof on the required domain. For $0 < \operatorname{Re} z < 1$ the Euler integrals give $\Gamma(z)\Gamma(1-z) = \int_0^\infty t^{z-1}/(1+t) dt$. On a keyhole contour with argument in $(0, 2\pi)$, the circular terms vanish with bounds $O(r^{\operatorname{Re} z})$ at zero and $O(R^{\operatorname{Re} z-1})$ at infinity. The residue at -1 is $e^{i\pi(z-1)}$; the two radial integrals therefore give $(1 - e^{2\pi iz}) \int_0^\infty t^{z-1}/(1+t) dt = 2\pi i e^{i\pi(z-1)}$. Their ratio is $\pi/\sin(\pi z)$. At $z = 1/4 + iy/2$ this proves

$$|\Gamma(1/4 + iy/2)|^2 |\Gamma(3/4 - iy/2)|^2 = \frac{2\pi^2}{\cosh(\pi y)}.$$

The unrotated Euler integral bounds the second squared factor by $\Gamma(3/4)^2$, proving $d\sigma/dy \geq C_L/\cosh(\pi y) \geq C_L e^{-\pi|y|}$.

Gamma recurrence, proved by one integration by parts in Euler's integral, gives the exact original-source identity

$$dm_s(y) = \beta_s \prod_{j=0}^{\ell_s-1} (y^2 + (2j + \frac{1}{2})^2) d\sigma(y). \quad (\text{OSP12})$$

Indeed $b/2 = \ell_s + 1/4$, and each recurrence factor before squaring and clearing denominators is $j + 1/4 + iy/2$; its squared modulus is $[y^2 + (2j + 1/2)^2]/4$. The factor $4^{-\ell_s}$ together with the prefactor in OSP2 is exactly $\beta_s/(2\pi)$, because $b = 2\ell_s + 1/2$. This proves every factor in OSP12. For $s = 1$ its product is empty and $\beta_1 = 1$. In particular, on the full interval $q \leq y \leq 2q$,

$$\frac{dm_s}{dy}(y) \geq \beta_s q^{2\ell_s} C_L e^{-2\pi q}. \quad (\text{OSP13})$$

The source mass-to-lower-density factor obeys the uniform finite bound

$$\frac{M_s}{\beta_s q^{2\ell_s}} = \sqrt{2} \Gamma(b) q^{-2\ell_s} \leq \sqrt{2\pi} \quad (s = 1, k). \quad (\text{OSP14})$$

To prove the inequality, use $\Gamma(b) = \sqrt{\pi} \prod_{j=0}^{2\ell_s-1} (j + 1/2) \leq \sqrt{\pi} b^{2\ell_s}$ and $b \leq q$. All source masses have thus been retained before taking this exact ratio.

32.2 A complete coefficient-space estimate for the inner interval

Let $p \in \mathbb{C}[y]$ have degree at most d , and let $0 < T \leq q/2$. Then

$$\sup_{|y| \leq T} |p(y)|^2 \leq \frac{(d+1)^2 10^{2d}}{q} \int_q^{2q} |p(u)|^2 du. \quad (\text{OSP15})$$

This controls every coefficient direction; no assumption on zeros, leading coefficient, or extremizers of p is made.

For proof define the Legendre polynomial $L_j(x) = (2^j j!)^{-1} (d/dx)^j (x^2 - 1)^j$. Its leading coefficient is $(2j)!/[2^j (j!)^2]$. Integrating by parts j times, with all lower boundary derivatives zero, proves orthogonality to lower powers and

$$\int_{-1}^1 L_j(x)^2 dx = \frac{(2j)!}{2^{2j} (j!)^2} \int_{-1}^1 (1 - x^2)^j dx = \frac{2}{2j + 1}.$$

For the middle integral set $x = 2t - 1$; repeated integration by parts gives $\int_0^1 t^j (1 - t)^j dt = (j!)^2 / (2j + 1)!$. Consequently $\sqrt{2j + 1} L_j(2z - 3)$, $0 \leq j \leq d$, form an orthonormal basis for degree-at-most- d polynomials on $[1, 2]$. Leibniz's rule also gives the explicit identity

$$L_j(x) = 2^{-j} \sum_{h=0}^j \binom{j}{h}^2 (x - 1)^{j-h} (x + 1)^h.$$

For $|x| \leq 4$ its modulus is at most $(5/2)^j \sum_h \binom{j}{h}^2 \leq (5/2)^j 4^j = 10^j$; the sum inequality follows by comparing with $(\sum_h \binom{j}{h})^2 = 4^j$. Expanding $p(qz)$ in the displayed orthonormal basis and applying Cauchy-Schwarz gives, for $|z| \leq T/q \leq 1/2$,

$$|p(qz)|^2 \leq \sum_{j=0}^d (2j + 1) 10^{2j} \int_1^2 |p(qv)|^2 dv \leq (d + 1)^2 10^{2d} \int_1^2 |p(qv)|^2 dv.$$

Here $|2z - 3| \leq 4$ and $\sum_{j=0}^d (2j + 1) = (d + 1)^2$. Changing variables proves OSP15, including $d = 0$.

32.3 The exact original-root map and a finite full-norm comparison

In this subsection P is either of the complete polynomials $P_k, P_{k'}$ in OSP3, and $Q = \deg P$ is respectively q, q' . For $d \geq 0$ retain the two injections into the same original polynomial Hilbert space

$$\begin{aligned} \mathcal{J}_P : \mathbb{C}[y]_{\leq d} &\longrightarrow \mathbb{C}[y]_{\leq Q+d}, & p &\longmapsto Pp, \\ \mathcal{J}_{0,Q} : \mathbb{C}[y]_{\leq d} &\longrightarrow \mathbb{C}[y]_{\leq Q+d}, & p &\longmapsto y^Q p. \end{aligned} \tag{OSP16}$$

Their codomain norm is precisely m_s in OSP2. If $P(y) = \sum_{a=0}^Q P_a y^a$, set $P_a = 0$ outside this range. In increasing coefficient frames their matrices are

$$(\mathcal{J}_P)_{rj} = P_{r-j}, \quad (\mathcal{J}_{0,Q})_{rj} = \mathbf{1}_{\{r=Q+j\}}, \quad 0 \leq r \leq Q + d, \quad 0 \leq j \leq d. \tag{OSP17}$$

Thus the exact isomorphism between their images is $y^Q p \mapsto Pp$, with inverse $Pp \mapsto y^Q p$. It preserves the original coefficient vector p . The first matrix contains every actual root coefficient; none is replaced inside that map. Returning to S or S' multiplies the relevant relation columns by the actual phase i^Q and applies the triangular coordinate pullback with diagonal entries i^j . Each has the stated exact inverse and determinant of modulus one.

Suppose

$$R < T \leq q/2. \tag{OSP18}$$

Define the finite positive constants

$$\begin{aligned} \eta_{Q,T} &= \frac{QR^2}{T^2 - R^2}, \\ \kappa_{Q,d,T,s} &= \frac{M_s}{\beta_s q^{2\ell_s} C_L} \frac{(d + 1)^2}{q} e^{2\pi q} 10^{2d} \left(\frac{T^2 + R^2}{q^2} \right)^Q, \\ e_{Q,d,T,s} &= \eta_{Q,T} + \log(1 + \kappa_{Q,d,T,s}). \end{aligned} \tag{OSP19}$$

Then every $p \in \mathbb{C}[y]_{\leq d}$ satisfies

$$e^{-e_{Q,d,T,s}} \int_{\mathbb{R}} |y|^{2Q} |p(y)|^2 dm_s(y) \leq \int_{\mathbb{R}} |P(y)p(y)|^2 dm_s(y) \leq e^{e_{Q,d,T,s}} \int_{\mathbb{R}} |y|^{2Q} |p(y)|^2 dm_s(y). \quad (\text{OSP20})$$

Here is a proof retaining the entire inner region. Pair the actual roots as $\omega, -\omega$. On $|y| \geq T$, each paired quotient is $1 - \omega^2/y^2$, so

$$\left| \log \frac{|P(y)|^2}{|y|^{2Q}} \right| \leq -Q \log(1 - R^2/T^2) \leq \frac{QR^2}{T^2 - R^2} = \eta_{Q,T}. \quad (\text{OSP21})$$

The first inequality uses $1 - u \leq |1 - z| \leq 1 + u$ for $|z| \leq u < 1$ on each of $Q/2$ pairs. The second follows by integrating $1/(1 - u) \leq 1/(1 - u_0)$ on $0 \leq u \leq u_0 = R^2/T^2$. For every real y , the same pairing gives $|P(y)|^2 \leq (y^2 + R^2)^Q$. Write

$$A_B = \int_q^{2q} |y|^{2Q} |p(y)|^2 dm_s(y).$$

For nonzero p this number is positive, and OSP13 gives $A_B \geq q^{2Q} \beta_s q^{2\ell_s} C_L e^{-2\pi q} \int_q^{2q} |p(y)|^2 dy$. OSP15 and the exact full mass OSP8 imply

$$\begin{aligned} \int_{|y| < T} |P(y)p(y)|^2 dm_s(y) &\leq M_s(T^2 + R^2)^Q \sup_{|y| \leq T} |p(y)|^2 \leq \kappa_{Q,d,T,s} A_B, \\ \int_{|y| < T} |y|^{2Q} |p(y)|^2 dm_s(y) &\leq M_s T^{2Q} \sup_{|y| \leq T} |p(y)|^2 \leq \kappa_{Q,d,T,s} A_B. \end{aligned} \quad (\text{OSP22})$$

Let A_{out} and B_{out} be the integrals outside $(-T, T)$ with factors $|y|^{2Q}$ and $|P|^2$, respectively. Since $[q, 2q]$ is outside that interval, $A_B \leq A_{\text{out}}$. OSP21 gives $e^{-\eta} A_{\text{out}} \leq B_{\text{out}} \leq e^{\eta} A_{\text{out}}$. OSP22 bounds both inner contributions by κA_{out} . Therefore the full reference norm is at most $(1 + \kappa) A_{\text{out}}$ and at least A_{out} , while the full actual norm is at least $e^{-\eta} A_{\text{out}}$ and at most $(e^{\eta} + \kappa) A_{\text{out}} \leq e^{\eta} (1 + \kappa) A_{\text{out}}$. These four inequalities prove OSP20. The zero polynomial satisfies it directly.

32.4 Uniform bounded logarithmic cost at every required dimension

Choose the fixed number and cutoff

$$\epsilon = 2^{-32}, \quad T = \epsilon q,$$

and impose the explicit finite conditions

$$k \geq 29, \quad k \sqrt{\delta^2 + \gamma^2} \leq \frac{\epsilon q}{2}. \quad (\text{OSP23})$$

In particular $T \geq 2R$, as required for a uniform margin in OSP21. These conditions hold eventually for every fixed original δ, γ ; the sufficient bound $k \geq \max\{29, 2\sqrt{\delta^2 + \gamma^2}/\epsilon\}$ follows from $q \geq k^2$. The first condition implies $q' \geq q/2$: $2(k - 7)^2 - (k + 1)^2 = k^2 - 30k + 97$ is positive at $k = 29$ and is increasing thereafter.

For $Q \in \{q', q\}$ and every $0 \leq d \leq 2q$, set

$$\begin{aligned} K_q &= \frac{\sqrt{2\pi}}{C_L} \frac{(2q + 1)^2}{q} e^{-2q}, \\ \mathcal{E}_k &= \frac{qR^2}{\epsilon^2 q^2 - R^2} + \log(1 + K_q). \end{aligned} \quad (\text{OSP24})$$

Then OSP19 satisfies $e_{Q,d,T,s} \leq \mathcal{E}_k$ for both original source orders. To verify this uniformly, OSP14, $T^2 + R^2 \leq (5/4)\epsilon^2 q^2$, $Q \geq q/2$ and $d \leq 2q$ give

$$\kappa_{Q,d,T,s} \leq \frac{\sqrt{2\pi}}{C_L} \frac{(2q + 1)^2}{q} \exp\left(q\left\{2\pi + 4 \log 10 + \frac{1}{2} \log(5/4) + \log \epsilon\right\}\right). \quad (\text{OSP25})$$

The expression in braces is less than -2 . Indeed $\pi < 4$, $\log 10 < 4 \log 2$, $\log(5/4) < 1/4$, and $\log 2 > 2/3$ give an upper bound $8 - 16 \log 2 + 1/8 < -61/24 < -2$. For completeness, $\pi < 4$ follows from $\pi/4 = \int_0^1 (1+x^2)^{-1} dx < 1$; and the substitution $x = (1+t)/(1-t)$ in $\log 2 = \int_1^2 dx/x$ gives $\log 2 = 2 \int_0^{1/3} (1-t^2)^{-1} dt > 2/3$. The logarithm bound follows by integrating $1/(1+x) < 1$. This proves OSP25 with K_q as stated. Also

$$0 \leq \mathcal{E}_k \leq \frac{4(\delta^2 + \gamma^2)}{3\epsilon^2} + \log(1 + K_q). \quad (\text{OSP26})$$

Here $R^2 \leq \epsilon^2 q^2/4$ and $R^2/q \leq \delta^2 + \gamma^2$. Thus the logarithmic norm cost is bounded at fixed original parameters, uniformly over all these coefficient degrees and both unchanged source orders.

32.5 Every relation determinant and the exact outer quotient

For a nonzero polynomial P and $a \geq 0$ define in the unchanged monomial frame

$$\begin{aligned} D_a(P; s) &= \det \left[\int_{\mathbb{R}} y^{i+j} |P(y)|^2 dm_s(y) \right]_{0 \leq i, j < a}, \\ \mathcal{D}_a(Q; s) &= \det \left[\int_{\mathbb{R}} y^{2Q+i+j} dm_s(y) \right]_{0 \leq i, j < a}, \quad D_0 = \mathcal{D}_0 = 1. \end{aligned} \quad (\text{OSP27})$$

All nonempty Grams are positive definite, because a nonzero polynomial has positive squared modulus on an open subinterval of the strictly positive original measure. In the notation of OSP17 their exact matrices are $J_P^* H_{Q+a-1}^{(s)} J_P$ and $J_{0,Q}^* H_{Q+a-1}^{(s)} J_{0,Q}$, where $H_{Q+a-1}^{(s)}$ is the complete original moment Gram. Consequently OSP20–26 prove for $1 \leq a \leq 2q+1$

$$|\log D_a(P; s) - \log \mathcal{D}_a(Q; s)| \leq a \mathcal{E}_k, \quad P = P_k \text{ or } P_{k'}. \quad (\text{OSP28})$$

To see the determinant step exactly, conjugate the actual Gram by the inverse positive square root of the reference Gram. OSP20 places every eigenvalue of the resulting positive Hermitian matrix in $[e^{-\mathcal{E}_k}, e^{\mathcal{E}_k}]$. The logarithm of its determinant is the sum of its a eigenvalue logarithms, proving OSP28. This is the full exterior determinant estimate, including every polynomial coefficient direction. The estimate also holds at $a = 0$ with both sides zero.

For $t \in \{0, 1, q, q+1\}$, let $L = t + \Delta - 1$. The actual quotient has source $\mathbb{C}[y]_{\leq L}$ in measure $|P_{k'}(y)|^2 dm_s(y)$, target the increasing remainder frame of $\mathbb{C}[y]/(P_{\partial,k})$, and its minimum Gram $Q_{\partial,L}^{(s)}$. Define a comparison quotient with source $\mathbb{C}[y]_{\leq L}$ in measure $|y|^{2q'} dm_s(y)$, target the increasing remainder frame of $\mathbb{C}[y]/(y^\Delta)$, and minimum Gram $Q_{0,L}^{(s)}$. Its source measure still has order s and the full mass factor in OSP2.

For precision, the exact map relating the coefficient presentations is the degree-preserving isomorphism

$$\mathcal{U}_L(y^j) = \begin{cases} y^j, & 0 \leq j < \Delta, \\ P_{\partial,k}(y) y^{j-\Delta}, & \Delta \leq j \leq L. \end{cases} \quad (\text{OSP29})$$

Its coefficient matrix has diagonal entries one and no terms of degree greater than that of its input column. It is therefore invertible with determinant one. If $R_{\partial,L}$ is division remainder modulo $P_{\partial,k}$ and $R_{0,L}$ is division remainder modulo y^Δ , then $R_{\partial,L} \mathcal{U}_L = R_{0,L}$: the first Δ basis vectors map to their stated remainders, and every later vector maps into the actual ideal. This proves the precise morphism between the quotient coefficient spaces; no equality of their metrics is assumed.

Using the original monic relation frames on the two sources gives the exact identities

$$\det Q_{\partial,t+\Delta-1}^{(s)} = \frac{D_{t+\Delta}(P_{k'}; s)}{D_t(P_k; s)}, \quad \det Q_{0,t+\Delta-1}^{(s)} = \frac{\mathcal{D}_{t+\Delta}(q'; s)}{\mathcal{D}_t(q; s)}. \quad (\text{OSP30})$$

For proof, append to $1, y, \dots, y^{\Delta-1}$ the ideal frame $P_{\partial,k}, yP_{\partial,k}, \dots, y^{t-1}P_{\partial,k}$. Its source coefficient determinant is one by OSP29. The relation Gram is $D_t(P_k; s)$ because $P_{k'}P_{\partial,k} = P_k$. Completing

the square in the ideal coordinates gives the minimum quotient Gram as the Schur complement; the determinant of the full source Gram is the product of that determinant and the relation determinant. This proves the first identity. For the second repeat the same calculation with y^Δ : $y^{q'} y^\Delta = y^q$, with the original source order s . The same proof includes $t = 0$ using the empty relation block.

OSP28 applies to every index here, since $t + \Delta \leq q + \Delta + 1 \leq 2q + 1$. Define the exact endpoint difference $\varepsilon_t^{(s)}$ by

$$\log \det Q_{\partial, t+\Delta-1}^{(s)} = \log \mathcal{D}_{t+\Delta}(q'; s) - \log \mathcal{D}_t(q; s) + \varepsilon_t^{(s)}.$$

Then, separately at each of the four endpoints,

$$|\varepsilon_t^{(s)}| \leq (2t + \Delta) \mathcal{E}_k. \quad (\text{OSP31})$$

In particular set

$$\begin{aligned} \mathcal{F}_{0,k}^{(s)} &= \log \mathcal{D}_\Delta(q'; s) + \log \mathcal{D}_{\Delta+1}(q'; s) - \log \mathcal{D}_{q+\Delta}(q'; s) - \log \mathcal{D}_{q+\Delta+1}(q'; s) \\ &\quad - \log \mathcal{D}_1(q; s) + \log \mathcal{D}_q(q; s) + \log \mathcal{D}_{q+1}(q; s). \end{aligned} \quad (\text{OSP32})$$

The actual four-endpoint quantity of CTV13 therefore obeys the proved finite estimate

$$\boxed{\left| \mathcal{F}_{\partial,k}^{(s)} - \mathcal{F}_{0,k}^{(s)} \right| \leq 4(q + \Delta + 1) \mathcal{E}_k = O_{\delta,\gamma}(q) = o(kq), \quad s = 1, k.} \quad (\text{OSP33})$$

Indeed its exact error is $\varepsilon_0 + \varepsilon_1 - \varepsilon_q - \varepsilon_{q+1}$; summing OSP31 gives $\Delta + (\Delta + 2) + (2q + \Delta) + (2q + \Delta + 2) = 4(q + \Delta + 1)$. OSP26 and $\Delta < q$ prove the last two estimates. This calculation uses all four source degrees $\Delta - 1, \Delta, q + \Delta - 1, q + \Delta$ and every source and relation determinant in OSP30.

32.6 The resulting same-source moment target

The moments in OSP27 are completely specified by the original analytic identity OSP9:

$$\mu_j^{(s)} = \int y^j dm_s(y) = M_s \frac{d^j}{dz^j} (\cos z)^{-s/2} \Big|_{z=0}, \quad \mathcal{D}_a(Q; s) = \det[\mu_{2Q+i+j}^{(s)}]_{0 \leq i, j < a}. \quad (\text{OSP34})$$

The differentiated integral is justified by OSP7, or directly by the finite exponential moments on a slightly larger interval. Odd moments vanish because the density is even. Reordering rows and columns together by parity therefore proves the exact further factorization

$$\mathcal{D}_a(Q; s) = \det[\mu_{2(Q+i+j)}^{(s)}]_{0 \leq i, j < \lceil a/2 \rceil} \det[\mu_{2(Q+1+i+j)}^{(s)}]_{0 \leq i, j < \lfloor a/2 \rfloor}. \quad (\text{OSP35})$$

There is no permutation sign, since the same permutation is applied to both rows and columns. Empty factors are one. In OSP32 the coefficient of $\log M_s$ is $\Delta + (\Delta + 1) - (q + \Delta) - (q + \Delta + 1) - 1 + q + (q + 1) = 0$. This is an exact cancellation only after all original moments and determinant dimensions have been retained.

Equation OSP33 proves an error-controlled comparison with the exact finite determinant difference OSP32. It does not obtain a first variation by differentiating a leading q^2 law. In particular no numerical value of the limit of $\mathcal{F}_{0,k}^{(s)}/(kq)$ is asserted by this comparison.

33 The outer-root and actual residual bounds in the original arithmetic return

Retain the complete original data of CEP, CTV and ROQ. In particular $m = 1$, $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = 16k - 48$, $r = 8k - 16$ and $j = 8k - 32$. The quartet, full amplitude unit, period and branch are the specified original ones on the proved five-orbit locus. The source orders remain $s \in \{1, k\}$. Every matrix below is the actual matrix in those chapters, with its original mass and coefficient frame.

33.1 The two finite error endpoints retain their signs

Write

$$\mathcal{N}_+ = \{q - 1, q\}, \quad \mathcal{N}_- = \{2q - 1, 2q\}, \quad \epsilon_{A,k}^{(s)} = 2 \sum_{N \in \mathcal{N}_+ \cup \mathcal{N}_-} E_N^{(s)}. \quad (\text{OAR1})$$

Here $E_N^{(s)}$ is the explicit full conductor constant of CEP35. The nonnegative constants $L_N^{\text{tail}}, U_N^{\text{tail}}$ are exactly CTV18 in the same source order and with the original q', δ, γ . Define

$$\begin{aligned} a_{k,s}^- &= v \left(\sum_{N \in \mathcal{N}_+} L_N^{\text{tail}} + \sum_{N \in \mathcal{N}_-} U_N^{\text{tail}} \right), \\ a_{k,s}^+ &= v \left(\sum_{N \in \mathcal{N}_+} U_N^{\text{tail}} + \sum_{N \in \mathcal{N}_-} L_N^{\text{tail}} \right), \\ e_{k,s}^- &= \epsilon_{A,k}^{(s)} + a_{k,s}^-, \quad e_{k,s}^+ = \epsilon_{A,k}^{(s)} + a_{k,s}^+. \end{aligned} \quad (\text{OAR2})$$

These are finite quantities for every original period in the stated domain. For each fixed original period, both $e_{k,s}^\pm = o(kq)$. Indeed CEP36 proves this for $\epsilon_{A,k}^{(s)}$ and CTV18–20 proves it for each of the four tail contributions, uniformly over the possible indices $0 \leq v \leq 80$ and both original source orders. When $v = 0$, both $a_{k,s}^\pm$ are exactly zero.

Set $\Phi_{k,s} = F_{\Xi,k}^{(s)}$ and retain the full outer quotient $\mathcal{F}_{\partial,k}^{(s)}$ of CTV11–13. The exact identity is

$$\begin{aligned} F_K^{(s)} &= \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s} + \mathfrak{e}_{k,s}, \\ \mathfrak{e}_{k,s} &= \mathcal{E}_{A,k}^{(s)} + \sum_{N \in \mathcal{N}_+} \mathcal{R}_{N,v}^{(s)} - \sum_{N \in \mathcal{N}_-} \mathcal{R}_{N,v}^{(s)}, \\ -e_{k,s}^- &\leq \mathfrak{e}_{k,s} \leq e_{k,s}^+. \end{aligned} \quad (\text{OAR3})$$

The equality is CTV22, keeping its original two errors. CTV19 says $-vL_N^{\text{tail}} \leq \mathcal{R}_{N,v}^{(s)} \leq vU_N^{\text{tail}}$. At a positive endpoint this contributes lower bound $-vL_N^{\text{tail}}$ and upper bound vU_N^{tail} ; at a negative endpoint it contributes lower bound $-vU_N^{\text{tail}}$ and upper bound vL_N^{tail} . Adding these four inequalities and the bounds $-\epsilon_{A,k}^{(s)} \leq \mathcal{E}_{A,k}^{(s)} \leq \epsilon_{A,k}^{(s)}$ proves each sign of OAR3.

33.2 Original positive increments give explicit kernel caps

The full value of $\Phi_{k,s}$ is ROQ19's finite sum

$$\Phi_{k,s} = \log(1 + \eta_{q-1}) + 2 \sum_{N=q}^{2q-2} \log(1 + \eta_N) + \log(1 + \eta_{2q-1}), \quad \lambda_s = \log(1 + \eta_{q-1}) \geq 0. \quad (\text{OAR4})$$

Every η_N here is ROQ17's original constrained column, including the complete conductor–observation cross covariance and its positive denominator. No new observation is selected. ROQ21 gives, in the original AKS notation,

$$\lambda_s \leq \Phi_{k,s} \leq V_s := j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - \lambda_s. \quad (\text{OAR5})$$

Retain the complete earlier kernel interval $L_s \leq F_K^{(s)} \leq U_s$ of AKJ2/MRI4e, with its pointwise minimum of the KAF and AKS bounds. Define two explicit quantities

$$\begin{aligned} L_s^\partial &= \max \left\{ L_s, \mathcal{F}_{\partial,k}^{(s)} - e_{k,s}^- - V_s \right\}, \\ U_s^\partial &= \min \left\{ U_s, \mathcal{F}_{\partial,k}^{(s)} + e_{k,s}^+ - \lambda_s \right\}. \end{aligned} \quad (\text{OAR6})$$

Then

$$\begin{aligned} L_s^\partial &\leq F_K^{(s)} \leq U_s^\partial, \\ \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s} - e_{k,s}^- &\leq F_K^{(s)} \leq \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s} + e_{k,s}^+. \end{aligned} \quad (\text{OAR7})$$

For the first line substitute both ends of OAR5 into OAR3 and intersect with the retained interval. The second line substitutes the full finite value OAR4 into OAR3. Intersecting these intervals preserves every bound. Both contain the actual value, so their intersection is nonempty. The dimension j is used only in the proved finite comparison OAR5. It assigns no leading determinant. Each additional evaluated positive term of OAR4 can be included in the upper subtraction in OAR6 by the same inequality.

33.3 The correlated arithmetic kernel and boundary

Keep the original arithmetic total $\mathcal{B}_k^{\text{ar}}$, the scalar δ_k^σ and the four source-difference constants $C_K^\pm, C_B^\pm$ of MRI4f. They satisfy, for the actual $d = F_K^{\text{ar}} - F_K^{(1)}$,

$$\max\{-C_K^-, \delta_k^\sigma - C_B^+\} \leq d \leq \min\{C_K^+, \delta_k^\sigma + C_B^-\}. \quad (\text{OAR8})$$

This is the pair of original kernel bounds together with the boundary bounds, transferred by $d + (F_B^{\text{ar}} - F_B^{(1)}) = \delta_k^\sigma$. It retains their common scalar and signs. Define

$$\begin{aligned} L_{\text{ar}}^\partial &= \max\{L_{\text{ar}}, L_1^\partial - C_K^-, L_1^\partial + \delta_k^\sigma - C_B^+\}, \\ U_{\text{ar}}^\partial &= \min\{U_{\text{ar}}, U_1^\partial + C_K^+, U_1^\partial + \delta_k^\sigma + C_B^-\}. \end{aligned} \quad (\text{OAR9})$$

Here $[L_{\text{ar}}, U_{\text{ar}}]$ is the complete interval of MRI4f. Adding OAR8 to the source-1 first line of OAR7 proves

$$\begin{aligned} L_{\text{ar}}^\partial &\leq F_K^{\text{ar}} \leq U_{\text{ar}}^\partial, \\ \mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^\partial &\leq F_B^{\text{ar}} \leq \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^\partial. \end{aligned} \quad (\text{OAR10})$$

The second line subtracts the first from the exact original identity $F_K^{\text{ar}} + F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}}$. It uses the same arithmetic total at both endpoints. If the second line of OAR7 is used instead, exactly the same addition and subtraction yield the further interval containing the full residual sum. Thus every exact positive residual subtraction is available to the original arithmetic kernel and its correlated boundary.

33.4 The signed mixed return with its complete finite error

Write $\mathfrak{S}_k^{\text{mix}} = F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)}$. The original BRI6d identity is $\mathfrak{S}_k^{\text{mix}} = F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma$. Consequently OAR3 gives the exact current expression

$$\begin{aligned} \mathfrak{S}_k^{\text{mix}} &= \mathcal{F}_{\partial,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \mathfrak{e}_{k,k}, \\ -e_{k,k}^- &\leq \mathfrak{e}_{k,k} \leq e_{k,k}^+, \quad e_{k,k}^\pm = o(kq). \end{aligned} \quad (\text{OAR11})$$

This substitution proves every sign, keeping the original Gamma return and arithmetic scalar. Addition of their unchanged values to the source- k bounds OAR7 yields the corresponding finite intervals for this same signed quantity. The full original four-endpoint metrics, outer-root polynomial, weighted source, period-dependent certificates and graph denominator remain in OAR11 through their proved maps CTV and ROQ. The next leading-scale calculation is the difference of these two explicit quotient returns.

34 The same-source Gamma moment comparison in the arithmetic return

Retain every original parameter, matrix and error in OAR1–11. The complete OSP comparison supplies an additional bound directly on the actual outer-root quotient. Set $\epsilon = 2^{-32}$ and retain its explicit range

$$k \geq 29, \quad k\sqrt{\delta^2 + \gamma^2} \leq \epsilon(k+1)^2/2. \quad (\text{PMR1})$$

This range contains all sufficiently large original $k = 4l + 1$ for each fixed original quartet: $k \geq \max\{29, 2\sqrt{\delta^2 + \gamma^2}/\epsilon\}$ is sufficient because $(k+1)^2 \geq k^2$. For the remaining finite values the full earlier OAR estimates remain valid without using this comparison.

Keep OSP24's nonnegative $\mathcal{E}_k$ and put

$$c_k^\circ = 4(q + \Delta + 1)\mathcal{E}_k, \quad e_{k,s}^{0,-} = e_{k,s}^- + c_k^\circ, \quad e_{k,s}^{0,+} = e_{k,s}^+ + c_k^\circ. \quad (\text{PMR2})$$

OSP26 and OSP33 prove $c_k^\circ = O_{\delta,\gamma}(q) = o(kq)$ on PMR1, uniformly for the unchanged source orders $s = 1, k$. The actual outer quotient differs from OSP32's exact seven original Gamma moment determinants by at most c_k° . Consequently the full identity and signed bounds are

$$\begin{aligned} F_K^{(s)} &= \mathcal{F}_{0,k}^{(s)} - \Phi_{k,s} + \mathfrak{e}_{k,s}^0, \\ \mathfrak{e}_{k,s}^0 &= \mathfrak{e}_{k,s} + \mathcal{F}_{\partial,k}^{(s)} - \mathcal{F}_{0,k}^{(s)}, \\ -e_{k,s}^{0,-} &\leq \mathfrak{e}_{k,s}^0 \leq e_{k,s}^{0,+}, \quad e_{k,s}^{0,\pm} = o(kq). \end{aligned} \quad (\text{PMR3})$$

The first two equalities are OAR3 with the displayed exact difference added and subtracted. The two bounds follow by adding the interval $[-c_k^\circ, c_k^\circ]$ of OSP33 to the unequal OAR3 endpoints. No metric equality is substituted for the proved OSP coefficient maps and determinant comparison. The residual sum $\Phi_{k,s}$ retains all actual period data.

Let V_s and λ_s be OAR4–5's exact residual upper bound and first positive increment. Define

$$\begin{aligned} L_s^0 &= \max\{L_s^\partial, \mathcal{F}_{0,k}^{(s)} - e_{k,s}^{0,-} - V_s\}, \\ U_s^0 &= \min\{U_s^\partial, \mathcal{F}_{0,k}^{(s)} + e_{k,s}^{0,+} - \lambda_s\}. \end{aligned} \quad (\text{PMR4})$$

Then $L_s^0 \leq F_K^{(s)} \leq U_s^0$. To prove this, insert $\lambda_s \leq \Phi_{k,s} \leq V_s$ into PMR3 and intersect with OAR7. Using the full finite sum for $\Phi_{k,s}$ in PMR3 gives the sharper interval centred at $\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}$. Both retain the same unequal finite errors of PMR2.

The exact arithmetic correlations of OAR8 give

$$\begin{aligned} L_{\text{ar}}^0 &= \max\{L_{\text{ar}}^\partial, L_1^0 - C_K^-, L_1^0 + \delta_k^\sigma - C_B^+\}, \\ U_{\text{ar}}^0 &= \min\{U_{\text{ar}}^\partial, U_1^0 + C_K^+, U_1^0 + \delta_k^\sigma + C_B^-\}. \end{aligned} \quad (\text{PMR5})$$

Adding those correlations to PMR4 at $s = 1$ proves $L_{\text{ar}}^0 \leq F_K^{\text{ar}} \leq U_{\text{ar}}^0$. Subtracting from the exact same arithmetic total gives

$$\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^0 \leq F_B^{\text{ar}} \leq \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^0. \quad (\text{PMR6})$$

Thus the actual kernel and boundary retain their common scalar and both original source comparisons.

Finally the original signed-return identity BRI6d and PMR3 at source order k give

$$\begin{aligned} F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} &= \mathcal{F}_{0,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \mathfrak{e}_{k,k}^0, \\ -e_{k,k}^{0,-} &\leq \mathfrak{e}_{k,k}^0 \leq e_{k,k}^{0,+}. \end{aligned} \quad (\text{PMR7})$$

This follows by substituting into $F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} = F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma$; every sign is retained. The entire original same-source moment expression is OSP32–35. The next asymptotic calculation must evaluate it with the actual residual sum and an error small compared with kq .

The exact finite endpoints under this comparison. The interval transport in PMR4–6 preserves the already retained outer-root endpoints exactly:

$$L_s^0 = L_s^\partial, \quad U_s^0 = U_s^\partial, \quad L_{\text{ar}}^0 = L_{\text{ar}}^\partial, \quad U_{\text{ar}}^0 = U_{\text{ar}}^\partial. \quad (\text{PMR8})$$

For proof, OSP33 gives $\mathcal{F}_{0,k}^{(s)} - c_k^\circ \leq \mathcal{F}_{\partial,k}^{(s)} \leq \mathcal{F}_{0,k}^{(s)} + c_k^\circ$. The new lower candidate in PMR4 is therefore at most $\mathcal{F}_{\partial,k}^{(s)} - e_{k,s}^- - V_s$, which is at most L_s^∂ by OAR6. The new upper candidate is at least $\mathcal{F}_{\partial,k}^{(s)} + e_{k,s}^+ - \lambda_s$, which is at least U_s^∂ . Taking the displayed maximum and minimum proves the source equalities. Substitution in PMR5 and the defining maxima and minima OAR9 proves the arithmetic equalities; subtracting from the same total proves equality of the boundary endpoints as well. Thus this comparison supplies the exact same-source moment target and controlled error; a further numerical improvement must use sharper evaluated moment or residual information. The preceding ROQ/OAR bounds remain the proved finite intervals throughout this transport.

35 Exact equilibrium for the square-root logarithmic potential

Throughout this section $\alpha > 0$ and $\beta > 0$. On $(0, \infty)$ set

$$V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha \log x, \quad \mathcal{E}_{\alpha,\beta}(\mu) = \int V_{\alpha,\beta}(x) d\mu(x) - \iint \log|x-y| d\mu(x)d\mu(y). \quad (\text{EIQ1})$$

The measures considered initially are probability measures for which $\int V_{\alpha,\beta} d\mu$ is finite. Because this potential is bounded below and tends to $+\infty$ at both ends, this implies $\int(\sqrt{x} + |\log x|) d\mu < \infty$. The positive part of $\log|x-y|$ is then integrable, by $\log^+|x-y| \leq \log(1+x) + \log(1+y)$. Thus the energy is well defined with value in $\mathbb{R} \cup \{+\infty\}$. All comparisons below also apply to the usual extended energy defined by integrating the kernel $\frac{1}{2}V_{\alpha,\beta}(x) + \frac{1}{2}V_{\alpha,\beta}(y) - \log|x-y|$: the kernel is bounded below, and finite energy for this kernel implies the same moment conditions. To see the latter assertion without cancellation, put $g(x) = \frac{1}{2}V_{\alpha,\beta}(x) - \log(1+x)$. The kernel is at least $g(x) + g(y)$; g is bounded below and controls positive constant multiples of both $\sqrt{x}$ near infinity and $|\log x|$ near zero.

35.1 Existence and uniqueness of the two endpoints

For $0 < k < 1$ define the complete elliptic integrals by their actual real integrals

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2 \sin^2 \theta}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1-k^2 \sin^2 \theta} d\theta. \quad (\text{EIQ2})$$

There is exactly one $k \in (0, 1)$ satisfying

$$\frac{\sqrt{1-k^2}K(k)}{E(k)} = \frac{\alpha}{\alpha+2}. \quad (\text{EIQ3})$$

For this k put

$$r = \sqrt{1-k^2}, \quad v = \frac{(\alpha+2)\pi}{\beta E(k)}, \quad u = rv, \quad a = u^2, \quad b = v^2. \quad (\text{EIQ4})$$

In particular $0 < a < b < \infty$, and the endpoint equations, with all constants retained, are

$$uK(k) = \frac{\alpha\pi}{\beta}, \quad vE(k) = \frac{(\alpha+2)\pi}{\beta}, \quad k^2 = 1 - \frac{u^2}{v^2}. \quad (\text{EIQ5})$$

Here is a proof of the assertion of uniqueness. Differentiation of the integrals in (EIQ2), and integration of the derivative of $\sin \theta \cos \theta / \sqrt{1-k^2 \sin^2 \theta}$ between its two zero endpoint values, give

$$E'(k) = \frac{E(k) - K(k)}{k}, \quad K'(k) = \frac{E(k)}{k(1-k^2)} - \frac{K(k)}{k}.$$

For completeness the second identity is equivalent, after differentiation under the integral sign, to

$$\int_0^{\pi/2} \left\{ \frac{k^2 \sin^2 \theta}{(1-k^2 \sin^2 \theta)^{3/2}} - \frac{1-k^2 \sin^2 \theta}{(1-k^2)\sqrt{1-k^2 \sin^2 \theta}} + \frac{1}{\sqrt{1-k^2 \sin^2 \theta}} \right\} d\theta = 0;$$

the expression in braces is $-k^2(1-k^2)^{-1} d(\sin \theta \cos \theta / \sqrt{1-k^2 \sin^2 \theta})/d\theta$. With $t = E(k)/K(k)$, the integral definitions show strictly $r^2 < t < 1$. Consequently, if $f(k) = rK(k)/E(k)$, then

$$\frac{kf'(k)}{f(k)} = \frac{t}{r^2} + \frac{1}{t} - 1 - \frac{1}{r^2} = \frac{(t-1)(t-r^2)}{r^2 t} < 0. \quad (\text{EIQ6})$$

At $k = 0$ the continuous limiting value is $f(0) = 1$. As $k \uparrow 1$, the integrand of $rK(k)$ tends to zero at every $\theta < \pi/2$ and is bounded by 1, while $E(k) \rightarrow 1$ by dominated convergence. Hence

$f(k) \rightarrow 0$. This proves (EIQ3), including existence. It also proves smooth dependence of k, u, v, a, b on (α, β) in $(0, \infty)^2$, by the nonzero derivative in (EIQ6).

Let $m = (a + b)/2$, $R_s = \sqrt{(a + s)(b + s)}$ for $s \geq 0$, and $R_0 = uv$. Substitution $t = a \cos^2 \theta + b \sin^2 \theta$ gives, directly from (EIQ5),

$$\int_a^b \frac{V'_{\alpha, \beta}(t)}{\sqrt{(b-t)(t-a)}} dt = 0, \quad \int_a^b \frac{tV'_{\alpha, \beta}(t)}{\sqrt{(b-t)(t-a)}} dt = 2\pi. \quad (\text{EIQ7})$$

Indeed the three integrals with numerators $t^{-1/2}, t^{-1}, t^{1/2}$ are respectively $2K(k)/v$, $\pi/(uv)$, and $2vE(k)$, and $V'(t) = \beta/(2\sqrt{t}) - \alpha/t$.

35.2 A positive density and its exact Cauchy transform

The elementary Stieltjes integral, obtained by $s = xz^2$, is

$$\int_0^\infty \frac{ds}{\sqrt{s}(x+s)} = \frac{\pi}{\sqrt{x}} \quad (x > 0). \quad (\text{EIQ8})$$

Using this identity in (EIQ7) and applying Tonelli to its positive constituents gives

$$\frac{\alpha}{R_0} = \frac{\beta}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}R_s}, \quad \int_0^\infty \frac{1 - s/R_s}{\sqrt{s}} ds = 2vE(k). \quad (\text{EIQ9})$$

Both integrals converge; their integrands are $O(s^{-1/2})$ at zero and $O(s^{-3/2})$ at infinity. The second formula also follows by averaging $\sqrt{t} = \pi^{-1} \int_0^\infty s^{-1/2} t/(t+s) ds$ against the arcsine probability measure on $[a, b]$.

Define for every $x > 0$

$$\begin{aligned} h(x) &= \frac{\alpha}{xR_0} - \frac{\beta}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}(x+s)R_s} \\ &= \frac{\beta}{2\pi x} \int_0^\infty \frac{\sqrt{s}}{(x+s)R_s} ds > 0. \end{aligned} \quad (\text{EIQ10})$$

The second equality is exactly (EIQ9), with no change of the potential. All integrals in (EIQ10) converge absolutely and locally uniformly in $x > 0$. The candidate density on the original positive half-line is

$$\rho_{\alpha, \beta}(x) = \begin{cases} \frac{\sqrt{(b-x)(x-a)}}{2\pi} h(x), & a < x < b, \\ 0, & x \notin (a, b). \end{cases} \quad (\text{EIQ11})$$

It is strictly positive on (a, b) and vanishes continuously with the square-root factor at each endpoint. Positivity here follows from the last integral in (EIQ10), not from a one-interval ansatz.

The following elementary transform computes its mass and resolvent. Choose

$$R(z) = \sqrt{(z-a)(z-b)}$$

on $\mathbb{C} \setminus [a, b]$ with $R(z)/z \rightarrow 1$ at infinity. Thus $R(x) > 0$ for $x > b$, $R(x) < 0$ for $x < a$, and $R(x + i0) = i\sqrt{(b-x)(x-a)}$ on (a, b) . Direct substitution $t = m + (b-a)\cos\theta/2$, followed by $w = \tan(\theta/2)$, proves

$$\frac{1}{\pi} \int_a^b \frac{dt}{(z-t)\sqrt{(b-t)(t-a)}} = \frac{1}{R(z)}.$$

For real $z > b$, the substitution reduces the integral to $\pi^{-1} \int_0^\infty 2dw/[(z-b) + (z-a)w^2]$; the identity elsewhere follows by analytic continuation. Polynomial division then gives

$$\frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{z-t} dt = z - m - R(z), \quad \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{t+s} dt = m + s - R_s. \quad (\text{EIQ12})$$

For example the first identity follows from $(b-t)(t-a) = -(t-z)^2 + 2(m-z)(t-z) - (z-a)(z-b)$, the preceding arcsine transform, and the arcsine first moment m . Partial fractions in these two identities yield

$$H_s(z) := \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{(t+s)(z-t)} dt = 1 - \frac{R_s + R(z)}{z+s}. \quad (\text{EIQ13})$$

The formula at $z = -s$ is interpreted by its removable limit.

The mass of (EIQ11), computed using (EIQ12), is

$$\begin{aligned} \int_a^b \rho_{\alpha,\beta}(t) dt &= \frac{\alpha}{2R_0}(m - R_0) - \frac{\beta}{4\pi} \int_0^\infty \frac{(m+s)/R_s - 1}{\sqrt{s}} ds \\ &= -\frac{\alpha}{2} + \frac{\beta}{4\pi} \int_0^\infty \frac{1-s/R_s}{\sqrt{s}} ds = -\frac{\alpha}{2} + \frac{\beta v E(k)}{2\pi} = 1. \end{aligned} \quad (\text{EIQ14})$$

In particular this is a probability density.

Define its Cauchy transform with the explicit sign convention $G(z) = \int_a^b \rho_{\alpha,\beta}(t)/(z-t) dt$. For z outside both $[a, b]$ and $(-\infty, 0]$, formulas (EIQ10) and (EIQ13) give

$$\begin{aligned} G(z) &= \frac{\alpha}{2R_0} H_0(z) - \frac{\beta}{4\pi} \int_0^\infty \frac{H_s(z)}{\sqrt{s} R_s} ds \\ &= -\frac{\alpha}{2z} + \frac{\beta}{4\sqrt{z}} - \frac{R(z)h(z)}{2} = \frac{V'_{\alpha,\beta}(z) - R(z)h(z)}{2}. \end{aligned} \quad (\text{EIQ15})$$

Here the positive square root is continued with its negative-real-axis cut, and $h(z)$ is the first integral formula of (EIQ10). Every integral exchange is absolutely convergent, uniformly on compact subsets of this domain: near zero the auxiliary integrand is $O(s^{-1/2})$, and near infinity the original integral defining H_s is $O(s^{-1})$. The constant terms on the second line cancel by (EIQ9), and (EIQ8) supplies the term $\beta/(4\sqrt{z})$.

35.3 The Euler equality, its strict exterior inequality, and uniqueness

Set

$$U(x) = V_{\alpha,\beta}(x) - 2 \int_a^b \log|x-t| \rho_{\alpha,\beta}(t) dt \quad (x > 0). \quad (\text{EIQ16})$$

This function is continuous on $(0, \infty)$. The integral is continuous because the density is bounded, compactly supported, and the integral of $|\log|x-t||$ over an interval of length ε tends to zero uniformly in its centre. In the interior of $[a, b]$, differentiation of the logarithmic potential is the principal-value integral $\text{PV} \int \rho(t)/(x-t) dt$. This follows by subtracting $\rho(x)$ on a symmetric interval about x ; the remaining numerator is $O(|t-x|)$ because the density is continuously differentiable locally in the interior. The real part of (EIQ15) on the upper edge of the interval is precisely this principal value. Indeed subtracting $\rho(x)$ in $\int \rho(t)/(x+i\varepsilon-t) dt$ gives the same limit, while the constant term has real part equal to its elementary principal value. Since $R(x+i0)$ is purely imaginary and $h(x)$ is real, (EIQ15) proves $U'(x) = 0$ on (a, b) . There is therefore a real constant ℓ such that

$$U(x) = \ell \quad (a \leq x \leq b). \quad (\text{EIQ17})$$

For positive x outside this interval, (EIQ15) instead gives the exact real derivative

$$U'(x) = R(x)h(x) \begin{cases} < 0, & 0 < x < a, \\ > 0, & x > b. \end{cases} \quad (\text{EIQ18})$$

Combining (EIQ17), continuity, and these signs proves

$$U(x) > \ell \quad \text{for } x \in (0, a) \cup (b, \infty). \quad (\text{EIQ19})$$

This verifies the entire exterior inequality on the original domain, including the interval next to zero.

Here is a complete positivity argument for the energy of the difference of two measures. If ν is a finite signed real measure of total mass zero for which $|\log|x-y||$ is integrable against $|\nu| \otimes |\nu|$, the scalar identity

$$-\log d = \frac{1}{2} \int_0^\infty \frac{e^{-td^2} - e^{-t}}{t} dt \quad (d > 0) \quad (\text{EIQ20})$$

follows by differentiating in d and evaluating at $d = 1$. One half of the integral over any finite subinterval of $(0, \infty)$ has the same sign as $-\log d$ and absolute value at most $|\log d|$. Consequently dominated convergence and the zero total mass give

$$-\iint \log|x-y| d\nu(x)d\nu(y) = \frac{1}{2} \int_0^\infty \frac{1}{t} \iint e^{-t(x-y)^2} d\nu(x)d\nu(y) dt \geq 0. \quad (\text{EIQ21})$$

The nonnegative sign is proved, rather than assumed, by the Gaussian Fourier identity

$$\iint e^{-t(x-y)^2} d\nu(x)d\nu(y) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-\xi^2/(4t)} |\widehat{\nu}(\xi)|^2 d\xi, \quad \widehat{\nu}(\xi) = \int e^{i\xi x} d\nu(x). \quad (\text{EIQ22})$$

The scalar Gaussian Fourier identity follows by completing the square in the Gaussian integral (or by differentiating its Fourier transform and its value at zero); its integration against the finite measures is justified by absolute integrability. Equality in (EIQ21) forces $\widehat{\nu} = 0$: the right side of (EIQ22) is strictly positive for every $t > 0$ if this continuous Fourier transform has a nonzero value. Finally $\widehat{\nu} = 0$ implies $\nu = 0$ without an extra uniqueness assumption. Convolve ν with a centred Gaussian approximate identity. Its value at every point is zero by that same Fourier integral. For every compactly supported continuous test function, its convolution with these Gaussians converges uniformly to the test function; integration against ν then proves that ν annihilates every such function.

Let μ be any probability measure of finite energy in (EIQ1), and write $\mu_*(dx) = \rho_{\alpha,\beta}(x) dx$ and $\nu = \mu - \mu_*$. The integrability needed for (EIQ21) holds. The $\mu \otimes \mu$ term has it by finite energy and the already established logarithmic moment; the $\mu_* \otimes \mu_*$ term has it by bounded compact support; and the cross term has it because the negative logarithmic singularity integrated against the bounded density is uniformly bounded, while the positive part is controlled by $\log(1+x) + \log(1+y)$. Expanding the energy therefore yields exactly

$$\mathcal{E}_{\alpha,\beta}(\mu) - \mathcal{E}_{\alpha,\beta}(\mu_*) = \int (U(x) - \ell) d\mu(x) - \iint \log|x-y| d\nu(x)d\nu(y) \geq 0. \quad (\text{EIQ23})$$

The first term is nonnegative by (EIQ17)–(EIQ19), and the second is strictly positive unless $\mu = \mu_*$. Infinite energy competitors cannot improve this finite value. Thus (EIQ11) is the unique global minimizing probability measure, with exactly the support $[a, b]$.

35.4 An explicit endpoint integral for the logarithmic moment

Retain m, R_s, u, v as defined above, and define

$$C = \frac{(u+v)^2}{4}, \quad \eta = \frac{v-u}{v+u}, \quad \zeta_s = \frac{b-a}{2(m+s+R_s)} \quad (s \geq 0). \quad (\text{EIQ24})$$

Thus $0 < \zeta_s \leq \eta < 1$ and $\zeta_0 = \eta$. For every $s \geq 0$ put

$$L_s = (m+s-R_s) \log C - (m-R_0) - 2R_s \log(1-\eta\zeta_s). \quad (\text{EIQ25})$$

Then the logarithmic moment of the equilibrium measure is exactly

$$\mathcal{L}(\alpha, \beta) := \int_a^b \log x \rho_{\alpha,\beta}(x) dx = \frac{\beta}{4\pi} \int_0^\infty \frac{L_0 - L_s}{\sqrt{s} R_s} ds. \quad (\text{EIQ26})$$

This is a convergent one-dimensional endpoint integral of explicitly given real elementary functions; its only endpoints are those specified uniquely in (EIQ3)–(EIQ5).

Here is a direct derivation, including the constants in (EIQ25). Set $h_0 = (b-a)/2$ and substitute $t = m + h_0 \cos \theta$ in arcsine integrals. The identities

$$t = C(1 + 2\eta \cos \theta + \eta^2), \quad t + s = \frac{m + s + R_s}{2}(1 + 2\zeta_s \cos \theta + \zeta_s^2)$$

give the absolutely convergent Fourier series

$$\log t = \log C + 2 \sum_{j \geq 1} \frac{(-1)^{j+1} \eta^j}{j} \cos(j\theta), \quad \frac{1}{t+s} = \frac{1}{R_s} \left(1 + 2 \sum_{j \geq 1} (-\zeta_s)^j \cos(j\theta) \right).$$

Orthogonality on $[0, \pi]$ gives

$$\begin{aligned} \frac{1}{\pi} \int_a^b \frac{\log t}{\sqrt{(b-t)(t-a)}} dt &= \log C, \\ \frac{1}{\pi} \int_a^b \frac{t \log t}{\sqrt{(b-t)(t-a)}} dt &= m \log C + (m - R_0), \\ \frac{1}{\pi} \int_a^b \frac{\log t}{(t+s)\sqrt{(b-t)(t-a)}} dt &= \frac{\log C + 2 \log(1 - \eta \zeta_s)}{R_s}. \end{aligned} \tag{EIQ27}$$

The second identity uses $h_0 \eta = m - R_0$. Polynomial division

$$\frac{(b-t)(t-a)}{t+s} = -t + 2m + s - \frac{R_s^2}{t+s}$$

now proves, exactly,

$$L_s = \frac{1}{\pi} \int_a^b \frac{\log t \sqrt{(b-t)(t-a)}}{t+s} dt. \tag{EIQ28}$$

Using the positive integral expression for h in (EIQ10), and the identity $1/[t(t+s)] = (1/t - 1/(t+s))/s$, yields (EIQ26). The exchanges are absolutely convergent: $\log t$ is bounded on $[a, b]$, $L_0 - L_s = O(s)$ at zero by (EIQ28), and $L_s = O(s^{-1})$ at infinity. Thus the integrand in (EIQ26) is $O(s^{1/2})$ at zero and $O(s^{-3/2})$ at infinity.

35.5 Parameter continuity and the variational derivative

The density and its logarithmic moment depend smoothly on $(\alpha, \beta) \in (0, \infty)^2$ in the following precise sense. On a compact parameter neighbourhood, (EIQ3)–(EIQ6) put a, b in a fixed compact subset of $(0, \infty)$, with $b - a$ bounded below by a positive number. The integral for h and each of its parameter and x derivatives converge uniformly for x in this fixed compact set: their absolute integrands are bounded by constants times $s^{1/2}$ near zero and $s^{-3/2}$ near infinity. Parameter derivatives of the endpoints are bounded on a smaller compact neighbourhood. Under the fixed-interval coordinate $x = a + (b-a)y$, $0 \leq y \leq 1$, the measure is exactly

$$\rho_{\alpha, \beta}(x) dx = \frac{(b-a)^2}{2\pi} \sqrt{y(1-y)} h(a + (b-a)y) dy. \tag{EIQ29}$$

All parameter derivatives of its coefficient are bounded by a constant times $\sqrt{y(1-y)}$. The same is true after multiplication by $\log(a + (b-a)y)$ or by $\sqrt{a + (b-a)y}$. Differentiation under this integral proves smoothness, in particular continuity, of both moments on the entire positive parameter quadrant.

Define $E_{\min}(\alpha, \beta) = \mathcal{E}_{\alpha, \beta}(\mu_{\alpha, \beta})$, where the measure is the unique minimizer proved above. Testing the minimizer for a parameter pair against the neighbouring potential, and then reversing the two tests, gives for $h > 0$

$$-\mathcal{L}(\alpha + h, \beta) \leq \frac{E_{\min}(\alpha + h, \beta) - E_{\min}(\alpha, \beta)}{h} \leq -\mathcal{L}(\alpha, \beta).$$

The corresponding inequalities for $h < 0$ and the proved continuity therefore give

$$\partial_{\alpha} E_{\min}(\alpha, \beta) = -\mathcal{L}(\alpha, \beta), \quad \partial_{\beta} E_{\min}(\alpha, \beta) = \int \sqrt{x} d\mu_{\alpha, \beta}(x). \quad (\text{EIQ30})$$

No unproved differentiability of a minimizing selection is being used. For the opposite convention $F(\alpha, \beta) = \sup_{\mu} \{ \iint \log |x - y| d\mu(x) d\mu(y) - \int V_{\alpha, \beta} d\mu \} = -E_{\min}(\alpha, \beta)$, the exact derivative is therefore $\partial_{\alpha} F(\alpha, \beta) = \mathcal{L}(\alpha, \beta)$.

The second moment in (EIQ30) has an exact value. Push the minimizing measure forward by the explicit dilation $x \mapsto c^2 x$, $c > 0$. Writing $M = \int \sqrt{x} d\mu_{\alpha, \beta}(x)$, its energy minus the energy at $c = 1$ is precisely $\beta(c - 1)M - 2(\alpha + 1) \log c$. Its derivative at the minimizing value $c = 1$ is zero. Hence

$$\int \sqrt{x} d\mu_{\alpha, \beta}(x) = \frac{2(\alpha + 1)}{\beta}. \quad (\text{EIQ31})$$

The same explicit dilation, or (EIQ3)–(EIQ5), proves that the map $x \mapsto (\pi/\beta)^2 x$ pushes $\mu_{\alpha, \pi}$ to $\mu_{\alpha, \beta}$, including the density and its two endpoints. Consequently

$$\mathcal{L}(\alpha, \beta) = \mathcal{L}(\alpha, \pi) + 2 \log(\pi/\beta). \quad (\text{EIQ32})$$

For the particular original limiting potential $V(x) = \pi\sqrt{x} - 2 \log x$, the substitutions are exactly $\alpha = 2$, $\beta = \pi$. Equations (EIQ5), (EIQ10), and (EIQ26) become

$$uK(k) = 2, \quad vE(k) = 4, \quad h(x) = \frac{1}{2x} \int_0^{\infty} \frac{\sqrt{s}}{(x+s)R_s} ds, \quad \mathcal{L}(2, \pi) = \frac{1}{4} \int_0^{\infty} \frac{L_0 - L_s}{\sqrt{s}R_s} ds. \quad (\text{EIQ33})$$

In particular the preceding proof includes positivity, unit mass, the full Euler inequality, the unique global minimizer, and the parameter continuity needed at these actual parameters.

36 A uniform finite partition estimate from the original equilibrium

QLG1. Exact measures and parameter domain. Fix the compact parameter rectangle

$$\mathcal{K} = [3/2, 5/2] \times [\pi/2, 3\pi/2], \quad V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha \log x \quad (x > 0). \quad (\text{QLG1})$$

The equilibrium proved in EIQ1–33 has density

$$\rho(x) = \frac{\beta\sqrt{(b-x)(x-a)}}{4\pi^2 x} \int_0^\infty \frac{\sqrt{s} ds}{(x+s)\sqrt{(a+s)(b+s)}} \quad (a < x < b), \quad \rho(x) = 0 \quad (x \notin (a, b)), \quad (\text{QLG2})$$

where the exact endpoints are those of EIQ3–5. In particular $0 < a < b < \infty$, the density is bounded and continuous, positive in the interior, and has integral one. Write

$$d\mu(x) = \rho(x) dx, \quad P(z) = \int_a^b \log|z-t| d\mu(t) \quad (z \in \mathbb{C}), \quad I(\mu) = \iint \log|x-y| d\mu(x) d\mu(y),$$

$$F(\alpha, \beta) = I(\mu) - \int V_{\alpha,\beta} d\mu. \quad (\text{QLG3})$$

The original Euler equality and full exterior inequality in EIQ17–19 give a real number ℓ for which

$$D(x) = V_{\alpha,\beta}(x) - 2P(x) - \ell \geq 0 \quad (x > 0), \quad D(x) = 0 \quad (a \leq x \leq b). \quad (\text{QLG4})$$

Integrating the equality on the support gives the exact sign identities

$$\ell = \int V_{\alpha,\beta} d\mu - 2I(\mu), \quad F(\alpha, \beta) = -\ell - I(\mu). \quad (\text{QLG5})$$

The two integration measures treated below are, separately,

$$d\lambda_0(x) = dx, \quad d\lambda_\omega(x) = d\omega(x) = \frac{1}{2}x^{-1/2}e^{-\sqrt{x}} dx. \quad (\text{QLG6})$$

The map $x = r^2$ gives $\int d\omega = \int_0^\infty e^{-r} dr = 1$. For either of these measures, with its stated density w_λ , define the literal finite integral

$$J_n^\lambda(\alpha, \beta) = \frac{1}{n!} \int_{(0,\infty)^n} \exp\left(2 \sum_{i < j} \log|x_i - x_j| - n \sum_{i=1}^n V_{\alpha,\beta}(x_i)\right) \prod_{i=1}^n d\lambda(x_i), \quad n \geq 1. \quad (\text{QLG7})$$

No source measure, scale, factorial, or exponent has been absorbed into F . The unweighted integral without the Heine factor is exactly $n!J_n^\lambda$.

QLG2. Constants proved finite on the whole rectangle. Let

$$a_* = \min_{(\alpha,\beta) \in \mathcal{K}} a(\alpha, \beta) > 0, \quad B_* = \max_{(\alpha,\beta) \in \mathcal{K}} b(\alpha, \beta) < \infty, \quad \beta_- = \pi/2, \quad \beta_+ = 3\pi/2. \quad (\text{QLG8})$$

These are explicit compact-parameter extrema of the uniquely specified endpoint integrals EIQ3–5. Their asserted strict inequalities follow from the continuous positive endpoints proved in EIQ6 and compactness of $\mathcal{K}$. Thus their definition requires no existence or regularity assumption about an unspecified minimizing measure. For $x \in [a, b]$, the integral in QLG2 satisfies

$$\int_0^\infty \frac{\sqrt{s} ds}{(x+s)\sqrt{(a+s)(b+s)}} \leq \int_0^\infty \frac{\sqrt{s} ds}{(a_*+s)^2} = \frac{\pi}{2\sqrt{a_*}}.$$

For the last equality substitute $s = a_* r^2$, followed by $r = \tan \theta$; the resulting integral is

$$2a_*^{-1/2} \int_0^{\pi/2} \sin^2 \theta \, d\theta.$$

Since $\sqrt{(b-x)(x-a)} \leq (b-a)/2 \leq B_*/2$, it follows that

$$0 \leq \rho(x) \leq M_* := \frac{\beta_+ B_*}{16\pi a_*^{3/2}} \quad \text{for all real } x \text{ and all parameters in } \mathcal{K}. \quad (\text{QLG9})$$

In particular $B_* M_* \geq 1$, by the mass-one identity and support length at most B_* .

The entropy relative to either stated measure is

$$H_\lambda(\mu) = \int_a^b \rho(x) \log \frac{\rho(x)}{w_\lambda(x)} \, dx, \quad 0 \log 0 = 0. \quad (\text{QLG10})$$

It is absolutely integrable: the function $u|\log u|$ is bounded on $[0, M_*]$, and $\log w_\lambda$ is bounded on the fixed compact interval $[a_*, B_*]$. In the two cases its upper bounds are

$$H_{\lambda_0}(\mu) \leq \log M_*, \quad H_{\lambda_\omega}(\mu) \leq \log M_* + \log 2 + \frac{1}{2} \log B_* + \sqrt{B_*}. \quad (\text{QLG11})$$

All pair logarithmic energies on this support are absolutely integrable. For the negative part, integration against a density bounded by M_* is bounded above by $M_* \int_{-1}^1 -\log |s| \, ds = 2M_*$, uniformly in the centre. For the positive part the bound $\log^+ B_*$ suffices.

Put

$$T_\lambda = \int_0^\infty e^{-D(x)} \, d\lambda(x). \quad (\text{QLG12})$$

For $\lambda = \lambda_\omega$, QLG4 and the mass of ω give $0 < T_{\lambda_\omega} \leq 1$. For the Lebesgue measure a finite uniform bound can also be displayed entirely. For every $x > 0$,

$$P(x) \leq \log(1+x) + \log(1+B_*), \quad P(x) \geq -2M_*.$$

The lower bound uses only the negative part estimate just proved. Evaluating QLG4 at any point of $[a, b]$ therefore yields

$$\ell \leq L_* := \beta_+ \sqrt{B_*} + \frac{5}{2} \max\{|\log a_*|, |\log B_*|\} + 4M_*.$$

Consequently

$$T_{\lambda_0} \leq e^{L_*} (1+B_*)^2 \int_0^\infty (1+x)^2 (1+x^{5/2}) e^{-\beta_- \sqrt{x}} \, dx =: T_* < \infty. \quad (\text{QLG13})$$

Here $x^\alpha \leq 1 + x^{5/2}$ for all parameters in $\mathcal{K}$. The integral in QLG13 equals, by $x = r^2$ and elementary repeated integration by parts,

$$2 \left(\frac{1!}{\beta_-^2} + \frac{2 \cdot 3!}{\beta_-^4} + \frac{5!}{\beta_-^6} + \frac{6!}{\beta_-^7} + \frac{2 \cdot 8!}{\beta_-^9} + \frac{10!}{\beta_-^{11}} \right). \quad (\text{QLG14})$$

QLG3. The planar logarithmic comparison, with its proof. We use the plane only to construct an exact comparison of logarithmic kernels; the original particles and potential remain on $(0, \infty)$. If η is a finite signed measure of compact support on $\mathbb{R}^2$, has total mass zero, and $\iint |\log |z-w|| \, d|\eta|(z) d|\eta|(w) < \infty$, then

$$I(\eta) := \iint \log |z-w| \, d\eta(z) d\eta(w) \leq 0. \quad (\text{QLG15})$$

To prove this, for $d > 0$ integrate the elementary identity

$$-\log d = \frac{1}{2} \int_0^\infty \frac{e^{-td^2} - e^{-t}}{t} dt.$$

It follows by differentiation in d and evaluation at $d = 1$. On any finite subinterval of $(0, \infty)$, the integral on the right has the same sign as $-\log d$ and absolute value at most $|\log d|$. The stipulated logarithmic integrability therefore permits dominated passage to the full integral against $d\eta^2$. The constant term vanishes by the zero mass. The elementary Gaussian Fourier integral in each of the two real coordinates gives

$$\begin{aligned} -I(\eta) &= \frac{1}{2} \int_0^\infty \frac{dt}{t} \iint e^{-t|z-w|^2} d\eta(z) d\eta(w), \\ \iint e^{-t|z-w|^2} d\eta(z) d\eta(w) &= \frac{1}{4\pi t} \int_{\mathbb{R}^2} e^{-|\xi|^2/(4t)} \left| \int e^{i\xi \cdot z} d\eta(z) \right|^2 d\xi \geq 0. \end{aligned} \quad (\text{QLG16})$$

For completeness the scalar Fourier identity is obtained by differentiating $\int e^{-u^2/(4t)} e^{isu} du$ with respect to s , integrating by parts in u , and using its value $\sqrt{4\pi t}$ at zero. The product of the two scalar identities has the displayed factor $(4\pi t)^{-1}$. All finite Gaussian integrations against the measures are absolutely convergent. This proves QLG15 without requiring an equilibrium theorem on the plane.

Let $c_{x,\varepsilon}$ be probability arc measure on the circle $z = x + \varepsilon e^{i\theta}$, where the real centre x is arbitrary and $\varepsilon > 0$. Its exact logarithmic potential is

$$\int \log |z - w| dc_{x,\varepsilon}(z) = \log \max\{\varepsilon, |x - w|\}. \quad (\text{QLG17})$$

Indeed, if $|x - w| > \varepsilon$, expand $\log |1 - re^{i\theta}| = -\sum_{j \geq 1} r^j \cos(j\theta)/j$ with $r = \varepsilon/|x - w| < 1$; all nonconstant terms average to zero. If $|x - w| < \varepsilon$, use the same expansion with $r = |x - w|/\varepsilon$. At equality the assertion is the elementary integral $\int_0^{2\pi} \log |1 - e^{i\theta}| d\theta = 0$. To verify this last integral directly, write $|1 - e^{i\theta}| = 2 \sin(\theta/2)$, and put $A = \int_0^{\pi/2} \log(\sin u) du$. Reflection gives the same integral for $\log \cos u$; the double-angle identity then gives $2A = -(\pi/2) \log 2 + A$, whence $A = -(\pi/2) \log 2$. The endpoint logarithms are integrable, for example from $2u/\pi \leq \sin u \leq u$ on $[0, \pi/2]$.

For finite centres x, y , the logarithmic kernel is absolutely integrable against $c_{x,\varepsilon} \otimes c_{y,\varepsilon}$. Its positive part is bounded by a finite constant depending on the centres and ε . Applying QLG17 first to this bounded positive part minus the negative part, Tonelli's theorem and the finite value $\log \max\{\varepsilon, |x - w|\}$ show that the negative part also has finite double integral. This argument includes identical and tangent circles. The same argument proves absolute integrability of the circle- μ cross kernel, using compactness of the support of μ ; the μ - μ term was proved above. In particular QLG15 applies to a finite mixture of these circles minus μ .

Two consequences of QLG17, retaining both self and cross terms, are

$$I(c_{x,\varepsilon}) = \log \varepsilon, \quad I(c_{x,\varepsilon}, c_{y,\varepsilon}) \geq \log \max\{\varepsilon, |x - y|\} \geq \log |x - y|. \quad (\text{QLG18})$$

The first equality applies QLG17 at w on its own circle. For the second inequality, the first circle average is $\log \max\{\varepsilon, |x - w|\} \geq \log |x - w|$; averaging the right side on the second circle and applying QLG17 again gives its asserted value. When $x = y$ the last lower bound is understood in the extended sense.

QLG4. A finite upper estimate on the original particle domain. For fixed particles $x_1, \dots, x_n > 0$, put $\nu_\varepsilon = n^{-1} \sum_i c_{x_i, \varepsilon}$. By QLG18 and QLG15, respectively,

$$2 \sum_{i < j} \log |x_i - x_j| \leq n^2 I(\nu_\varepsilon) - n \log \varepsilon, \quad I(\nu_\varepsilon) \leq 2I(\nu_\varepsilon, \mu) - I(\mu). \quad (\text{QLG19})$$

There is no potential evaluated at a complex particle in these identities. The exact comparison back to the original real potential is

$$0 \leq \int P(z) dc_{x,\varepsilon}(z) - P(x) = \int_{|t-x|<\varepsilon} \log \frac{\varepsilon}{|t-x|} \rho(t) dt \leq 2M_*\varepsilon. \quad (\text{QLG20})$$

The last constant is the explicit integral $M_* \int_{-\varepsilon}^{\varepsilon} \log(\varepsilon/|s|) ds = 2M_*\varepsilon$. Combining QLG19–20 and then substituting QLG4–5 proves, for every original particle configuration,

$$2 \sum_{i<j} \log |x_i - x_j| - n \sum_i V_{\alpha,\beta}(x_i) \leq n^2 F - n \sum_i D(x_i) - n \log \varepsilon + 4M_* n^2 \varepsilon. \quad (\text{QLG21})$$

The coefficient $4M_*$ comes from twice the logarithmic potential and the bound $2M_*\varepsilon$; the coefficient of the self energy is $-n \log \varepsilon$. Take $\varepsilon = 1/n$. Since $D \geq 0$ and $n \geq 1$, $\int e^{-nD} d\lambda \leq T_\lambda$. Integrating QLG21 gives the finite positive upper bound

$$\log J_n^\lambda \leq n^2 F + 4M_* n + n \log T_\lambda + n \log n - \log(n!). \quad (\text{QLG22})$$

In particular the original integrals in QLG7 are finite. The elementary inequality $\log(n!) \geq \int_1^n \log t dt = n \log n - n + 1$ also proves

$$\log J_n^\lambda \leq n^2 F + n(1 + 4M_* + \log T_\lambda) - 1. \quad (\text{QLG23})$$

The inequality for $n!$ follows by monotonicity of $\log t$ on each interval $[j-1, j]$, for $j = 2, \dots, n$, and is exact at $n = 1$.

QLG5. Quantile lower estimate and the exact factorial cancellation. Because ρ is continuous and positive on (a, b) , there are unique endpoints

$$a = t_0 < t_1 < \dots < t_n = b, \quad \int_{t_{j-1}}^{t_j} \rho(x) dx = \frac{1}{n}, \quad \Delta_j = t_j - t_{j-1} > 0, \quad p_j(x) = n\rho(x)\mathbf{1}_{(t_{j-1}, t_j)}(x). \quad (\text{QLG24})$$

The endpoints have measure zero. Each $p_j(x)dx$ is a probability measure with bounded density, on a compact interval of positive length. Every pair logarithmic integral below is therefore absolutely integrable, including self integrals and integrals over touching intervals, by the same bounded-density estimate used after QLG11. Also $\int p_j |\log(p_j/w_\lambda)| dx < \infty$, since $p_j \leq nM_*$ and $\log w_\lambda$ is bounded on the support.

The product cell $\prod_{j=1}^n (t_{j-1}, t_j)$ and its $n!$ coordinate permutations are disjoint. Symmetry of the integrand and the product integration measure in QLG7 implies that the Heine factor $1/n!$ cancels these $n!$ equal integrals exactly. Restricting to their union therefore gives

$$J_n^\lambda \geq \int_{\prod_j (t_{j-1}, t_j)} \exp \left(2 \sum_{i<j} \log |x_i - x_j| - n \sum_i V(x_i) \right) \prod_j w_\lambda(x_j) dx_j. \quad (\text{QLG25})$$

On this cell use the probability density $\prod_j p_j(x_j)$. The logarithm of the integrand relative to this probability is integrable in absolute value, by the preceding estimates and boundedness of V on $[a, b]$. The elementary convex inequality $e^u \geq e^v(1 + u - v)$, integrated with v equal to the expectation of this logarithm, proves Jensen's lower bound directly. Since $\sum_j p_j = n\rho$, its pair and potential contributions are exactly

$$\sum_{i \neq j} I(p_i, p_j) - n \sum_j \int V p_j dx = n^2 F - \sum_j I(p_j).$$

Here $I(p_i, p_j) = \iint \log |x - y| p_i(x) p_j(y) dx dy$ and $I(p_j) = I(p_j, p_j)$. Its entropy contribution is exactly

$$\sum_j \int p_j \log \frac{p_j}{w_\lambda} dx = n \log n + n H_\lambda(\mu). \quad (\text{QLG26})$$

Moreover $|x - y| \leq \Delta_j$ on the j -th cell, whence $I(p_j) \leq \log \Delta_j$. Concavity of the logarithm, proved for example by its negative second derivative, and $\sum_j \Delta_j = b - a$ now give

$$\sum_j I(p_j) \leq \sum_j \log \Delta_j \leq n \log \frac{b - a}{n}. \quad (\text{QLG27})$$

Thus the $n \log n$ in QLG26 cancels that in QLG27, and the complete lower bound is

$$\log J_n^\lambda(\alpha, \beta) \geq n^2 F(\alpha, \beta) - n \{ \log(b - a) + H_\lambda(\mu) \}. \quad (\text{QLG28})$$

This also proves strict positivity of QLG7. No lower bound on the density at an endpoint or on a quantile interval length has been used.

QLG6. The uniform order- n theorem. For every $(\alpha, \beta) \in \mathcal{K}$, every integer $n \geq 1$, and each of the two measures QLG6, QLG22 and QLG28 are the explicit two-sided finite estimate

$$-n \{ \log(b - a) + H_\lambda(\mu) \} \leq \log J_n^\lambda - n^2 F \leq 4M_* n + n \log T_\lambda + n \log n - \log(n!). \quad (\text{QLG29})$$

For entirely parameter-uniform constants set

$$\begin{aligned} L_0 &= \log(B_* M_*), & L_\omega &= \log(B_* M_*) + \log 2 + \frac{1}{2} \log B_* + \sqrt{B_*}, \\ C_0 &= \max\{0, L_0, 1 + 4M_* + \log^+ T_*\}, & C_\omega &= \max\{0, L_\omega, 1 + 4M_*\}. \end{aligned} \quad (\text{QLG30})$$

These constants are finite by QLG8–14. Since $b - a \leq B_*$, QLG11 bounds the lower side of QLG29 by $-nL_\lambda$; QLG12–14 and QLG23 bound its upper side by nC_λ . Consequently

$$\left| \log J_n^{\lambda_0}(\alpha, \beta) - n^2 F(\alpha, \beta) \right| \leq C_0 n, \quad \left| \log J_n^{\lambda_\omega}(\alpha, \beta) - n^2 F(\alpha, \beta) \right| \leq C_\omega n. \quad (\text{QLG31})$$

For the integral Z_n^λ without the Heine factor, the exact relation $\log Z_n^\lambda = \log J_n^\lambda + \log(n!)$ gives

$$\left| \log Z_n^\lambda - n^2 F - \log(n!) \right| \leq C_\lambda n, \quad \left| \log Z_n^\lambda - n^2 F \right| \leq n \log n + C_\lambda n. \quad (\text{QLG32})$$

The theorem therefore retains the distinction between the original Heine determinant integral and the integral with labelled particles. The gain to order n in QLG31 follows from the proved circle comparison and the quantile-cell cancellation, with the same exact potential, positive-half-line particle domain, and equilibrium of EI1–33 throughout.

37 The leading first variation of the actual outer-strip quotient

Retain every original object and both original source orders of OSP1–35 and CTV1–22. In addition to the explicit finite conditions OSP23, take $k \geq 257$. This is an eventual threshold on the same fixed original parameters. Put

$$\vartheta = \pi/2, \quad \ell_s = (s-1)/4, \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)} \quad (s=1, k). \quad (\text{OFV1})$$

The symbol ϑ is the fixed Gamma tail rate; it is not either original source centre.

We use the equilibrium of EIQ1–33 on its original positive half-line:

$$\begin{aligned} V_{\alpha,\beta}(x) &= \beta\sqrt{x} - \alpha \log x, \quad \mu_{\alpha,\beta}(dx) = \rho_{\alpha,\beta}(x) dx, \\ F(\alpha, \beta) &= \iint \log|x-y| d\mu_{\alpha,\beta}(x) d\mu_{\alpha,\beta}(y) - \int V_{\alpha,\beta} d\mu_{\alpha,\beta}. \end{aligned} \quad (\text{OFV2})$$

Here ρ , its two endpoints, the Euler equality and the entire exterior inequality are exactly the explicit proved EIQ formulas. There is no new equilibrium assumption. EIQ29–30 proves smoothness and $F_\alpha(\alpha, \beta) = \int \log x d\mu_{\alpha,\beta}(x)$. QLG31, with its complete circle and quantile proofs, gives

$$|\log J_n(\alpha, \beta) - n^2 F(\alpha, \beta)| \leq C_0 n, \quad J_n = \frac{1}{n!} \int_{(0,\infty)^n} \prod_{i < j} (x_i - x_j)^2 \prod_i e^{-n V_{\alpha,\beta}(x_i)} dx_i \quad (\text{OFV3})$$

for every $n \geq 1$ and every $(\alpha, \beta) \in [3/2, 5/2] \times [\pi/2, 3\pi/2]$. The constant C_0 is the explicit finite constant QLG30. We use precisely this uniform finite error, including the factor $1/n!$; the conclusion below is not a derivative of an earlier unquantified q^2 limit.

37.1 Exact original-order recurrence and a small metric error

For $a \geq 0$, define

$$\mathcal{D}_a(Q; s) = \det \left[\int_{\mathbb{R}} y^{2Q+i+j} dm_s(y) \right]_{0 \leq i, j < a}, \quad \mathcal{A}_a(Q; \vartheta) = \det \left[\int_{\mathbb{R}} y^{2Q+i+j} e^{-\vartheta|y|} dy \right]_{0 \leq i, j < a}, \quad (\text{OFV4})$$

with empty determinants one. The Q used here is an integer; all the instances below have $Q \geq q/2$. The original recurrence OSP12 gives exactly

$$|y|^{2Q} dm_s(y) = \beta_s |y|^{2Q} \prod_{j=0}^{\ell_s-1} (y^2 + t_j^2) d\sigma(y), \quad t_j = 2j + \frac{1}{2}. \quad (\text{OFV5})$$

For each actual source centre $c_0 \in \{k/2, k/2 - 4\}$, the coordinate polynomial of this multiplier is $f_{\ell_s, c_0}(S) = \prod_j (S - c_0 - t_j)$. Its exact pullback satisfies $f_{\ell_s, c_0}(c_0 + iy) = \prod_j (iy - t_j) = i^{\ell_s} \prod_j (y + it_j)$. Thus multiplication by $y^Q \prod_{j=0}^{\ell_s-1} (y + it_j)$, together with the retained phase i^{ℓ_s} , is the exact coefficient map into the original one-factor Gamma source. Its squared modulus is the displayed density multiplier, with no change to any t_j or to β_s .

Set $T = \epsilon q$, $\epsilon = 2^{-32}$ as in OSP23, and

$$S_{\ell_s} = \sum_{j=0}^{\ell_s-1} t_j^2 = \frac{\ell_s(16\ell_s^2 - 12\ell_s - 1)}{12}, \quad r_{\ell_s} = \frac{S_{\ell_s}}{T^2} + \log(1 + K_q), \quad (\text{OFV6})$$

where K_q is exactly OSP24. At $\ell_s = 0$ we may and do take $r_0 = 0$ instead, since the source identity is exact. For $Q \in \{q', q\}$, $0 \leq a \leq 2q + 1$,

$$0 \leq \log \mathcal{D}_a(Q; s) - a \log \beta_s - \log \mathcal{D}_a(Q + \ell_s; 1) \leq ar_{\ell_s}. \quad (\text{OFV7})$$

Here is a proof of the full-form comparison underlying OFV7. For every real y , $|y|^{2Q} \prod_j (y^2 + t_j^2) \geq |y|^{2(Q+\ell_s)}$. On $|y| \geq T$, the logarithm of the ratio is at most $\sum_j t_j^2/T^2 = S_{\ell_s}/T^2$, by $\log(1+u) \leq u$. For $|y| < T$, its numerator is at most $(T^2 + (k/2)^2)^{Q+\ell_s}$. OSP15 on the complete coefficient space of degree at most $a-1$, OSP13 with source $s=1$, and mass $M_1 = \sqrt{2\pi}$ therefore bound its inner quadratic form by K_q times the reference quadratic form on $[q, 2q]$. This is exactly the calculation OSP22–25 with exponent $Q + \ell_s \geq q/2$, degree $a-1 \leq 2q$, and root bound $k/2 \leq R \leq T/2$. No upper bound on that exponent is needed in OSP25, since its base $(5/4)\epsilon^2$ is less than one. The full upper ratio is consequently at most $e^{S_{\ell_s}/T^2} + K_q \leq e^{r_{\ell_s}}$. The lower full ratio is at least one by the pointwise lower inequality. The positive-Gram eigenvalue argument OSP28 now proves OFV7 in the original coefficient frames. Since $S_{\ell_s} \leq \ell_s k^2/4$ and $q \geq k^2$, $r_{\ell_s} = O(1/k)$, with its stated fixed cutoff constant.

37.2 A uniform finite Gamma-to-exponential determinant estimate

Put $h = 1/q$, and define the actual finite constants

$$L_h = C_L(\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi}(\sin h)^{-1/2}. \quad (\text{OFV8})$$

Then the original one-factor Gamma density satisfies on the entire real line

$$L_h e^{-(\vartheta+h)|y|} \leq \frac{d\sigma}{dy}(y) \leq U_h e^{-(\vartheta-h)|y|}. \quad (\text{OFV9})$$

Indeed apply OSP7 to $\Gamma(1/4 + iy/2)$ with $\theta = \pi/2 - h$ for the upper bound. For the lower bound apply it to $\Gamma(3/4 - iy/2)$ in the exact reflection product proved before OSP11. The upper bound on that squared factor is $\Gamma(3/4)^2(\sin h)^{-3/2}e^{-(\vartheta-h)|y|}$. After division, the identity and $\cosh(\pi y) \leq e^{\pi|y|}$ give exactly the lower side of OFV9. In particular both inequalities hold near zero as well as in the tails.

The exact change of variables $y = \vartheta^{-1}u$ in OFV4 gives

$$\mathcal{A}_a(Q; \vartheta) = \vartheta^{-a(2Q+a)} \mathcal{A}_a(Q; 1). \quad (\text{OFV10})$$

The exponent follows from a moment Jacobians, $2Q$ powers in each, and the row and column powers $\sum_{i=0}^{a-1} i + \sum_{j=0}^{a-1} j = a(a-1)$. Applying OFV9 as full quadratic-form inequalities, and then using OFV10, proves the separate finite bounds

$$\begin{aligned} a \log L_h - a(2Q+a) \log(1+h/\vartheta) &\leq \log \mathcal{D}_a(Q; 1) - \log \mathcal{A}_a(Q; \vartheta) \\ &\leq a \log U_h - a(2Q+a) \log(1-h/\vartheta). \end{aligned} \quad (\text{OFV11})$$

Thus the absolute difference is at most

$$B_a(Q) = a \max\{|\log L_h|, |\log U_h|\} + a(2Q+a) \frac{h}{\vartheta - h}. \quad (\text{OFV12})$$

Here $-\log(1-u) \leq u/(1-u)$ and $\log(1+u) \leq u$ justify the second summand. Since $2h/\pi \leq \sin h \leq h$ for $0 < h \leq \pi/2$, the first summand is $O(a \log q)$. The sine inequalities follow from concavity and the endpoint chord, and by integrating $\cos t \leq 1$. For all the present $a \leq 2q+1$, $Q \leq q + \ell_s \leq 2q$, OFV12 is $O(q \log q)$ uniformly in a, Q . Combining OFV7 and OFV11 gives the error-controlled identity

$$\log \mathcal{D}_a(Q; s) = a \log \beta_s + \log \mathcal{A}_a(Q + \ell_s; \vartheta) + R_{a,Q,s}, \quad |R_{a,Q,s}| \leq a r_{\ell_s} + B_a(Q + \ell_s). \quad (\text{OFV13})$$

This is a comparison on complete original polynomial coefficient spaces, with the order shift proved by the literal multiplier OFV5; it is not an assigned replacement of the original source order.

37.3 Direct low-dimension estimates, outside the compact equilibrium range

Define

$$g(Q) = 2Q \log(2Q/\vartheta) - 2Q. \quad (\text{OFV14})$$

For $Q \geq q/2$ and $1 \leq a \leq \Delta + 1$ we prove

$$\log \mathcal{A}_a(Q; \vartheta) = ag(Q) + O(a^2 \log(Q + a) + a \log(Q + a)), \quad (\text{OFV15})$$

with constants independent of k, Q, a in these ranges. There is no use of OFV3 here; its compact parameter rectangle does not contain these low-dimension ratios.

For the lower bound take the full interval $[y_0, y_0 + 1]$, $y_0 = 2Q/\vartheta$. On it $y^{2Q} e^{-\vartheta y} \geq e^{g(Q) - \vartheta}$. Consequently its moment Gram is bounded below by $e^{g(Q) - \vartheta}$ times the unweighted monomial Gram on that interval. Translation has a triangular coefficient matrix with diagonal one, so the latter Gram determinant equals the one on $[0, 1]$. The Rodrigues norms proved in OSP15 show that it is exactly

$$\prod_{j=0}^{a-1} \frac{1}{(2j+1) \binom{2j}{j}^2}.$$

Indeed $\binom{2j}{j}$ is the leading coefficient of $L_j(2x-1)$, whose squared norm is $1/(2j+1)$. Using $\binom{2j}{j} \leq 4^j$ therefore proves

$$\log \mathcal{A}_a(Q; \vartheta) \geq ag(Q) - a\vartheta - a \log(2a) - 2a(a-1) \log 2. \quad (\text{OFV16})$$

For the upper bound, Gram–Schmidt shows that a positive Gram determinant is at most the product of the squared norms of its displayed vectors: each orthogonal component has norm no greater than its original vector. The exact monomial moments give

$$\log \mathcal{A}_a(Q; \vartheta) \leq \sum_{j=0}^{a-1} \log \left(\frac{2(2(Q+j))!}{\vartheta^{2(Q+j)+1}} \right). \quad (\text{OFV17})$$

For every positive integer m , integrating the increasing function $\log x$ between successive integers proves $m \log m - m + 1 \leq \log(m!) \leq m \log m - m + 1 + \log m$. Each summand in OFV17 is consequently at most $g(Q+j) + |\log(2/\vartheta) + 1| + \log(2(Q+j))$. Furthermore $g'(u) = 2 \log(2u/\vartheta)$, so $g(Q+j) - g(Q) \leq 2j \log(2(Q+a)/\vartheta)$ in the present range. This proves the explicit upper bound

$$\begin{aligned} \log \mathcal{A}_a(Q; \vartheta) &\leq ag(Q) + a\{|\log(2/\vartheta) + 1| + \log(2(Q+a))\} \\ &\quad + a(a-1) \log(2(Q+a)/\vartheta). \end{aligned} \quad (\text{OFV18})$$

OFV16 and OFV18 prove OFV15 with its complete lower and upper errors. They also apply at $a = 1$.

37.4 High-dimension partitions and their controlled first variation

For an integer a write $n_e = \lceil a/2 \rceil$ and $n_o = \lfloor a/2 \rfloor$. The two exact changes of variables $y = \sqrt{x}$ and $y = -\sqrt{x}$ give, with their two positive Jacobians, the parity identity

$$\mathcal{A}_a(Q; \vartheta) = Z_{n_e}(Q - \tfrac{1}{2}; \vartheta) Z_{n_o}(Q + \tfrac{1}{2}; \vartheta), \quad (\text{OFV19})$$

where

$$Z_n(A; \vartheta) = \det \left[\int_0^\infty x^{A+i+j} e^{-\vartheta \sqrt{x}} dx \right]_{0 \leq i, j < n}.$$

The two branches add to the density $x^{-1/2} e^{-\vartheta \sqrt{x}}$; thus no factor two is missing from OFV19. The odd block has the additional factor x . Expanding the two Vandermonde determinants and integrating each finite permutation sum proves the exact Heine identity for this moment determinant:

$$Z_n(A; \vartheta) = \frac{1}{n!} \int_{(0, \infty)^n} \prod_{i < j} (x_i - x_j)^2 \prod_i x_i^A e^{-\vartheta \sqrt{x_i}} dx_i.$$

Changing variables $x_i = n^2 u_i$ now gives

$$Z_n(A; \vartheta) = n^{2n(A+n)} J_n(A/n, \vartheta). \quad (\text{OFV20})$$

There are $2n(A+1)$ powers from the weights and Jacobians and $2n(n-1)$ from the squared Vandermonde, proving the exponent.

For the high instances we use exactly

$$(a, Q) \in \{(q, q + \ell_s), (q+1, q + \ell_s), (q + \Delta, q - \Delta + \ell_s), (q + \Delta + 1, q - \Delta + \ell_s)\}. \quad (\text{OFV21})$$

Since $k \geq 257$, $\Delta/q \leq 16/k < 1/16$ and $\ell_s/q \leq 1/(4k) < 1/64$. For both parity blocks, elementary rounding therefore gives

$$\frac{15q}{32} \leq n \leq \frac{9q}{16}, \quad \frac{29q}{32} \leq A \leq \frac{33q}{32}.$$

The first inequalities use $q \geq 32$ to absorb the possible rounding by $1/2$ and the extra endpoint 1. Hence $29/18 \leq A/n \leq 11/5$, contained strictly in $[3/2, 5/2]$. Every partition in OFV20 is thus in the proved parameter rectangle OFV3, with $\beta = \vartheta = \pi/2$.

Define the smooth explicit function

$$\mathcal{H}(a, Q) = a(2Q + a) \log(a/2) + \frac{a^2}{2} F(2Q/a, \vartheta) \quad (a > 0, Q > 0). \quad (\text{OFV22})$$

Equations OFV3 and OFV19–20 give uniformly over OFV21

$$\log \mathcal{A}_a(Q; \vartheta) = \mathcal{H}(a, Q) + O(q \log q). \quad (\text{OFV23})$$

To verify the rounding error rather than omit it, set $G(n, A) = 2n(A+n) \log n + n^2 F(A/n, \vartheta)$. Each actual pair (n, A) differs from $(a/2, Q)$ by at most $1/2$ in each coordinate. On all connecting segments n is bounded above and below by positive constant multiples of q , while A/n remains in $[3/2, 5/2]$. Its derivatives are

$$G_A = 2n \log n + n F_\alpha, \quad G_n = 2(A+2n) \log n + 2(A+n) + 2nF - A F_\alpha.$$

They are $O(q \log q)$ there, since the exact EIQ functions F, F_α are bounded on that compact interval. The mean value formula bounds each rounding error by $O(q \log q)$. The two QLG errors sum to at most $C_0 a = O(q)$, and $2G(a/2, Q) = \mathcal{H}(a, Q)$, proving OFV23.

The derivative needed for the first variation is computed without altering the direction of the two original degree changes:

$$(\partial_a - \partial_Q) \mathcal{H}(a, Q) = 2Q \log(a/2) + (2Q + a) + aF(2Q/a, \vartheta) - (Q + a)F_\alpha(2Q/a, \vartheta). \quad (\text{OFV24})$$

EIQ29–30 proves that F_α is smooth; in particular $F, F_\alpha, F_{\alpha\alpha}$ are bounded on the compact interval just used. Differentiating the right side of OFV24 shows that its two first partial derivatives are $O(\log q)$ on the segment $a = q + u$, $Q = q + \ell_s - u$, $0 \leq u \leq \Delta$ and on its connections to (q, q) . Every term follows by the product rule; the derivatives of $2Q/a$ are $2/a$ and $-2Q/a^2$, both $O(1/q)$. The mean value formula and then integration in u consequently give

$$\begin{aligned} & \mathcal{H}(q + \Delta, q - \Delta + \ell_s) - \mathcal{H}(q, q + \ell_s) \\ &= \Delta q \{2 \log(q/2) + 3 + F(2, \vartheta) - 2F_\alpha(2, \vartheta)\} + O(\Delta(\Delta + \ell_s) \log q). \end{aligned} \quad (\text{OFV25})$$

This remainder is $O(q \log q)$ because $\Delta = O(k)$, $\ell_s = O(k)$ and $q = (k+1)^2$. The additional endpoint $a+1$ in OFV21 changes $\mathcal{H}$ by $O(q \log q)$, directly from its displayed derivatives. Thus all rounding and endpoint changes have been quantified.

37.5 All seven determinants and the resulting leading constant

Replace the seven moment determinants in OSP32 by the exact exponential determinants in OFV13, with their respective $Q + \ell_s$. Their $a \log \beta_s$ terms cancel because their signed dimensions are exactly

$$\Delta + (\Delta + 1) - (q + \Delta) - (q + \Delta + 1) - 1 + q + (q + 1) = 0. \quad (\text{OFV26})$$

The sum of their absolute finite errors in OFV13 is $O(q \log q)$, since the sum of their dimensions is $4(q + \Delta + 1)$, $r_{\ell_s} = O(1/k)$, and OFV12 applies uniformly with $Q + \ell_s \leq 2q$. The low terms, using OFV15–18, are

$$\begin{aligned} \log \mathcal{A}_\Delta(q - \Delta + \ell_s; \vartheta) + \log \mathcal{A}_{\Delta+1}(q - \Delta + \ell_s; \vartheta) - \log \mathcal{A}_1(q + \ell_s; \vartheta) \\ = 2\Delta g(q) + O(q \log q) = 4\Delta q \{\log(2q/\vartheta) - 1\} + O(q \log q). \end{aligned} \quad (\text{OFV27})$$

Indeed replacing each $g(q - \Delta + \ell_s)$ or $g(q + \ell_s)$ by $g(q)$ costs at most $O(((2\Delta + 1)(\Delta + \ell_s) + \ell_s) \log q) = O(q \log q)$, using $g'(u) = 2 \log(2u/\vartheta)$ and the actual degree ranges. The individual low determinant errors are $O((\Delta + 1)^2 \log q) = O(q \log q)$.

The remaining four high terms are, by OFV23–25,

$$\begin{aligned} \log \mathcal{A}_q(q + \ell_s; \vartheta) + \log \mathcal{A}_{q+1}(q + \ell_s; \vartheta) \\ - \log \mathcal{A}_{q+\Delta}(q - \Delta + \ell_s; \vartheta) - \log \mathcal{A}_{q+\Delta+1}(q - \Delta + \ell_s; \vartheta) \\ = -2\Delta q \{2 \log(q/2) + 3 + F(2, \vartheta) - 2F_\alpha(2, \vartheta)\} + O(q \log q). \end{aligned} \quad (\text{OFV28})$$

Combining OFV27 and OFV28 gives the complete constant

$$\mathfrak{C}_\partial = 4 \log(4/\vartheta) - 10 - 2F(2, \vartheta) + 4F_\alpha(2, \vartheta), \quad \vartheta = \pi/2, \quad (\text{OFV29})$$

with every logarithmic scale term canceled by direct algebra. OSP33 now proves the actual original-root asymptotic

$$\boxed{\mathcal{F}_{\partial, k}^{(s)} = \Delta q \mathfrak{C}_\partial + O_{\delta, \gamma}(q \log q), \quad \frac{\mathcal{F}_{\partial, k}^{(s)}}{kq} \longrightarrow 16 \mathfrak{C}_\partial, \quad s = 1, k.} \quad (\text{OFV30})$$

All constants in the error depend only on the fixed original parameters, the fixed cutoff, and the proved compact equilibrium constants. The two original source orders have been treated through their exact recurrence maps and full mass factors. Since $\Delta/k = 16 - 48/k$ and $\log q/k \rightarrow 0$, the displayed limit follows from its quantified remainder.

An equivalent expression uses the logarithmic energy of precisely the original EIQ limiting potential, $V(x) = \pi\sqrt{x} - 2 \log x$:

$$I_{2, \pi} = \iint \log |x - y| d\mu_{2, \pi}(x) d\mu_{2, \pi}(y), \quad \mathfrak{C}_\partial = 4 \log(4/\pi) + 2 - 2I_{2, \pi}. \quad (\text{OFV31})$$

To prove it, EIQ31 gives $\vartheta \int \sqrt{x} d\mu_{2, \vartheta} = 6$, so $F(2, \vartheta) = I_{2, \vartheta} - 6 + 2F_\alpha(2, \vartheta)$. Substituting in OFV29 yields $4 \log(4/\vartheta) + 2 - 2I_{2, \vartheta}$. The proved EIQ32 dilation $x \mapsto 4x$ carries $\mu_{2, \pi}$ to $\mu_{2, \pi/2}$. Its exact logarithmic energy change is $I_{2, \pi/2} = I_{2, \pi} + \log 4$, proving OFV31. This energy is a finite explicit integral of the fully specified EIQ density; its logarithmic singularity is integrable because that density is bounded and compactly supported.

Finally, the exact quotient metrics decrease when the source degree increases: each smaller-degree lift remains an allowed lift in the larger source with exactly the same norm. Hence each low-degree quotient Gram in CTV13 dominates the corresponding high-degree Gram on the identical target frame. Positive conjugation shows their determinant ratios are at least one. It follows that $\mathcal{F}_{\partial, k}^{(s)} \geq 0$, and OFV30 therefore also proves $\mathfrak{C}_\partial \geq 0$. The already proved finite tail bounds CTV18–20, with $0 \leq v \leq 80$, further give

$$\mathcal{T}_k^{(s)} = \Delta q \mathfrak{C}_\partial + O_{\delta, \gamma}(q \log q), \quad s = 1, k, \quad (\text{OFV32})$$

uniformly over the possible actual first-moment indices. No leading value of the separate original observation quotient is assigned here.

38 Fixed coefficient control of the original residual module

Retain the complete actual simple-quartet period matrix $H = F^{-1}\Pi U\mathcal{E}_{\text{prim}}$, the original substitution $t = (z_0w_0, z_0w_1, z_1w_0, z_1w_1)^T$, and $Q(x) = Q_0(H^{-1}x)$, where $Q_0(t) = t_{++}t_{--} - t_{+-}t_{-+}$. The original action is $gf(x) = f(D^{-1}x)$ with $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$ and $\zeta = e^{2\pi i/5}$. At the fixed actual period on the proved five-member orbit locus retain the entire polynomials $Q_a = g^a Q$ and $\mathcal{A} = \prod_{a=1}^4 Q_a$. Their coefficients and all phases are unchanged. The degree- k calculations below use $k = 4l + 1 \geq 9$ and

$$q = (k+1)^2, \quad q' = (k-7)^2, \quad \Delta = 16k - 48, \quad j = 8k - 32, \quad r = 8k - 16. \quad (\text{FMC1})$$

All constants introduced below are determined by this one fixed actual coefficient datum. They are independent of k and of the original Gamma cutoff. A fixed presentation will supply the residual frame in place of dividing new selected minors at each degree.

38.1 A fixed free module containing the entire invariant image

Let $P = \mathbb{C}[x_1, x_2, x_3, x_4]$ and introduce the specified graded ring injection

$$T = \mathbb{C}[y_1, y_2, y_3, y_4] \longrightarrow P, \quad y_i \longmapsto x_i^5, \quad \deg y_i = 5. \quad (\text{FMC2})$$

It is injective because distinct monomials have distinct exponent vectors after multiplying each exponent by five. Put $\mathcal{E} = \{a \in \mathbb{Z}^4 : 0 \leq a_i \leq 4\}$ and let e_a be a free T -basis element of degree $|a|$. The exact isomorphism is

$$\mathcal{P} := \bigoplus_{a \in \mathcal{E}} T e_a \xrightarrow{\iota} P, \quad y^\nu e_a \longmapsto x^{5\nu+a}. \quad (\text{FMC3})$$

Every nonnegative exponent vector has a unique coordinatewise quotient and remainder on division by five. This proves surjectivity, injectivity, preservation of degree, and the explicit inverse. In particular this is a rank-625 free module, with coefficient ℓ^1 norm preserved exactly by ι . The ring and monomial coefficients in P have not been rescaled.

Let $\mathcal{E}_0 = \{a \in \mathcal{E} : a_1 + 2a_2 + 3a_3 + 4a_4 \equiv 0 \pmod{5}\}$. The full original invariant algebra, regarded here as a T -module, is

$$\mathcal{J} = P^{(g)} = \iota \left(\bigoplus_{a \in \mathcal{E}_0} T e_a \right), \quad |\mathcal{E}_0| = 125. \quad (\text{FMC4})$$

Indeed the monomial $x^{5\nu+a}$ has character $\zeta^{-(a_1+2a_2+3a_3+4a_4)}$. The diagonal action fixes a polynomial precisely when each nonzero monomial has character one. For each of the 5^3 choices of the first three residues there is exactly one a_4 since 4 is invertible modulo five. This proves the count and the displayed equality. These pure fifth powers are four of the fourteen original generators OCF25; this is a new exact module presentation of the same entire invariant algebra.

Define two homogeneous T -submodules of $\mathcal{P}$ by

$$\begin{aligned} \mathcal{H} &= \iota^{-1}(QP + \mathcal{A}P), \\ \mathcal{S} &= \mathcal{H} + \bigoplus_{a \in \mathcal{E}_0} T e_a. \end{aligned} \quad (\text{FMC5})$$

They have the fixed finite generating columns $\iota^{-1}(Qx^a)$ and $\iota^{-1}(\mathcal{A}x^a)$ for $a \in \mathcal{E}$, followed for $\mathcal{S}$ by the 125 columns e_a , $a \in \mathcal{E}_0$. The first two lists have respectively homogeneous degrees $|a| + 2$ and $|a| + 8$, all at most 24. To obtain every matrix entry explicitly, expand either of the full original polynomials as $f(x) = \sum_b f_b x^b$ and use

$$\iota^{-1}(f x^a) = \sum_b f_b y^{[(a+b)/5]} e_{(a+b) \bmod 5}. \quad (\text{FMC6})$$

Both quotient and remainder are coordinatewise. Formula FMC3 verifies this identity term by term. Since P is free on the x^a , these columns generate precisely the two modules in FMC5; no later degree is needed to define a new relation matrix.

The full coefficient substitution and conductor quotient give the exact graded maps

$$\mathcal{P}/\mathcal{H} \xrightarrow{\iota, \psi} \mathcal{B}/(AB) =: \mathcal{L}, \quad \mathcal{S}/\mathcal{H} \xrightarrow{\sim} \overline{\mathcal{J}}, \quad A = \psi_8(\mathcal{A}). \quad (\text{FMC7})$$

Here $\psi f = f(Ht)$, and $\overline{\mathcal{J}}$ is the image of the original invariant algebra in $\mathcal{L}$, exactly OCF16. OCF3 proves $\ker \psi = QP$ and its surjectivity in every degree. If $\psi f = Ab$, lift b to a polynomial h using that surjectivity; then $f - Ah \in QP$. Conversely every $QP + \mathcal{A}P$ element maps to zero. This proves the first isomorphism. FMC4–5 show that its restriction has precisely the second image and kernel. Thus the quotient in FMC7 is the complete original conductor quotient, and its indicated submodule is the actual residual certificate image of ROQ4, with no new invariant premise.

38.2 Finite reduction data with only fixed actual denominators

The next construction specifies fixed data for both modules in FMC5. Order their module monomials $y^\nu e_a$ first lexicographically by $\nu_1, \nu_2, \nu_3, \nu_4$, and then by a fixed index $\kappa_a \in \{0, \dots, 624\}$ listing a lexicographically. Multiplication by a y -monomial preserves this order. It is a well-order: in a decreasing infinite sequence the first coordinate can decrease only finitely often; after it is fixed the same applies successively to the other three coordinates and then the finite index. For a nonzero column f , retain its full leading coefficient $\lambda_f \neq 0$ and its leading module monomial m_f . No leading coefficient is deleted from the stored column.

Start with the finite nonzero homogeneous generators of FMC5–6. Division of a column repeatedly cancels its largest divisible module monomial $cy^\nu e_a$ using the first available column f for which $m_f = y^\mu e_a$ divides it: subtract $(c/\lambda_f)y^{\nu-\mu}f$ and record that exact multiple. A largest term divisible by none of the current leading monomials is put into the remainder and removed from the current dividend. Each operation strictly decreases the largest term still being processed, so this division terminates. Remainders have no monomial divisible by the stored leading monomials.

For every pair f, h whose leading positions agree, put $m = \text{lcm}(m_f, m_h)$ in that position and form the full column

$$\text{Sp}(f, h) = \frac{m/m_f}{\lambda_f} f - \frac{m/m_h}{\lambda_h} h. \quad (\text{FMC8})$$

Divide it as just described and append its nonzero remainder to the list. Reconsider all new pairs. Track every appended column as a T -linear combination of the original generators throughout these operations. Every column and recorded certificate remains homogeneous: the two terms in FMC8 have the same weighted degree, because their leading position agrees and every input column is homogeneous in the shifted grading of FMC3.

This algorithm stops after finitely many appended columns. Here is the needed finiteness argument. Every infinite sequence in $\mathbb{N}^4$ has two terms in their original order with all coordinates weakly increasing. To prove it, an infinite sequence of nonnegative integers has either an infinitely repeated value or an increasing infinite subsequence; use this successively for all four coordinates. For module monomials first take a subsequence with one common position. Consequently every infinite sequence of module monomials has an earlier one dividing a later one. Each appended remainder's leading monomial is divisible by none of the previous leading monomials. An infinite sequence of appendages would contradict the proved property.

At termination every element of the generated module has leading monomial divisible by one of the stored leading monomials. For completeness, write an element as a finite sum of monomial multiples of the stored columns. Among all such representations choose one for which the largest leading module monomial of a summand is smallest, and among these choose one with the fewest summands at that monomial. These choices exist by the well-order and the positive integers. If cancellation of those largest terms occurs, two summands at the same module position can be

combined using a multiple of FMC8, with its exact two leading coefficients. Explicitly, if their common leading monomial is M and their leading coefficients are a, b , their sum is

$$a \frac{M}{\text{lcm}(m_f, m_h)} \text{Sp}(f, h) + (a + b) \frac{M/m_h}{\lambda_h} h.$$

This follows by substituting FMC8; it retains exactly both original summands and every coefficient. The recorded division of this S-column, together with its appended remainder when one was needed, expresses it in stored columns with all leading multiples strictly below its canceled common monomial. Replacing the two summands by this expression either decreases the largest monomial or decreases the number attaining it; combining their coefficients if necessary gives the latter alternative. Both contradict the chosen representation. Thus the largest remaining monomial is an actual leading monomial of the element, and it is divisible by one of the stored ones, as claimed. This also proves that division of a module element has zero remainder.

Write $\mathcal{G}_H$ and $\mathcal{G}_S$ for the two resulting fixed finite lists for FMC5. The preceding argument proves their full reduction property. A column supported outside the respective leading module is called a standard remainder. Such a remainder represents each quotient class uniquely: two representatives of one class differ by a module element, and a nonzero standard difference would violate the proved leading-monomial property. Denote these exact linear remainder maps by NF_H and NF_S . Their linearity follows from uniqueness and linearity of quotienting.

All data constructed here depend only on the actual matrices in FMC6. Every operation is an addition, multiplication, or division by one of the nonzero actual leading coefficients encountered in this finite calculation. Thus the complete list of denominators is finite and independent of k . Its precise coefficient chart records each tested coefficient's vanishing or nonvanishing and every selected leading monomial. For every actual period exactly its corresponding decisions give the above terminating calculation. This does not posit a numerical test for equality of unevaluated periods, nor does it assert a lower bound for a later sequence of minors. The exact stored coefficients and their nonzero leading entries define the constants that follow.

38.3 A fixed weight bounds every homogeneous reduction path

The finite reductions admit a linear-in-degree depth bound. Let $D_0 \geq 1$ be the largest of one and all absolute differences between corresponding y -exponents in a leading and a nonleading term of any column in $\mathcal{G}_H \cup \mathcal{G}_S$. Empty tail lists contribute zero. Set

$$B_0 = D_0 + 626, \quad \mathfrak{w}(y^\nu e_a) = B_0^4 \nu_1 + B_0^3 \nu_2 + B_0^2 \nu_3 + B_0 \nu_4 + \kappa_a, \quad L_k = B_0^4 \lfloor k/5 \rfloor + 624. \quad (\text{FMC9})$$

Every nonleading term of either fixed list has weight at least one less than its leading term. If the y -exponents agree, this follows from the position order. Otherwise let their first differing coordinate have weight B_0^m , $1 \leq m \leq 4$. Its exponent drops by at least one, while the later differences have absolute value at most D_0 , and the position difference has absolute value at most 624. The total weight drop is at least

$$B_0^m - D_0 \sum_{h=1}^{m-1} B_0^h - 624 \geq \frac{625 B_0^m}{D_0 + 625} - 624 > 1. \quad (\text{FMC10})$$

The middle inequality follows by bounding the geometric sum by $B_0^m/(B_0 - 1)$; the last follows from $B_0^m \geq B_0 = D_0 + 626$. Multiplication by a monomial adds the same weight to the leading and all its replacement terms, so the same drop holds in every division step. A homogeneous degree- k monomial has $5|\nu| + |a| = k$, hence weight between zero and L_k . Every branch of its reduction tree consequently has length at most L_k .

For each of the two lists define the complete coefficient constant

$$C_H = \max \left\{ 1, \max_{g \in \mathcal{G}_H} \sum_{m \neq m_g} |[m]g/\lambda_g| \right\}, \quad C_S = \max \left\{ 1, \max_{g \in \mathcal{G}_S} \sum_{m \neq m_g} |[m]g/\lambda_g| \right\}. \quad (\text{FMC11})$$

The maximum over an empty list is zero. They are finite actual numbers because the lists are finite and their displayed leading coefficients are nonzero. For every homogeneous degree- k column,

$$\|\text{NF}_H f\|_1 \leq C_H^{L_k} \|f\|_1, \quad \|\text{NF}_S f\|_1 \leq C_S^{L_k} \|f\|_1. \quad (\text{FMC12})$$

To prove these inequalities start with one monomial of coefficient one. A reduction replaces its leading term by the complete tail with total absolute coefficient at most the indicated constant. Along each branch the nonnegative integer weight drops at least one. Induction on that weight bounds the total absolute coefficient of all final leaves by $C^{\mathfrak{w}(m)}$: a standard leaf contributes one, and an internal node contributes at most C times the bound at weight one lower. Cancellation between branches can only decrease the coefficient ℓ^1 norm. Linearity, the triangle inequality, and $\mathfrak{w}(m) \leq L_k$ prove FMC12 for every input.

There is equally explicit control of the recorded original-generator certificates. For each stored $g \in \mathcal{G}_S$ let c_g be its complete certificate in the fixed original generating list of $\mathcal{S}$ in FMC5–6. Define

$$D_S = \max \left\{ 1, \max_{g \in \mathcal{G}_S} \|c_g\|_1 / |\lambda_g| \right\}. \quad (\text{FMC13})$$

At a reduction node with input coefficient c , the certificate contribution has norm at most $D_S |c|$. At depth h the total absolute coefficient entering all nodes is at most C_S^h times the input norm, by the same tree proof. Summing through depth L_k therefore proves that a certificate of $f - \text{NF}_S f$ in the original generators has norm at most $D_S (L_k + 1) C_S^{L_k} \|f\|_1$. All certificate entries retain the original homogeneous grading. Their denominators are among the same finite fixed reduction data; no degree- k linear system is inverted.

38.4 The two exact maps back to the original conductor strip

Let $\mathbf{N}_k$ and $\mathbf{S}_k$ be the original OCF strip remainder and its coordinate insertion, with leading conductor coefficient $a_* = a_{r_* s_*}$. Define

$$A_0 = \max \left\{ 1, \sum_{(a,b) \neq (r_*, s_*)} |a_{ab}/a_*| \right\}. \quad (\text{FMC14})$$

Their exact coefficient bounds are $\|\mathbf{N}_k\|_1 \leq A_0^{10k}$ and $\|\mathbf{S}_k\|_1 = 1$. To prove the first, give the biform monomial $e_{ab}^{(k)}$ the weight $9a + b \in [0, 10k]$. Every other nonzero term of the full A has smaller weight than its lexicographically largest term: a decrease in the first index outweighs at most eight increases in the second index, and a decrease in only the second index also drops the weight. Every translated OCF6 subtraction therefore has weight drop at least one. Its complete tail has coefficient norm at most A_0 . The reduction-tree proof of FMC12 now applies with depth $10k$. The insertion bound is immediate from its unchanged coordinate coefficients.

Put

$$h_+ = \max \{ 1, \max_i \sum_a |\mathbf{H}_{ia}| \}, \quad h_- = \max \{ 1, \max_a \sum_i |(\mathbf{H}^{-1})_{ai}| \}, \quad h_* = \max \{ h_+, h_- \}. \quad (\text{FMC15})$$

Substitution $x = \mathbf{H}t$ on degree- k polynomial coefficients has ℓ^1 norm at most h_+^k . Expanding each product of linear forms and using the multinomial theorem proves this for each monomial; the triangle inequality proves it for their sum. The Segre substitution adds coefficients on identical biform monomials and so has norm at most one. Conversely choose the following exact lift of $e_{ab}^{(k)}$: with $h = \min(a, b)$ take

$$t_{++}^h t_{+-}^{a-h} t_{-+}^{b-h} t_{--}^{k-a-b+h}, \quad t = \mathbf{H}^{-1}x. \quad (\text{FMC16})$$

All exponents are nonnegative since $0 \leq a, b \leq k$, and their sum is k . Direct substitution gives the required biform monomial. This lift, extended linearly and denoted $\mathcal{L}_k$, has norm at most h_-^k by the same complete multinomial estimate.

Let $\mathcal{M}_{H,k}$ be the set of degree- k standard module monomials for $\mathcal{G}_H$. Its coefficient space is the exact degree- k quotient model of FMC7. Define

$$\begin{aligned} U_k f &= \mathbf{N}_k \psi \iota f, & f &\in \mathbb{C}^{\mathcal{M}_{H,k}}, \\ V_k w &= \mathbf{NF}_H \iota^{-1} \mathcal{L}_k \mathbf{S}_k w, & w &\in W_k. \end{aligned} \quad (\text{FMC17})$$

These are inverse maps. For the first composition, $\mathbf{NF}_H$ changes a lift only by $\mathcal{H}$, which maps to zero in the original strip by FMC7, and $\psi \mathcal{L}_k = I$ together with $\mathbf{N}_k \mathbf{S}_k = I$ proves $U_k V_k = I$. For the opposite composition, $\mathcal{L}_k \mathbf{S}_k \mathbf{N}_k \psi \iota f - \iota f$ maps to zero in the strip, so it belongs to $QP + \mathcal{AP}$ by FMC7. Its free-module difference belongs to $\mathcal{H}$ and is killed by the unique remainder. The original f is already standard. Hence $V_k U_k = I$. The bounds already proved give

$$\|U_k\|_1 \leq h_+^k A_0^{10k}, \quad \|V_k\|_1 \leq h_-^k C_H^{L_k}. \quad (\text{FMC18})$$

These bounds use only the displayed original coefficients and the two fixed homogeneous relation lists.

38.5 An identity pivot block for the entire residual image

Since $\mathcal{H} \subset \mathcal{S}$, every leading monomial of $\mathcal{H}$ is a leading monomial of an element of $\mathcal{S}$. The proved reduction property for $\mathcal{G}_S$ then implies inclusion of their leading modules. Consequently

$$\mathcal{M}_{S,k} \subset \mathcal{M}_{H,k}, \quad \mathcal{U}_k = \mathcal{M}_{H,k} \setminus \mathcal{M}_{S,k}, \quad |\mathcal{M}_{H,k}| = \Delta, \quad |\mathcal{M}_{S,k}| = r, \quad |\mathcal{U}_k| = j. \quad (\text{FMC19})$$

The first count is FMC7 and OCF8. The quotient by $\mathcal{S}$ is exactly the original conductor quotient modulo its invariant image, which has dimension r by OCF22–24 and OKM3–5. Subtracting proves the last count. These are counts of actual standard monomials for the fixed relation lists, not independent generic ranks.

Order $\mathcal{M}_{H,k}$ by listing $\mathcal{U}_k$ first and $\mathcal{M}_{S,k}$ second, with the specified monomial order within each list. For every $u \in \mathcal{U}_k$ put $z_u = u - \mathbf{NF}_S(u)$. Both terms are H -standard, and their difference belongs to $\mathcal{S}$. The j resulting columns form a basis of $\mathcal{S}_k/\mathcal{H}_k$ in this quotient model. Indeed their coordinates on $\mathcal{U}_k$ are the distinct unit vectors. For a standard column in $\mathcal{S}$, the remainder under $\mathbf{NF}_S$ is zero. Its coordinates on $\mathcal{M}_{S,k}$ must therefore be the negatives of the remainders of its $\mathcal{U}_k$ terms. It is exactly the stated linear combination of the z_u . This proves spanning and independence, including all tails of every column.

Write B_k for the matrix with columns $\mathbf{NF}_S(u)$ in the second standard set. The full exact graph frame and inverse are

$$P_k = \begin{pmatrix} I_j & 0 \\ -B_k & I_r \end{pmatrix}, \quad P_k^{-1} = \begin{pmatrix} I_j & 0 \\ B_k & I_r \end{pmatrix}, \quad \det P_k = 1, \quad \|P_k\|_1, \|P_k^{-1}\|_1 \leq 1 + C_S^{L_k}. \quad (\text{FMC20})$$

The two matrix identities follow by multiplication and block triangularity, and the norm bound follows from FMC12 applied to each entire column. All entries of B_k come from fixed reductions; no minor is selected or divided at degree k .

Each z_u has an actual invariant certificate. The recorded reduction certificate of $u - \mathbf{NF}_S(u)$ from FMC13 expresses its polynomial representative as

$$\iota z_u = Qh_u + Aa_u + f_u, \quad f_u \in \mathcal{J}_k, \quad \|f_u\|_{1,x} \leq D_S(L_k + 1)C_S^{L_k}. \quad (\text{FMC21})$$

To obtain the three displayed pieces, collect the certificate terms corresponding respectively to Qx^a , Ax^a , and the 125 invariant basis elements x^a . Every coefficient is a polynomial in the $y_i = x_i^5$. The last terms therefore belong to $\mathcal{J}$, and homogeneity gives exactly degree k . FMC3 preserves coefficient norm, and discarding the first two certificate lists when bounding only the last cannot increase their total ℓ^1 norm. FMC13 proves the bound. Thus $U_k z_u$ is exactly $\mathbf{N}_k \psi f_u$, an original invariant restriction, with the entire original periods in its coefficients.

Define the full original strip frame, its residual columns and their explicit left inverse by

$$T_k = U_k P_k, \quad Z_k^{\text{fix}} = T_k E_j, \quad L_{Z,k} = E_j^\top P_k^{-1} V_k, \quad L_{Z,k} Z_k^{\text{fix}} = I_j. \quad (\text{FMC22})$$

Here E_j inserts the first j coordinates into the new ordered standard basis, while all output strip coordinates retain their original OCF order. FMC17 and FMC20 prove the inverse formula $T_k^{-1} = P_k^{-1} V_k$ and the last identity. By FMC21 these Z_k^{fix} columns span precisely the original $\bar{J}_k$. They provide all actual residual coefficients with a left inverse obtained from fixed relations.

For the original coefficient Euclidean norms set the explicit number

$$M_k = 2\sqrt{\Delta} h_*^k A_0^{10k} (C_H C_S)^{L_k} \geq 1. \quad (\text{FMC23})$$

Then

$$\|T_k\|_2, \|T_k^{-1}\|_2, \|Z_k^{\text{fix}}\|_2, \|L_{Z,k}\|_2 \leq M_k, \quad M_k^{-1} \|v\|_2 \leq \|Z_k^{\text{fix}} v\|_2 \leq M_k \|v\|_2. \quad (\text{FMC24})$$

For the first two bounds, FMC18–20 give respectively the column-sum bounds $h_+^k A_0^{10k} (1 + C_S^{L_k})$ and $(1 + C_S^{L_k}) h_-^k C_H^{L_k}$. For a matrix with Δ input coordinates, its Euclidean norm is at most $\sqrt{\Delta}$ times its coefficient column-sum norm: use $\|Av\|_2 \leq \|Av\|_1 \leq \|A\|_1 \|v\|_1 \leq \sqrt{\Delta} \|A\|_1 \|v\|_2$. All constants are at least one, proving the two stated bounds. The coordinate insertion and restriction have Euclidean norm one, so FMC22 proves the remaining operator bounds. Its left-inverse identity then proves the lower bound on Z_k^{fix} .

For every $0 \leq t \leq \Delta$, singular values now give

$$\log^+ \left\| \bigwedge^t T_k \right\|_2, \log^+ \left\| \bigwedge^t T_k^{-1} \right\|_2 \leq t \log M_k \leq \Delta \log M_k = O_{\text{actual}}(k^2) = o(kq). \quad (\text{FMC25})$$

Indeed every singular value of both square matrices is at most M_k , and exterior singular values are their products. This last fact follows by applying a singular-value decomposition: unitary factors preserve exterior norms, and the diagonal factor multiplies each coordinate wedge by the corresponding product. Explicitly

$$\log M_k = \log 2 + \frac{1}{2} \log \Delta + k \log h_* + 10k \log A_0 + L_k \log(C_H C_S). \quad (\text{FMC26})$$

All data except k, Δ, L_k are fixed actual numbers, $L_k \leq B_0^4 k/5 + 624$, $\Delta = 16k - 48$, and $kq = k(k+1)^2$. These formulas prove the last two statements of FMC25 directly. The same bounds hold for transpose inverse maps, whose singular values are unchanged. For $0 \leq t \leq j$, FMC24 also puts every singular value of Z_k^{fix} between M_k^{-1} and M_k , so the same exterior bounds apply to its map onto its image and its inverse there.

38.6 Exact relation to every earlier certificate frame

Let Z_k^{old} be the full actual OCF/ROQ certificate frame, with its selected invertible $j \times j$ row block $(Z_k^{\text{old}})_I$. Both frames have exactly the same column space, as proved in FMC21–22. Their exact transition and inverse are

$$A_k = ((Z_k^{\text{old}})_I)^{-1} (Z_k^{\text{fix}})_I, \quad A_k^{-1} = L_{Z,k} Z_k^{\text{old}}, \quad Z_k^{\text{fix}} = Z_k^{\text{old}} A_k. \quad (\text{FMC27})$$

Each new column has a unique expansion in the old independent columns; restriction to the old pivot rows gives the first formula. Applying $L_{Z,k}$ gives the second and both inverse identities. Thus this is the exact coefficient morphism to the earlier frame, including its actual pivot determinant. No bound for that old determinant is needed or asserted by FMC24–26. The new frame and its proved bounds use only the fixed data of FMC5–23.

The two residual maps on V_A therefore obey $\Xi_k^{\text{fix}} = A_k^\top \Xi_k^{\text{old}}$ in their respective coordinates, since the maps in the original strip dual are the ordinary transposes of their certificate matrices.

For every original source and cutoff their minimum quotient metrics satisfy the exact congruence and determinant formulas

$$Q_{\Xi,N}^{\text{fix}} = \overline{A_k}^{-1} Q_{\Xi,N}^{\text{old}} A_k^{-\text{T}}, \quad \log \det Q_{\Xi,N}^{\text{fix}} = \log \det Q_{\Xi,N}^{\text{old}} - 2 \log |\det A_k|. \quad (\text{FMC28})$$

The constraints on a lift are transformed by precisely A_k^{T} ; substituting the inverse coordinates in the same attained minimum proves the congruence, and determinants give the second formula. The scalar is independent of N and of source order. Its coefficient is exactly $1 + 1 - 1 - 1 = 0$ in the original four-endpoint return. Thus this return agrees identically with the entire earlier ROQ/RCA return, even if an earlier selected pivot is small at some degrees.

The complete inherited period Gram is transported by the same exact map at each endpoint. Let L_{old} be the primary right section ROQ6 and let L_{fix} be any primary right section of Ξ_k^{fix} inside V_A , for example the first j columns of FMC30 below. Both L_{fix} and $L_{\text{old}} A_k^{-\text{T}}$ have residual image I_j under Ξ_k^{fix} , so their difference has columns in the original $K = \ker \Lambda$. Consequently

$$\Lambda L_{\text{fix}} = \Lambda L_{\text{old}} A_k^{-\text{T}}, \quad P_{\text{per}}^{\text{fix}} = \overline{A_k}^{-1} P_{\text{per}}^{\text{old}} A_k^{-\text{T}}, \quad \frac{\det Q_{\Xi,N}^{\text{fix}}}{\det P_{\text{per}}^{\text{fix}}} = \frac{\det Q_{\Xi,N}^{\text{old}}}{\det P_{\text{per}}^{\text{old}}}.$$

The first equality preserves the literal boundary vectors and their full cyclic coefficient map. Taking their Gram with the unchanged inherited metric ROQ9 proves the second equality. Taking both determinants together with FMC28 proves the last equality pointwise, before taking any endpoint difference.

38.7 The unchanged Gamma source in the fixed degree model

Retain $c = k/2$, the original nodes $\beta_{ab} = c + (2a - k)\delta + i(2b - k)\gamma$, and $\omega_{ab} = (\beta_{ab} - c)/i$. For each original source $s \in \{1, k\}$ retain its complete recurrence and masses

$$p_0 = 1, \quad p_1 = y, \quad p_{n+1} = yp_n - n(s/2 + n - 1)p_{n-1}, \quad (\text{FMC29})$$

$$h_n = (2\pi)^{s/2} n! (s/2)_n, \quad (U_N)_{(a,b),n} = p_n(\omega_{ab})/\sqrt{h_n}, \quad G_N = (U_N U_N^*)^{-1}.$$

Here N remains one of $q - 1, q, 2q - 1, 2q$. These are exactly the original primary quotient metrics of CEP21–22 and ROQ10, with every source mass and coordinate phase present.

The complete primary frame of V_A , in residual coordinates first and common-kernel coordinates second, is

$$J_k = N_k^{\text{T}} T_k^{-\text{T}}, \quad \Xi_k^{\text{fix}} J_k = E_j^{\text{T}}, \quad K = \text{im}(J_k E_r). \quad (\text{FMC30})$$

Here E_r inserts the last r coordinates. For a primary vector $x \in V_A$, ROQ3 gives its strip dual coordinate $z = S_k^{\text{T}} x$. The new residual map is $(Z_k^{\text{fix}})^{\text{T}} z = E_j^{\text{T}} T_k^{\text{T}} z$. Substitution of $z = T_k^{-\text{T}} y$ proves the second identity, and its kernel is the last coordinate block, proving the third. Thus this full frame has precisely the original common kernel; it changes no observation. Its left inverse on V_A is $T_k^{\text{T}} S_k^{\text{T}}$. The original coefficient bounds additionally give

$$\|J_k\|_2 \leq \sqrt{q} A_0^{10k} M_k =: \widehat{M}_k, \quad \|T_k^{\text{T}} S_k^{\text{T}}\|_2 \leq M_k. \quad (\text{FMC31})$$

Indeed FMC14 bounds the norm of N_k by its column-sum norm times $\sqrt{q}$, transpose preserves Euclidean norm, and FMC24 applies to $T_k^{-\text{T}}$. The insertion transpose is a coordinate restriction of norm one. Both logarithms in FMC31 are $O_{\text{actual}}(k + \log k)$ with their full displayed constants.

The exact source Gram in this specified degree model, and its actual residual quotient, are

$$\mathcal{H}_N = J_k^* G_N J_k = \overline{T_k}^{-1} \overline{N_k} G_N N_k^{\text{T}} T_k^{-\text{T}}, \quad (\text{FMC32})$$

$$\mathcal{H}_N = \begin{pmatrix} H_{11,N} & H_{12,N} \\ H_{21,N} & H_{22,N} \end{pmatrix}, \quad Q_{\Xi,N}^{\text{fix}} = H_{11,N} - H_{12,N} H_{22,N}^{-1} H_{21,N}.$$

The first formula is direct substitution of the actual primary frame. Since $G_N > 0$ and J_k is injective, $\mathcal{H}_N > 0$; therefore $H_{22,N} > 0$. For prescribed first coordinate y , the second coordinate minimizing the full quadratic form is $-H_{22,N}^{-1}H_{21,N}y$. Completing the square, or multiplying out this substitution, gives exactly the displayed Schur complement. This is the original minimum on $\Xi_k^{\text{fix}}x = y$ inside V_A , so it equals FMC28 and the full ROQ metric. All four blocks, including every off-diagonal entry, are retained. The original minimum source lift of the full frame is explicitly $U_N^*G_NJ_k$, whose Gram is $\mathcal{H}_N$ by $U_NU_N^* = G_N^{-1}$. It contains the entire original polynomial recurrence and remainder at each stated cutoff.

There is a useful quantitative determinant comparison which does not replace that Gamma form by a coefficient form. In strip dual coordinates write $H_{V,N} = \overline{\mathbf{N}}_k G_N \mathbf{N}_k^T$ and $Z = Z_k^{\text{fix}}$. Define the actual coefficient quotient form and the relative determinant by

$$Q_{\text{coeff}} = (Z^T \overline{Z})^{-1}, \quad \Phi_{\Xi,N} = \log \det Q_{\Xi,N}^{\text{fix}} - \log \det Q_{\text{coeff}}. \quad (\text{FMC33})$$

The first is the minimum quotient of the original coefficient Euclidean norm in the fixed OCF strip, with the identical constraint $Z^T z = y$. The elementary minimum proof of ROQ11 gives both this formula and $Q_{\Xi,N}^{\text{fix}} = (Z^T H_{V,N}^{-1} \overline{Z})^{-1}$. Hence the full, directly calculable expression is

$$\begin{aligned} \Phi_{\Xi,N} &= \log \det(Z^T \overline{Z}) - \log \det(Z^T H_{V,N}^{-1} \overline{Z}), \\ |\log \det Q_{\text{coeff}}| &\leq 2j \log M_k = O_{\text{actual}}(k^2) = o(kq), \\ F_{\Xi,k}^{(s)} &= \Phi_{\Xi,q-1} + \Phi_{\Xi,q} - \Phi_{\Xi,2q-1} - \Phi_{\Xi,2q}. \end{aligned} \quad (\text{FMC34})$$

FMC24 puts the singular values of Z between M_k^{-1} and M_k . The positive eigenvalues of $Z^T \overline{Z}$ are their squares, proving the determinant bound by summing j logarithms. The first identity follows by the two exact inverse Gram formulas. The final identity follows by canceling the fixed $\log \det Q_{\text{coeff}}$ with the four original signs and then using FMC28. Thus all coefficient denominators and frame-volume effects in this specified construction are controlled below the kq scale, while the original Gamma compression remains fully visible in its exact determinant.

For the kernel frame $I_k = J_k E_r$, FMC31 likewise gives the concrete coefficient-Gram bounds

$$M_k^{-2r} \leq \det(I_k^* I_k) \leq \widehat{M}_k^{2r}, \quad |\log \det(I_k^* I_k)| \leq 2r \log \widehat{M}_k = O_{\text{actual}}(k^2) = o(kq). \quad (\text{FMC35})$$

Its left inverse on its image is $E_r^T T_k^T S_k^T$, with norm at most M_k ; FMC31 bounds its forward norm. The singular-value argument just given proves FMC35. Define the exact primary-coordinate projection and determinant by

$$P_{K,\text{prim}} = I_k(I_k^* I_k)^{-1} I_k^*, \quad D_{K,N}^{\text{prim}} = [z^r] \det(I_q + z P_{K,\text{prim}} G_N P_{K,\text{prim}}).$$

Multiplication proves that this projection is Hermitian, idempotent, and has exactly the original primary-coordinate K as its image. For the determinant proof take $U = I_k(I_k^* I_k)^{-1/2}$, so $U^* U = I_r$, and extend its columns to an orthonormal basis of $\mathbb{C}^q$. In that basis the projected metric has block $U^* G_N U$ and a zero block. Its indicated determinant coefficient is $\det(U^* G_N U)$, and substitution of the full Gram square root therefore proves

$$\det(I_k^* G_N I_k) = \det(I_k^* I_k) D_{K,N}^{\text{prim}}.$$

The precise map to AKS12–14's original polynomial-coefficient projector is the ordered Vandermonde: put $I_{k,\text{poly}} = V_\beta^{-1} I_k$ and $G_{N,\text{poly}} = V_\beta^* G_N V_\beta$. Their compressed Gram is identically $I_k^* G_N I_k$ by multiplication. Applying the determinant proof just given in these polynomial coordinates proves the exact bridge

$$D_{K,N}^{\text{AKS}} = \frac{\det(I_k^* I_k)}{\det(I_{k,\text{poly}}^* I_{k,\text{poly}})} D_{K,N}^{\text{prim}}.$$

Here the polynomial projector has image the actual coefficient kernel and thus agrees with AKS12's projector. The displayed positive factor is independent of N and source order and cancels exactly with the four endpoint signs. FMC35 bounds its primary-coordinate numerator; the polynomial denominator is retained as the full displayed Vandermonde Gram. No bound for that different denominator follows from FMC35 alone.

FMC25 is a coefficient-space estimate for specified exact maps. Its relation to the original Gamma norms is the complete congruence and Schur calculation FMC29–34, not an assertion that those maps are Gamma isometries. In particular no upper or lower leading value for the remaining Gamma compression is inferred from their coefficient condition numbers. The new result supplies a fixed degree presentation, actual invariant certificates, bounded forward and inverse exterior maps, and the exact original determinant on that model, with every fixed denominator specified and every source order and endpoint retained.

38.8 The exact map to the original polynomial degree flag

The homogeneous degree k in FMC3 is the original tensor degree. Its precise map to the original polynomial-in- S model is the ordered evaluation isomorphism, not an identification of the two degree indices. Retain $\chi_k(S) = \prod_{a,b=0}^k (S - \beta_{ab})$ and the original Vandermonde $(V_\beta)_{(a,b),d} = \beta_{ab}^d$, $0 \leq d < q$. For a fixed-model primal coordinate column $y \in \mathbb{C}^\Delta$ the exact polynomial representative is

$$p_y(S) = \sum_{a,b=0}^k (J_k y)_{ab} \frac{\chi_k(S)}{(S - \beta_{ab}) \chi'_k(\beta_{ab})}, \quad [p_y]_{\text{coeff}} = V_\beta^{-1} J_k y. \quad (\text{FMC36})$$

All denominators are nonzero because the original nodes are distinct. Evaluation at each node proves the displayed inverse and hence the equality of the two formulas. It includes the complete roots, their order, and their actual centre. Thus the source degree- N metric in FMC32 is equivalently $(V_\beta^{-1} J_k)^* (V_\beta^* G_N V_\beta) (V_\beta^{-1} J_k)$, identically $J_k^* G_N J_k$ by multiplication. This proves the full typed relation to the original coefficient Gamma model.

There is an exact further map to the degree-adapted conductor basis AKJ17. Retain all actual conductor moments μ_d , their first nonzero index $v \leq 80$, the literal $B_q(\partial_S)$ of AKJ14, and $\mathcal{J}_v S^d = d! S^{d+v} / (d+v)!$. The ordered polynomial columns of $P_{A,k}$ are

$$1, S, \dots, S^{v-1}, \quad P_h = \frac{(q' + h + v)!}{(q' + h)!} B_q(\partial_S)^{-1} \mathcal{J}_v(\chi_{k'} S^h), \quad 0 \leq h \leq \Delta - v - 1. \quad (\text{FMC37})$$

Their distinct monic degrees are $0, \dots, v-1$ and $q' + v, \dots, q-1$. AKJ14–17 proves directly that they form a basis of this same V_A , using the complete conductor derivative and its finite nilpotent inverse. Define $B_{A,k}(p)$ on V_A by successively subtracting the coefficient of the highest remaining degree times its matching column in FMC37, down through the high list and then through the low monomials. Each subtraction uses leading coefficient one. Spanning and the distinct degree pivots show that it ends at zero and records the unique coordinate column of p ; this also proves its linearity and both inverse identities on V_A . Consequently the precise transition is

$$R_{A,k} = S_k^\top V_\beta P_{A,k}, \quad R_{A,k}^{-1} z = B_{A,k}(V_\beta^{-1} \mathbf{N}_k^\top z), \quad (\text{FMC38})$$

$$V_\beta^{-1} J_k = P_{A,k} R_{A,k}^{-1} T_k^{-\top}, \quad \Xi_{A,k}^{\text{fix}} = E_j^\top T_k^\top R_{A,k}.$$

The columns $V_\beta P_{A,k}$ lie in V_A . Therefore ROQ3 gives $\mathbf{N}_k^\top R_{A,k} = V_\beta P_{A,k}$, which proves the first inverse formula by applying the unique degree coordinates just constructed. Both sides are invertible maps between the displayed Δ -dimensional coefficient spaces. Substituting FMC30 gives the third equality, and evaluating the new residual map on these same primary columns gives the last. Thus its degree-prefix ranks give exactly $\dim(K \cap \mathbb{C}[S]_{\leq d})$ by subtracting that prefix rank from the number of columns of FMC37 of degree at most d , as in AKJ18. Every original factorial,

finite inverse, primary denominator and degree pivot is visible in these maps. The subleading coefficient-frame estimate proved in FMC25 pertains to the fixed strip module maps; FMC36–38 retain the separate polynomial interpolation and its full Gamma metric exactly.

39 The evaluated outer contribution in the original mixed return

Retain the actual simple quartet, original period on the proved five-orbit locus, full amplitude unit, branch, and original source orders $s = 1, k$ of OAR and OFV. All arithmetic statements retain the original domain $|u| \geq R_*$ of AMT; the actual period is fixed as $k = 4l + 1$ grows. Here $q = (k + 1)^2$, $\Delta = 16k - 48$, and every finite use of OFV retains its stated threshold, including OSP23. Write

$$C_{\partial} = \mathfrak{C}_{\partial} = 4 \log(4/\pi) + 2 - 2I_{2,\pi}, \quad I_{2,\pi} = \iint \log |x - y| d\mu_{2,\pi}(x) d\mu_{2,\pi}(y). \quad (\text{OVR1})$$

The measure here is the full EIQ density with its uniquely specified endpoints. OFV proves this exact expression and $C_{\partial} \geq 0$. No value of the original residual observation is inserted into it.

39.1 The exact remainder and the sourcewise leading identity

Define the actual outer remainder and retain OAR3's full conductor and monic-tail remainder by

$$b_{k,s} = \mathcal{F}_{\partial,k}^{(s)} - \Delta q C_{\partial}, \quad R_{k,s} = b_{k,s} + \mathfrak{e}_{k,s}. \quad (\text{OVR2})$$

Thus $b_{k,s} = O_{\delta,\gamma}(q \log q)$ by OFV30. The second summand is precisely the signed finite expression OAR3, with $-e_{k,s}^- \leq \mathfrak{e}_{k,s} \leq e_{k,s}^+$ and $e_{k,s}^{\pm} = o(kq)$. In particular $R_{k,s} = o(kq)$, because $\log q/k \rightarrow 0$. Its two exact finite error endpoints remain $b_{k,s} - e_{k,s}^-$ and $b_{k,s} + e_{k,s}^+$; their signs are not replaced by a symmetry assumption. Substitution into OAR3 proves

$$\begin{aligned} F_K^{(s)} + \Phi_{k,s} &= \Delta q C_{\partial} + R_{k,s}, \\ \frac{F_K^{(s)}}{kq} + \frac{\Phi_{k,s}}{kq} &= 16C_{\partial} - \frac{48C_{\partial}}{k} + \frac{R_{k,s}}{kq} \longrightarrow 16C_{\partial}, \quad s = 1, k. \end{aligned} \quad (\text{OVR3})$$

Here $\Phi_{k,s}$ is exactly ROQ19's positive sum of constrained covariance increments, built from the original period certificate matrix D and the complete conductor cross covariance. Its coefficient maps, minimum sections, inherited period metric and graph denominator are those already proved in ROQ; OVR3 changes none of them.

39.2 Transfer between the two original Gamma sources

Let $d_k = F_K^{(k)} - F_K^{(1)}$ denote the actual difference. AMT44–46 proves $d_k = o(kq)$, with its original finite two-sided source-comparison bounds. Subtracting the two exact lines OVR3 gives

$$\Phi_{k,k} - \Phi_{k,1} = R_{k,k} - R_{k,1} - d_k, \quad \frac{\Phi_{k,k} - \Phi_{k,1}}{kq} \longrightarrow 0. \quad (\text{OVR4})$$

This proves the transfer using the actual two residual observations, whose individual metrics retain their respective original sources. There is no asserted equality between those metrics. The equality in OVR4 is the exact scalar consequence of their common original-kernel restriction maps and the proved determinant identities.

39.3 Arithmetic kernel and its correlated boundary

Put $d_k^{\text{ar}} = F_K^{\text{ar}} - F_K^{(1)}$. This is the actual quantity bounded in OAR8; AMT38 and AMT46 prove $d_k^{\text{ar}} = o(kq)$. Keeping its full value first gives

$$\begin{aligned} F_K^{\text{ar}} &= \Delta q C_{\partial} - \Phi_{k,1} + R_{k,1} + d_k^{\text{ar}}, \\ F_B^{\text{ar}} &= \mathcal{B}_k^{\text{ar}} - \Delta q C_{\partial} + \Phi_{k,1} - R_{k,1} - d_k^{\text{ar}}. \end{aligned} \quad (\text{OVR5})$$

The first equality adds the literal definition of d_k^{ar} to OVR3 at $s = 1$. The second subtracts the first from $F_K^{\text{ar}} + F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}}$. In particular the same arithmetic total and the same two finite errors

occur with opposite signs in the boundary. OAR8 supplies the unchanged correlated finite bounds on d_k^{ar} . Dividing these exact equalities gives

$$\frac{F_K^{\text{ar}} + \Phi_{k,1}}{kq} \longrightarrow 16C_\partial, \quad \frac{\mathcal{B}_k^{\text{ar}} - F_B^{\text{ar}} + \Phi_{k,1}}{kq} \longrightarrow 16C_\partial. \quad (\text{OVR6})$$

Thus evaluation of the residual sum in either original Gamma order passes through to the same original arithmetic kernel and boundary.

39.4 The signed return retains its original Gamma correction

The exact signed quantity of OAR11 is $\mathfrak{S}_k^{\text{mix}} = F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)}$. Applying OVR3 at $s = k$ to that identity yields

$$\mathfrak{S}_k^{\text{mix}} + \Phi_{k,k} = \Delta q C_\partial + \mathcal{H}_k + \delta_k^\sigma + R_{k,k}. \quad (\text{OVR7})$$

All finite terms are the original ones. AMT48–49, with the retained total correction estimates used there, proves $(\mathcal{H}_k + \delta_k^\sigma)/(kq) \rightarrow -C_\Gamma/4$, where

$$C_\Gamma = 2 \int \log x \, d\mu_{2,\pi}(x) - 4(2 \log 2 - 1) > 0.$$

The equilibrium and source maps are exactly the EIQ/GEL providers used in OFV and AMT. Hence OVR7 and OVR4 prove both limits

$$\frac{\mathfrak{S}_k^{\text{mix}} + \Phi_{k,k}}{kq} \longrightarrow 16C_\partial - \frac{C_\Gamma}{4}, \quad \frac{\mathfrak{S}_k^{\text{mix}} + \Phi_{k,1}}{kq} \longrightarrow 16C_\partial - \frac{C_\Gamma}{4}. \quad (\text{OVR8})$$

The two constants are explicit integrals of the same proved density, respectively its full logarithmic pair energy and its logarithmic moment. No identity between these distinct integrals is presumed.

39.5 Bounds on the remaining residual coefficient

Each $F_K^{(s)}$ is nonnegative: the original quotient metrics decrease with increasing source degree, and restriction to the identical kernel preserves the two Gram inequalities at the paired endpoints. Their determinant ratios are consequently at least one, as proved in AMT15. ROQ19 gives $\Phi_{k,s} \geq 0$. Thus OVR3 bounds both nonnegative sequences after division by kq . Let $\kappa_* = 32 \log(25 \log 3 / (4\pi))$ be the retained AKS asymptotic kernel ceiling on this actual locus, applied in both original orders. The actual residual cap OAR5, proved by ROQ21, is $\Phi_{k,s} \leq j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - \lambda_s$, where $j = 8k - 32$ and $\lambda_s \geq 0$ is the original first increment. AKS31 gives $\limsup j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1})/(kq) \leq \kappa_*$ in both original source orders: its same explicit logarithmic comparison constants apply, and $j/k \rightarrow 8$ just as $r/k \rightarrow 8$. Thus both the original kernel and residual sequences have the same proved upper ceiling. Combining their two ceilings with OVR3 gives

$$\begin{aligned} \max\{0, 16C_\partial - \kappa_*\} &\leq \liminf \frac{F_K^{(s)}}{kq} \leq \limsup \frac{F_K^{(s)}}{kq} \leq \min\{\kappa_*, 16C_\partial\}, \\ \max\{0, 16C_\partial - \kappa_*\} &\leq \liminf \frac{\Phi_{k,s}}{kq} \leq \limsup \frac{\Phi_{k,s}}{kq} \leq \min\{\kappa_*, 16C_\partial\}. \end{aligned} \quad (\text{OVR9})$$

For each line, subtract the other bounded nonnegative sequence from the convergent sum in OVR3. More explicitly, that sum differs from $16C_\partial$ by a sequence tending to zero; taking its upper and lower eventual bounds proves $\liminf(\Phi/(kq)) = 16C_\partial - \limsup(F_K/(kq))$ and $\limsup(\Phi/(kq)) = 16C_\partial - \liminf(F_K/(kq))$. The same argument with the two sequences interchanged proves the first line, including its lower bound. The two first positive increments also survive in the finite common

upper bound. In the original AKS notation let d_1 be the first section-corrected kernel increment of AKS29, in the same source order s . AKS28 and OAR5 give

$$\begin{aligned}\Delta q C_{\partial} + R_{k,s} &= F_K^{(s)} + \Phi_{k,s} \\ &\leq (r+j)(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1 - \lambda_s \\ &= \Delta(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1 - \lambda_s.\end{aligned}\tag{OVR10}$$

The last equality is the exact rank identity $(8k-16) + (8k-32) = 16k-48$; both subtractions retain their actual minimum-section definitions. Adding the eventual upper bounds already proved and using OVR3 also proves the explicit consequence

$$0 \leq 16C_{\partial} \leq 2\kappa_*, \quad 0 \leq C_{\partial} \leq \frac{\kappa_*}{8}.\tag{OVR11}$$

Indeed for every positive ε , both sequences are eventually at most $\kappa_* + \varepsilon$, so their limiting sum is at most $2\kappa_* + 2\varepsilon$; let ε decrease to zero. Their nonnegativity proves the lower bound independently. OVR4 and AMT46 carry the same limiting bounds to the two original sources and the arithmetic kernel. The outstanding coefficient is therefore the explicit original positive residual sum; no convergence or numerical value for that sequence has been assumed.